

Pathway Analysis Report

This report contains the pathway analysis results for the submitted sample ". Analysis was performed against Reactome version 95 on 19/12/2025. The web link to these results is:

<https://reactome.org/PathwayBrowser/#/ANALYSIS=MjAyNTEyMTkwOTAxMjJfNTM0OA%3D%3D>

Please keep in mind that analysis results are temporarily stored on our server. The storage period depends on usage of the service but is at least 7 days. As a result, please note that this URL is only valid for a limited time period and it might have expired.

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1. Introduction

Reactome is a curated database of pathways and reactions in human biology. Reactions can be considered as pathway 'steps'. Reactome defines a 'reaction' as any event in biology that changes the state of a biological molecule. Binding, activation, translocation, degradation and classical biochemical events involving a catalyst are all reactions. Information in the database is authored by expert biologists, entered and maintained by Reactome's team of curators and editorial staff. Reactome content frequently cross-references other resources e.g. NCBI, Ensembl, UniProt, KEGG (Gene and Compound), ChEBI, PubMed and GO. Orthologous reactions inferred from annotation for Homo sapiens are available for 14 non-human species including mouse, rat, chicken, puffer fish, worm, fly and yeast. Pathways are represented by simple diagrams following an SBGN-like format.

Reactome's annotated data describe reactions possible if all annotated proteins and small molecules were present and active simultaneously in a cell. By overlaying an experimental dataset on these annotations, a user can perform a pathway over-representation analysis. By overlaying quantitative expression data or time series, a user can visualize the extent of change in affected pathways and its progression. A binomial test is used to calculate the probability shown for each result, and the p-values are corrected for the multiple testing (Benjamini-Hochberg procedure) that arises from evaluating the submitted list of identifiers against every pathway.

To learn more about our Pathway Analysis, please have a look at our relevant publications:

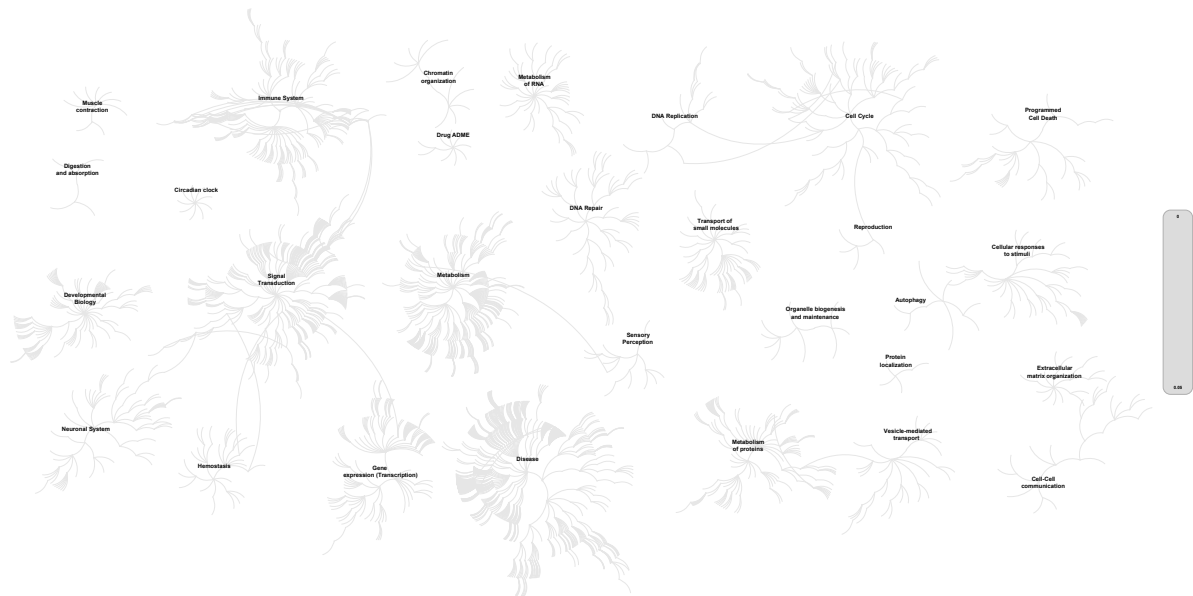
Fabregat A, Sidiropoulos K, Garapati P, Gillespie M, Hausmann K, Haw R, ... D'Eustachio P (2016). The reactome pathway knowledgebase. *Nucleic Acids Research*, 44(D1), D481–D487. <https://doi.org/10.1093/nar/gkv1351>. 

Fabregat A, Sidiropoulos K, Viteri G, Forner O, Marin-Garcia P, Arnau V, ... Hermjakob H (2017). Reactome pathway analysis: a high-performance in-memory approach. *BMC Bioinformatics*, 18. 

2. Properties

- This is an **overrepresentation** analysis: A statistical (hypergeometric distribution) test that determines whether certain Reactome pathways are over-represented (enriched) in the submitted data. It answers the question 'Does my list contain more proteins for pathway X than would be expected by chance?' This test produces a probability score, which is corrected for false discovery rate using the Benjamini-Hochberg method. [↗](#)
- 7464 out of 11842 identifiers in the sample were found in Reactome, where 2719 pathways were hit by at least one of them.
- All non-human identifiers have been converted to their human equivalent. [↗](#)
- IntAct interactors were included to increase the analysis background. This greatly increases the size of Reactome pathways, which maximises the chances of matching your submitted identifiers to the expanded pathway, but will include interactors that have not undergone manual curation by Reactome and may include interactors that have no biological significance, or unexplained relevance.
- This report is filtered to show only results for species 'Homo sapiens' and resource 'all resources'.
- The unique ID for this analysis (token) is MjAyNTEyMTkwOTAxMjJfNTM0OA%3D%3D. This ID is valid for at least 7 days in Reactome's server. Use it to access Reactome services with your data.

3. Genome-wide overview



This figure shows a genome-wide overview of the results of your pathway analysis. Reactome pathways are arranged in a hierarchy. The center of each of the circular "bursts" is the root of one top-level pathway, for example "DNA Repair". Each step away from the center represents the next level lower in the pathway hierarchy. The color code denotes over-representation of that pathway in your input dataset. Light grey signifies pathways which are not significantly over-represented.

4. Most significant pathways

The following table shows the 25 most relevant pathways sorted by p-value.

Pathway name	Entities				Reactions	
	found	ratio	p-value	FDR*	found	ratio
SRP-dependent cotranslational protein targeting to membrane	103 / 119	0.005	0.145	1	5 / 5	3.15e-04
Formation of the ternary complex, and subsequently, the 43S complex	49 / 54	0.002	0.154	1	3 / 3	1.89e-04
Collagen chain trimerization	39 / 44	0.002	0.224	1	24 / 28	0.002
Formation of a pool of free 40S subunits	100 / 113	0.005	0.23	1	2 / 2	1.26e-04
Interaction between L1 and Ankyrins	29 / 33	0.001	0.276	1	4 / 4	2.52e-04
Vif-mediated degradation of APOBEC3G	39 / 44	0.002	0.276	1	4 / 4	2.52e-04
Defective B4GALT7 causes EDS, progeroid type	19 / 21	8.46e-04	0.282	1	1 / 1	6.30e-05
Defective B3GALT6 causes EDSP2 and SEMDJL1	19 / 21	8.46e-04	0.282	1	1 / 1	6.30e-05
ZNF598 and the Ribosome-associated Quality Trigger (RQT) complex dissociate a ribosome stalled on a no-go mRNA	92 / 109	0.004	0.291	1	4 / 4	2.52e-04
RHOQ GTPase cycle	53 / 63	0.003	0.296	1	5 / 5	3.15e-04
LGI-ADAM interactions	13 / 14	5.64e-04	0.296	1	5 / 5	3.15e-04
Loss of Function of TGFBR1 in Cancer	7 / 7	2.82e-04	0.303	1	2 / 2	1.26e-04
Defective B3GALT6 causes PpS	36 / 41	0.002	0.309	1	1 / 1	6.30e-05
Abortive elongation of HIV-1 transcript in the absence of Tat	21 / 24	9.67e-04	0.32	1	2 / 2	1.26e-04
TGFBR1 KD Mutants in Cancer	6 / 6	2.42e-04	0.324	1	1 / 1	6.30e-05
cGMP effects	16 / 18	7.25e-04	0.327	1	4 / 4	2.52e-04
HIV elongation arrest and recovery	29 / 34	0.001	0.33	1	3 / 3	1.89e-04
Pausing and recovery of HIV elongation	29 / 34	0.001	0.33	1	2 / 2	1.26e-04
Tat-mediated HIV elongation arrest and recovery	28 / 33	0.001	0.343	1	3 / 3	1.89e-04
Pausing and recovery of Tat-mediated HIV elongation	28 / 33	0.001	0.343	1	2 / 2	1.26e-04
CDC42 GTPase cycle	128 / 159	0.006	0.35	1	6 / 6	3.78e-04
Defective B3GAT3 causes JDSSDHD	19 / 22	8.86e-04	0.351	1	1 / 1	6.30e-05
Defective EXT1 causes exostoses 1, TRPS2 and CHDS	14 / 16	6.44e-04	0.363	1	3 / 3	1.89e-04

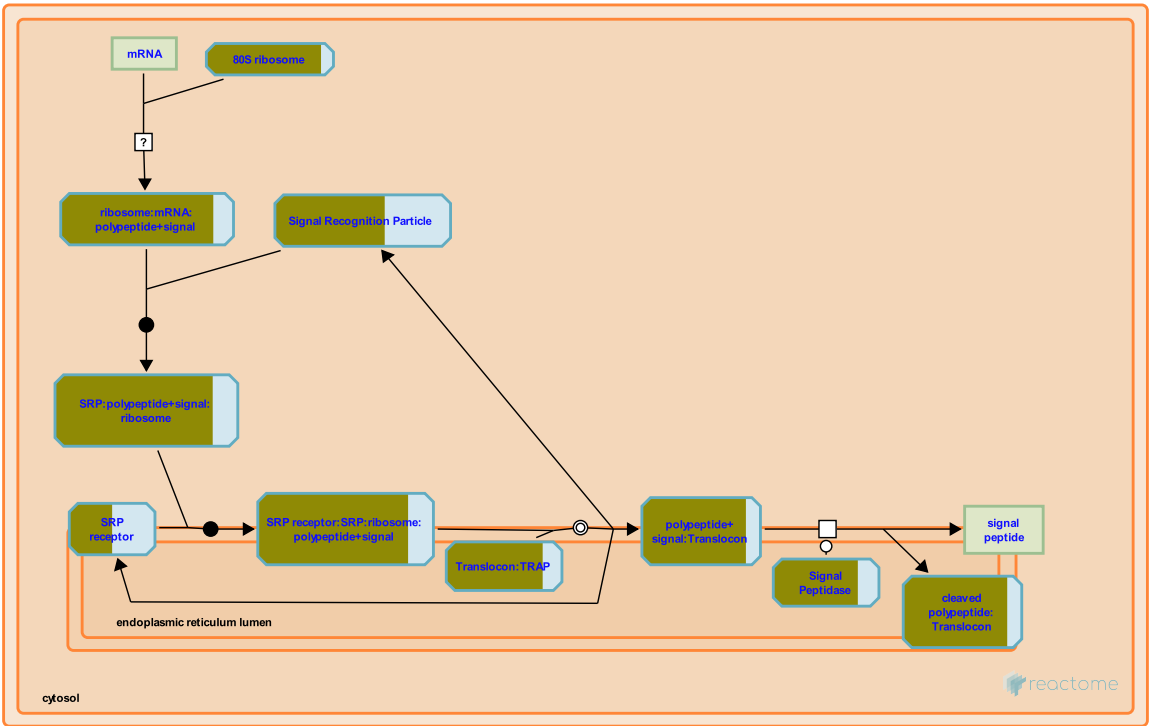
Pathway name	Entities				Reactions	
	found	ratio	p-value	FDR*	found	ratio
Defective EXT2 causes exostoses 2	14 / 16	6.44e-04	0.363	1	3 / 3	1.89e-04
Antigen processing: Ub, ATP-independent proteasomal degradation	18 / 21	8.46e-04	0.368	1	2 / 2	1.26e-04

* False Discovery Rate

5. Pathways details

For every pathway of the most significant pathways, we present its diagram, as well as a short summary, its bibliography and the list of inputs found in it.

1. SRP-dependent cotranslational protein targeting to membrane (R-HSA-1799339)



Cellular compartments: cytosol, endoplasmic reticulum lumen, endoplasmic reticulum membrane.

The process for translation of a protein destined for the endoplasmic reticulum (ER) branches from the canonical cytosolic translation process at the point when a nascent polypeptide containing a hydrophobic signal sequence is exposed on the surface of the cytosolic ribosome:mRNA:peptide complex. The signal sequence mediates the interaction of this complex with a cytosolic signal recognition particle (SRP) to form a complex which in turn docks with an SRP receptor complex on the ER membrane. There the ribosome complex is transferred from the SRP complex to a translocon complex embedded in the ER membrane and reoriented so that the nascent polypeptide protrudes through a pore in the translocon into the ER lumen. Translation, which had been halted by SRP binding, now resumes, the signal peptide is cleaved from the polypeptide, and elongation proceeds, with the growing polypeptide oriented into the ER lumen.

References

Edit history

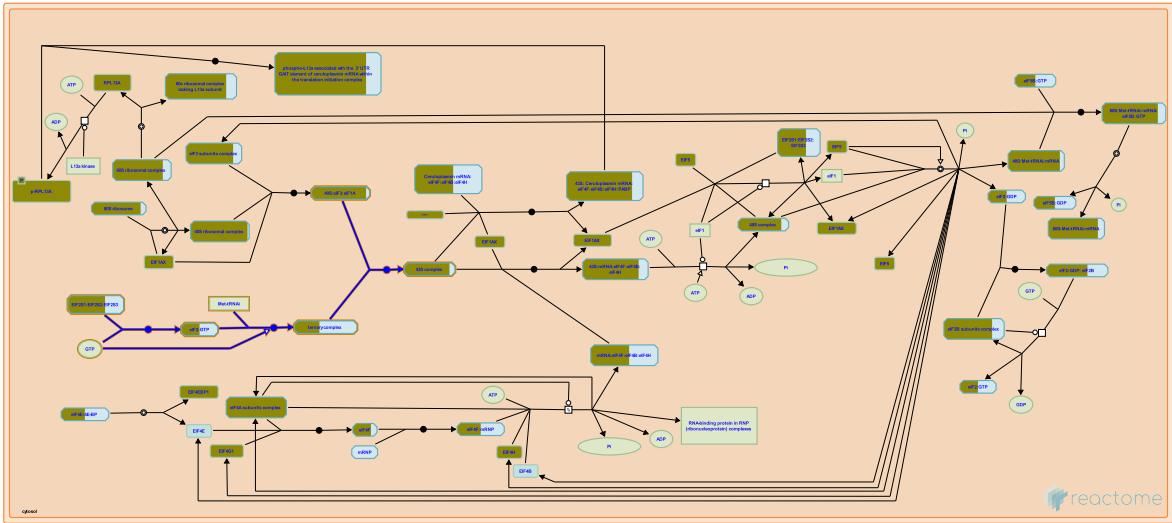
Date	Action	Author
2008-11-20	Authored	May B, Gopinathrao G

Date	Action	Author
2008-12-02	Reviewed	Matthews L, D'Eustachio P, Gillespie ME
2011-10-22	Revised	D'Eustachio P
2011-10-23	Edited	D'Eustachio P
2011-10-23	Created	D'Eustachio P

107 submitted entities found in this pathway, mapping to 131 Reactome entities

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
DDOST	P39656	FAU	P62861	NAB1	P08865
RPL10	P27635, Q96L21	RPL10A	P62906	RPL11	P62913
RPL12	P30050	RPL13	P26373, P40429	RPL13A	P40429
RPL14	P50914	RPL15	P61313	RPL17	P18621
RPL18	Q07020	RPL18A	Q02543	RPL19	P84098
RPL21	P46778	RPL22	P35268	RPL23	P62829
RPL23A	P62750	RPL24	P83731	RPL26	P61254, Q9UNX3
RPL26L1	Q9UNX3	RPL27	P61353	RPL27A	P46776, P61353
RPL28	P46776, P46779	RPL29	P47914	RPL3	P39023, Q92901
RPL30	P62888	RPL31	P62899	RPL32	P62888, P62910
RPL34	P49207, P62899	RPL35	P42766	RPL35A	P18077, P42766
RPL36	Q9Y3U8	RPL36A	P83881, Q969Q0	RPL36AL	Q969Q0
RPL37	P61927	RPL37A	P18077, P61513	RPL38	P63173
RPL4	P36578	RPL5	P46777	RPL6	Q02878
RPL7	P18124	RPL7A	P18124, P62424	RPL8	P62424, P62917
RPLP0	P05388	RPLP1	P05386	RPLP2	P05387
RPN1	P04843	RPN2	P04844	RPS10	P46783
RPS11	P62280	RPS12	P25398	RPS13	P62277
RPS14	P62263	RPS15	P62841	RPS15A	P62244
RPS16	P62249	RPS17	P08708	RPS18	P62269
RPS19	P39019	RPS2	P08865, P15880, P46782	RPS20	P60866
RPS21	P62841, P63220	RPS23	P62266	RPS24	P62244, P62847
RPS25	P62851	RPS26	P62854	RPS27	P42677, Q71UM5
RPS27A	P62979, P62987, Q71UM5	RPS27L	Q71UM5	RPS28	P62857
RPS29	P62273	RPS3	P23396	RPS3A	P61247
RPS4X	P62701, Q8TD47	RPS4Y1	P22090, Q8TD47	RPS5	P46782
RPS6	P62753	RPS7	P62081	RPS8	P62241
RPS9	P46781	RPSA	P08865	RRP1	P05386
SEC11C	Q9BY50	SEC61A1	P61619	SEC61B	P60468
SEC61G	P60059	SOS1	P42766	SPC25	Q15005
SPCS1	Q9Y6A9	SPCS2	Q15005	SPCS3	P61009
SRP14	P37108	SRP19	P09132	SRP68	Q9UHB9
SRP72	O76094	SRP9	P49458	SRPRB	Q9Y5M8
SSR2	P43308	SSR3	Q9UNL2	SSR4	P51571
TRAM1	Q15629	UBA52	P62979, P62987	UBB	P62979, P62987
UBC	P62987	WBP1	P39656		

2. Formation of the ternary complex, and subsequently, the 43S complex (R-HSA-72695)



Cellular compartments: cytosol.

Binding of the methionyl-tRNA initiator to the active eIF2:GTP complex results in the formation of the ternary complex. Subsequently, this Met-tRNA_i:eIF2:GTP (ternary) complex binds to the complex formed by the 40S subunit, eIF3 and eIF1A, to form the 43S complex.

References

Peterson DT, Merrick WC & Safer B (1979). Binding and release of radiolabeled eukaryotic initiation factors 2 and 3 during 80 S initiation complex formation. J Biol Chem, 254, 2509-16. [↗](#)

Safer B, Adams SL, Anderson WF & Merrick WC (1976). Binding of MET-TRNA_f and GTP to homogeneous initiation factor MP. J Biol Chem, 250, 9076-82. [↗](#)

Trachsel H, Erni B, Schreier MH & Staehelin T (1978). Initiation of mammalian protein synthesis. II. The assembly of the initiation complex with purified initiation factors. J Mol Biol, 116, 755-67. [↗](#)

McCarthy JE & Tuite M (1990). *New insights into an old problem: ternary complex (Met-tRNA_f:eIF:GTP) formation in animal cells.*, Post-Transcriptional Control of Gene Expression, 521-526.

Benne R & Hershey JW (1978). The mechanism of action of protein synthesis initiation factors from rabbit reticulocytes. J Biol Chem, 253, 3078-87. [↗](#)

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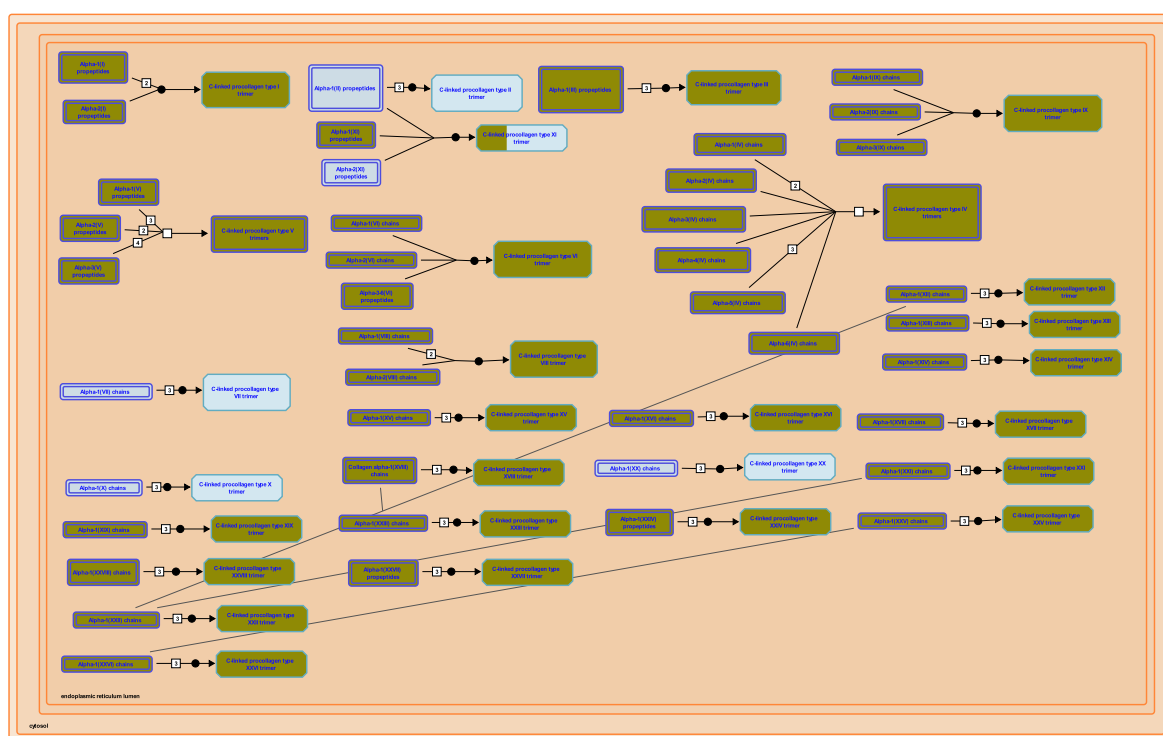
Date	Action	Author
2025-11-15	Modified	Weiser JD

52 submitted entities found in this pathway, mapping to 60 Reactome entities

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
DNMT1	O75821	EIF1AY	P47813	EIF2A	P05198
EIF2S2	P20042	EIF3A	Q14152	EIF3B	P55884
EIF3C	Q99613	EIF3D	O15371	EIF3F	O00303

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
EIF3G	O75821	EIF3I	Q13347	EIF3J	O75822
EIF3K	Q9UBQ5	EIF3L	Q9Y262	EIF3M	Q7L2H7
FAU	P62861	NAB1	P08865	RPS10	P46783
RPS11	P62280	RPS12	P25398	RPS13	P62277
RPS14	P62263	RPS15	P62841	RPS15A	P62244
RPS16	P62249	RPS17	P08708	RPS18	P62269
RPS19	P39019	RPS2	P08865, P15880, P46782	RPS20	P60866
RPS21	P62841, P63220	RPS23	P62266	RPS24	P62244, P62847
RPS25	P62851	RPS26	P62854	RPS27	P42677, Q71UM5
RPS27A	P62979, Q71UM5	RPS27L	Q71UM5	RPS28	P62857
RPS29	P62273	RPS3	P23396	RPS3A	P61247
RPS4X	P62701, Q8TD47	RPS4Y1	P22090, Q8TD47	RPS5	P46782
RPS6	P62753	RPS7	P62081	RPS8	P62241
RPS9	P46781	RPSA	P08865	UBA52	P62979
UBB	P62979				

3. Collagen chain trimerization ([R-HSA-8948216](#))



reactome

The C-propeptides of collagen propeptide chains are essential for the association of three peptide chains into a trimeric but non-helical procollagen. This initial binding event determines the composition of the trimer, brings the individual chains into the correct register and initiates formation of the triple helix at the C-terminus, which then proceeds towards the N-terminus in a zipper-like fashion (Engel & Prockop 1991). Most early refolding studies were performed with collagen type III, which contains a disulfide linkage at the C-terminus of its triple helix (Bächinger et al. 1978, Bruckner et al. 1978) that acts as a permanent linker even after removal of the non-collagenous domains.

Mutations within the C-propeptides further suggest that they are crucial for the correct interaction of the three polypeptide chains and for subsequent correct folding (refs. in Boudko et al. 2011).

References

Byers PH, Click EM, Harper E & Bornstein P (1975). Interchain disulfide bonds in procollagen are located in a large nontriple-helical COOH-terminal domain. *Proc Natl Acad Sci U S A*, 72, 3009-13

Bächinger HP, Brückner P, Timpl R & Engel J (1978). The role of cis-trans isomerization of peptide bonds in the coil leads to and comes from triple helix conversion of collagen. *Eur J Biochem*, 90, 605-13. 

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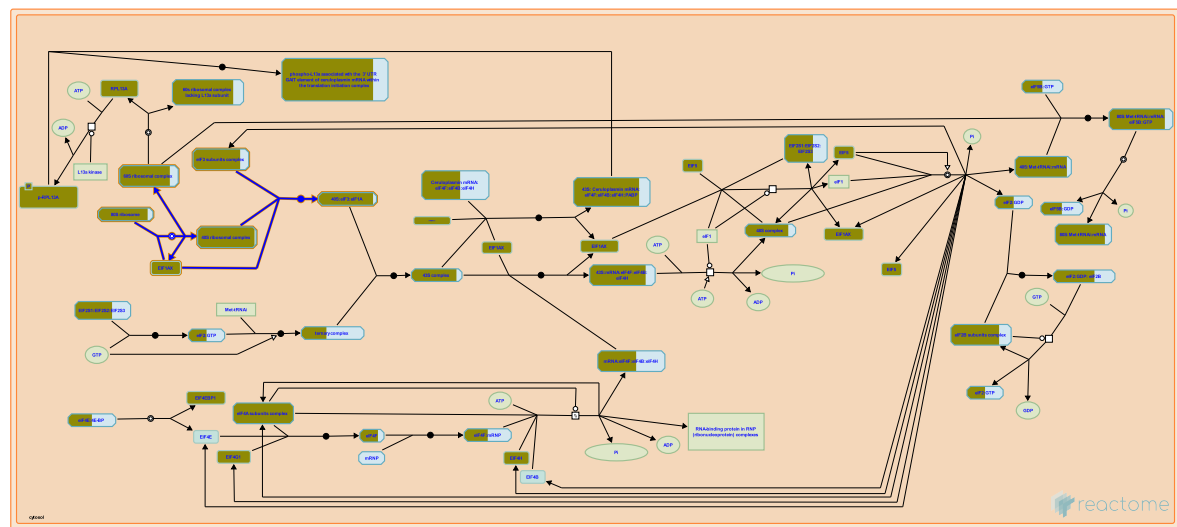
Date	Action	Author
2012-04-11	Authored	Jupe S
2012-05-24	Reviewed	Canty-Laird EG
2016-11-03	Edited	Jupe S

Date	Action	Author
2016-11-11	Created	Jupe S
2025-11-15	Modified	Weiser JD

39 submitted entities found in this pathway, mapping to 46 Reactome entities

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COL11A1	P12107	COL12A1	Q99715	COL13A1	Q5TAT6
COL14A1	Q05707	COL15A1	P39059, Q5TAT6	COL16A1	Q07092
COL17A1	Q9UMD9	COL18A1	P39060, P53420	COL19A1	Q14993, Q9BXS0
COL1A1	P02452	COL1A2	P08123	COL21A1	Q96P44
COL22A1	Q8NFW1	COL23A1	Q86Y22	COL24A1	Q17RW2
COL25A1	Q9BXS0	COL26A1	Q86Y22, Q96A83	COL27A1	Q8IZC6
COL28A1	Q2UY09	COL3A1	P02461	COL4A1	P02462, Q14055
COL4A2	P08572	COL4A3	Q01955	COL4A4	P53420
COL4A5	P29400	COL4A6	Q14031	COL5A1	P20908
COL5A2	P05997	COL5A3	P25940	COL6A1	P12109
COL6A2	P12110	COL6A3	A8TX70, P12111	COL6A6	A6NMZ7
COL8A1	P27658	COL8A2	P25067	COL9A1	P20849
COL9A2	Q14055, Q9UMD9	COL9A3	Q14050	EMID1	P39060

4. Formation of a pool of free 40S subunits (R-HSA-72689)



Cellular compartments: cytosol.

The 80S ribosome dissociates into free 40S (small) and 60S (large) ribosomal subunits. Each ribosomal subunit is constituted by several individual ribosomal proteins and rRNA.

References

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Date	Action	Author
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98 submitted entities found in this pathway, mapping to 122 Reactome entities

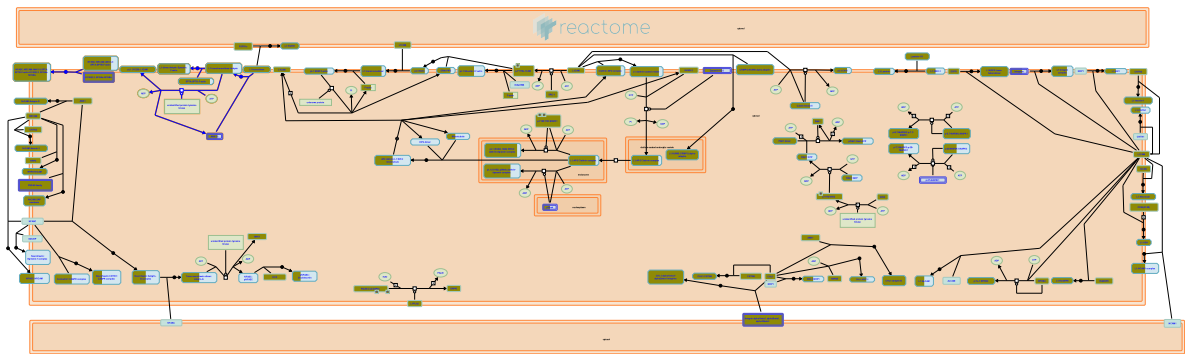
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DNMT1	O75821	EIF1AY	P47813	EIF3A	Q14152
EIF3B	P55884	EIF3C	Q99613	EIF3D	O15371
EIF3F	O00303	EIF3G	O75821	EIF3I	Q13347
EIF3J	O75822	EIF3K	Q9UBQ5	EIF3L	Q9Y262
EIF3M	Q7L2H7	FAU	P62861	NAB1	P08865
RPL10	P27635, Q96L21	RPL10A	P62906	RPL11	P62913
RPL12	P30050	RPL13	P26373, P40429	RPL13A	P40429
RPL14	P50914	RPL15	P61313	RPL17	P18621
RPL18	Q07020	RPL18A	Q02543	RPL19	P84098
RPL21	P46778	RPL22	P35268	RPL23	P62829
RPL23A	P62750	RPL24	P83731	RPL26	P61254, Q9UNX3
RPL26L1	Q9UNX3	RPL27	P61353	RPL27A	P46776, P61353
RPL28	P46776, P46779	RPL29	P47914	RPL3	P39023, Q92901
RPL30	P62888	RPL31	P62899	RPL32	P62888, P62910
RPL34	P49207, P62899	RPL35	P42766	RPL35A	P18077, P42766
RPL36	Q9Y3U8	RPL36A	P83881, Q969Q0	RPL36AL	Q969Q0
RPL37	P61927	RPL37A	P18077, P61513	RPL38	P63173
RPL4	P36578	RPL5	P46777	RPL6	Q02878
RPL7	P18124	RPL7A	P18124, P62424	RPL8	P62424, P62917
RPLP0	P05388	RPLP1	P05386	RPLP2	P05387
RPS10	P46783	RPS11	P62280	RPS12	P25398

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
RPS13	P62277	RPS14	P62263	RPS15	P62841
RPS15A	P62244	RPS16	P62249	RPS17	P08708
RPS18	P62269	RPS19	P39019	RPS2	P08865, P15880, P46782
RPS20	P60866	RPS21	P62841, P63220	RPS23	P62266
RPS24	P62244, P62847	RPS25	P62851	RPS26	P62854
RPS27	P42677, Q71UM5	RPS27A	P62979, P62987, Q71UM5	RPS27L	Q71UM5
RPS28	P62857	RPS29	P62273	RPS3	P23396
RPS3A	P61247	RPS4X	P62701, Q8TD47	RPS4Y1	P22090, Q8TD47
RPS5	P46782	RPS6	P62753	RPS7	P62081
RPS8	P62241	RPS9	P46781	RPSA	P08865
RRP1	P05386	SOS1	P42766	UBA52	P62979, P62987
UBB	P62979, P62987	UBC	P62987		

Interactors found in this pathway (7)

Input	UniProt Id	Interacts with	Input	UniProt Id	Interacts with
EIF1AY	O14602	P47813	EIF3A	Q14152	P47813
EIF3C	Q99613	P47813	EIF5	P55010	P47813
IPO13	O94829	P47813	NCR3LG1	Q68D85	P47813
RAN	P62826	P47813			

5. Interaction between L1 and Ankyrins (R-HSA-445095)



Ankyrins are a family of adaptor proteins that couple membrane proteins such as voltage gated Na⁺ channels and the Na⁺/K⁺ anion exchanger to the spectrin actin cytoskeleton. Ankyrins are encoded by three genes (ankyrin-G, -B and -R) of which ankyrin-G and -B are the major forms expressed in the developing nervous system. Ankyrins bind to the cytoplasmic domain of L1 CAMs and couple them and ion channel proteins, to the spectrin cytoskeleton. This binding enhances the homophilic adhesive activity of L1 and reduces its mobility within the plasma membrane. L1 interaction with ankyrin mediates branching and synaptogenesis of cortical inhibitory neurons.

References

Hortsch M, Nagaraj K & Godenschwege TA (2009). The interaction between L1-type proteins and ankyrins--a master switch for L1-type CAM function. *Cell Mol Biol Lett*, 14, 57-69. [🔗](#)

Herron LR, Hill M, Davey F & Gunn-Moore FJ (2009). The intracellular interactions of the L1 family of cell adhesion molecules. *Biochem J*, 419, 519-31. [🔗](#)

Nagaraj K & Hortsch M (2006). Phosphorylation of L1-type cell-adhesion molecules--ankyrins away! . *Trends Biochem Sci*, 31, 544-6. [🔗](#)

Guan H & Maness PF (2010). Perisomatic GABAergic Innervation in Prefrontal Cortex Is Regulated by Ankyrin Interaction with the L1 Cell Adhesion Molecule. *Cereb Cortex*. [🔗](#)

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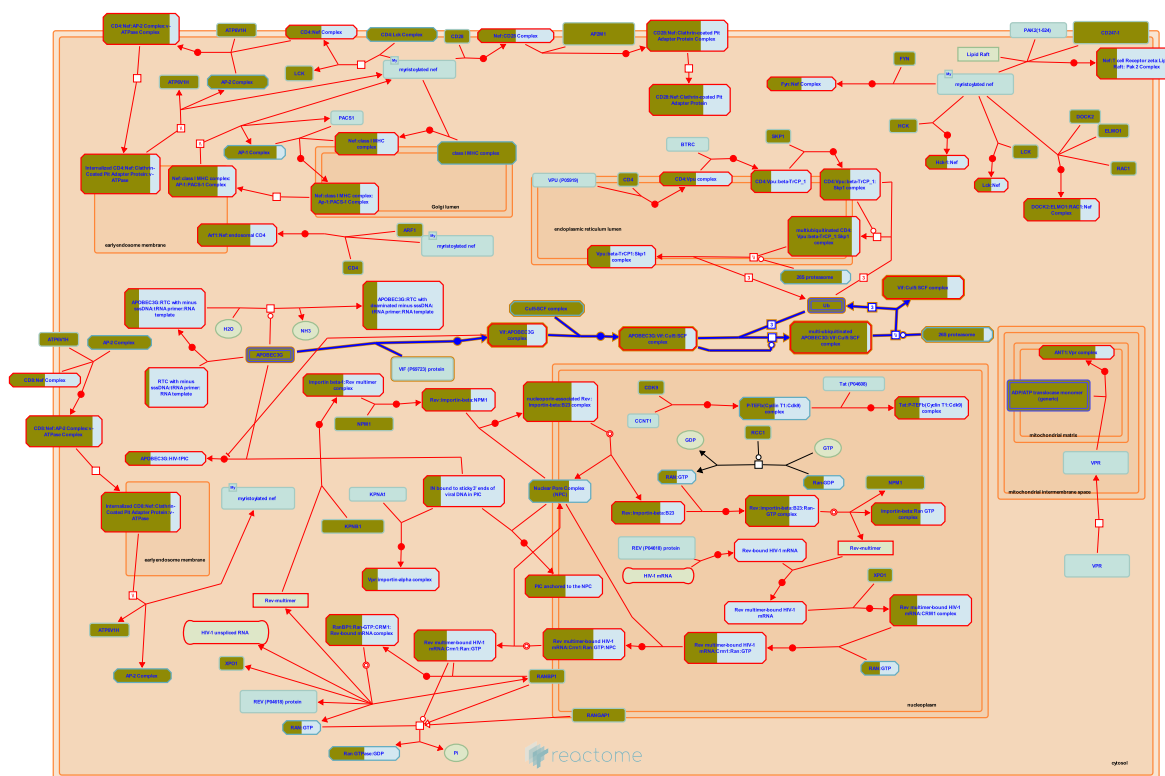
Date	Action	Author
2008-07-30	Edited	Garapati P V
2008-07-30	Authored	Garapati P V
2009-10-27	Created	Garapati P V
2010-02-16	Reviewed	Maness PF
2025-11-15	Modified	Weiser JD

29 submitted entities found in this pathway, mapping to 32 Reactome entities

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
ACTB	P60709, P63261	ACTG1	P60709, P63261	ANK3	Q12955
KCNQ2	O43526	KCNQ3	O43525	L1CAM	P32004
NRCAM	Q92823	SCN10A	Q9Y5Y9	SCN11A	Q9UI33
SCN1A	P35498	SCN1B	Q07699	SCN2A	Q99250
SCN2B	O60939	SCN3A	Q99250, Q9NY46	SCN3B	Q9NY72

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
SCN4A	P35499	SCN4B	Q8IWT1	SCN5A	Q14524
SCN7A	Q01118	SCN8A	Q9UQD0	SCN9A	Q15858
SPTA1	P02549	SPTAN1	Q13813	SPTB	P11277
SPTBN1	Q01082	SPTBN2	O15020	SPTBN4	Q9H254
SPTBN5	Q9NRC6	TNKS2	P16157		

6. Vif-mediated degradation of APOBEC3G (R-HSA-180585)



Cellular compartments: cytosol.

Diseases: Human immunodeficiency virus infectious disease.

The HIV-1 accessory protein Vif (Viral infectivity factor) is required for the efficient infection of primary cell populations (e.g., lymphocytes and macrophages) and 'non-permissive' cell lines. Vif neutralises the host DNA editing enzyme, APOBEC3G, in the producer cell. Indeed, in the absence of a functional Vif, APOBEC3G is selectively incorporated into the budding virions and in the next cycle of infection leads to the deamination of deoxycytidines (dC) within the minus-strand cDNA during reverse transcription (Sheehy et al 2003; Li et al., 2005 ; Stopak et al. 2003).

Deamination changes cytidine to uracil and thus results in G to A transitions and stop codons in the provirus. The aberrant cDNAs produced in the infected cell can either be integrated in form of non-functional proviruses or degraded. Vif counteracts the antiviral activity of APOBEC3G by associating directly with it and promoting its polyubiquitination and degradation by the 26S proteasome.

Vif binds APOBEC3G and recruits it into an E3 ubiquitin-enzyme complex composed by the cytoplasmic proteins Cullin5, Rbx, ElonginC and ElonginB (Yu et al., 2003) . Thus, in the presence of Vif, APOBEC3G incorporation into the virion is minimal.

References

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Edit history

Date	Action	Author
2006-05-16	Authored	Matthews L
2006-05-23	Created	Matthews L
2007-01-30	Edited	Matthews L
2007-01-31	Reviewed	Mulder L, Simon V
2023-10-12	Modified	Weiser JD

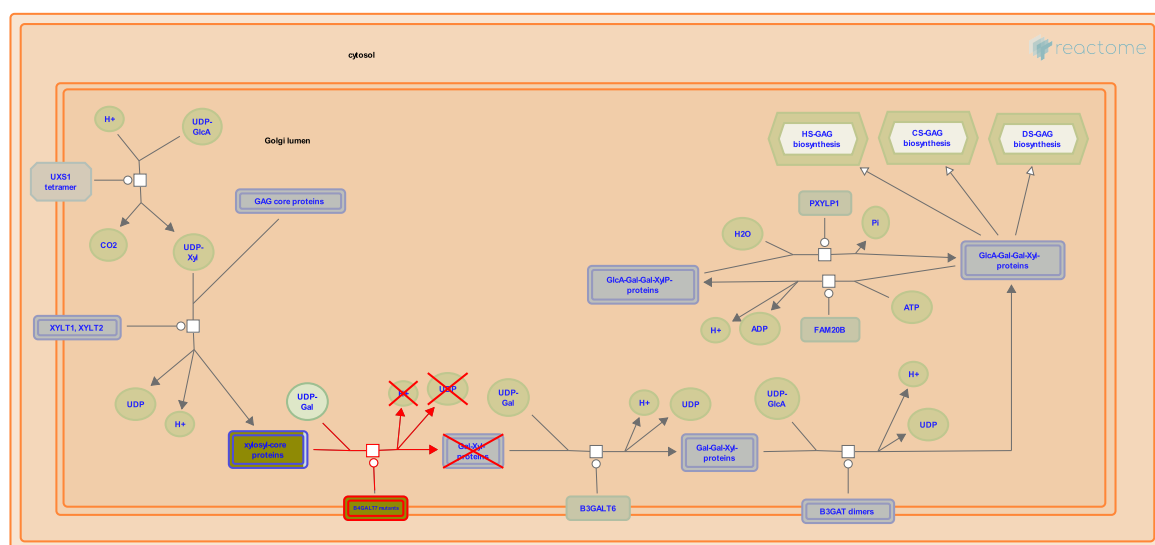
39 submitted entities found in this pathway, mapping to 45 Reactome entities

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
ADRM1	Q16186	APOBEC3G	Q9HC16-1, Q9HC16-3	CUL5	Q93034
PSMA1	P25786	PSMA3	P25788	PSMA4	P25789
PSMA5	P28066	PSMA6	P60900	PSMA7	O14818
PSMB1	P20618	PSMB2	P49721	PSMB3	P49720
PSMB4	P28070	PSMB5	P28074	PSMB6	P28072
PSMB7	Q99436	PSMC1	P62191	PSMC2	P35998
PSMC3	P17980	PSMC4	P43686	PSMC5	P62195
PSMC6	P62333	PSMD1	Q99460	PSMD11	O00231
PSMD13	Q9UNM6	PSMD2	Q13200	PSMD3	O43242
PSMD6	Q15008	PSMD7	P51665	PSMD8	P48556
RBX1	P62877	RPN1	Q13200	RPN2	Q99460
RPS27A	P62979, P62987	TCEB1	Q15369	TCEB2	Q15370
UBA52	P62979, P62987	UBB	P0CG47, P62979, P62987	UBC	P0CG48, P62987

Interactors found in this pathway (1)

Input	UniProt Id	Interacts with	Input	UniProt Id	Interacts with
APOBEC3G	Q9HC16	P69723			

7. Defective B4GALT7 causes EDS, progeroid type ([R-HSA-3560783](#))



Diseases: Ehlers-Danlos syndrome.

Ehlers–Danlos syndrome (EDS) is a group of inherited connective tissue disorders, caused by a defect in the synthesis of collagen types I or III. Abnormal collagen renders connective tissues more elastic. The severity of the mutation can vary from mild to life-threatening. There is no cure and treatment is supportive, including close monitoring of the digestive, excretory and particularly the cardiovascular systems. Defective B4GALT7, a galactosyltransferase important in proteoglycan synthesis, causes the progeroid variant of EDS (MIM:130070). Features include an aged appearance, developmental delay, short stature, generalized osteopenia, defective wound healing, hypermobile joints, hypotonic muscles, and loose but elastic skin (Okajima et al. 1999).

References

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Edit history

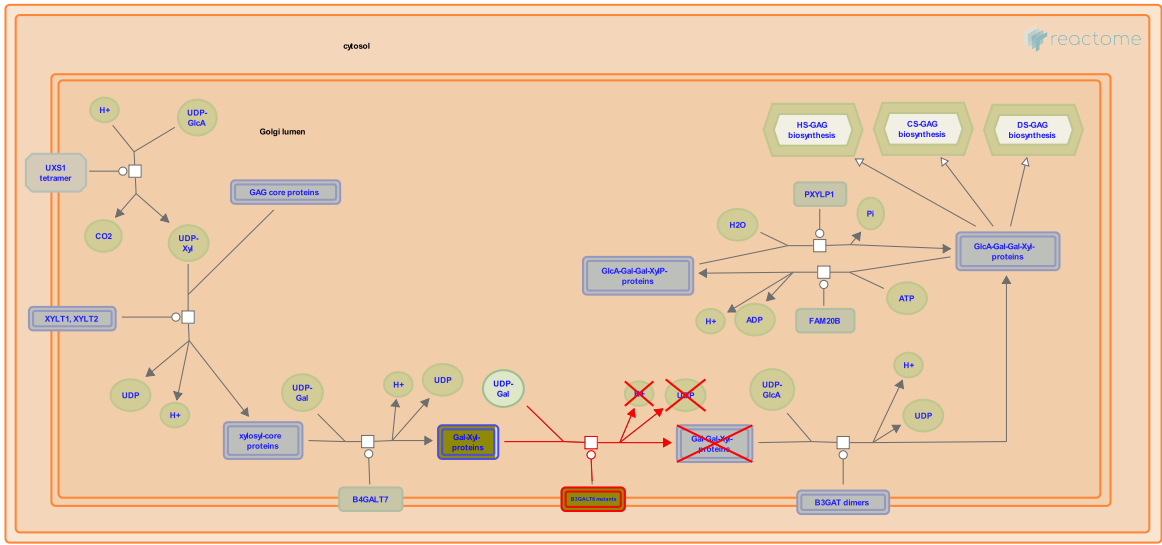
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2013-05-21	Edited	Jassal B
2013-05-21	Authored	Jassal B
2013-05-21	Created	Jassal B
2014-07-09	Reviewed	Spillmann D
2025-06-02	Modified	Stephan R

19 submitted entities found in this pathway, mapping to 19 Reactome entities

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
AGRN	O00468	B4GALT7	Q9UBV7	BCAN	Q96GW7
BGN	P21810	CSPG4	Q6UVK1	DCN	P07585
GPC1	P35052	GPC2	Q8N158	GPC3	P51654

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
GPC4	O75487	GPC5	P78333	GPC6	Q9Y625
HSPG2	P98160	NCAN	O14594	SDC1	P18827
SDC2	P34741	SDC3	O75056	SDC4	P31431
VCAN	P13611				

8. Defective B3GALT6 causes EDSP2 and SEMDJL1 ([R-HSA-4420332](#))



Diseases: Ehlers-Danlos syndrome, spondyloepimetaphyseal dysplasia.

The biosynthesis of dermatan sulfate/chondroitin sulfate and heparin/heparan sulfate glycosaminoglycans (GAGs) starts with the formation of a tetrasaccharide linker sequence attached to the core protein. Beta-1,3-galactosyltransferase 6 (B3GALT6) is one of the critical enzymes involved in the formation of this linker sequence. Defects in B3GALT6 causes Ehlers-Danlos syndrome progeroid type 2 (EDSP2; MIM:615349), a severe disorder resulting in a broad spectrum of skeletal, connective tissue and wound healing problems. Defects in B3GALT6 can also cause spondyloepimetaphyseal dysplasia with joint laxity type 1 (SEMDJL1; MIM:271640), characterised by spinal deformity and lax joints, especially of the hands and respiratory compromise resulting in early death (Nakajima et al. 2013, Malfait et al. 2013).

References

Malfait F, Kariminejad A, Van Damme T, Gauche C, Syx D, Merhi-Soussi F, ... De Paepe A (2013). Defective Initiation of Glycosaminoglycan Synthesis due to B3GALT6 Mutations Causes a Pleiotropic Ehlers-Danlos Syndrome-like Connective Tissue Disorder. *Am. J. Hum. Genet.* [\[Link\]](#)

Nakajima M, Mizumoto S, Miyake N, Kogawa R, Iida A, Ito H, ... Ikegawa S (2013). Mutations in B3GALT6, which Encodes a Glycosaminoglycan Linker Region Enzyme, Cause a Spectrum of Skeletal and Connective Tissue Disorders. *Am. J. Hum. Genet.* [\[Link\]](#)

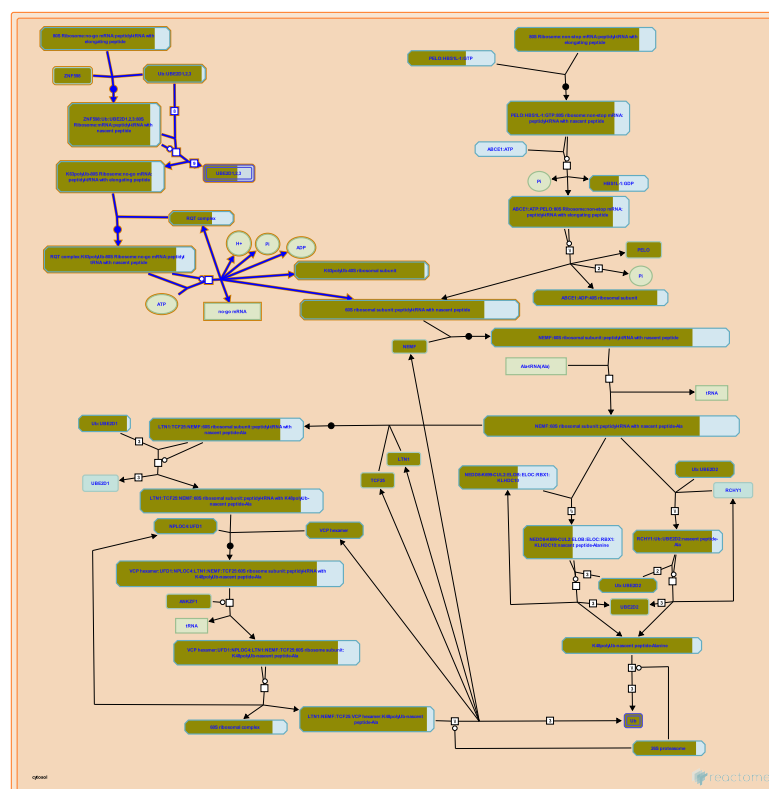
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2013-09-03	Authored	Jassal B
2013-09-03	Created	Jassal B
2014-07-09	Reviewed	Spillmann D
2025-06-02	Modified	Stephan R

19 submitted entities found in this pathway, mapping to 19 Reactome entities

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
AGRN	O00468	B3GALT6	Q96L58	BCAN	Q96GW7
BGN	P21810	CSPG4	Q6UVK1	DCN	P07585
GPC1	P35052	GPC2	Q8N158	GPC3	P51654
GPC4	O75487	GPC5	P78333	GPC6	Q9Y625
HSPG2	P98160	NCAN	O14594	SDC1	P18827
SDC2	P34741	SDC3	O75056	SDC4	P31431
VCAN	P13611				

9. ZNF598 and the Ribosome-associated Quality Trigger (RQT) complex dissociate a ribosome stalled on a no-go mRNA (R-HSA-9954716)



Due to features such as damaged nucleotides, strong secondary structure, presence of stretches of more than four uninterrupted AAA codons in a mRNA coding region (designated no-go mRNAs) a ribosome translating an mRNA can stall and cause collisions with the ribosomes trailing it on the mRNA (reviewed in Joazeiro 2017, Eisenack and Trentini 2022, Filbeck et al. 2022, McGirr et al. 2025). In cases in which the ribosome stalls internally on the mRNA and a 3' region of the mRNA protrudes from the ribosome, ZNF598, a ubiquitin E3 ligase that is the homolog of yeast HEL2, binds the ribosome (Garzia et al. 2017, Juskiewicz and Hegde 2017, Sundaramoorthy et al. 2017) and ubiquitinates the 40S subunit ribosomal proteins eS10 (RPS10) at residues K138 and K139 and uS10 (RPS20) at residues K4 and K8 to initiate splitting of the 40S and 60S ribosomal subunits (Juskiewicz and Hegde 2017, Sundaramoorthy et al. 2017, Garzia et al. 2017, Juskiewicz et al. 2018, Narita et al. 2022, Miścicka et al. 2024, also inferred from yeast homologs in Matsuo et al. 2017, Sitron et al. 2017, Ikeuchi et al. 2019, reviewed in Ford et al. 2024). ZAK also interacts with the collided ribosome at the 18S helix through its c-terminal domain, which triggers ZAK autophosphorylation and phosphorylation of p38/JNK in a MAPK cascade known as the ribotoxic stress response (Wu et al. 2020, Vind et al. 2020, Stoneley et al. 2022).

The ASCC2 subunit (Narita et al. 2022) of the ribosome quality control trigger complex (RQT complex, ASCC2:ASCC3:TRIP4) binds K63-linked polyubiquitin conjugated to the 40S protein uS10 (Hashimoto et al. 2020, Juskiewicz et al. 2020, Narita et al. 2022, and inferred from yeast homologs in Matsuo et al. 2023). The ASCC3 subunit of the RQT complex splits stalled 80S ribosomes with K63-polyubiquitinated uS10 into 60S and 40S subunits (Hashimoto et al. 2020, Juskiewicz et al. 2020, Narita et al. 2022, Miścicka et al. 2024) apparently by exerting a pulling force on the mRNA (inferred from the yeast homolog Slh1 in Best et al. 2023). The peptidyl-tRNA remains bound in the 60S subunit, with the tRNA positioned in the P site. The problematic mRNA dissociates after splitting and is thought to be degraded at this time. In the case of collided yeast ribosomes, the mRNA is first endonucleolytically cleaved and the cleavage products are exonucleolytically degraded by XRN1 and the exosome (Ikeuchi et al. 2019).

References

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Edit history

Date	Action	Author
2025-06-06	Edited	May B
2025-06-06	Authored	May B
2025-06-08	Created	May B
2025-08-09	Reviewed	Bennett EJ, Ford PW
2025-11-07	Reviewed	McGirr T, Jafarnejad SM

90 submitted entities found in this pathway, mapping to 116 Reactome entities

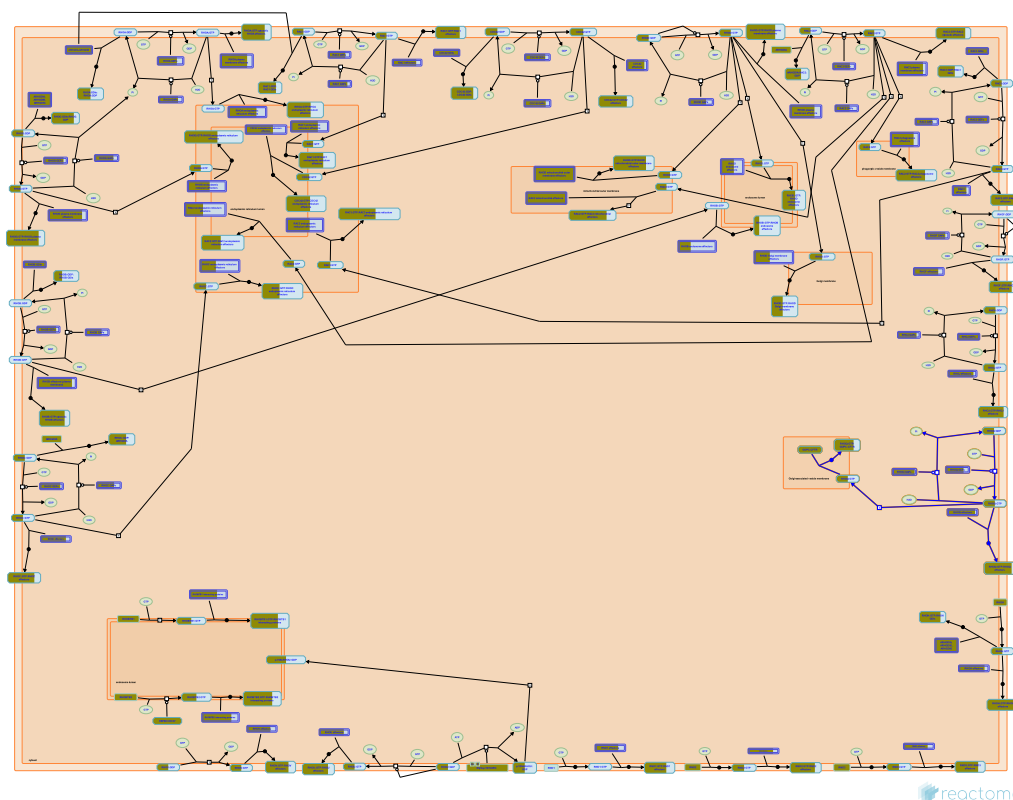
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NAB1	P08865	RPL10	P27635, Q96L21	RPL10A	P62906
RPL11	P62913	RPL12	P30050	RPL13	P26373, P40429
RPL13A	P40429	RPL14	P50914	RPL15	P61313
RPL17	P18621	RPL18	Q07020	RPL18A	Q02543
RPL19	P84098	RPL21	P46778	RPL22	P35268
RPL23	P62829	RPL23A	P62750	RPL24	P83731
RPL26	P61254, Q9UNX3	RPL26L1	Q9UNX3	RPL27	P61353
RPL27A	P46776, P61353	RPL28	P46776, P46779	RPL29	P47914

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
RPL3	P39023, Q92901	RPL30	P62888	RPL31	P62899
RPL32	P62888, P62910	RPL34	P49207, P62899	RPL35	P42766
RPL35A	P18077, P42766	RPL36	Q9Y3U8	RPL36A	P83881, Q969Q0
RPL36AL	Q969Q0	RPL37	P61927	RPL37A	P18077, P61513
RPL38	P63173	RPL4	P36578	RPL5	P46777
RPL6	Q02878	RPL7	P18124	RPL7A	P18124, P62424
RPL8	P62424, P62917	RPLP0	P05388	RPLP1	P05386
RPLP2	P05387	RPS10	P46783	RPS11	P62280
RPS12	P25398	RPS13	P62277	RPS14	P62263
RPS15	P62841	RPS15A	P62244	RPS16	P62249
RPS17	P08708	RPS18	P62269	RPS19	P39019
RPS2	P08865, P15880, P46782	RPS20	P60866	RPS21	P62841, P63220
RPS23	P62266	RPS24	P62244, P62847	RPS25	P62851
RPS26	P62854	RPS27	P42677, Q71UM5	RPS27A	P62979, P62987, Q71UM5
RPS27L	Q71UM5	RPS28	P62857	RPS29	P62273
RPS3	P23396	RPS3A	P61247	RPS4X	P62701, Q8TD47
RPS4Y1	P22090, Q8TD47	RPS5	P46782	RPS6	P62753
RPS7	P62081	RPS8	P62241	RPS9	P46781
RPSA	P08865	RRP1	P05386	SOS1	P42766
UBA52	P62979, P62987	UBB	P0CG47, P62979, P62987	UBC	P0CG48, P62987
UBE2D2	P62837	UBE2D3	P61077	ZNF598	Q86UK7

Interactors found in this pathway (2)

Input	UniProt Id	Interacts with	Input	UniProt Id	Interacts with
GNB2L1	P63244	Q86UK7	YWHAЕ	P62258	Q86UK7

10. RHOQ GTPase cycle (R-HSA-9013406)



This pathway catalogues RHOQ (also known as TC10) guanine nucleotide exchange factors (GEFs), GTPase activator proteins (GAPs), and RHOQ effectors. No GDP dissociation inhibitors (GDIs) have been shown to interact with RHOQ. Two GDIs, ARHGDIA (also known as Rho-GDI alpha) (Michaelson et al. 2001; Murphy et al. 2001) and ARHGDIB (also known as D4-GDI) (Zhang et al. 2009), were specifically shown not to interact with RHOQ. RHOQ, together with RHOJ (TCL), belongs to the CDC42 subfamily of RHO GTPases, with the three family members sharing 65-85% of homology (Murphy et al. 1999, Vignal et al. 2000, Donnelly et al. 2014; Shi et al. 2016). RHOQ is highly activated on exocytosing vesicles and recycling endosomes. RHOQ GTPase activity is necessary for fusion of vesicles with the plasma membrane (Kawase et al. 2006, Donnelly et al. 2014). RHOQ is required for insulin-stimulated glucose uptake (Kanzaki and Pessin 2003; Saltiel and Pessin 2003) and is involved in insulin signaling in adipocytes (Satoh 2014). While some studies reported RHOQ involvement in neurite outgrowth (Abe et al. 2003, Gonzalez-Billault et al. 2012) and regulation of membrane addition during axon formation (Dupraz et al. 2009, Gonzalez-Billault et al. 2012), other studies failed to identify RHOQ as a regulator of neurite outgrowth (Murphy et al. 1999, Murphy et al. 2001).

References

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Shi TT, Li G & Xiao HT (2016). The Role of RhoJ in Endothelial Cell Biology and Tumor Pathology. Biomed Res Int, 2016, 6386412. [🔗](#)

Donnelly SK, Bravo-Cordero JJ & Hodgson L (2014). Rho GTPase isoforms in cell motility: Don't fret, we have FRET. Cell Adh Migr, 8, 526-34. [🔗](#)

Satoh T (2014). Rho GTPases in insulin-stimulated glucose uptake. Small GTPases, 5, e28102. [🔗](#)

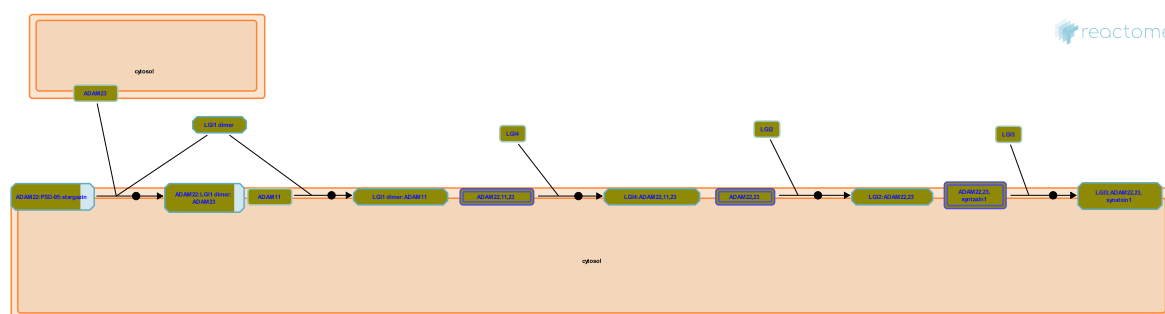
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2020-07-14	Authored	Orlic-Milacic M
2021-02-05	Reviewed	Fort P
2021-02-25	Edited	Orlic-Milacic M
2021-05-17	Reviewed	Das ME
2021-06-10	Revised	Orlic-Milacic M
2021-06-10	Edited	Orlic-Milacic M
2025-11-15	Modified	Weiser JD

54 submitted entities found in this pathway, mapping to 55 Reactome entities

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ABCC2	P13569	ARHGAP1	Q07960	ARHGAP10	Q5T5U3
ARHGAP17	Q68EM7	ARHGAP26	Q9UNA1	ARHGAP32	A7KAX9
ARHGAP33	A7KAX9, O14559	ARHGAP5	Q13017	ARHGEF7	Q14155
ARL13B	Q3SXY8	CAV1	Q03135	CDC42	P60953
CDC42BPA	Q5VT25	CDC42BPB	Q9Y5S2	CDC42EP1	Q00587
CDC42EP2	O14613	CDC42EP3	Q9UKI2	CDC42EP4	Q9H3Q1
CPNE8	Q86YQ8	DEPDC1B	Q8WUY9	DIAPH3	Q9NSV4
DLC1	Q96QB1	FNBP1	Q96RU3	GFOD1	Q9NXC2
GIT1	Q9Y2X7	GIT2	Q14161	GJA1	P17302
GOPC	Q9HD26	IQGAP1	P46940	ITSN1	Q15811
JUP	P14923	LAMTOR1	Q6IAA8	MPP7	Q5T2T1
OBSL1	Q5VST9	OPHN1	O60890	PAK1	Q13153
PAK3	Q13153	PLEKHG3	A1L390	PREX1	Q8TCU6
RAB7A	P51149	RHOQ	P17081	SCRIB	Q14160
SLC1A5	Q15758	SLC4A7	Q9Y6M7	SNAP23	O00161
SRGAP2	O75044	STEAP3	Q658P3	STOM	P27105
SYDE1	Q6ZW31	TFRC	P02786	TRIP10	Q15642
VAMP3	Q15836	VANGL1	Q8TAA9	WASL	O00401

11. LGI-ADAM interactions (R-HSA-5682910)



Cellular compartments: extracellular region.

Synapse formation and maturation require multiple interactions between presynaptic and postsynaptic neurons. These interactions are mediated by a diverse set of synaptogenic proteins (Kegel et al. 2013, Siddiqui & Craig 2011). Initial synapse formation needs both the binding of secreted proteins to presynaptic and postsynaptic receptors, and the direct binding between presynaptic and postsynaptic transmembrane proteins. One class of molecules that plays an important role in cellular interactions in nervous system development and function is the leucine-rich glioma inactivated (LGI) protein family. These are secreted synaptogenic proteins consisting of an LRR (leucine-rich repeat) domain and an epilepsy-associated or EPTP (epitempin) domain (Gu et al. 2002). Both protein domains are generally involved in protein-protein interactions. Genetic and biochemical evidence suggests that the mechanism of action of LGI proteins involves binding to a subset of cell surface receptors belonging to the ADAM (a disintegrin and metalloproteinase) family, i.e. ADAM11, ADAM22 and ADAM23. These interactions play crucial role in the development and function of the vertebrate nervous system mainly mediating synaptic transmission and myelination (Kegel et al. 2013, Novak 2004, Seals & Courtneidge 2003).

References

Kegel L, Aunin E, Meijer D & Bermingham JR (2013). LGI proteins in the nervous system. *ASN Neuro*, 5, 167-81. [🔗](#)

Edit history

Date	Action	Author
2015-03-11	Edited	Garapati P V
2015-03-11	Authored	Garapati P V
2015-03-11	Created	Garapati P V
2015-04-20	Reviewed	Meijer D
2025-11-15	Modified	Weiser JD

13 submitted entities found in this pathway, mapping to 13 Reactome entities

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
ADAM11	O75078	ADAM22	Q9P0K1	ADAM23	O75077
CACNG3	O60359	CACNG4	Q9UBN1	CACNG8	Q8WXS5
DLG4	P78352	LGI1	O95970	LGI2	Q8N0V4
LGI3	Q8N145	LGI4	Q8N135	STX1A	Q16623

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
STX1B	P61266				

12. Loss of Function of TGFBR1 in Cancer ([R-HSA-3656534](#))

TGFBR1 LBD Mutants in Cancer



TGFBR1 KD Mutants in Cancer

Diseases: cancer.

TGF-beta receptor 1 (TGFBR1) loss-of-function is a less frequent mechanism for inactivation of TGF-beta signaling in cancer compared to SMAD4 and TGFBR2 inactivation. Genomic deletion of TGFBR1 locus has been reported in pancreatic cancer (Goggins et al. 1998), biliary duct cancer (Goggins et al. 1998) and lymphoma (Schiemann et al. 1999), while loss-of-function mutations have been reported in breast (Chen et al. 1998) and ovarian cancer (Chen et al. 2001), metastatic head-and-neck cancer (Chen et al. 2001), and in Ferguson-Smith tumors (multiple self-healing squamous epithelioma - MSSE) (Goudie et al. 2011). Loss-of-function mutations mainly affect the ligand-binding extracellular domain of TGFBR1 and the kinase domain of TGFBR1 (Goudie et al. 2011). In the mouse model of colorectal cancer, *Tgfr1* haploinsufficiency cooperates with *Apc* haploinsufficiency in the development of intestinal tumors (Zeng et al. 2009).

References

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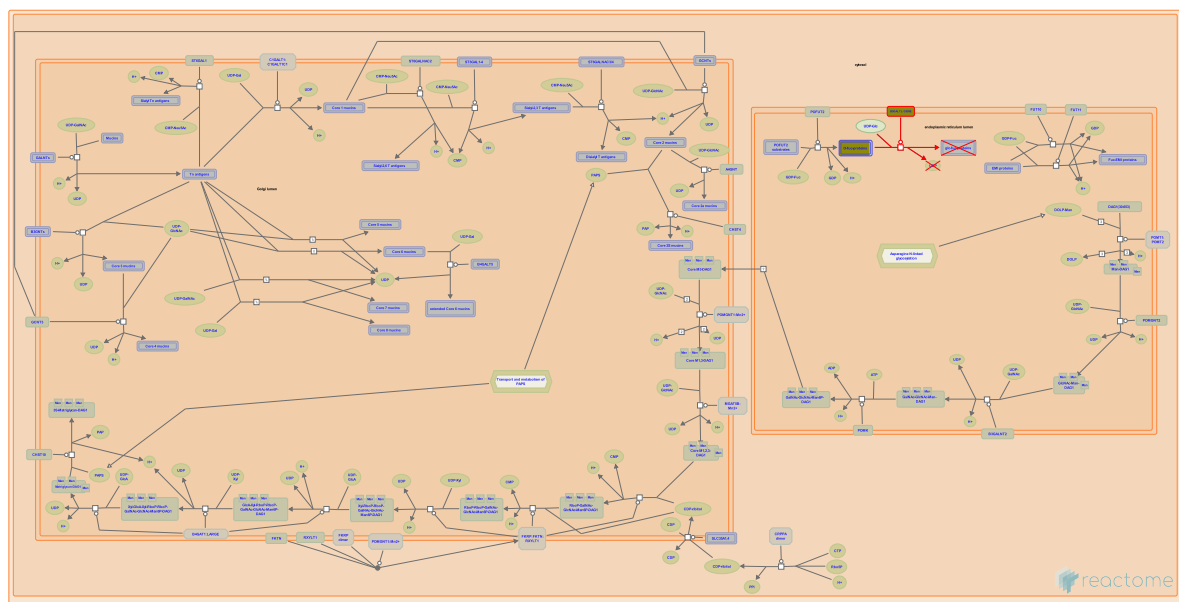
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2013-08-08	Edited	Orlic-Milacic M
2013-08-08	Reviewed	Meyer S, Akhurst RJ
2013-08-08	Authored	Meyer S, Akhurst RJ, Orlic-Milacic M
2023-10-12	Modified	Weiser JD

8 submitted entities found in this pathway, mapping to 8 Reactome entities

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FKBP1A	P62942	FPR1	P62942	SMAD2	Q15796
SMAD3	P84022	TGFB1	P01137	TGFBR1	P36897
TGFBR2	P37173	ZFYVE9	O95405-1		

13. Defective B3GALTL causes PpS (R-HSA-5083635)



Diseases: eye disease, orofacial cleft.

Human beta-1,3-glucosyltransferase like protein (B3GALTL, HGNC Approved Gene Symbol: B3GLCT; MIM:610308; CAZy family GT31), localised on the ER membrane, glucosylates O-fucosylated proteins. The resultant glc-beta-1,3-fuc disaccharide modification on thrombospondin type 1 repeat (TSR1) domain-containing proteins is thought to assist in the secretion of many of these proteins from the ER lumen, and mediate an ER quality-control mechanism of folded TSRs (Vasudevan et al. 2015). Defects in B3GALTL can cause Peters plus syndrome (PpS; MIM:261540), an autosomal recessive disorder characterised by anterior eye chamber defects, short stature, delay in growth and mental developmental and cleft lip and/or palate (Heinonen & Maki 2009).

References

Heinonen TY & Maki M (2009). Peters'-plus syndrome is a congenital disorder of glycosylation caused by a defect in the beta1,3-glucosyltransferase that modifies thrombospondin type 1 repeats. *Ann. Med.*, 41, 2-10. [🔗](#)

Vasudevan D, Takeuchi H, Johar SS, Majerus E & Haltiwanger RS (2015). Peters plus syndrome mutations disrupt a noncanonical ER quality-control mechanism. *Curr. Biol.*, 25, 286-95. [🔗](#)

Edit history

Date	Action	Author
2013-11-07	Edited	Jassal B
2013-11-07	Authored	Jassal B
2013-11-07	Created	Jassal B
2015-12-18	Reviewed	Hansen L, Joshi HJ
2023-10-12	Modified	Weiser JD

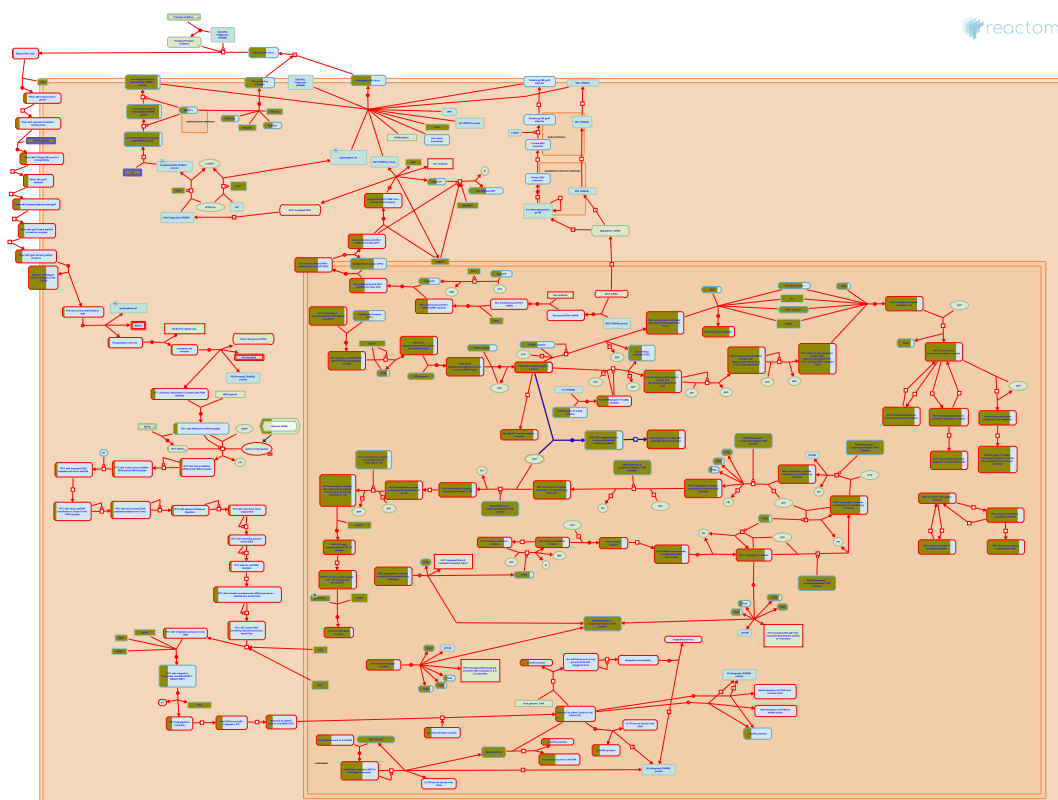
35 submitted entities found in this pathway, mapping to 35 Reactome entities

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ADAMTS1	Q9UHI8	ADAMTS10	Q9H324	ADAMTS12	P58397
ADAMTS13	Q76LX8	ADAMTS14	Q8WXS8	ADAMTS15	Q8TE58
ADAMTS16	Q8TE57	ADAMTS17	Q8TE56	ADAMTS18	Q8TE60
ADAMTS19	Q8TE59	ADAMTS2	O95450	ADAMTS3	O15072
ADAMTS4	O75173	ADAMTS5	Q9UNA0	ADAMTS6	Q9UKP5
ADAMTS7	Q9UKP4	ADAMTS8	Q9UP79	ADAMTS9	Q9P2N4
ADAMTSL1	Q8N6G6	ADAMTSL2	Q86TH1	ADAMTSL3	P82987
ADAMTSL4	Q6UY14	ADAMTSL5	Q6ZMM2	B3GLCT	Q6Y288
SBSPON	Q8IVN8	SEMA5A	Q13591	SPON1	Q9HCB6
SPON2	Q9BUD6	SSPO	A2VEC9	THBS1	P07996
THBS2	P35442	THSD1	Q9NS62	THSD4	Q6ZMP0
THSD7A	Q9UPZ6	THSD7B	Q9C0I4		

Interactors found in this pathway (3)

Input	UniProt Id	Interacts with	Input	UniProt Id	Interacts with
ADAMTS18	Q8TE60	Q6Y288	C6	P13671	Q6Y288
THBS2	P35442	Q6Y288			

14. Abortive elongation of HIV-1 transcript in the absence of Tat (R-HSA-167242)



Cellular compartments: nucleoplasm.

Diseases: Human immunodeficiency virus infectious disease.

This event was inferred from the corresponding Reactome human Pol II transcription elongation event. The details specific to HIV-1 transcription elongation are described below. In the absence of the HIV-1 Tat protein, the RNA Pol II complexes associated with the HIV-1 template are non-processive. RNA Pol II is arrested after promoter clearance by the negative transcriptional elongation factors DSIF and NELF as occurs during early elongation of endogenous templates (Wada et al, 1998; Yamaguchi et al. 1999). This arrest cannot be overcome by P-TEFb mediated phosphorylation in the absence of Tat however, and elongation aborts resulting in the accumulation of short transcripts (Kao et al., 1987).

References

- Kao SY, Calman AF, Luciw PA & Peterlin BM (1987). Anti-termination of transcription within the long terminal repeat of HIV-1 by tat gene product. *Nature*, 330, 489-93. [🔗](#)
- Wada T, Takagi T, Yamaguchi Y, Ferdous A, Imai T, Hirose S, ... Handa H (1998). DSIF, a novel transcription elongation factor that regulates RNA polymerase II processivity, is composed of human Spt4 and Spt5 homologs. *Genes Dev*, 12, 343-56. [🔗](#)
- Yamaguchi Y, Takagi T, Wada T, Yano K, Furuya A, Sugimoto S, ... Handa H (1999). NELF, a multisubunit complex containing RD, cooperates with DSIF to repress RNA polymerase II elongation. *Cell*, 97, 41-51. [🔗](#)

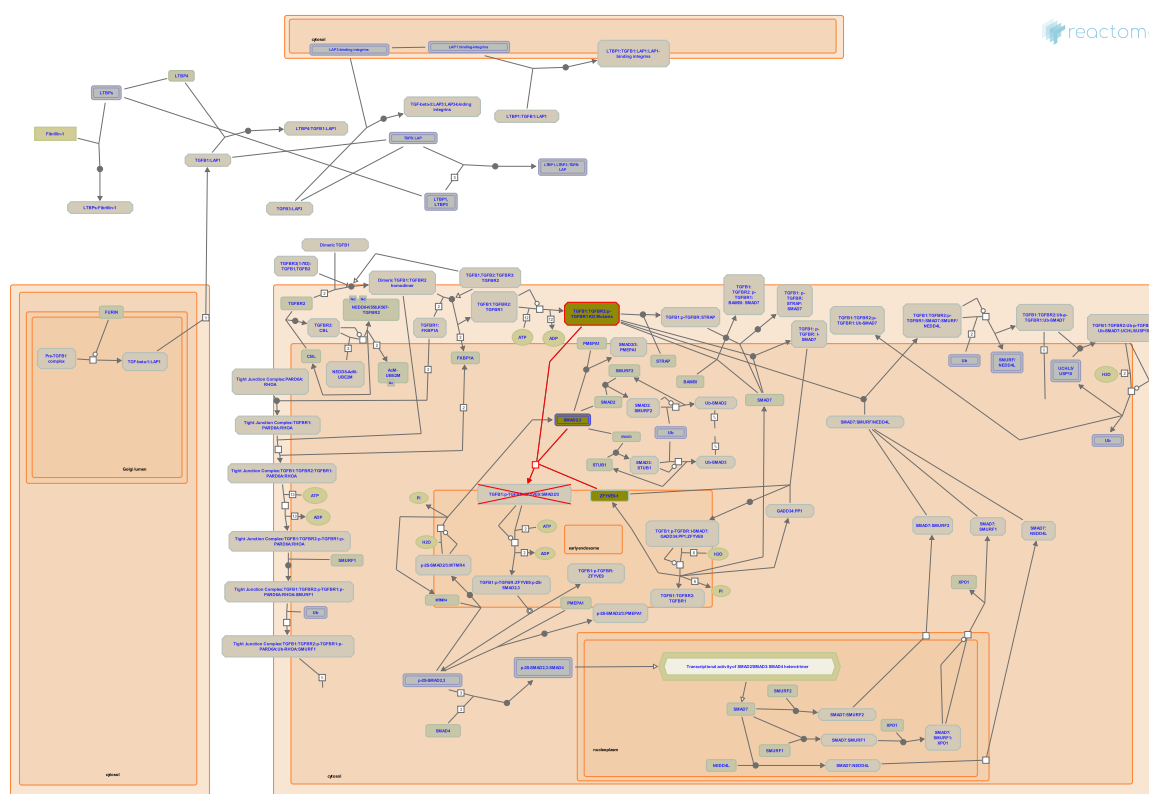
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Date	Action	Author
2005-10-14	Created	Matthews L
2023-10-12	Modified	Weiser JD

21 submitted entities found in this pathway, mapping to 21 Reactome entities

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
CTDP1	Q9Y5B0	GTF2F1	P35269	GTF2F2	P13984
NCBP2	P52298	NELFB	Q8WX92	NELFCD	Q8IXH7
NELFE	P18615	POLR2A	P24928	POLR2B	P30876
POLR2C	P19387	POLR2D	O15514	POLR2E	P19388
POLR2F	P61218	POLR2G	P62487	POLR2H	P52434
POLR2I	P36954	POLR2J	P52435	POLR2K	P53803
POLR2L	P62875	SUPT4H1	P63272	SUPT5H	O00267

15. TGFBR1 KD Mutants in Cancer (R-HSA-3656532)



Diseases: cancer.

Mutations in the kinase domain (KD) of TGF-beta receptor 1 (TGFBR1) have been found in Ferguson-Smith tumor i.e. multiple self-healing squamous epithelioma - MSSE (Goudie et al. 2011), breast cancer (Chen et al. 1998), ovarian cancer (Chen et al. 2001) and head-and-neck cancer (Chen et al. 2001). KD mutations reported in MSSE are nonsense and frameshift mutations that cause premature termination of TGFBR1 translation, resulting in truncated receptors that lack substantial portions of the kinase domain, or cause nonsense-mediated decay of mutant transcripts. A splice site KD mutation c.806-2A>C is predicted to result in the skipping of exon 5 and the absence of KD amino acid residues 269-324 from the mutant receptor. The splice site mutant is expressed at the cell surface but unresponsive to TGF-beta stimulation (Goudie et al. 2004).

TGFBR1 KD mutations reported in breast, ovarian and head-and-neck cancer are missense mutations, and it appears that these mutant proteins are partially functional but that their catalytic activity or protein stability is decreased (Chen et al. 1998, Chen et al. 2001a and b). These mutants are not shown.

References

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Chen T, Carter D, Garrigue-Antar L & Reiss M (1998). Transforming growth factor beta type I receptor kinase mutant associated with metastatic breast cancer. *Cancer Res.*, 58, 4805-10. [↗](#)

Chen T, Yan W, Wells RG, Rimm DL, McNiff J, Leffell D & Reiss M (2001). Novel inactivating mutations of transforming growth factor-beta type I receptor gene in head-and-neck cancer metastases. *Int. J. Cancer*, 93, 653-61. [↗](#)

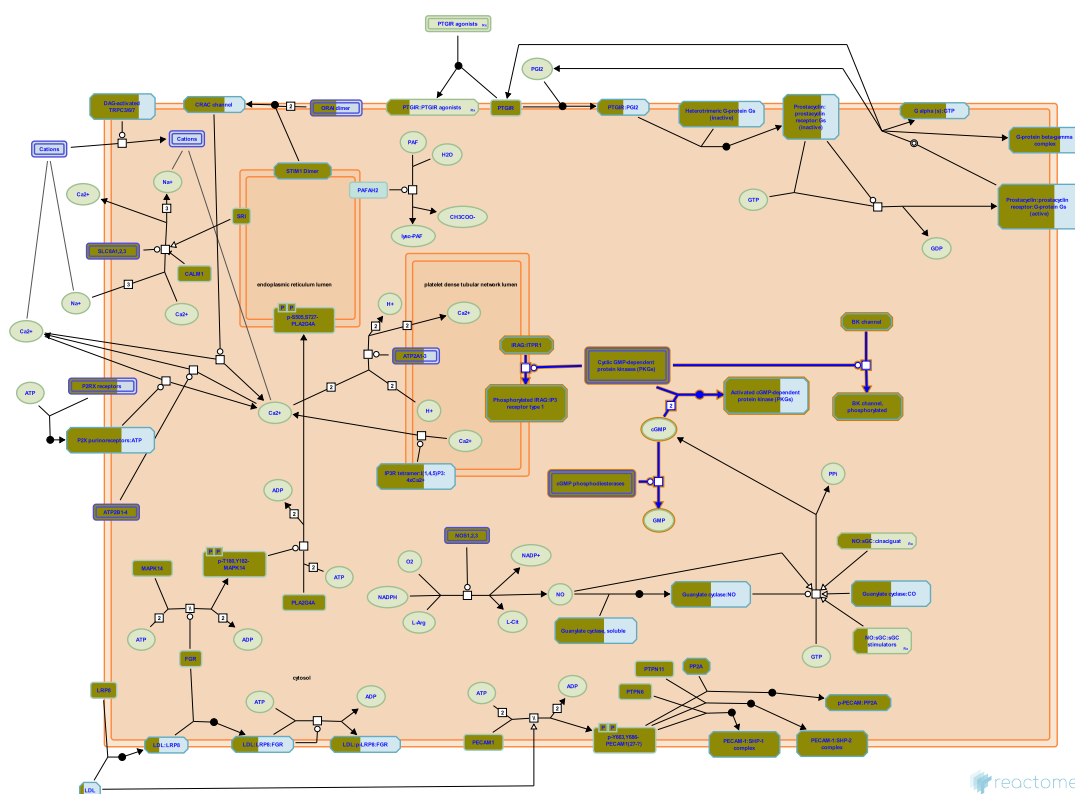
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Date	Action	Author
2013-06-05	Created	Orlic-Milacic M
2013-08-08	Edited	Orlic-Milacic M
2013-08-08	Reviewed	Meyer S, Akhurst RJ
2013-08-08	Authored	Meyer S, Akhurst RJ, Orlic-Milacic M

6 submitted entities found in this pathway, mapping to 6 Reactome entities

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
SMAD2	Q15796	SMAD3	P84022	TGFB1	P01137
TGFBR1	P36897	TGFBR2	P37173	ZFYVE9	O95405-1

16. cGMP effects (R-HSA-418457)



Cellular compartments: cytosol.

Cyclic guanosine monophosphate (cGMP) is an important secondary messenger synthesized by guanylate cyclases. cGMP has effects on phosphodiesterases (PDE), ion-gated channels, and the cGMP-dependent protein kinases (cGK, Protein Kinase G or PKG). It is involved in regulation of several physiological functions including vasodilation, platelet aggregation and neurotransmission. Elevation of intracellular cGMP activates PKG (Haslam et al. 1999) which regulates several intracellular molecules and pathways including the vasodilator-stimulated phosphoprotein (VASP) (Halbrugge et al. 1990) and the ERK pathway (Hood and Granger 1998, Li et al. 2001). cGMP mediates nitric oxide (NO)-induced vascular smooth muscle relaxation (Furchgott and Vanhoutte 1989). Phosphodiesterase 5 (PDE5) hydrolyzes cGMP; the PDE5 inhibitor sildenafil (Viagra) increases intracellular cGMP and thereby can be used as a treatment for erectile dysfunction (Corbin and Francis 1999). The role of the cGMP and PKG in platelet activation was controversial as increases in platelet cGMP levels were observed in response to both platelet agonists (thrombin, ADP or collagen) and inhibitors (NO donors such as sodium nitroprusside), but it is currently accepted that PKG inhibits platelet activation (Haslam et al. 1999). Consistent with this, nitric oxide (NO) donors that inhibit platelet activation enhance intracellular cGMP (Haslam et al. 1999). cGMP also plays an important stimulatory role in GPIb-IX-mediated platelet activation. Platelet responses to cGMP have been proposed to be biphasic, consisting of an early stimulatory response that promotes platelet activation followed by a delayed platelet inhibition that serves to limit the size of platelet aggregates (Li et al 2003).

References

Münzel T, Feil R, Mülsch A, Lohmann SM, Hofmann F & Walter U (2003). Physiology and pathophysiology of vascular signaling controlled by guanosine 3',5'-cyclic monophosphate-dependent protein kinase [corrected]. *Circulation*, 108, 2172-83. <https://doi.org/10.1161/01.CIR.0000081111.06663.8F>

Warner TD, Mitchell JA, Sheng H & Murad F (1994). Effects of cyclic GMP on smooth muscle relaxation. Adv Pharmacol, 26, 171-94. [🔗](#)

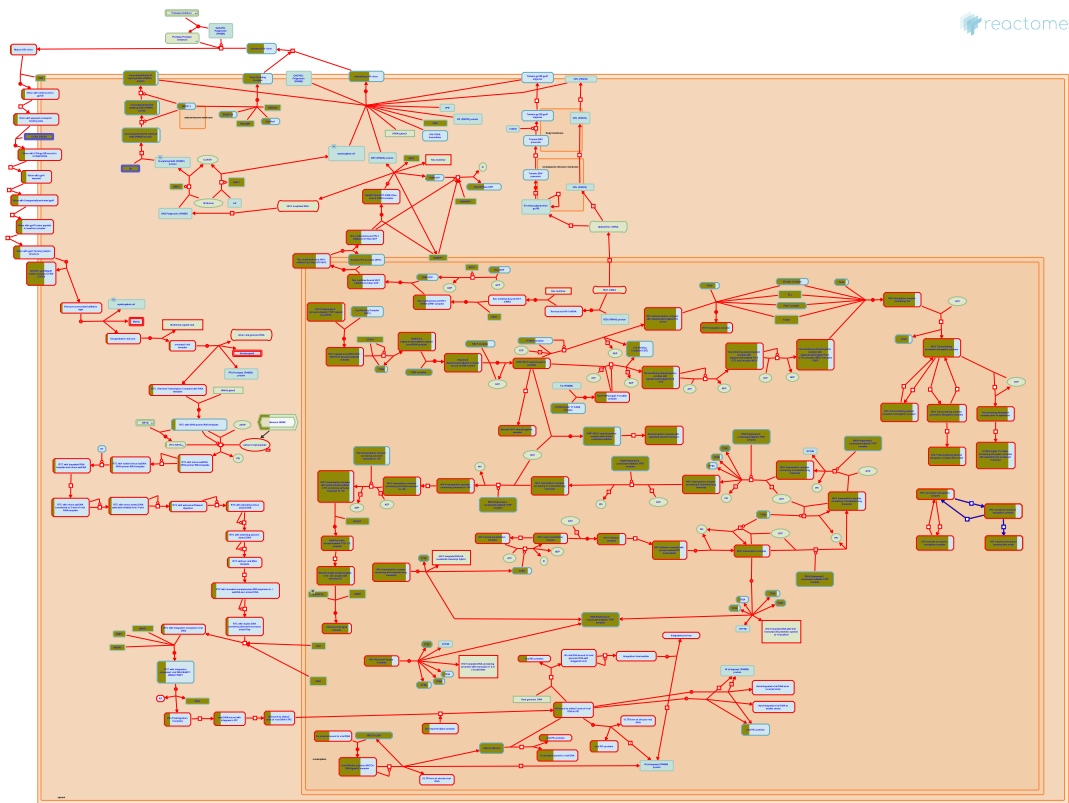
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Date	Action	Author
2009-04-22	Created	Jupe S
2009-06-03	Authored	Akkerman JW
2010-06-07	Edited	Jupe S
2010-06-07	Reviewed	Kunapuli SP
2025-11-15	Modified	Weiser JD

16 submitted entities found in this pathway, mapping to 17 Reactome entities

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
ITPR1	Q14643	KCNMA1	Q12791	KCNMB1	Q16558
KCNMB3	Q9NPA1, Q9Y691	KCNMB4	Q86W47	MRV11	Q9Y6F6
PDE10A	Q9Y233	PDE11A	Q9HCR9	PDE1A	P54750
PDE1B	Q01064	PDE1C	Q01064	PDE2A	O00408
PDE5A	O76074	PDE9A	O76083	PRKG1	Q13976-1
PRKG2	Q13237				

17. HIV elongation arrest and recovery (R-HSA-167287)



Cellular compartments: nucleoplasm.

Diseases: Human immunodeficiency virus infectious disease.

RNA Pol II arrest is believed to be a result of irreversible backsliding of the enzyme by ~7-14 nucleotides. TFIIS reactivates arrested RNA Pol II by promoting the excision of nascent transcript ~7-14 nucleotides upstream of the 3' end.

References

Edit history

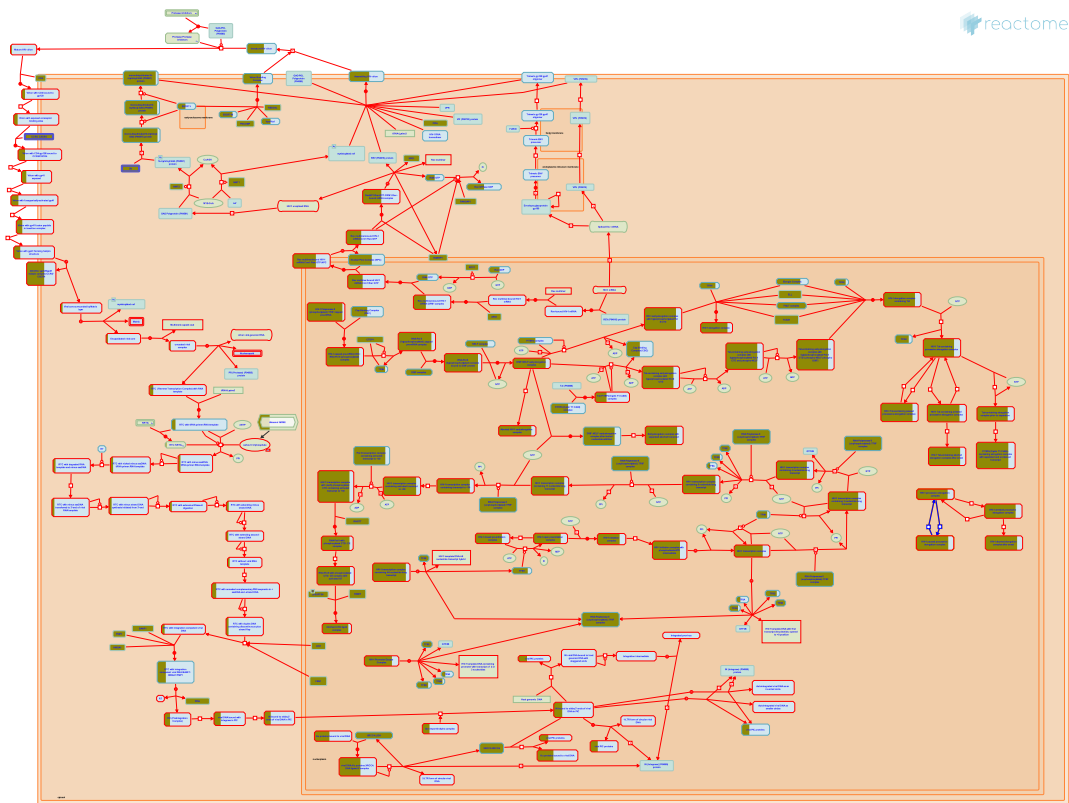
Date	Action	Author
2005-10-14	Created	Matthews L
2023-10-12	Modified	Weiser JD

29 submitted entities found in this pathway, mapping to 29 Reactome entities

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CCNT2	O60583	CDK9	P50750	CTDP1	Q9Y5B0
ELL	P55199	GTF2F1	P35269	GTF2F2	P13984
NELFB	Q8WX92	NELFCD	Q8IXH7	NELFE	P18615
POLR2A	P24928	POLR2B	P30876	POLR2C	P19387
POLR2D	O15514	POLR2E	P19388	POLR2F	P61218
POLR2G	P62487	POLR2H	P52434	POLR2I	P36954
POLR2J	P52435	POLR2K	P53803	POLR2L	P62875
SSRP1	Q08945	SUPT16H	Q9Y5B9	SUPT4H1	P63272
SUPT5H	O00267	TCEA1	P23193	TCEB1	Q15369

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
TCEB2	Q15370	TCEB3	Q14241		

18. Pausing and recovery of HIV elongation (R-HSA-167290)



Cellular compartments: nucleoplasm.

Diseases: Human immunodeficiency virus infectious disease.

After Pol II pauses by back tracking 2 -4 nucleotides on the HIV-1 template, elongation of the HIV-1 transcript resumes.

References

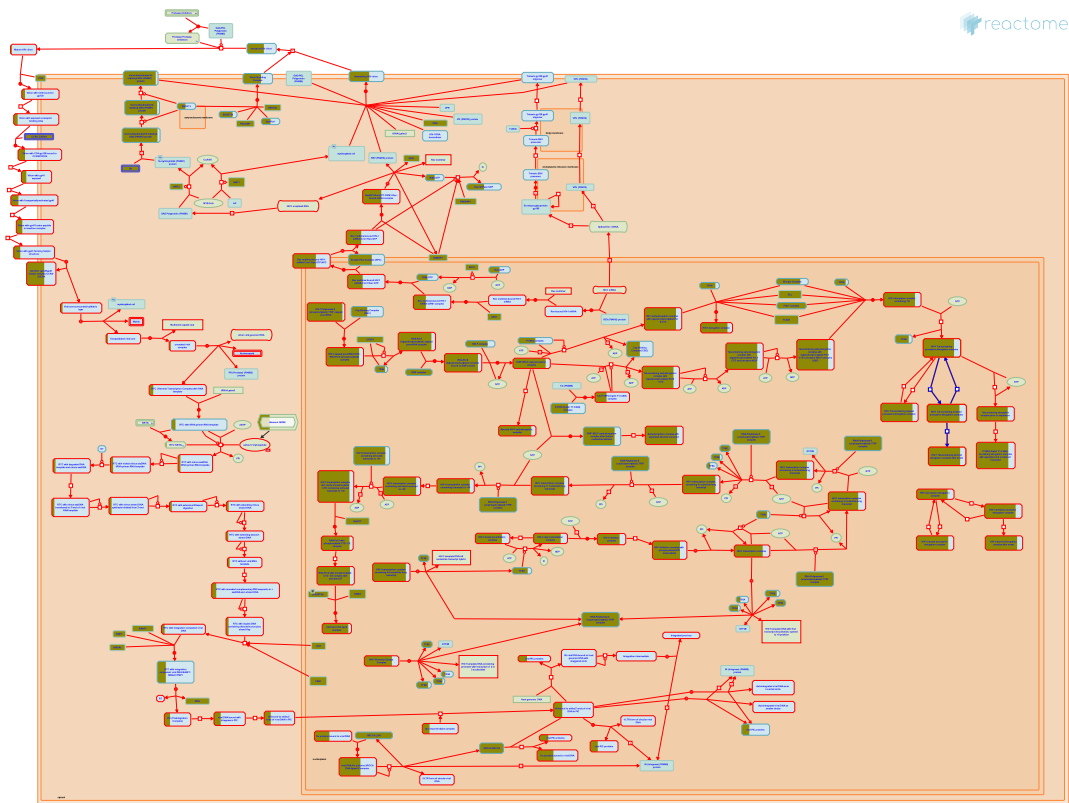
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Date	Action	Author
2005-10-14	Created	Matthews L
2023-10-12	Modified	Weiser JD

29 submitted entities found in this pathway, mapping to 29 Reactome entities

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
CCNT2	O60583	CDK9	P50750	CTDP1	Q9Y5B0
ELL	P55199	GTF2F1	P35269	GTF2F2	P13984
NELFB	Q8WX92	NELFCD	Q8IXH7	NELFE	P18615
POLR2A	P24928	POLR2B	P30876	POLR2C	P19387
POLR2D	O15514	POLR2E	P19388	POLR2F	P61218
POLR2G	P62487	POLR2H	P52434	POLR2I	P36954
POLR2J	P52435	POLR2K	P53803	POLR2L	P62875
SSRP1	Q08945	SUPT16H	Q9Y5B9	SUPT4H1	P63272
SUPT5H	O00267	TCEA1	P23193	TCEB1	Q15369
TCEB2	Q15370	TCEB3	Q14241		

19. Tat-mediated HIV elongation arrest and recovery (R-HSA-167243)



Cellular compartments: nucleoplasm.

Diseases: Human immunodeficiency virus infectious disease.

RNA Pol II arrest is believed to be a result of irreversible backsliding of the enzyme by ~7-14 nucleotides. TFIIS reactivates arrested RNA Pol II by promoting the excision of nascent transcript ~7-14 nucleotides upstream of the 3' end.

References

Edit history

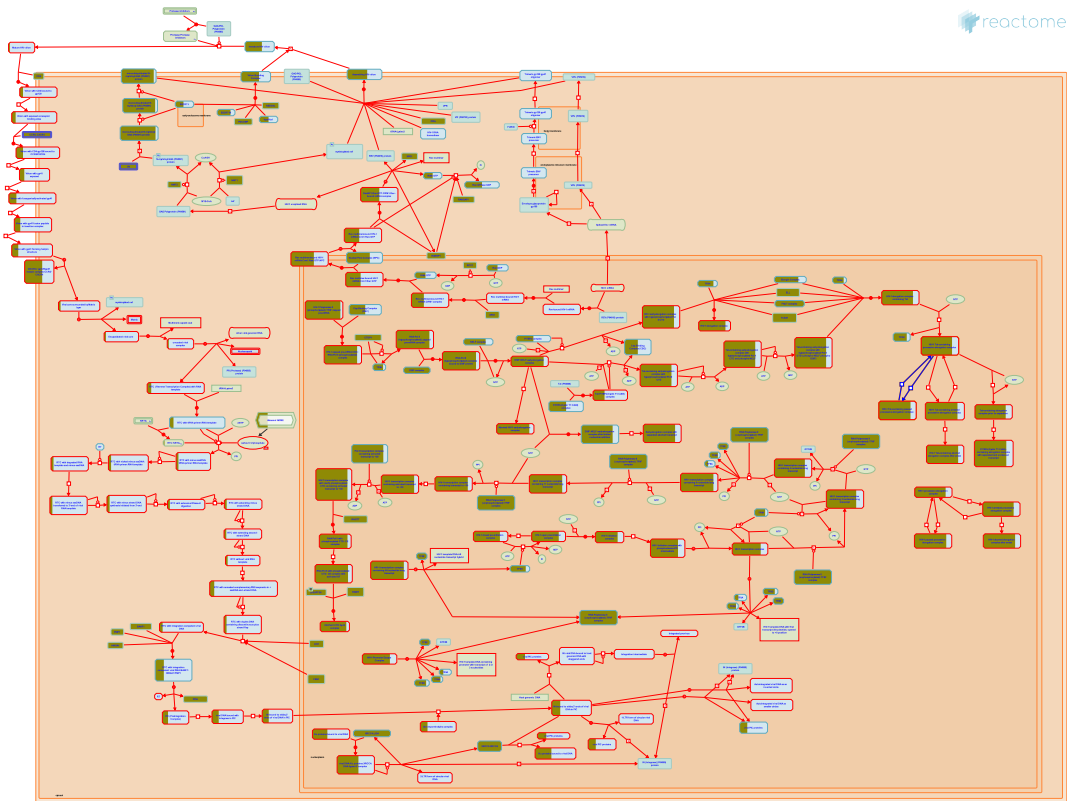
Date	Action	Author
2005-10-14	Created	Matthews L
2023-10-12	Modified	Weiser JD

28 submitted entities found in this pathway, mapping to 28 Reactome entities

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
CDK9	P50750	CTDP1	Q9Y5B0	ELL	P55199
GTF2F1	P35269	GTF2F2	P13984	NELFB	Q8WX92
NELFCD	Q8IXH7	NELFE	P18615	POLR2A	P24928
POLR2B	P30876	POLR2C	P19387	POLR2D	O15514
POLR2E	P19388	POLR2F	P61218	POLR2G	P62487
POLR2H	P52434	POLR2I	P36954	POLR2J	P52435
POLR2K	P53803	POLR2L	P62875	SSRP1	Q08945
SUPT16H	Q9Y5B9	SUPT4H1	P63272	SUPT5H	O00267
TCEA1	P23193	TCEB1	Q15369	TCEB2	Q15370

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
TCEB3	Q14241				

20. Pausing and recovery of Tat-mediated HIV elongation (R-HSA-167238)



Cellular compartments: nucleoplasm.

Diseases: Human immunodeficiency virus infectious disease.

After Pol II pauses by back tracking 2 -4 nucleotides on the HIV-1 template, elongation of the HIV-1 transcript resumes.

References

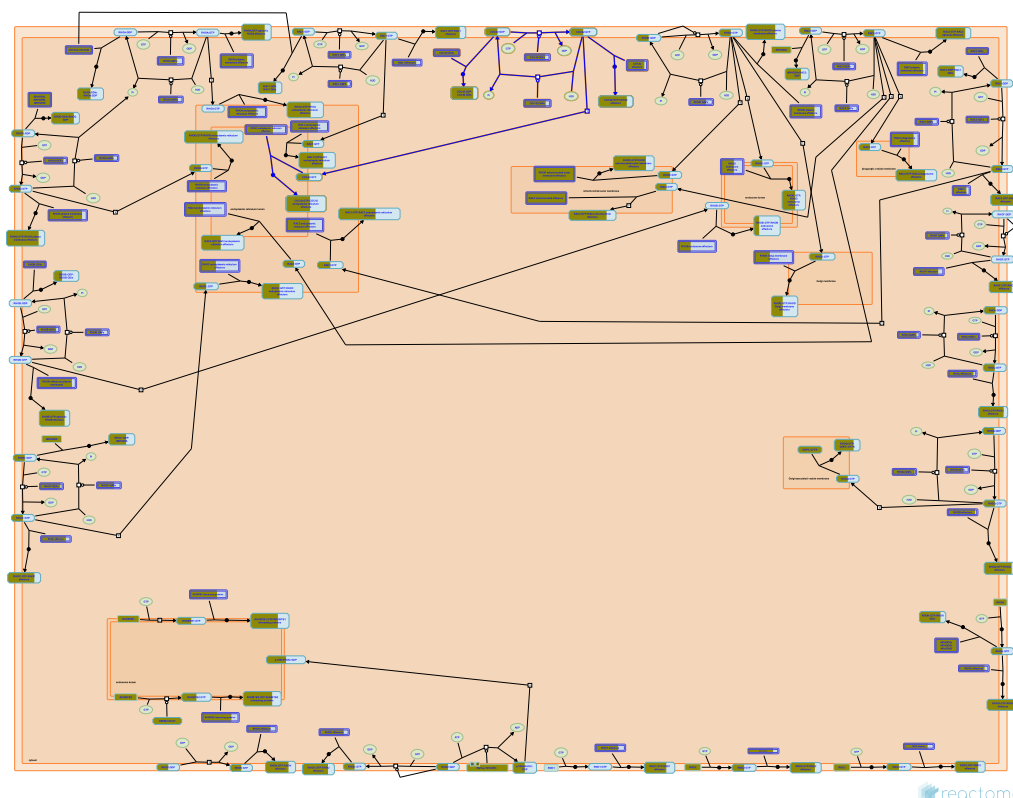
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Date	Action	Author
2005-10-14	Created	Matthews L
2023-10-12	Modified	Weiser JD

28 submitted entities found in this pathway, mapping to 28 Reactome entities

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
CDK9	P50750	CTDP1	Q9Y5B0	ELL	P55199
GTF2F1	P35269	GTF2F2	P13984	NELFB	Q8WX92
NELFCD	Q8IXH7	NELFE	P18615	POLR2A	P24928
POLR2B	P30876	POLR2C	P19387	POLR2D	O15514
POLR2E	P19388	POLR2F	P61218	POLR2G	P62487
POLR2H	P52434	POLR2I	P36954	POLR2J	P52435
POLR2K	P53803	POLR2L	P62875	SSRP1	Q08945
SUPT16H	Q9Y5B9	SUPT4H1	P63272	SUPT5H	O00267
TCEA1	P23193	TCEB1	Q15369	TCEB2	Q15370
TCEB3	Q14241				

21. CDC42 GTPase cycle (R-HSA-9013148)



This pathway catalogues CDC42 guanine nucleotide exchange factors (GEFs), GTPase activator proteins (GAPs), GDP dissociation inhibitors (GDIs) and CDC42 effectors. CDC42 is one of the three best characterized RHO GTPases, the other two being RHOA and RAC1. By regulating the cytoskeleton, CDC42 regulates cell polarity across different species, from yeast to humans (Pichaud et al. 2019, Woods and Lew 2019). CDC42 is an essential regulator of polarized morphogenesis in epithelial cells, where it coordinates formation of the apical membrane and lumen formation, as well as junction maturation (Pichaud et al. 2019). CDC42 plays a role in cell-to-cell adhesion and cell cycle regulation (Xiao et al. 2018). CDC42 takes part in the regulation of membrane trafficking. Dysfunction of several CDC42-specific GEFs has been shown to impair intracellular trafficking (Egorov and Polishchuk 2017). CDC42 participates in insulin synthesis and secretion and contributes to the pathogenesis of insulin resistance and diabetic nephropathy (Huang et al. 2019). CDC42 is often dysregulated in cancer because a number of GEFs and GEF activators that act upstream of RAC1 and CDC42 are known oncogenes (Aguilar et al. 2017; Maldonado et al. 2018; Zhang et al. 2019; Maldonado et al. 2020). CDC4 promotes cancer cell proliferation, survival, invasion, migration and metastasis (Xiao et al. 2018), especially under hyperglycemia (Huang et al. 2019).

References

- Maldonado MDM, Medina JI, Velazquez L & Dharmawardhane S (2020). Targeting Rac and Cdc42 GEFs in Metastatic Cancer. *Front Cell Dev Biol*, 8, 201. [↗](#)
- Pichaud F, Walther RF & Nunes de Almeida F (2019). Regulation of Cdc42 and its effectors in epithelial morphogenesis. *J. Cell. Sci.*, 132. [↗](#)
- Zhang Y, Li J, Lai XN, Jiao XQ, Xiong JP & Xiong LX (2019). Focus on Cdc42 in Breast Cancer: New Insights, Target Therapy Development and Non-Coding RNAs. *Cells*, 8. [↗](#)

Huang QY, Lai XN, Qian XL, Lv LC, Li J, Duan J, ... Xiong LX (2019). Cdc42: A Novel Regulator of Insulin Secretion and Diabetes-Associated Diseases. *Int J Mol Sci*, 20. [↗](#)

Maldonado MDM & Dharmawardhane S (2018). Targeting Rac and Cdc42 GTPases in Cancer. *Cancer Res.*, 78, 3101-3111. [↗](#)

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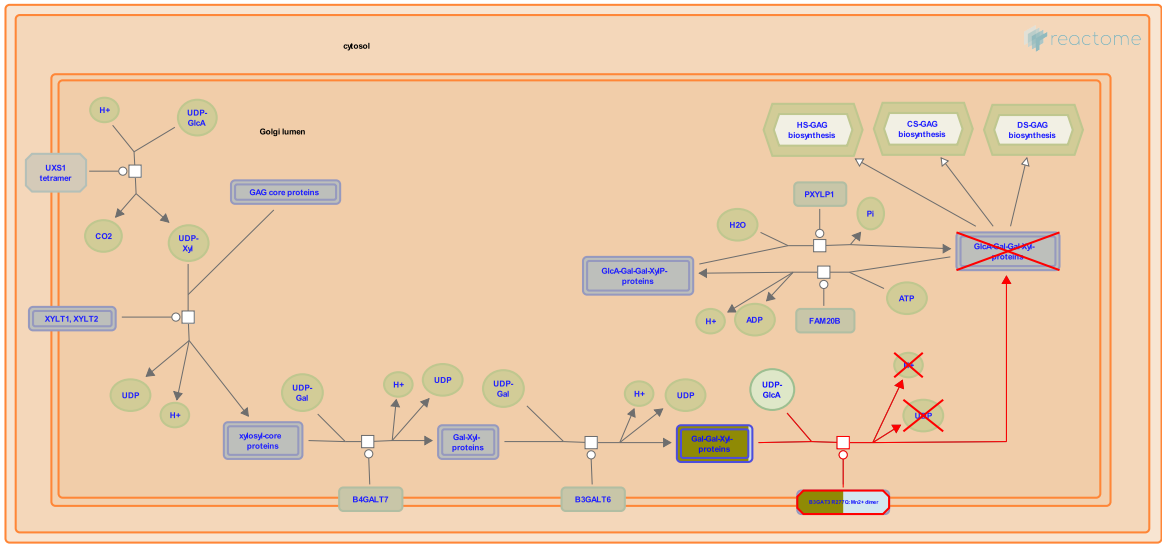
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2017-07-24	Created	Orlic-Milacic M
2020-07-14	Authored	Rothfels K, Orlic-Milacic M
2021-02-05	Reviewed	Fort P
2021-02-25	Edited	Orlic-Milacic M
2021-05-17	Reviewed	Das ME
2021-06-10	Revised	Orlic-Milacic M
2021-06-10	Edited	Orlic-Milacic M
2025-11-15	Modified	Weiser JD

129 submitted entities found in this pathway, mapping to 134 Reactome entities

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ARAP1	Q96P48	ARAP2	Q8WZ64	ARAP3	Q8WWN8
ARFGAP1	Q9NP61	ARFGAP2	Q8N6H7	ARFGAP3	Q9NP61
ARHGAP1	Q07960	ARHGAP10	A1A4S6, Q5T5U3	ARHGAP11B	Q3KRB8
ARHGAP17	Q68EM7	ARHGAP20	Q9P2F6	ARHGAP24	Q8N264
ARHGAP26	Q9UNA1	ARHGAP27	Q6ZUM4	ARHGAP29	Q52LW3
ARHGAP30	Q7Z6I6	ARHGAP31	Q2M1Z3	ARHGAP32	A7KAX9
ARHGAP33	A7KAX9, O14559	ARHGAP4	P98171	ARHGAP42	A6NI28
ARHGAP44	Q17R89	ARHGAP5	Q13017	ARHGAP9	Q9BRR9
ARHGDIA	P52565	ARHGDIB	P52566	ARHGDIG	Q99819
ARHGEF10	O15013	ARHGEF11	O15085	ARHGEF12	Q9NZN5
ARHGEF16	Q5VV41	ARHGEF26	Q96DR7	ARHGEF4	Q9NR80
ARHGEF6	Q15052	ARHGEF7	Q14155	CAV1	Q03135
CDC42	P60953	CDC42BPA	Q5VT25	CDC42BPB	Q9Y5S2
CDC42EP1	Q00587	CDC42EP2	O14613	CDC42EP3	Q9UKI2
CDC42EP4	Q9H3Q1	CDC42EP5	Q6NZY7	CDC42SE2	Q9NRR3
CHN1	P15882	CPNE8	Q86YQ8	DEF6	Q9H4E7
DEPDC1B	Q8WUY9	DIAPH3	Q9NSV4	DLC1	Q96QB1
DNMBP	Q6XZF7	DOCK10	Q96BY6	DOCK11	Q5JSL3
DOCK6	Q96HP0	DOCK7	Q96N67	DOCK8	Q8NF50
DOCK9	Q9BZ29	ECT2	Q9H8V3	FAM13B	Q9NYF5
FARP1	Q9Y4F1	FGD1	P98174	FGD2	Q7Z6J4
FGD3	Q5JSP0	FGD4	Q96M96	FHOD3	Q8IVF7
FMNL1	O95466	FMNL2	Q96PY5	FMNL3	Q8IVF7
FNBP1	Q96RU3	FNBP1L	Q5T0N5	GIT1	Q9Y2X7
GIT2	Q14161	GMIP	Q9P107	GNA13	Q14344
HMHA1	Q92619	IQGAP1	P46940	IQGAP2	Q13576
ITSN1	Q15811	JUP	P14923	KCTD3	Q9Y597

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
KTN1	Q86UP2	LAMTOR1	Q6IAA8	MAP3K11	Q16584
MCF2	P10911	MCF2L	O15068	MYO9B	Q13459
NGEF	Q8N5V2	OPHN1	O60890	PAK1	Q13153
PAK3	O75914, Q13153	PAK7	Q9P286	PARD6A	Q9NPB6
PIK3R1	P27986	PLD1	Q13393	PLEKHG1	Q9ULL1
PLEKHG2	Q9H7P9	PLEKHG3	A1L390	PLEKHG4	Q58EX7
PREX1	Q8TCU6	PREX2	Q70Z35	RAB7A	P51149
RASGRF2	O14827	SCRIB	Q14160	SH3PXD2A	Q5TCZ1
SHKBP1	Q8TBC3	SNAP23	O00161	SRGAP1	Q7Z6B7
SRGAP2	O43295, O75044	SRGAP3	O43295	STARD13	Q9Y3M8
STARD8	Q92502	STEAP3	Q658P3	STOM	P27105
SYDE1	Q6ZW31	TAGAP	Q8N103	TFRC	P02786
TIAM1	Q13009	TMEM133	A6NI28	TMPO	P42167
TRIO	O75962, Q96PX9	VAMP3	Q15836	VANGL1	Q8TAA9
VAV2	P52735	WAS	P42768	WASL	O00401
WDR91	A4D1P6	WIPF1	O43516	YKT6	O15498

22. Defective B3GAT3 causes JDSSDHD (R-HSA-3560801)



Diseases: congenital heart defect, Larsen syndrome.

Galactosylgalactosylxylosylprotein 3-beta-glucuronosyltransferases1, 2 and 3 (B3GAT1-3) are involved in forming the linker tetrasaccharide present in heparan sulfate and chondroitin sulfate. Defects in B3GAT3 cause multiple joint dislocations, short stature, craniofacial dysmorphism, and congenital heart defects (JDSSDHD; MIM:245600). This is an autosomal recessive disease characterized by dysmorphic facies, elbow, hip and knee dislocations, clubfeet, short stature and cardiovascular defects (Steel & Kohl 1972, Bonaventure et al. 1992, Baasanjav et al. 2011). JDSSDHD has phenotypic similarities to Larsen syndrome (Larsen et al. 1950).

References

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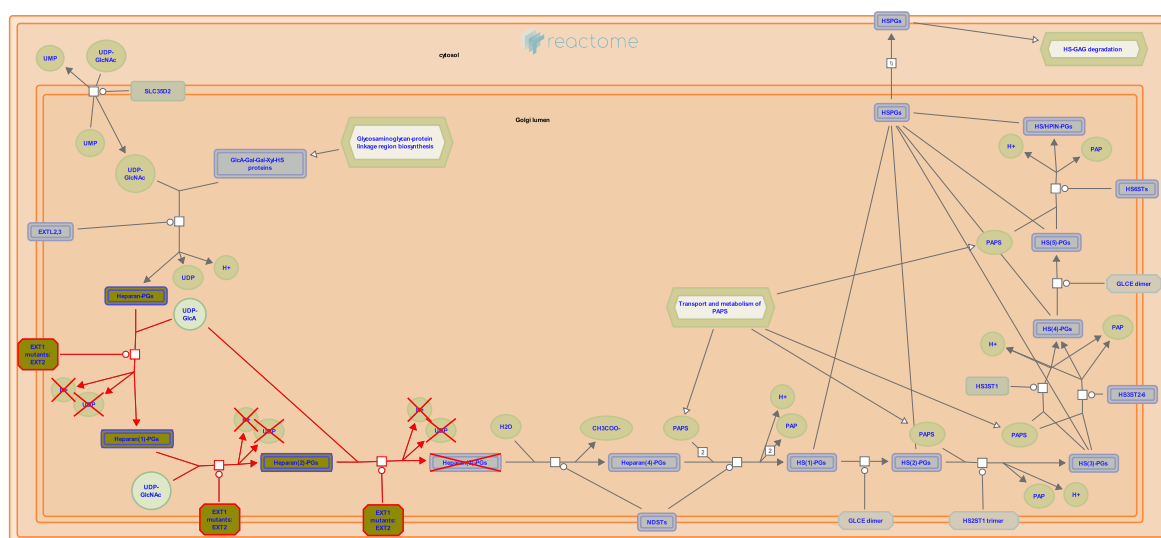
Edit history

Date	Action	Author
2013-05-21	Edited	Jassal B
2013-05-21	Authored	Jassal B
2013-05-21	Created	Jassal B
2014-07-09	Reviewed	Spillmann D
2025-06-02	Modified	Stephan R

19 submitted entities found in this pathway, mapping to 19 Reactome entities

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
AGRN	O00468	B3GAT3	O94766	BCAN	Q96GW7
BGN	P21810	CSPG4	Q6UVK1	DCN	P07585
GPC1	P35052	GPC2	Q8N158	GPC3	P51654
GPC4	O75487	GPC5	P78333	GPC6	Q9Y625
HSPG2	P98160	NCAN	O14594	SDC1	P18827
SDC2	P34741	SDC3	O75056	SDC4	P31431
VCAN	P13611				

23. Defective EXT1 causes exostoses 1, TRPS2 and CHDS ([R-HSA-3656253](#))



Diseases: hereditary multiple exostoses.

Heparan sulfate (HS) is involved in regulating various body functions during development, homeostasis and pathology including blood clotting, angiogenesis and metastasis of cancer cells. Exostosin 1 and 2 (EXT1 and 2) glycosyltransferases are required to form HS. They are able to transfer N-acetylglucosamine (GlcNAc) and glucuronate (GlcA) to HS during its synthesis. The functional form of these enzymes appears to be a complex of the two located on the Golgi membrane. Defects in either EXT1 or EXT2 can cause hereditary multiple exostoses 1 (Petersen 1989) and 2 (McGaughran et al. 1995) respectively (MIM:133700 and MIM:133701), autosomal dominant disorders characterized by multiple projections of bone capped by cartilage resulting in deformed legs, forearms and hands. Trichorhinophalangeal syndrome, type II (TRPS2 aka Langer-Giedion syndrome, LGS) is a disorder that combines the clinical features of trichorhinophalangeal syndrome type I (TRPS1, MIM:190350) and multiple exostoses type I, caused by mutations in the TRPS1 and EXT1 genes, respectively (Langer et al. 1984, Ludecke et al. 1995). Defects in EXT1 may also be responsible for chondrosarcoma (CHDS; MIM:215300) (Schajowicz & Bessone 1967, Hecht et al. 1995).

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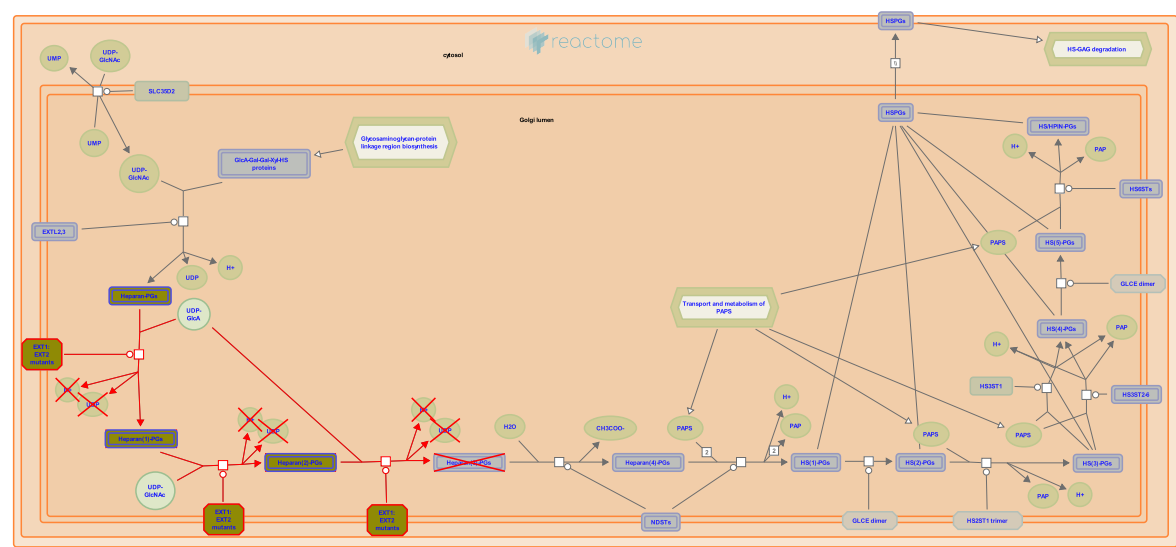
Edit history

Date	Action	Author
2013-05-31	Edited	Jassal B
2013-05-31	Authored	Jassal B
2013-05-31	Created	Jassal B
2014-07-09	Reviewed	Spillmann D
2025-05-30	Reviewed	Matthews L
2025-06-02	Modified	Stephan R

15 submitted entities found in this pathway, mapping to 15 Reactome entities

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
ACCS	Q93063	AGRN	O00468	EXT1	Q16394
EXT2	Q93063	GPC1	P35052	GPC2	Q8N158
GPC3	P51654	GPC4	O75487	GPC5	P78333
GPC6	Q9Y625	HSPG2	P98160	SDC1	P18827
SDC2	P34741	SDC3	O75056	SDC4	P31431

24. Defective EXT2 causes exostoses 2 ([R-HSA-3656237](#))



Diseases: hereditary multiple exostoses.

Heparan sulfate (HS) is involved in regulating various body functions during development, homeostasis and pathology including blood clotting, angiogenesis and metastasis of cancer cells. Exostosin 1 and 2 (EXT1 and 2) glycosyltransferases are required to form HS. They are able to transfer N-acetylglucosamine (GlcNAc) and glucuronate (GlcA) to HS during its synthesis. The functional form of these enzymes appears to be a complex of the two located on the Golgi membrane. Defects in either EXT1 or EXT2 can cause hereditary multiple exostoses 1 (Petersen 1989) and 2 (McGaughran et al. 1995) respectively (MIM:133700 and MIM:133701), autosomal dominant disorders characterised by multiple projections of bone capped by cartilage resulting in deformed legs, forearms and hands.

References

Peterson HA (1989). Multiple hereditary osteochondromata. Clin. Orthop. Relat. Res., 222-30. [🔗](#)

McGaughran JM, Ward HB & Evans DG (1995). WAGR syndrome and multiple exostoses in a patient with del(11)(p11.2p14.2). J. Med. Genet., 32, 823-4. [🔗](#)

Edit history

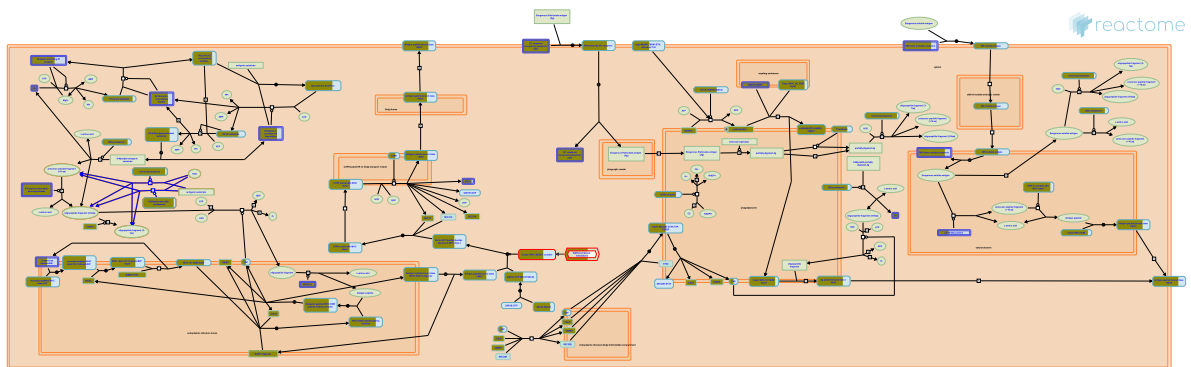
Date	Action	Author
2013-05-31	Edited	Jassal B
2013-05-31	Authored	Jassal B
2013-05-31	Created	Jassal B
2014-07-09	Reviewed	Spillmann D
2025-05-30	Reviewed	Matthews L
2025-06-02	Modified	Stephan R

15 submitted entities found in this pathway, mapping to 15 Reactome entities

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
ACCS	Q93063	AGRN	O00468	EXT1	Q16394

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
EXT2	Q93063	GPC1	P35052	GPC2	Q8N158
GPC3	P51654	GPC4	O75487	GPC5	P78333
GPC6	Q9Y625	HSPG2	P98160	SDC1	P18827
SDC2	P34741	SDC3	O75056	SDC4	P31431

25. Antigen processing: Ub, ATP-independent proteasomal degradation ([R-HSA-9912633](#))



Intracellular foreign or aberrant host proteins are cleaved into peptide fragments of a precise size, such that they can be loaded on to class I major histocompatibility complex (MHC I) molecules and presented externally to cytotoxic CD8⁺ T lymphocytes (CTLs) (reviewed by Embgenbroich M & Burgdorf S 2018). In the cytosol, CTL-recognized peptides are generated by proteasomes, which exist as a mixture of compositionally distinct complexes (reviewed by Kloetzel PM 2004; Habib JA et al., 2022). The proteasome complex consists of 20S proteolytic core particle (CP) associated with a regulatory particle (RP) such as 19S RP or PA28 RP (Zhao J et al. 2022; Adolf F et al., 2024; reviewed by Watanabe A et al., 2022). The catalytic subunits PSMB6 (Ø1), PSMB7 (Ø2), and PSMB5 (Ø5) within 20S CP exhibit caspase-like, trypsin-like, and chymotrypsin-like activities, respectively (Vigneron N et al. 2015; Rut W & Drag M 2016). During peptide bond cleavage, threonine from the active site of the catalytic Ø-subunit binds to the N-terminal peptide fragment forming an unstable acyl-enzyme intermediate, where a peptide fragment remains attached to the proteasome. Subsequently, a water molecule hydrolyzes the bond between the peptide and the proteasome, resulting in the release of the peptide's C-terminal end (Mishto M et al. 2012; Ebstein F et al. 2016; Soh WT et al., 2024). Either one or two regulatory particles (RP) can attach to the 20S CP (Zhao J et al. 2022; Adolf F et al., 2024; reviewed by Watanabe A et al., 2022). The 19S RP binds 20S CP to form the 26S proteasome, which degrades both foreign and self-proteins in the ubiquitin (Ub)- and ATP-dependent manner (Zhu Y et al., 2018; Dong Y et al., 2019; Zhang S et al., 2022; reviewed by Sahu I & Glickman MH 2021a). In addition, a subset of intracellular proteins, especially intrinsically disordered proteins, can directly bind to 20S CP for Ub- and ATP-independent proteasomal degradation (Makaros Y et al., 2023; Mamrosh JL et al., 2023; Pepelnjak M et al., 2024; reviewed by Sahu I & Glickman MH 2021b; Bialek W et al., 2023). Unlike 19S, the PA28 regulatory particle does not have a ubiquitin receptor; binding of PA28 to the 20S CP promotes Ub-, ATP-independent protein breakdown. PA28 proteasomes are thought to be important for rapid degradation of misfolded proteins under conditions of oxidative stress or degradation of proteins with intrinsically disordered regions (Habib JA et al., 2020; Chen J et al., 2021; Zhao J et al. 2022; reviewed by Thomas T et al. 2023). Various immune cells, such as T cells, B cells, and antigen-presenting cells, constitutively express a specialized form of proteasomes, known as immunoproteasomes, in which the catalytic subunits PSMB6 (Ø1), PSMB7 (Ø2), and PSMB5 (Ø5) of 20S CP are substituted with PSMB9 (Ø1i), PSMB10 (Ø2i), and PSMB8 (Ø5i), respectively (Shin EC et al., 2006; Bai M et al., 2014; Santos R et al., 2017). This variation of the core particle is known as 20S iCP. Immunoproteasomes utilize the PA28 regulatory particle to degrade antigens (Lesne J et al., 2020; Chen J et al., 2021). Under inflammatory conditions, the expression of the catalytic and PA28 regulatory subunits of immunoproteasome is induced by cytokines like interferon-gamma (IFNØ), type I interferons, and tumor necrosis factor alpha (TNF-Ø). Immunoproteasomes possess enhanced chymotrypsin- and trypsin-like activities, alongside with the reduced caspase-like activity. These distinct enzymatic properties of immunoproteasomes generate substrate cleavage patterns that enhance loading of peptides onto the class I MHC for immune presentation to CTL (reviewed by Tomko RJ and Hochstrasser M 2013).

This Reactome module describes Ub-independent antigen processing by PA28-20S CP and PA28-20S iCP proteasomes.

References

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Edit history

Date	Action	Author
2024-06-10	Created	Shamovsky V
2024-06-17	Authored	Shamovsky V
2025-02-05	Edited	Shamovsky V
2025-11-15	Modified	Weiser JD

17 submitted entities found in this pathway, mapping to 18 Reactome entities

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
PSMA1	P25786	PSMA3	P25788	PSMA4	P25789
PSMA5	P28066	PSMA6	P60900	PSMA7	O14818
PSMB1	P20618	PSMB2	P49721	PSMB3	P49720
PSMB4	P28070	PSMB5	P28074	PSMB6	P28072
PSMB7	P40306, Q99436	PSMB8	P28062	PSMB9	P28065
PSME2	Q9UL46	PSME3	Q06323		

6. Identifiers found

Below is a list of the input identifiers that have been found or mapped to an equivalent element in Reactome, classified by resource.

7464 of the submitted entities were found, mapping to 8626 Reactome entities

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AACS	Q86V21	AADAT	Q8N5Z0	AAK1	Q2M2I8
AAMP	Q13685	AANAT	Q16613	AARS	P49588
AASDHPPT	Q9NRN7	AASS	Q9UDR5	ABCA1	O95477
ABCA10	Q8WWZ4	ABCA13	Q86UQ4	ABCA3	Q99758
ABCA4	P78363	ABCA5	Q8IUA7, Q8WWZ7	ABCA6	Q8N139
ABCA7	Q8IZY2	ABCA8	O94911	ABCA9	Q8IUA7, Q8WWZ4
ABCB1	P21439	ABCB10	Q9NRK6	ABCB4	P21439
ABCB7	O75027	ABCB8	Q9NUT2	ABCC1	P33527
ABCC10	Q5T3U5	ABCC2	P13569	ABCC3	O15438
ABCC4	O15439	ABCC6	O95255	ABCC9	O60706
ABCD1	P33897	ABCD2	P33897	ABCD3	P28288
ABCD4	O14678	ABCF1	Q8NE71	ABCF2	Q9UG63
ABCG1	P45844	ABCG2	Q9UNQ0	ABCG4	P45844
ABHD10	Q9NUJ1	ABHD17A	Q96GS6	ABHD17C	Q6PCB6
ABHD3	Q8WU67	ABHD4	Q8TB40	ABI1	Q8IZP0
ABI2	Q9NYB9	ABL2	P42684	ABLIM2	Q6H8Q1
ABLIM3	O94929	ABO	P16442	ACAA1	P09110
ACAA2	P42765	ACACB	O00763	ACAD10	Q6JQN1
ACAD8	Q9UKU7	ACADL	P28330	ACADS	P16219
ACADSB	P45954	ACAN	P16112	ACAT1	P24752
ACAT2	Q9BWD1	ACBD6	Q9BR61	ACCS	Q93063
ACD	Q96AP0	ACE	P06276	ACER2	Q5QJU3
ACER3	Q9NUN7	ACHE	P22303	ACIN1	Q9UKV3
ACKR1	Q16570	ACKR3	P25106	ACLY	P53396
ACO1	Q99798	ACO2	Q99798	ACOT11	Q8WXI4
ACOT12	Q8WYK0	ACOT2	P49753	ACOT7	O00154
ACOT8	O14734	ACOT9	Q9Y305	ACOX1	O15254
ACOX2	Q99424	ACOXL	Q9NUZ1	ACP1	O14561
ACP2	Q9BT43	ACP6	Q9NPH0	ACPP	P15309
ACSBG1	Q96GR2	ACSF2	Q96CM8	ACSF3	Q4G176
ACSL1	P33121, Q9ULC5	ACSL4	O60488	ACSL5	Q9ULC5
ACSM5	Q6NUN0	ACSS1	Q9NUB1	ACSS2	Q9NR19
ACSS3	Q9H6R3	ACTA1	P68133	ACTA2	P62736
ACTB	P60709, P63261	ACTG1	P60709, P63261	ACTG2	P63267
ACTL6A	O96019	ACTL6B	O94805	ACTN1	P12814
ACTN2	P35609	ACTN4	O43707	ACTR10	Q9NZ32
ACTR1A	P61163	ACTR1B	P42025	ACTR2	P61160
ACTR5	Q9H9F9	ACTR8	Q9H981	ACVR1B	P36896
ACVR1C	Q8NER5	ACVR2A	P27037	ACVR2B	Q13705
ACVRL1	P37023	ACY3	Q96HD9	ADA	P00813

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ADAL	Q6DHV7	ADAM10	O14672	ADAM11	O75078
ADAM12	O43184	ADAM17	P78536	ADAM22	Q9P0K1
ADAM23	O75077	ADAM33	Q9BZ11	ADAM9	Q99965
ADAMTS1	Q9UHI8	ADAMTS10	Q9H324	ADAMTS12	P58397
ADAMTS13	Q76LX8	ADAMTS14	Q8WXS8	ADAMTS15	Q8TE58
ADAMTS16	Q8TE57	ADAMTS17	Q8TE56	ADAMTS18	Q8TE60
ADAMTS19	Q8TE59	ADAMTS2	O95450	ADAMTS3	O15072
ADAMTS4	O75173	ADAMTS5	Q9UNA0	ADAMTS6	Q9UKP5
ADAMTS7	Q9UKP4	ADAMTS8	Q9UP79	ADAMTS9	Q9P2N4
ADAMTSL1	Q8N6G6	ADAMTSL2	Q86TH1	ADAMTSL3	P82987
ADAMTSL4	Q6UY14	ADAMTSL5	Q6ZMM2	ADAP1	O75689
ADARB1	P78563	ADAT2	Q7Z6V5	ADCK3	Q8NI60
ADCY1	Q08828	ADCY2	Q08462	ADCY3	O60266
ADCY4	Q8NFM4	ADCY8	P40145	ADCY9	O60503
ADCYAP1	P18509	ADCYAP1R1	P41586	ADD1	P35611
ADD2	P35612	ADD3	Q9UEY8	ADGRE1	Q14246
ADGRE2	Q9UHX3	ADGRE5	P48960	ADGRF5	Q8IZF2
ADGRG6	Q86SQ4	ADGRV1	Q8WXG9	ADH1A	P07327
ADH1B	P00325, P28332	ADH1C	P00325, P00326	ADH5	P08319, P11766
ADHFE1	Q8IWW8	ADI1	Q9BV57	ADIPOQ	Q15848
ADIPOR1	Q96A54	ADIPOR2	Q86V24	ADIRF	Q15847
ADK	P55263	ADM	P35318	ADM2	Q7Z4H4
ADO	Q96SZ5	ADORA2B	P29275	ADPRHL2	Q9NX46
ADRA1A	P25100, P35348	ADRA1B	P35368	ADRA1D	P25100
ADRB2	P07550	ADRBK1	P25098	ADRBK2	P35626
ADRM1	Q16186	ADSL	P30566	AEBP2	Q6ZN18
AES	Q08117	AFF4	Q9UHB7	AFG3L2	Q9Y4W6
AGAP2	Q99490	AGBL1	Q96MI9	AGBL2	Q5U5Z8
AGBL3	Q8NEM8	AGBL4	Q5VU57	AGBL5	Q8NDL9
AGFG1	P52594	AGMO	Q6ZNB7	AGO2	Q9H9G7, Q9UKV8
AGO3	Q9H9G7, Q9UL18	AGPAT1	O15120	AGPAT4	Q9NRZ5
AGPAT6	Q86UL3	AGPAT9	Q53EU6	AGPS	O00116
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AKR7A2	O43488, O95154	AKT1	P31749	AKT1S1	Q96B36
AKT3	Q9Y243	ALAS1	P13196	ALAS2	P22557
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ALDH1B1	P30837	ALDH1L1	O75891	ALDH2	P05091
ALDH3B1	P43353	ALDH5A1	P51649	ALDH6A1	Q02252
ALDH7A1	P49419	ALDH8A1	Q9H2A2	ALDH9A1	P49189

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ALG8	Q9BVK2	ALG9	Q9H6U8	ALK	Q9UM73
ALKBH2	Q6NS38	ALMS1	Q8TCU4	ALOX15B	O15296
ALOX5	P09917	ALOX5AP	P20292	ALOXE3	Q9BYJ1
ALPK1	Q96QP1	ALPL	P05186	ALS2CL	Q60I27
ALYREF	Q86V81	AMBRA1	Q9C0C7	AMD1	Q01432
AMDHD2	Q9Y303	AMER1	Q5JTC6	AMH	P03971
AMICA1	Q86YT9	AMIGO2	Q86SJ2	AMOT	Q4VCS5-1
AMOTL1	Q8IY63	AMPD2	Q01433	AMPD3	Q01432
AMPH	Q9UBW5	AMY2B	P19961	ANAPC1	Q9H1A4
ANAPC10	Q9UM13	ANAPC11	Q9NYG5	ANAPC13	Q9BS18
ANAPC15	P60006	ANAPC2	Q9UJX6	ANAPC4	Q9UJX5
ANAPC7	Q9UJX3	ANGPT1	Q15389	ANGPT2	O15123
ANGPTL4	Q9BY76	ANK3	Q12955	ANKH	Q9HCJ1
ANKRD26	Q9UPS8	ANKRD27	Q96NW4	ANKRD28	O15084
ANKRD44	Q9H672	ANKZF1	Q9H8Y5	ANO1	Q5XXA6
ANO10	Q9NW15	ANO2	Q9NQ90	ANO3	Q9BYT9
ANO4	Q32M45	ANO5	Q75V66	ANO6	Q4KMQ2
ANO8	Q9HCE9	ANOS1	P23352	ANP32A	P39687
ANPEP	P15144	ANTXR1	Q9H6X2-1, Q9H6X2-2	ANTXR2	P58335-1, P58335-4
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AP1S1	P56377, P61966	AP1S2	P56377, Q96PC3	AP1S3	Q96PC3
AP2A1	O95782	AP2A2	O94973	AP2B1	P63010
AP2M1	Q96CW1	AP2S1	P53680	AP3B1	O00203
AP3M1	Q9Y2T2	AP4B1	Q9Y6B7	AP4E1	Q9UPM8
AP4M1	O00189	AP4S1	Q9Y587	APAF1	O14727
APBA1	Q02410	APBA2	Q99767	APBA3	O96018
APBB1	O00213	APBB1IP	Q7Z5R6	APC2	Q9UJX6
APEH	P13798	APEX1	P27695	APH1A	Q96BI3
APH1B	Q8WW43	APIP	Q96GX9	APITD1	Q8N2Z9
APLN	Q9ULZ1	APLP2	Q06481	APOA1BP	Q8NCW5
APOB	P04114	APOBEC3C	Q9NRW3	APOBEC3G	Q9HC16-1, Q9HC16-3
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APOL1	O14791	APOO	Q9BUR5	APOOL	Q6UXV4
APP	P05067	APPL1	Q9UKG1	APPL2	Q06481
APRT	P07741	AQP1	P29972	AQP11	Q8NBQ7
AQP3	Q92482	AQP4	P55087	AQP7	O14520
AQP9	O14520, O43315	AQR	O60306	AR	P10275
ARAP1	Q96P48	ARAP2	Q8WZ64	ARAP3	Q8WWN8
ARC	Q7LC44	ARCN1	P48444	AREG	P01135, P15514
AREL1	O15033	ARF1	P18085	ARF3	P61204
ARF4	P18085	ARF5	P84077	ARFGAP1	Q9NP61
ARFGAP2	Q8N6H7	ARFGAP3	Q9NP61	ARFGEF2	Q9Y6D5
ARFIP2	P53365	ARFRP1	Q13795	ARG1	P05089
ARG2	P78540	ARHGAP1	Q07960	ARHGAP10	Q5T5U3

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
ARHGAP11B	Q3KRB8	ARHGAP12	Q8IWW6	ARHGAP15	Q53QZ3
ARHGAP17	Q68EM7	ARHGAP18	Q8N392	ARHGAP19	Q14CB8
ARHGAP20	Q9P2F6	ARHGAP24	Q8N264	ARHGAP25	P42331
ARHGAP26	Q9UNA1	ARHGAP27	Q6ZUM4	ARHGAP28	Q9P2N2
ARHGAP29	Q52LW3	ARHGAP30	Q7Z6I6	ARHGAP31	Q2M1Z3
ARHGAP32	A7KAX9	ARHGAP33	A7KAX9, O14559	ARHGAP4	P98171
ARHGAP42	A6NI28	ARHGAP44	Q17R89	ARHGAP5	Q13017
ARHGAP6	O43182	ARHGAP9	Q9BRR9	ARHGDIA	P52565
ARHGDIB	P52566	ARHGDIG	Q99819	ARHGEF1	Q92888
ARHGEF10	O15013	ARHGEF10L	Q9HCE6	ARHGEF11	O15085
ARHGEF12	Q9NZN5	ARHGEF16	Q5VV41	ARHGEF17	Q96PE2
ARHGEF26	Q96DR7	ARHGEF28	Q8N1W1	ARHGEF3	Q9NR81
ARHGEF37	A1IGU5	ARHGEF4	Q9NR80	ARHGEF40	Q8TER5
ARHGEF6	Q15052	ARHGEF7	Q14155	ARID1B	Q8NFD5
ARID2	Q68CP9	ARID3A	Q99856	ARID4A	P29374
ARID4B	Q4LE39	ARID5B	Q14865	ARIH1	Q9Y4X5
ARL1	P40616	ARL13B	Q3SXY8	ARL2	P36404
ARL2BP	Q9Y2Y0	ARL3	P36405	ARL4C	P56559
ARL6	Q9H0F7	ARL8A	Q96BM9	ARMC10	Q9UH62
ARMC8	Q8IUR7	ARMCX3	Q9UH62	ARNT	P27540
ARNT2	Q9HBZ2	ARNTL	O00327	ARNTL2	Q8WYA1
ARPC1A	Q92747	ARPC1B	O15143	ARPC2	O15144
ARPC3	O15145	ARPC4	P59998	ARPC5	O15511
ARRB2	P32121	ARSA	O43681	ARSB	P15848
ARSD	P51689	ARSE	P51690	ARSG	Q96EG1
ARSI	Q5FYB1	ARSJ	Q5FYB0	ARSK	Q6UWY0
ART3	Q13508	ARV1	Q9H2C2	ASAH1	Q13510
ASAH2	Q9NR71	ASAP1	Q9ULH1	ASB1	Q9Y576
ASB12	Q8WXI3, Q8WXX4	ASB13	Q8WXX3	ASB15	Q8WXX1, Q9Y575
ASB2	Q96Q27	ASB3	A6NK59, Q9Y575	ASB4	Q9Y574
ASB5	Q8WWX0	ASB6	Q9NWX5	ASB7	Q8WXX1, Q9H672
ASB8	Q9H765	ASB9	Q96DX5	ASCC1	Q8N9N2
ASCC2	Q9H1I8	ASCC3	Q8N3C0	ASF1A	Q9Y294
ASGR1	P07306	ASH1L	Q9NR48	ASH2L	Q9UBL3
ASIC1	P78348	ASIC2	Q16515	ASIC4	Q96FT7
ASL	P04424	ASNA1	O43681	ASNS	P08243
ASPA	P45381, Q86U10	ASPG	Q86U10	ASPH	Q12797
ASPN	Q9BXN1	ASRGL1	Q7L266	ASS1	P00966
ASXL2	Q76L83	ATAD1	Q8NBU5	ATAD2	Q6PL18
ATAD3A	Q5T9A4	ATAD3B	Q5T9A4	ATAT1	Q5SQI0
ATF1	P18846	ATF2	P15336	ATF3	P18847
ATF4	P18848	ATF5	Q9Y2D1	ATF6	P18850
ATF6B	Q99941	ATF7IP	Q6VMQ6	ATG10	Q9H0Y0
ATG101	Q9BSB4	ATG12	O94817	ATG13	O75143
ATG14	Q6ZNE5	ATG16L2	Q676U5, Q8NAA4	ATG4B	Q96DT6, Q9Y4P1
ATG4C	Q96DT6	ATG4D	Q86TL0	ATG5	Q9H1Y0
ATG7	O95352	ATG9A	Q7Z3C6	ATG9B	Q674R7
ATIC	P31939	ATL2	Q8NHH9	ATM	Q13315
ATN1	P54259	ATOX1	O00244	ATP10A	O60312

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ATP10B	O94823	ATP10D	Q9P241	ATP11B	Q9Y2G3
ATP11C	Q8NB49	ATP13A1	Q9HD20	ATP13A2	Q9NQ11
ATP13A4	Q4VNC0, Q4VNC1	ATP1A1	P05023	ATP1A2	P50993, Q13733
ATP1A3	P13637	ATP1A4	Q13733	ATP1B1	P05026
ATP1B2	P14415	ATP2A2	P16615	ATP2A3	Q93084
ATP2B1	P20020	ATP2B2	Q01814-2	ATP2B3	Q16720
ATP2B4	P23634	ATP2C2	O75185	ATP5A1	P25705
ATP5B	P06576	ATP5C1	P36542	ATP5D	P30049
ATP5E	P56381	ATP5F1	P24539	ATP5G1	P05496
ATP5G2	P48201, Q06055	ATP5G3	P48201, Q06055	ATP5H	O75947
ATP5I	P56385	ATP5J	P18859	ATP5J2	P56134
ATP5L	O75964	ATP5O	P48047	ATP5SL	Q9NWX81
ATP6AP1	Q15904	ATP6AP2	O75787	ATP6V0A1	Q93050
ATP6V0A2	Q9Y487	ATP6V0B	Q99437	ATP6V0C	P27449
ATP6V0D1	P61421	ATP6V0E1	O15342	ATP6V0E2	Q8NHE4
ATP6V1A	P38606	ATP6V1B2	P21281	ATP6V1D	Q9Y5K8
ATP6V1E1	P36543	ATP6V1F	Q16864	ATP6V1G1	O75348, Q96LB4
ATP6V1G2	O95670	ATP6V1H	Q9UI12	ATP7A	Q04656
ATP8A1	Q9Y2Q0	ATP8A2	Q9NTI2	ATP8B1	O43520
ATP8B2	P98198	ATP8B4	Q8TF62	ATP9A	O75110
ATP9B	O43861	ATR	Q13535	ATRN1	Q9Y2I2
ATRX	P46100	ATXN3	P54252, Q9H3M9	ATXN7	O15265
ATXN7L3	Q14CW9	AUH	Q13825	AUP1	Q9Y679
AURKAIP1	Q9NWT8	AURKB	O14965	AUTS2	Q8WXX7
AVEN	Q9NQS1	AVPR1A	P37288	AXL	P30530
AZGP1	P25311	AZIN1	O14977	B2M	P61769
B3GALNT1	O75752	B3GALNT2	Q8NCR0	B3GALT1	Q9Y5Z6
B3GALT2	O43825	B3GALT5	Q9Y2C3	B3GALT6	Q96L58
B3GAT1	Q9P2W7	B3GAT2	Q9NPZ5	B3GAT3	O94766
B3GLCT	Q6Y288	B3GNT2	Q9NY97	B3GNT4	Q9C0J1
B3GNT5	Q9BYG0	B3GNT9	Q6UX72	B3GNTL1	Q67FW5
B4GALNT1	Q00973	B4GALT1	P15291	B4GALT2	O60909
B4GALT3	O60512	B4GALT4	O60513	B4GALT5	O43286
B4GALT6	Q9UBX8	B4GALT7	Q9UBV7	B4GAT1	O43505
B9D2	Q9BPU9	BABAM1	Q9NWX8	BACE1	P56817
BACH1	Q9BX63	BAD	Q92934	BAG1	Q99933
BAG2	O95816	BAG3	O95817	BAG4	O95429
BAG5	Q9UL15	BAG6	P46379	BAIAP2L1	Q9UHR4
BAK1	Q16611	BAMBI	Q13145	BANF1	O75531
BANP	Q8N9N5	BAP1	Q99496	BARD1	Q99728
BASP1	P80723	BAX	Q07812	BBC3	Q9BXH1
BBOX1	O75936	BBS10	Q8TAM1	BBS2	Q9BXC9
BBS9	Q3SYG4	BCAN	Q96GW7	BCAP31	P51572
BCAR1	P56945	BCAT1	P54687	BCAT2	O15382
BCDIN3D	Q7Z5W3	BCKDHB	P21953	BCKDK	O14874
BCL11A	Q9H165	BCL11B	Q9C0K0	BCL2	P10415
BCL2L1	Q07817	BCL2L11	O43521	BCL6	P41182
BCL7A	Q4VC05	BCL7B	Q9BQE9	BCL7C	Q8WUZ0
BCO2	Q9BYV7	BCS1L	Q9Y276	BDH1	Q02338

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
BDH2	Q9BUT1	BDNF	P23560	BDP1	A6H8Y1
BEAN1	O00142	BEGAIN	Q9BUH8	BEST1	O76090
BEST4	O76090, Q8NFU0	BET1	O15155	BET1L	Q9NYM9
BGN	P21810	BHLHE22	Q8NFIJ8	BHLHE40	O14503
BHLHE41	Q9C0J9	BHMT	Q93088	BHMT2	Q9H2M3
BICD1	Q96G01	BID	P55957	BIN1	O00422
BIN2	Q9UBW5	BIRC2	Q13489, Q13490	BIRC3	Q13489
BIRC6	Q9NR09	BIRC7	Q96CA5	BLMH	Q13867
BLNK	Q8WV28	BLOC1S1	P78537	BLOC1S4	Q9NUP1
BLVRA	P53004	BLVRB	P30043	BLZF1	Q9H2G9
BMF	Q96LC9	BMI1	P35226	BMP2K	Q2M2I8
BMP4	P12644	BMP7	P18075	BMPR1A	P36894
BMPR1B	O00238	BMPR2	Q13873	BMS1	Q14692
BMX	P51813	BNIP1	Q12981	BNIP2	Q12982
BNIP3L	O60238	BOC	Q9BWV1	BOD1	Q8NFC6, Q96IK1
BOLA2B	Q9H3K6	BOP1	Q14137	BORA	Q6PGQ7
BPGM	P07738	BPHL	Q86WA6	BPI	P17213
BPNT1	P49441	BRAF	P15056	BRCA1	P38398
BRCA2	P51587	BRD2	P25440	BRD4	O60885
BRD8	Q9H0E9	BRD9	Q9H8M2	BRE	Q9NXR7
BRF2	Q9HAW0	BRIP1	Q9P0J6	BRK1	Q8WUW1
BRMS1	Q9HCU9	BRPF1	P55201	BRPF3	Q9ULD4
BRWD1	Q9NSI6	BRWD3	Q9NSI6	BSG	P35613
BSN	Q9UPA5	BST1	Q10588	BST2	Q10589
BTBD6	Q96KE9	BTC	P35070	BTB	P43251
BTF3	O00478, P78410	BTG1	P62324	BTG2	P78543
BTK	Q06187	BTN2A1	Q7KYR7	BTN2A2	Q8WVV5
BTN3A1	O00481	BTN3A2	P78410	BTN3A3	O00478
BTNL9	Q6UXG8	BUB1	O60566	BUB3	O43684
BUD13	Q9BRD0	BUD31	P41223	BYSL	Q13895
BZRAP1	O95153	C10orf2	Q96RR1	C11orf73	Q53FT3
C12orf65	Q9H3J6	C12orf66	Q96MD2	C12orf73	Q69YU5
C14orf142	Q9BXV9	C14orf166	Q9Y224	C14orf169	Q9H6W3
C14orf2	P56378	C16orf62	Q7Z3J2	C17orf89	A1L188
C17orf96	A6NHQ4	C19orf70	Q5XKP0	C1GALT1	Q9NS00
C1QA	P02745	C1QB	P02746	C1QBP	Q07021
C1QC	P02747	C1R	P00738	C1S	P09871
C1orf112	Q9NSG2	C1orf35	Q9BU76	C2	P06681
C20orf24	Q9BUV8	C2CD3	Q4AC94	C2CD5	Q86YS7
C2orf44	Q9H6R7	C3	P01024	C3AR1	Q16581
C4A	P0C0L4	C4B	P0C0L4	C5	P01031
C5AR1	P21730	C5AR2	Q9P296	C6	P13671
C6orf120	Q7Z4R8	C7	P10643	C7orf60	Q1RMZ1
C9orf64	Q5T6V5	C9orf78	Q9NZ63	CA1	P00915
CA12	O43570	CA13	Q8N1Q1	CA14	Q9ULX7
CA2	P00918	CA3	P07451	CA5B	P35218, Q9Y2D0
CAB39	Q9Y376	CAB39L	Q9H9S4	CABIN1	Q9Y6J0
CABLES1	Q8TDN4	CABLES2	Q9BTV7	CABP1	Q15084
CACNA1A	O00555	CACNA1B	Q00975	CACNA1C	Q13936

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
CACNA1D	Q01668	CACNA1E	Q15878	CACNA1H	O95180
CACNA1I	Q9P0X4	CACNA2D1	Q9UIU0	CACNA2D2	Q9NY47
CACNA2D3	Q8IZS8	CACNB1	Q02641	CACNB2	Q08289
CACNB3	P54284	CACNB4	O00305	CACNG3	O60359
CACNG4	Q9UBN1	CACNG7	P62955	CACNG8	Q8WXS5
CACTIN	Q8WUQ7	CAD	O76075	CADM1	Q9BY67
CADM2	Q8N3J6	CADM3	Q8N126	CALB1	P05937
CALCA	P01258, P06881, P10092	CALCB	P01258, P10092	CALCR	P30988
CALCRL	Q16602	CALD1	Q05682	CALHM1	Q8IU99
CALM1	P0DP23	CALM2	P0DP24	CALM3	P0DP23
CALR	P27797	CALU	O43852	CALY	Q92560
CAMK2A	Q9UQM7	CAMK2B	Q13554	CAMK2D	Q13554, Q13557
CAMKK2	Q96RR4	CAMKMT	Q7Z624	CAMLG	P49069
CANT1	Q8WVQ1	CAP1	Q01518	CAPG	Q9BPX3
CAPN1	P07384	CAPN10	Q9HC96	CAPN11	Q9UMQ6
CAPN13	Q6MZZ7	CAPN2	P17655	CAPN6	Q9Y6Q1
CAPN7	Q9Y6W3	CAPNS1	P04632, Q96L46	CAPZA2	P47755
CAPZB	P47756	CARD11	Q9BXL7	CARD8	Q9C000
CARD9	Q9H257	CARKD	Q8IW45	CARM1	Q86X55
CARNMT1	Q8N4J0	CARNS1	A5YM72	CARS	P49589
CARS2	Q9HA77	CASC3	O15234	CASK	P68400
CASP1	P29466	CASP10	Q92851	CASP3	P42574
CASP4	P49662	CASP7	P55210	CASP8	Q14790
CASP8AP2	Q9UKL3	CASQ2	O14958	CAST	O15446
CAT	P04040	CATSPER2	Q96P56	CAV1	Q03135
CAV2	P51636	CBFA2T2	O43439	CBFB	Q13951
CBLB	Q13191	CBR3	O75828	CBR4	Q8N4T8
CBS	P35520	CBX1	P83916	CBX2	Q14781
CBX3	Q13185	CBX4	O00257	CBX5	P45973
CBX8	Q9HC52	CBY1	Q9Y3M2	CC2D2A	Q9P2K1
CCAR2	Q8N163	CCBL2	Q6YP21	CCDC115	Q96NT0
CCDC22	O60826	CCDC59	Q9P031	CCDC8	Q9H0W5
CCDC88C	Q9P219	CCDC94	Q9BW85	CCK	P06307
CCKAR	P32238	CCL13	Q99616	CCL2	P13500
CCL28	Q9NRJ3	CCL3	P10147	CCL4	P13236, P13501
CCL4L2	Q8NHW4	CCL5	P13501	CCNA1	P78396
CCNA2	P20248	CCND1	P24385	CCND2	P30279
CCND3	P30281	CCNE1	P24864	CCNF	P41002
CCNH	P51946	CCNL1	P49736	CCNT2	O60583
CCR1	P35228	CCR10	P46092	CCR2	P51681
CCR5	P51681	CCR9	P51686	CCRL2	O00421
CCT2	P78371	CCT3	P49368	CCT4	P50991
CCT5	P48643	CCT6A	P40227, Q92526	CCT7	Q99832
CCT8	P50990	CCZ1	P86791	CD101	Q93033
CD109	Q6YHK3	CD14	P08571	CD163	Q86VB7
CD177	Q8N6Q3	CD180	Q99467	CD2	P06729
CD200	P41217	CD200R1	Q8TD46	CD209	Q9NNX6
CD22	P20273	CD226	Q15762	CD24	P25063
CD244	Q9BZW8	CD247	P20963-1	CD274	Q9NZQ7

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
CD28	P10747	CD2AP	Q9Y5K6	CD2BP2	O95400
CD300A	Q9UGN4	CD300C	Q08708	CD300LB	A8K4G0
CD300LF	Q6UXZ3, Q8TDQ1	CD300LG	Q6UXG3	CD320	Q9NPF0
CD33	P20138	CD34	P28906-1	CD36	P16671
CD38	P28907	CD3E	P07766	CD3EAP	O15446
CD4	P01730	CD40	P25942-1	CD44	P16070
CD46	P15529	CD47	Q08722	CD48	Q96DU3
CD52	P31358	CD53	P19397	CD55	P08174
CD58	P19256	CD59	P13987	CD6	Q86VB7
CD63	P08962	CD68	P34810	CD72	P21854
CD74	P04233	CD79B	P40259	CD80	P33681
CD84	Q9UIB8	CD86	P42081	CD8A	P01732
CD8B	P10966	CD9	P21926	CD93	Q9NPNY3
CD96	P40200	CD99	P14209	CDA	P32320
CDC14A	Q9UNH5	CDC14B	O60729	CDC25A	P30304
CDC26	Q8NHZ8	CDC27	P30260	CDC34	P49427
CDC37	Q16543	CDC40	O60508	CDC42	P60953
CDC42BPA	Q5VT25	CDC42BPB	Q9Y5S2	CDC42EP1	Q00587
CDC42EP2	O14613	CDC42EP3	Q9UKI2	CDC42EP4	Q9H3Q1
CDC42EP5	Q6NZY7	CDC42SE2	Q9NRR3	CDC7	O00311
CDC73	Q6P1J9	CDH1	Q9UM11	CDH10	Q9Y6N8
CDH11	P55287, P55287-2	CDH12	P55289	CDH13	P55290
CDH15	P55291	CDH18	Q13634	CDH19	Q9H159
CDH2	P19022	CDH23	Q9H251	CDH24	Q86UP0
CDH3	P12830, P22223, P55291	CDH4	P55283	CDH5	P33151
CDH6	P55285, Q9ULB4	CDH7	Q9ULB5	CDH8	P55286
CDIPT	O14735	CDK1	P24941	CDK11B	P21127
CDK12	Q14004	CDK13	Q14004	CDK19	Q9BWU1
CDK2	P24941	CDK4	P11802	CDK5	Q00535
CDK5R1	Q15078	CDK5R2	Q13319	CDK5RAP2	Q96SN8
CDK6	Q00534	CDK8	P49336	CDK9	P50750
CDKAL1	Q5VV42	CDKN1A	P38936	CDKN1B	P46527
CDKN1C	P49918	CDKN2A	P42771	CDKN2B	P42771
CDKN2C	P42773	CDKN2D	P55273	CDON	Q4KMG0
CDS1	Q92903	CEACAM1	P13688, P31997	CEACAM8	P31997
CEBPA	P49715	CEBPB	P17676	CEBPD	P49716
CEBPG	P53567	CECR1	Q9NZK5	CEL	P19835
CEMIP	Q8WUJ3	CENPC	Q03188	CENPF	P49454
CENPJ	Q9HC77	CENPL	Q8N0S6	CENPO	Q9BU64
CENPP	Q6IPU0	CENPQ	Q7L2Z9	CENPT	Q96BT3
CEP131	Q9UPN4	CEP135	Q66GS9	CEP152	O94986
CEP162	Q5TB80	CEP164	Q9UPV0	CEP192	Q8TEP8
CEP290	O15078	CEP41	Q9BYV8	CEP63	Q96MT8
CEP70	Q8NHQ1	CEP83	Q9Y592	CEP89	Q96ST8
CEP97	Q8IW35	CEPT1	Q9Y6K0	CERK	Q8TCT0
CERS2	Q96G23	CERS5	Q8N5B7	CERS6	Q6ZMG9
CES1	P23141	CES2	P23141	CES3	P23141
CETN2	P41208	CFD	P00746	CFH	P08603, Q92496
CFI	P05156	CFL1	P23528	CFLAR	O15519-1, O15519-2

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
CH25H	O95992	CHAC1	Q9BUX1	CHADL	P46060
CHAT	P28329	CHCHD1	Q96BP2	CHCHD10	Q8WYQ3
CHCHD2	Q9Y6H1	CHCHD3	Q9NX63	CHCHD4	Q8N4Q1
CHCHD5	Q9BSY4	CHCHD7	Q9BUK0	CHD1	O14646
CHD1L	Q86WJ1	CHD3	Q12873	CHD4	Q14839
CHD5	O00258	CHD6	Q8TD26	CHD8	Q9HCK8
CHD9	Q3L8U1	CHEK1	O14757	CHEK2	O96017
CHERP	Q8IWX8	CHGA	P10645	CHGB	P05060
CHI3L1	P36222	CHID1	Q9BWS9	CHKA	P35790
CHKB	Q9Y259	CHL1	O00533	CHM	P24386, P26374
CHMP1A	Q9HD42	CHMP2A	O43633	CHMP2B	Q9UQN3
CHMP4B	Q96CF2, Q9H444	CHMP5	Q9NZZ3	CHMP6	Q96FZ7
CHMP7	Q8WUX9	CHN1	P15882	CHN2	P52757
CHPF	Q8IZ52, Q9P2E5	CHPF2	Q9P2E5	CHPT1	Q8WUD6
CHRD1	Q9BU40	CHRM2	P08172	CHRNA3	P32297
CHRNA4	P43681	CHRNA5	P30532	CHRNA6	Q15825
CHRNA7	P36544	CHRNA9	Q9UGM1	CHRNA2	P17787
CHRNA3	Q05901	CHRNA4	P30926	CHST1	O43916
CHST10	O43529	CHST11	Q9NPF2	CHST14	Q8NCH0
CHST3	Q7LGC8	CHST7	Q9NS84	CHST8	Q9H2A9
CHST9	Q7L1S5-2	CHSY1	Q86X52	CHSY3	Q70JA7, Q9P2E5
CHTF18	Q8WVB6	CHTF8	P0CG13	CHTOP	Q9Y3Y2
CIAO1	O76071	CIAPIN1	Q6FI81	CIB2	O75838
CIDEA	O60543	CIDEC	Q96AQ7	CIITA	P33076
CILP	O75339	CIR1	P38117	CIRH1A	Q969X6
CIT	O14578, O14578-3	CITED1	Q99966	CITED2	Q99967
CITED4	Q96RK1	CKAP4	Q07065	CKAP5	Q14008
CKB	P12277	CKM	P06732	CKMT1A	P12532
CKMT1B	P12532	CKMT2	P17540	CKS1B	P61024
CLASP1	Q7Z460	CLCN2	P51788	CLCN4	P51793
CLCN5	P51795	CLCNKB	P51800, P51801	CLDN1	O95832
CLDN11	O75508	CLDN14	O95500	CLDN15	P56746
CLDN16	Q9Y5I7	CLDN19	Q8N6F1	CLDN5	O00501
CLDN6	P56747	CLDN9	O95484	CLEC10A	Q8IUN9
CLEC12A	Q5QGZ9	CLEC2B	Q92478	CLEC2D	Q9UHP7
CLEC3B	P05452	CLEC4A	Q9UMR7	CLEC4E	Q9ULY5
CLEC5A	Q9NY25	CLEC7A	Q9BXN2	CLIC2	O15247
CLIC5	Q9NZA1	CLINT1	Q14677	CLIP3	Q96DZ5
CLOCK	O15516	CLP1	Q92989	CLPP	Q16740
CLTA	P09496	CLTB	P09497	CLTCL1	P53675
CLU	P10909	CLVS2	Q5SYC1	CMAS	Q8NFW8
CMBL	Q96DG6	CMC1	Q7Z7K0	CMC2	Q9NRP2
CMC4	P56277	CMKLR1	Q99788	CMTM6	Q9NX76
CNDP2	Q96KP4	CNGA1	P29973	CNGB1	Q14028
CNIH1	O95406	CNIH2	Q6PI25	CNIH3	Q8TBE1
CNKSRI	Q969H4	CNKSRI	Q8WXI2	CNN1	P51911
CNN2	Q99439	CNOT11	Q9UKZ1	CNOT2	Q9NZN8
CNOT3	O75175	CNOT4	O95628	CNOT6L	Q96LI5
CNOT7	Q9UIV1	CNOT8	Q9UFF9	CNPY3	Q9BT09

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
CNR1	P21554	CNTF	P26441	CNTFR	P26992
CNTN2	Q02246	CNTN3	Q9P232	CNTN4	Q8IWW2
CNTN6	Q9UQ52	CNTNAP1	P78357	CNTRL	Q7Z7A1
COA3	Q9Y2R0	COA6	Q5JTJ3	COASY	Q13057
COG1	Q8WTW3	COG2	Q14746	COG3	Q96JB2
COG4	Q9H9E3	COG5	Q9UP83	COG6	Q9Y2V7
COG7	P83436	COG8	Q96MW5	COL11A1	P12107
COL12A1	Q99715	COL13A1	Q5TAT6	COL14A1	Q05707
COL15A1	P39059, Q5TAT6	COL16A1	Q07092	COL17A1	Q9UMD9
COL18A1	P39060, P53420	COL19A1	Q14993, Q9BXS0	COL1A1	P02452
COL1A2	P08123	COL21A1	Q96P44	COL22A1	Q8NFW1
COL23A1	Q86Y22	COL24A1	Q17RW2	COL25A1	Q9BXS0
COL26A1	Q86Y22, Q96A83	COL27A1	Q8IZC6	COL28A1	Q2UY09
COL3A1	P02461	COL4A1	P02462, Q14055	COL4A2	P08572
COL4A3	Q01955	COL4A3BP	Q9Y5P4-2	COL4A4	P53420
COL4A5	P29400	COL4A6	Q14031	COL5A1	P20908
COL5A2	P05997	COL5A3	P25940	COL6A1	P12109
COL6A2	P12110	COL6A3	A8TX70, P12111	COL6A6	A6NMZ7
COL8A1	P27658	COL8A2	P25067	COL9A1	P20849
COL9A2	Q14055, Q9UMD9	COL9A3	Q14050	COLEC10	Q9Y6Z7
COLEC12	Q5KU26	COLGALT1	Q8NBJ5	COLGALT2	Q8IYK4
COMMD1	Q8N668	COMMD10	Q9Y6G5	COMMD2	Q86X83
COMMD3	Q9UBI1	COMMD4	Q9H0A8	COMMD5	Q9GZQ3
COMMD6	Q7Z4G1	COMMD7	Q86VX2	COMMD9	Q9P000
COMP	P49747	COMT	P21964	COPB1	P53618
COPB2	P35606	COPE	O14579	COPG2	Q9UBF2
COPRS	Q9NQ92	COPS2	P61201	COPS3	Q9UNS2
COPS4	Q9BT78	COPS5	Q92905	COPS6	Q7L5N1
COPS7A	Q9H9Q2, Q9UBW8	COPS7B	Q9H9Q2	COPS8	Q99627
COPZ1	P61923	COPZ2	Q9P299	COQ10A	Q96MF6
COQ2	Q96H96	COQ4	Q9Y3A0	COQ5	Q5HYK3
COQ6	Q9Y2Z9	COQ9	O75208	CORIN	P03951
CORO1A	P31146	COTL1	Q14019	COX14	Q96I36
COX16	P57103	COX18	Q8N8Q8	COX19	Q49B96
COX20	Q5RI15	COX4I1	P13073	COX5A	P20674
COX5B	P10606	COX6A1	P12074	COX6B1	P14854
COX6C	P09669	COX7A1	P24310	COX7A2	P14406
COX7A2L	O14548	COX7B	P24311	COX7C	P15954
COX8A	P10176	CP	P00450	CPA3	P15088
CPD	O75976	CPE	P16870	CPLX1	O14810
CPM	P14384	CPNE1	Q99829	CPNE3	O75131
CPNE6	O95741	CPNE7	Q9UBL6	CPNE8	Q86YQ8
CPPED1	Q9BRF8	CPS1	P31327	CPSF1	Q10570
CPSF2	Q9P2I0	CPSF3	Q9UKF6	CPSF3L	Q5TA45
CPSF4	O95639	CPSF6	Q16630-1, Q16630-2	CPT1A	P50416
CPT2	P23786	CPTP	Q5TA50	CR1	P17927
CRIL	P17927	CRABP1	P29762	CRABP2	P29373
CRACR2A	Q9BSW2	CRADD	P78560	CRAT	P43155
CRBN	Q96SW2	CRCP	O75575	CREB1	P16220-1

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
CREB3	O43889	CREB3L2	Q70SY1	CREBBP	Q92793
CREBRF	Q8IUR6	CREG1	O75629	CREM	Q03060-6
CRHBP	P24387	CRHR1	P34998	CRISPLD2	Q9H0B8
CRK	P46108	CRKL	P46109	CRLF1	O75462
CRLF2	Q9HC73	CRLS1	Q9UJA2	CRMP1	Q14194
CROCC2	Q9BV73	CROT	Q9UKG9	CRTAM	O95727
CRTAP	O75718	CRTC1	Q6UUV9	CRTC3	Q6UUV7
CRY1	Q16526	CRYAB	P02511	CRYL1	Q9Y2S2
CS	O75390	CSAD	Q9Y600	CSF1	P09603
CSF1R	P07333	CSF2RA	P15509	CSF2RB	P32927
CSF3R	Q99062	CSGALNACT1	Q8TDX6	CSGALNACT2	Q8N6G5
CSK	P41240	CSNK1D	P48730	CSNK1E	P49674
CSNK1G2	P78368	CSNK2A1	P68400	CSNK2A2	P19784
CSPG4	Q6UVK1	CSRP1	P21291	CSRP2BP	Q9H8E8
CST3	P01034	CSTA	P01040	CSTB	P04080
CSTF2	P33240	CTAGE5	Q96PC5	CTBP1	Q13363
CTBP2	P56545	CTDNEP1	O95476	CTDP1	Q9Y5B0
CTDSP2	O14595	CTF1	P33240	CTGF	P29279
CTNNA1	P35221	CTNNA2	P26232	CTNNBIP1	Q9NSA3
CTNNBL1	Q8WYA6	CTPS1	P17812, Q9NRF8	CTR9	Q6PD62
CTSA	P10619	CTSB	P07858	CTSC	P53634
CTSD	P07339	CTSF	Q9UBX1	CTSG	P08311
CTSH	P09668	CTSK	P43235	CTSO	P43235
CTSS	P25774	CTSZ	Q9UBR2	CTTN	Q14247
CTU1	Q7Z7A3	CTU2	Q2VPK5	CUBN	O60494
CUL3	Q13618	CUL5	Q93034	CUL7	Q14999
CUX1	Q13948	CWC15	Q9P013	CWC22	Q9HCG8
CWC27	Q6UX04	CWF19L2	Q2TBE0	CX3CL1	P78423
CX3CR1	P49238	CXADR	P78310	CXCL10	P02778
CXCL12	P48061	CXCL13	P10145	CXCL16	Q9H2A7
CXCL2	P19875	CXCL8	P10145	CXCL9	Q07325
CXCR4	P61073	CXXC1	Q9P0U4	CXXC4	Q9H2H0
CXXC5	Q7LFL8	CXorf21	Q9HAI6	CXorf56	Q9H5V9
CYB5A	P00167-1	CYB5B	P00167	CYB5R2	Q6BCY4
CYB5R3	P00387	CYBA	P13498	CYBB	P04839
CYBRD1	Q53TN4	CYC1	P08574	CYCS	P99999
CYFIP1	Q7L576	CYFIP2	Q96F07	CYGB	Q8WWM9
CYLD	Q9NQC7	CYP11A1	P05108	CYP19A1	P11511
CYP1A1	P04798	CYP1B1	Q16678	CYP24A1	Q07973
CYP26B1	Q9NR63	CYP27A1	Q02318	CYP2A13	Q16696
CYP2C8	P10632	CYP2J2	P51589	CYP2R1	Q6VVX0
CYP2U1	Q7Z449	CYP2W1	Q8TAV3	CYP39A1	Q9NYL5
CYP46A1	Q9Y6A2	CYP4A11	Q02928	CYP4B1	P13584
CYP4F11	Q9HBI6, Q9HCS2	CYP4F12	Q9HCS2	CYP4F3	Q6NT55
CYP4V2	Q6ZWL3	CYP51A1	Q16850	CYP7B1	O75881
CYR61	O00622	CYS1	Q717R9	CYSLTR1	Q9NS75, Q9Y271
CYSLTR2	Q9NS75	CYTH1	Q15438, Q99418	CYTH2	Q15438, Q99418
CYTH3	O43739	CYTH4	Q9UIA0	D2HGDH	Q8N465
DAB1	O75553	DAB2	P98082	DAD1	P61803

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
DAG1	Q14118	DAGLA	Q9Y4D2	DAGLB	Q8NCG7
DAP3	P51398	DAPK1	P53355	DAPK2	Q9UIK4
DAPK3	O43293	DAPP1	Q9UN19	DARS	P14868
DAXX	Q9UER7	DBF4	Q9UBU7	DBI	P07108
DBN1	Q16643	DBNL	Q9UJU6	DBP	Q10586
DBT	P49674	DCAF10	Q5QP82	DCAF11	Q8TEB1
DCAF13	Q9NV06	DCAF17	Q5H9S7	DCAF4	Q8WV16
DCAF6	Q58WW2	DCAF7	P61962	DCAKD	Q8WVC6
DCC	P43146	DCK	P27707	DCLRE1B	Q9H816
DCLRE1C	Q96SD1	DCN	P07585	DCP1A	Q9NPI6
DCP2	Q8IU60	DCT	P40126	DCTN1	Q14203-2
DCTN2	Q13561	DCTN3	O75935	DCTN5	Q9BTE1
DCTN6	O00399	DCTPP1	Q9H773	DCUN1D1	Q96GG9
DCUN1D4	Q92564	DCX	O43602	DCXR	Q7Z4W1
DDA1	Q9BW61	DDAH1	O94760	DDB1	Q16531
DDB2	Q92466	DDC	P20711	DDHD1	Q8NEL9
DDIT3	P35638	DDIT4	Q9NX09	DDO	Q86VW1
DDOST	P39656	DDR1	Q08345	DDR2	Q16832
DDRK1	Q96HY6	DDX1	Q92499	DDX11	Q96FC9
DDX17	Q92841	DDX20	Q9UHI6	DDX21	Q9NR30
DDX23	Q9BUQ8	DDX39A	O00148	DDX39B	Q13838
DDX41	Q7L591	DDX42	Q86XP3	DDX49	Q9Y6V7
DDX5	P17844	DDX52	Q9Y2R4	DDX58	O95786
DDX6	P26196	DECR1	Q16698	DECR2	Q9NUI1
DEF6	Q9H4E7	DEFA3	P59666	DEK	P51580
DENND1A	Q8TEH3	DENND1B	Q6P3S1	DENND2A	Q9ULE3
DENND2C	Q68D51	DENND2D	Q9H6A0	DENND3	A2RUS2
DENND4A	Q7Z401	DENND4B	O75064	DENND4C	Q5VZ89
DENND5A	Q6IQ26	DENND5B	Q6ZUT9	DENND6A	Q8IWF6
DENND6B	Q8NEG7	DEPDC1B	Q8WUY9	DEPDC5	O75140
DERA	Q9Y315	DERL1	Q9GZP9	DERL2	Q9GZP9
DES	P17661	DFFA	O00273	DFNA5	O60443
DFNB31	Q9P202	DGAT1	O75907	DGAT2	Q96PD6, Q96PD7
DGCR8	Q8WYQ5	DGKA	P23743, P52824	DGKB	Q9Y6T7
DGKD	Q16760	DGKE	P52429	DGKG	P49619
DGKH	Q86XP1	DGKI	O75912	DGKK	Q5KSL6
DGKQ	P52824	DGKZ	Q13574	DGUOK	Q16854
DHCR24	Q15392	DHCR7	Q9UBM7	DHDDS	Q86SQ9
DHH	O43323	DHPS	P49366	DHRS3	O75911
DHRS4	Q9BTZ2	DHRS9	Q9BPW9	DHRX	Q8N5I4
DHX16	O60231	DHX35	Q9H5Z1	DHX37	Q8IY37
DHX58	Q96C10	DHX8	Q14562	DHX9	Q08211
DIABLO	Q9NR28	DIAPH2	O60879	DIAPH3	Q9NSV4
DICER1	Q9UPY3	DIO2	Q92813	DIO3	P49895, P55073
DIS3L2	Q8IYB7	DISP2	A7MBM2	DKC1	O60832
DKK2	Q9UBU2	DLC1	Q96QB1	DLG1	Q12959, Q92796
DLG2	Q15700	DLG3	Q92796	DLG4	P78352
DLGAP1	O14490	DLGAP2	Q9P1A6	DLGAP3	O95886
DLGAP4	Q9Y2H0	DLK1	P80370	DLL1	O00548

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
DLL3	Q9NYJ7	DLST	P36957	DMAP1	Q9NPF5
DMC1	Q14565	DMD	P11532	DMGDH	Q9UI17
DMKN	Q6E0U4	DMPK	Q09013	DMTN	Q08495
DNA2	P51530	DNAJA1	P31689	DNAJA2	O60884
DNAJA4	Q8WW22	DNAJB11	Q9UBS4	DNAJB6	O75190
DNAJC10	Q8IXB1	DNAJC11	Q9NVH1	DNAJC13	O75165
DNAJC19	Q96DA6	DNAJC2	Q99543	DNAJC24	Q6P3W2
DNAJC27	O60266	DNAJC3	Q13217	DNAJC5	Q9H3Z4
DNAJC6	O75061	DNAJC7	Q99615	DNAJC8	O75937
DNAL4	O96015	DNASE1L1	P49184	DNASE2	O00115
DNER	Q8NFT8	DNM1	Q05193	DNM1L	O00429
DNM3	Q9UQ16	DNMBP	Q6XZF7	DNMT1	O75821
DNMT3B	Q9UBC3	DNP1	O43598	DNTTIP1	Q9H147
DOCK1	Q14185	DOCK10	Q96BY6	DOCK11	Q5JSL3
DOCK2	Q92608	DOCK4	Q8N1I0	DOCK5	Q9H7D0
DOCK6	Q96HP0	DOCK7	Q96N67	DOCK8	Q8NF50
DOCK9	Q9BZ29	DOHH	Q9BU89	DOK2	Q7L591
DOK3	Q7L591	DOK4	Q8TEW6	DOK5	Q9P104
DOK6	Q6PKX4	DOLK	Q9UPQ8	DOLPP1	Q86YN1
DOT1L	Q8TEK3	DPAGT1	Q9H3H5	DPEP2	Q9H4A9
DPF1	Q92782	DPF2	Q92785	DPF3	Q92784
DPH2	Q9BQC3	DPH3	Q96FX2	DPH6	Q7L8W6
DPH7	Q9BTV6	DPM1	O60762	DPM2	O94777
DPM3	Q9P2X0	DPP3	Q9NY33	DPP4	P27487
DPPA4	Q7L190	DPY30	Q9C005	DPYD	Q12882
DPYSL2	Q16555	DPYSL3	Q14195	DPYSL4	O14531
DPYSL5	Q9BPU6	DRAP1	Q14919	DRD2	P14416
DRG1	Q9Y295	DRP2	Q13474	DSC2	Q02487
DSC3	Q08554	DSCAM	O60469	DSCAML1	Q8TD84
DSCC1	Q9BVC3	DSE	Q9UL01	DSEL	Q8IZU8
DSP	P15924	DST	Q03001	DTNA	Q9Y4J8
DTNB	O60941	DTNBP1	Q96EV8	DTX1	Q86Y01
DTX2	Q86UW9	DTX3L	Q8TDB6	DTX4	Q9Y2E6
DTYMK	P23919	DUOX1	Q9NRD8, Q9NRD9	DUOXA1	Q1HG43
DUSP1	P28562	DUSP10	Q9Y6W6	DUSP16	Q9BY84
DUSP3	P51452	DUSP4	Q13115	DUSP6	Q16828
DUSP7	Q16829	DUSP8	Q13202	DUSP9	Q99956
DVL1	O14640	DVL3	Q92997	DXO	O77932
DYNC1I1	O14576	DYNC1I2	Q13409	DYNC1LI1	Q9Y6G9
DYNC2H1	Q8NCM8	DYNLL1	P63167	DYNLL2	Q96FJ2
DYNLRB1	Q9NP97	DYNLT1	P63172	DYRK1A	Q13627
DYRK3	Q92630	DYSF	Q8IUM7	DZIP1	Q86YF9
E2F4	Q16254	E2F6	O75461	EAF1	Q96JC9
EAF2	Q96CJ1	EBF1	Q9UH73	EBF2	Q9HAK2
EBI3	Q14213	EBNA1BP2	Q99848	EBP	Q15125
ECE1	P42892	ECE2	P0DPD6	ECH1	Q13011
ECHS1	P30084	ECI1	P42126	ECI2	O75521-2
ECM1	Q16610	ECSIT	Q9BQ95	ECT2	Q9H8V3
EDA	Q92838, Q92838-3	EDA2R	Q9HAV5	EDAR	Q9UNE0

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
EDARADD	Q8WWZ3	EDC3	Q96F86	EDC4	Q6P2E9
EDEM1	Q92611	EDEM2	Q9BV94	EDEM3	Q9BZQ6
EDIL3	O43854	EDN1	P05305	EDNRA	P25101
EDNRB	P24530	EEA1	Q15075	EED	O75530
EEF1A1	P68104	EEF1A1P5	Q5VTE0	EEF1A2	Q05639
EEF1B2	P24534	EEF1E1	O43324	EEF1G	P26641
EEF2	P13639	EEF2K	O00418	EEF2KMT	Q96G04
EEFSEC	P57772	EEPD1	Q7L9B9	EFEMP1	Q12805
EFEMP2	O95967, Q12805	EFNA2	O43921	EFNA3	P52797
EFNA5	P52803	EFNB1	P98172	EFNB3	Q15768
EFTUD2	Q15029	EGF	P01133	EGFR	P00533, P04626
EGLN1	Q9GZT9	EGLN2	Q96KS0	EGLN3	Q9H6Z9
EGR1	P18146	EGR2	P11161	EGR3	Q06889
EGR4	Q05215	EHD1	Q9H4M9	EHD2	Q9NZN3, Q9NZN4
EHD3	Q6NVY1	EHHADH	Q08426	EHMT1	Q9H9B1
EHMT2	Q96KQ7	EIF1AY	P47813	EIF2A	P05198
EIF2AK1	Q9BQI3	EIF2AK2	P19525	EIF2AK3	Q9NZJ5
EIF2AK4	Q9P2K8	EIF2B1	Q14232	EIF2B2	P49770
EIF2B4	Q9UI10	EIF2B5	Q13144	EIF2S2	P20042
EIF3A	Q14152	EIF3B	P55884	EIF3C	Q99613
EIF3D	O15371	EIF3F	O00303	EIF3G	O75821
EIF3I	Q13347	EIF3J	O75822	EIF3K	Q9UBQ5
EIF3L	Q9Y262	EIF3M	Q7L2H7	EIF4A1	P60842
EIF4A2	Q14240	EIF4A3	P38919	EIF4E2	O60573
EIF4EBP1	Q13541	EIF4G1	Q04637	EIF4G2	P78344
EIF4G3	O43432	EIF4H	Q15056	EIF5	P55010
EIF5A	P63241, Q9GZV4	EIF5A2	Q9GZV4	EIF5B	O60841
ELAC2	Q9BQ52	ELANE	P08246	ELAVL1	Q15717
ELAVL2	Q12926	ELF1	O43921	ELF2	P52799
ELK1	P19419	ELL	P55199	ELL2	O00472
ELMO1	Q92556	ELMO2	Q96JJ3	ELN	P15502
ELOVL2	Q9NXB9, Q9NYP7	ELOVL4	Q9GZR5	ELOVL5	Q9NYP7
ELOVL6	Q9H5J4	ELOVL7	A1L3X0	ELP3	Q9H9T3
ELP4	Q96EB1	ELP5	Q8TE02	ELP6	Q0PNE2
EMB	Q6PCB8	EMC3	Q9P0I2	EMC4	Q5J8M3
EMD	P50402	EME2	A4GXA9	EMID1	P39060
EMILIN1	Q9Y6C2	EMILIN2	Q9BXX0	EMILIN3	Q9NT22
EML4	Q9HC35	ENDOD1	O94919	ENGASE	Q8NFI3
ENO1	P06733	ENO2	P09104	ENO4	A6NNW6
ENOPH1	Q9UHY7	ENPEP	Q07075	ENPP1	P22413
ENPP2	Q13822	ENPP3	O14638	ENPP4	Q9Y6X5
ENPP6	Q6UWR7	ENSA	O43768	ENTPD1	P49961
ENTPD2	Q9Y5L3	ENTPD3	O75355	ENTPD5	O75356
ENTPD6	O75354	ENY2	Q9NPA8	EP300	Q09472
EP400	Q96L91	EPAS1	Q99814	EPB41	O43491, P11171
EPB41L1	Q9H4G0	EPB41L2	O43491	EPB41L3	Q9Y2J2
EPB41L5	Q9HCM4	EPC1	Q9H2F5	EPCAM	P16422
EPGN	Q6UW88	EPHA10	Q5JZY3	EPHA2	P29317
EPHA3	P29320	EPHA4	P54764	EPHA5	P54756

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
EPHA6	Q9UF33	EPHA7	Q15375	EPHA8	P29322
EPHB1	P54762	EPHB2	P29323	EPHB3	P54753
EPHB4	P54760	EPHB6	O15197	EPHX1	P07099
EPHX2	P34913	EPM2A	O95278	EPN1	Q9Y613
EPN2	O95208	EPRS	P07814	EPS15	P42566
EPS8	Q12929	EPSTI1	Q96J88	ERAL1	O75616
ERAP1	Q9NZ08	ERAP2	Q6P179	ERBB2	P04626
ERBB2IP	Q96RT1	ERBB3	P21860-1	ERBB4	Q15303-1, Q15303-2
ERC1	Q96FL8	ERCC1	P07992	ERCC2	P18074
ERCC3	P19447	ERCC5	P28715	ERF	P50548
ERG	Q12809	ERI1	Q8IV48	ERICH1-AS1	Q9P1A6
ERICH5	Q6P6B1	ERLIN1	O75477	ERN1	O75460
ERN2	O75460	ERO1A	Q96HE7	ERO1B	Q86YB8
ERP44	Q9BS26	ESAM	Q96AP7	ESCO1	Q5FWF5
ESD	P10768	ESR1	P03372, P03372-3, P03372-4	ESR2	Q92731
ESRP1	Q6NXG1	ESRRA	P11474	ESRRB	O95718
ESRRG	P62508	ESYT2	A0FGR8, A0FGR9	ESYT3	A0FGR9
ETFA	P13804	ETFB	P38117	ETFDH	Q16134
ETNK1	Q9HBU6	ETNK2	Q9NVF9	ETNPPL	Q8TBG4
ETS1	P14921	ETV4	P43268	ETV6	P41212
EVA1A	Q9H8M9	EVC	P57679	EVC2	Q86UK5
EVL	Q9UI08	EVPL	Q92817	EXOC1	Q9NV70
EXOC4	Q96A65	EXOC5	O00471	EXOC6	Q8TAG9
EXOC7	Q9UPT5	EXOSC1	Q9Y3B2	EXOSC2	Q13868
EXOSC3	Q9NQT5	EXOSC4	Q9NPD3	EXOSC5	Q9NQT4
EXOSC9	Q06265	EXT1	Q16394	EXT2	Q93063
EXTL1	Q92935	EXTL2	Q9UBQ6	EXTL3	O43909
EYA1	Q99502	EYA2	O00167	EYA3	Q99504
EYA4	O95677	EYS	Q13753	EZH1	Q92800
EZR	P15311	F10	P00742	F12	P00748
F13A1	P00488	F2R	P25116	F3	P13726
F5	P12259	F8	P00451	FA2H	Q7L5A8
FAAH	Q7L5A8	FAAH2	Q6GMR7	FAAP100	Q0VG06
FAAP20	Q6NZ36	FABP3	P05413, Q0Z7S8	FABP4	P15090
FABP5	Q01469	FABP6	P51161	FABP7	O15540
FADD	Q13158	FADS2	O95864	FAH	P16930
FAHD1	Q6P587	FAM120B	Q96EK7	FAM131B	Q86XD5
FAM134B	Q9H6L5	FAM135A	Q9P2D6	FAM13A	O94988
FAM13B	Q9NYF5	FAM150B	Q6UX46	FAM175A	Q6UWZ7
FAM195B	C9JLW8	FAM208A	Q9UK61	FAM20A	Q96MK3
FAM20C	Q8IXL6	FAM213B	Q8TBF2	FAM32A	Q9Y421
FAM3C	Q92520	FAM50A	Q14320	FAM73A	Q8NAN2
FAM73B	Q7L4E1	FAM91A1	Q658Y4	FAM96B	Q9Y3D0
FAN1	Q16658	FANCB	Q8NB91	FANCC	Q00597
FANCD2	Q9BXW9	FANCE	Q9HB96	FANCF	Q9NPI8
FANCG	O15287	FANCL	Q9NW38	FANCM	Q8IYD8
FAP	P27487	FAR1	Q8WVX9	FAR2	Q96K12
FARP1	Q9Y4F1	FARP2	O94887	FARS2	O95363
FARSA	Q9Y285	FARSB	Q9NSD9	FAS	P25445

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
FASN	P49327	FASTK	Q14296-4	FASTKD5	Q7L8L6
FAU	P62861	FBL	P22087	FBLIM1	Q8WUP2
FBLN1	P23142	FBLN2	P98095	FBLN5	Q12805, Q9UBX5
FBN1	P35555	FBN2	P35556, Q75N90	FBN3	Q75N90
FBP1	P09467	FBXL12	Q9NXXK8, Q9UKT6	FBXL13	Q8NEE6
FBXL14	Q8N1E6	FBXL15	Q9H469	FBXL16	Q8N461
FBXL17	Q9UF56	FBXL18	Q96ME1	FBXL19	Q6PCT2
FBXL20	Q96IG2	FBXL21	Q9UKT6	FBXL3	Q9UKT7
FBXL4	Q9UKA2	FBXL5	Q9UKA1	FBXL7	Q9UJT9
FBXO10	Q9UK96	FBXO11	Q86XK2	FBXO15	Q8NCQ5
FBXO17	Q96EF6	FBXO2	Q96EF6, Q9H4M3, Q9UK22	FBXO21	O94952
FBXO30	Q8TB52	FBXO31	Q5XUX0	FBXO32	Q969P5
FBXO4	Q9UKT5	FBXO41	Q8TF61	FBXO44	Q9NRD1
FBXO9	Q9UK97	FBXW4	P57775	FBXW5	Q969U6
FBXW7	Q969H0	FBXW8	Q8N3Y1	FBXW9	Q5XUX1
FCAR	P24071	FCER1A	P12319	FCER1G	P30273
FCGR1A	P12314	FCGR1B	Q92637	FCGR2A	P12318
FCGR2B	P12318	FCGR3A	O75015	FCHO1	O14526
FCHO2	Q0JRZ9	FCN1	O00602	FDFT1	P37268
FDPS	P14324	FECH	P22830	FEM1A	Q9BSK4
FEM1B	Q9UK73	FEM1C	Q96JP0	FEN1	P39748
FER	P16591	FERMT2	Q96AC1	FERMT3	Q86UX7
FES	P07332	FGD1	P98174	FGD2	Q7Z6J4
FGD3	Q5JSP0	FGD4	Q96M96	FGD5	Q6ZNL6
FGF1	P05230	FGF10	O15520	FGF11	Q92914
FGF12	P61328	FGF13	Q92913	FGF14	Q92915
FGF17	O60258-1	FGF18	O76093	FGF2	P09038
FGF4	P08620, P10767	FGF5	P12034-1	FGF7	P21781
FGF9	P31371	FGFBP2	Q9BYJ0	FGFBP3	Q8TAT2
FGFR1	P11362-1, P11362-19	FGFR1OP	O95684	FGFR1OP2	Q9NVK5
FGFR2	P21802-1, P21802-18, P21802-3, P21802-5	FGFR3	P22607-1, P22607-2	FGFRL1	Q8N441
FGL2	Q14314	FGR	P09769	FH	Q16595
FHL2	Q14192	FHOD3	Q8IVF7	FIGF	O43915
FIP1L1	Q6UN15	FIS1	Q9Y3D6	FITM2	A5D6W6, Q8N6M3
FKBP14	Q9NWM8	FKBP1A	P62942	FKBP1B	P68106
FKBP4	Q02790	FKBP5	Q13451	FKBP8	Q14318
FKBP9	O95302	FKBPL	Q9UIM3	FKRP	Q9H9S5
FKTN	O75072	FLAD1	Q8NFF5	FLCN	Q8NFG4
FLI1	Q01543	FLNA	P21333	FLNB	O75369
FLNC	Q14315	FLOT1	O75955	FLOT2	Q14254
FLRT2	O43155	FLT1	P17948	FLT3	P36888
FLT4	P35916	FLVCR1	Q9Y5Y0-1, Q9Y5Y0-2	FMN1	Q969G6
FMNL1	O95466	FMNL2	Q96PY5	FMNL3	Q8IVF7
FMO1	Q01740, Q99518	FMO2	Q99518	FMO3	P31513
FMO5	Q01740	FMOD	Q06828	FN1	P02751
FN3K	Q9H479	FN3KRP	Q9H479, Q9HA64	FNBP1	Q96RU3
FNBP1L	Q5T0N5	FNIP2	Q9P278	FNTA	P49354

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FNTB	P49356	FOLH1	Q04609	FOLR2	P14207
FOS	P01100	FOSB	P53539	FOSL1	P15407
FOXC1	Q12948	FOXC2	Q99958	FOXO3	Q9UJU5
FOXK2	Q01167	FOXO1	Q12778	FOXO3	O43524
FOXO4	P98177	FOXO6	A8MYZ6	FOXP1	Q9H334-8
FOXP2	O15409	FOXP4	Q9BZS1	FOXRED1	Q96CU9
FPGS	Q05932-1, Q05932-2	FPR1	P62942	FPR3	P25089
FRAT1	Q92837	FRAT2	O75474	FRK	P42685
FRMPD3	Q5JV73	FRS2	Q8WU20	FRS3	O43559
FSCN1	Q16658	FST	P19883	FSTL1	Q12841
FSTL3	O95633	FTCD	O95954	FTH1	P02794
FTL	P02792	FTO	Q9C0B1	FTSJ1	Q9UET6
FTSJ2	Q9UI43	FTSJ3	Q8IY81	FUK	Q8N0W3
FUNDC1	Q8IVP5	FUNDC2	P46736	FUS	P35637
FUT10	Q6P4F1	FUT8	Q9BYC5	FUT9	Q9Y231
FXN	Q16595	FXR1	P51114	FXYD1	O00168, P54710
FXYD3	Q14802	FXYD5	Q14802	FXYD6	Q9H0Q3
FXYD7	P58549, P59646	FYB	O15117	FYN	P06241
FZD1	Q9UP38	FZD10	Q9ULW2	FZD2	Q14332
FZD4	Q9ULV1	FZD5	Q13467	FZD6	O60353
FZD7	O75084	FZD8	Q9H461	FZR1	Q9UM11
G0S2	P27469	G3BP1	Q13283	G3BP2	Q9UN86
G6PC3	Q9BUM1	G6PD	P11413	GAB1	Q13480
GAB3	Q8WWW8	GABARAP	O95166	GABARAPL2	P60520
GABBR1	Q9UBS5	GABBR2	O75899	GABPA	Q06546
GABPB1	Q06547-1, Q06547-2, Q06547-3, Q06547-4	GABPB2	Q06547-1, Q06547-2, Q06547-3, Q06547-4	GABRA2	P47869
GABRA3	P34903	GABRA5	P31644	GABRB1	P18505
GABRB3	P28472	GABRG2	P18507	GABRG3	Q99928
GABRQ	Q9UN88	GABRR2	P28476	GADD45A	P24522
GADD45GIP1	Q8TAE8	GADL1	Q6ZQY3	GAK	O14976
GAL	P16278	GAL3ST1	Q99999	GALC	P54803
GALE	Q14376	GALM	Q96C23	GALNT1	Q10472
GALNT10	Q86SR1	GALNT12	Q8IXK2	GALNT14	Q96FL9
GALNT15	Q7Z4T8, Q8N3T1	GALNT16	Q8N428	GALNT18	Q6P9A2
GALNT2	Q10471	GALNT3	Q14435	GALNT5	Q7Z7M9
GALNT6	Q14435	GALNT7	Q86SF2	GALNT8	Q9NY28
GALNT9	Q9HCQ5	GALNTL6	Q49A17	GALR1	P47211
GAMT	Q14353	GAN	Q9H2C0	GANAB	Q14697
GAP43	P17677	GAPDH	P04406	GAPVD1	Q14C86
GARS	P41250	GART	P22102	GAS1	P54826
GAS2	O43903	GAS6	Q14393	GAS8	O95995
GATA2	P23769	GATA6	Q92908	GATAD2B	Q8WXI9
GATM	P50440	GBA2	Q9HCG7	GBA3	Q9H227
GBAS	O75323	GBE1	Q04446	GBP1	P32455, Q9H0R5
GBP2	P32456, Q9H0R5	GBP3	Q9H0R5	GBP4	Q96PP9
GBP5	Q96PP8	GCA	P28676	GCC1	Q96CN9
GCC2	Q8IWJ2	GCFC2	P16383	GCGR	P47871
GCH1	P30793	GCHFR	P30047	GCK	P35557

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
GCLC	P48506	GCLM	P48507	GCNT1	Q02742
GCNT4	Q9P109	GCSH	P23434	GDA	P16260
GDE1	Q9NZC3	GDI1	P50395	GDI2	P50395
GDNF	P39905	GDPD1	Q8ND04	GDPD5	Q8WTR4
GEM	P55040	GEMIN2	O14893	GEMIN4	P57678
GEMIN6	Q8WXD5	GEMIN7	Q9H840	GEMIN8	Q9NWZ8
GET4	Q7L5D6	GFAP	P14136	GFER	P55789
GFM1	Q96RP9	GFOD1	Q9NXC2	GFPT2	O94808
GFRA1	P56159	GFRA2	O00451	GFRA3	O60609
GGA1	Q9UJY4, Q9UJY5	GGA2	Q9UJY4	GGA3	Q9NZ52
GGCT	O75223	GGCX	P38435	GGH	Q92820
GGPS1	O95749	GGT1	P19440	GGT5	P36269
GGT7	Q9UJ14	GHDC	Q8N2G8	GHR	P15509
GID8	Q9NWU2	GIN2	Q9Y248	GIN3	Q9BRX5
GIP	P09681	GIPC1	O14908	GIPR	P48546
GIT1	Q9Y2X7	GIT2	Q14161	GJA1	P17302
GJA4	P35212	GJA5	P36382	GJB1	P08034
GJB2	P29033	GJB6	P29033	GJC1	P36383
GK	P32189, Q14409	GLB1	P16278	GLB1L2	Q8IW92, Q8NCI6
GLB1L3	Q8NCI6	GLCE	O94923	GLDC	P23378
GLG1	Q92896	GLI1	P08151	GLI2	P10070
GLI3	P10071	GLIPR1	P48060	GLMN	Q92990
GLO1	Q04760	GLP1R	P43220	GLRA2	P23416
GLRA3	O75311	GLRB	P48167	GLRX	P35754
GLRX3	O76003	GLRX5	Q86SX6	GLS	O94925
GLS2	O94925, Q9UI32	GLTP	Q9NZD2	GLTSCR1	Q9NZM4
GLTSCR1L	Q6AI39	GLUD1	P00367	GLUD2	P00367, P49448
GLUL	P15104	GLYAT	Q61B77	GM2A	P17900
GMDS	O60547	GMIP	Q9P107	GMPPA	Q96IJ6
GMPPB	Q9Y5P6	GMPR	P36959	GMPR2	P36959
GMPS	P49915	GNA11	P29992	GNA13	Q14344
GNA14	O95837, P29992	GNA15	P30679	GNAI1	P63096
GNAI3	P08754	GNAL	P38405	GNAO1	P09471
GNAQ	P50148	GNAS	P63092, Q5JWF2	GNAZ	P19086
GNB1	P62873	GNB2	P62879	GNB2L1	P63244
GNB4	Q9HAV0	GNB5	O14775	GNE	Q9Y223
GNG10	P50151	GNG11	P61952	GNG12	Q9UBI6
GNG2	P59768	GNG3	P63215	GNG4	P50150
GNG7	O60262	GNGT2	O14610	GNL3	Q9BVP2
GNPDA1	P46926, Q8TDQ7	GNPDA2	Q8TDQ7	GNRH1	P01148
GNRHR2	P30968, Q96P88	GNS	P15586	GOLGA2	Q08379
GOLGA4	Q13439	GOLGA7	Q7Z5G4	GOLIM4	O00461
GOLM1	Q8NBJ4	GOPC	Q9HD26	GORASP1	Q9BQQ3
GORASP2	Q9H8Y8	GOSR2	O14653	GOT1	P17174
GOT2	P00505	GPAA1	O43292	GPAM	Q9HCL2
GPAT2	Q6NUI2	GPC1	P35052	GPC2	Q8N158
GPC3	P51654	GPC4	O75487	GPC5	P78333
GPC6	Q9Y625	GPCPD1	Q9NPB8	GPD1	P21695, Q8N335
GPD1L	Q8N335	GPD2	P21695, P43304	GPHN	Q9NQX3

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
GPIHBP1	O43653, Q8IV16	GPKOW	Q92917	GPNMB	Q14956
GPR132	Q9UNW8	GPR161	Q8N6U8	GPR162	Q8IVL6
GPR17	Q13304	GPR176	Q14439	GPR183	P32249
GPR20	Q99678	GPR27	Q9NS67	GPR35	Q9HC97
GPR37	O15354	GPR37L1	O60883	GPR39	O43194
GPR65	Q8IYL9	GPR83	Q9NYM4	GPRIN1	Q7Z2K8
GPS1	Q13098	GPS2	Q13227	GPSM1	Q86YR5
GPSM2	P81274	GPT	P24298	GPX1	P07203
GPX2	P18283	GPX3	O75715, P22352	GPX4	P36969
GPX7	Q96SL4	GPX8	Q8TED1	GRAP	Q13588
GRAP2	O75791	GRAPL	Q13588	GRB10	Q13322
GRB14	Q14449	GRB2	P62993-1	GREB1	Q4ZG55
GREM1	O60565	GREM2	Q9H772	GRHPR	Q9UBQ7
GRIA1	P42261	GRIA2	P42262	GRIA3	P42263
GRIA4	P48058	GRIK1	P39086	GRIK2	Q13002
GRIK3	Q13003	GRIK4	Q16099	GRIK5	Q16478
GRIN1	Q05586	GRIN2A	Q12879	GRIN2B	Q13224
GRIN2D	O15399	GRIN3A	Q8TCU5	GRIP2	Q9C0E4
GRK5	P34947	GRK6	P43250	GRM1	Q13255
GRM2	Q14416	GRM3	Q14832	GRM4	Q14833
GRM7	Q14831	GRM8	O00222	GRN	P28799
GRPEL1	Q9HAV7	GRPEL2	Q8TAA5	GRXCR1	A8MXD5
GSDMD	P57764	GSK3A	P49841	GSN	P06396
GSPT1	P15170, Q8IYD1	GSPT2	Q8IYD1	GSS	P48637
GSTA4	O15217, P08263	GSTK1	Q9Y2Q3	GSTM1	P09488, P46439
GSTM2	P28161, P46439	GSTM3	P21266	GSTM5	P21266, P46439
GSTO2	P78417, Q9H4Y5	GSTP1	P09211	GSTZ1	O43708
GTF2A2	P52657	GTF2E1	P29083	GTF2E2	P29084
GTF2F1	P35269	GTF2F2	P13984	GTF2H1	P32780
GTF2H2	Q13888	GTF2H4	Q92759	GTF2H5	Q6ZYL4
GTF3A	Q92664	GTF3C1	Q12789	GTF3C2	Q8WUA4
GTF3C4	Q9UKN8	GTF3C5	Q9Y5Q8	GTF3C6	Q969F1
GTPBP1	Q9BX10	GTPBP2	Q9BX10	GTPBP3	Q969Y2
GUCA1A	P43080	GUCY1A2	P33402	GUCY1A3	Q02108
GUCY1B3	Q02153	GUCY2D	P51841, Q02846	GUK1	Q16774
GXYLT2	A0PJZ3	GYG1	P46976	GYG2	O15488
GYPC	P04921	GYS1	P13807	H1F0	P07305
H2AFJ	Q9BTM1	H2AFX	P16104	H2AFZ	P0C0S5
H3F3A	P84243	H3F3B	P84243	HAAO	P46952
HACD2	Q6Y1H2, Q9P035	HACD3	Q9P035	HACD4	Q5VWC8
HACE1	Q8IYU2	HACL1	Q9UJ83	HADH	P40939, Q16836
HADHA	P40939	HADHB	P55084	HAGH	Q16775-2
HAP1	P27695	HARS	P12081	HARS2	P49590
HAS2	Q92819	HAS3	O00219	HAT1	O14929
HAUS2	Q9NVX0	HAUS4	Q9H6D7	HAUS6	Q7Z4H7
HAVCR1	Q96D42	HBA1	P69905	HBA2	P69905
HBB	P68871	HBD	P02042	HBG2	P69892
HBP1	Q8IV16	HBQ1	P69905	HBSIL	Q9Y450-1
HCAR1	Q9BXC0	HCCS	P53701	HCFC1	P51610

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
HCFC2	Q9Y5Z7	HCK	P08631	HCN1	O60741
HCN2	Q9UL51	HCN3	Q9P1Z3	HCN4	Q9Y3Q4
HCRT1	O43613	HCRT2	O43614	HCST	Q9UBK5
HDAC1	Q13547, Q92769	HDAC10	Q969S8	HDAC11	Q96DB2
HDAC2	Q92769	HDAC3	O15379	HDAC4	P56524
HDAC5	Q9UQL6	HDAC6	Q9UBN7	HDAC7	Q8WUI4, Q9UKV0
HDAC8	Q9BY41	HDAC9	Q9UKV0	HDC	P19113
HDGF	P51858	HDHD3	P13716	HEATR1	Q9H583
HEBP1	Q9NRV9	HEBP2	Q9Y5Z4	HECTD1	Q9ULT8
HECTD2	Q5U5R9	HECTD3	Q5T447	HECW2	Q9P2P5
HELLS	Q9NRZ9	HENMT1	Q5T8I9	HEPH	Q9BQS7
HERC1	Q15751	HERC3	Q15034	HERC4	Q15034, Q5GLZ8
HERC5	Q9UII4	HERC6	Q8IVU3, Q9UII4	HERPUD1	Q15011
HES1	Q14469	HES7	Q9BYE0	HEXB	P07686
HEY1	Q9Y5J3	HEY2	Q9UBP5	HEYL	Q9NQ87
HFE	Q30201	HGF	P14210	HGFAC	Q04756
HGS	O14964	HHAT	Q5VTY9	HHEX	Q03014
HHIP	Q96QV1	HIBADH	P31937	HIBCH	Q6NVY1
HIC1	Q14526	HIF1A	Q16665	HIF1AN	Q9NWT6
HIF3A	Q9Y2N7	HIGD1C	A8MV81	HIGD2A	Q9BW72
HILPDA	Q9Y5L2	HINT1	P49773	HINT2	Q9BX68
HIP1R	O75146	HIPK1	Q86Z02	HIRA	P54198
HIST1H1B	P16401	HIST1H1D	P16402	HIST1H1E	P10412
HIST1H2AC	Q93077	HIST1H2AD	P20671	HIST1H2AE	P04908
HIST1H2AG	P0C0S8	HIST1H2AH	Q96KK5	HIST1H2BB	P33778
HIST1H2BD	P58876	HIST1H2BE	P62807, Q93079	HIST1H2BF	P62807
HIST1H2BG	P62807, Q93079	HIST1H2BH	P58876, Q93079	HIST1H2BI	P62807, Q93079
HIST1H2BJ	P06899	HIST1H2BK	O60814, Q99880	HIST1H2BL	Q99880
HIST1H2BO	P23527	HIST1H3A	P68431	HIST1H3B	P68431
HIST1H3D	P68431	HIST1H4A	P62805	HIST1H4B	P62805
HIST1H4C	P62805	HIST1H4D	P62805	HIST1H4E	P62805
HIST1H4F	P62805	HIST1H4H	P62805	HIST1H4J	P62805
HIST2H2AB	Q8IUE6	HIST2H2AC	Q16777	HIST2H2BE	P06899, Q16778
HIST2H2BF	Q5QNW6	HIST3H2A	Q7L7L0	HIST4H4	P62805
HIVEP3	Q5T1R4	HK1	P19367	HK2	P52789
HK3	P52790	HKDC1	P52789, Q2TB90	HKR1	P10072
HLA-A	P04439	HLA-B	P01889	HLA-C	P10321
HLA-DMA	P28067	HLA-DMB	P28068	HLA-DOA	P06340
HLA-DPA1	P20036	HLA-DPB1	P04440	HLA-DQA1	P01909
HLA-DQA2	P01906	HLA-DQB1	P01920, P05538	HLA-DQB2	P05538
HLA-DRA	P01903	HLA-DRB1	P01911	HLA-DRB5	Q30154
HLA-E	P13747	HLA-F	P30511	HLA-H	P01893
HLCS	P50747	HM13	Q8TCT9	HMBS	P08397
HMCN1	Q96DU3	HMCN2	Q15109	HMGA1	P17096
HMGA2	P52926	HMGB2	P26583	HMGCR	P04035
HMGCS1	P54868	HMHA1	Q92619	HMOX1	P09601
HMOX2	P30519	HNF1B	P35680	HNF4G	Q14541
HNMT	P50135	HNRNPA0	Q13151	HNRNPA1	P09651
HNRNPA2B1	P22626	HNRNPA3	P51991	HNRNPC	P07910

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
HNRNPD	Q14103	HNRNPDL	O14979	HNRNPH2	P55795
HNRNPK	P61978	HNRNPL	P14866	HNRNPM	P52272
HNRNPR	O43390	HNRNPU	Q00839	HNRNPUL1	Q9BUJ2
HOMER1	Q86YM7	HOMER3	Q9NSC5	HOPX	Q9BPY8
HOXA2	O43364	HOXA3	O43365	HOXB2	P14652
HOXB3	P14651	HOXC4	P09017	HOXD1	Q9GZZ0
HOXD3	P31249	HP	P00738	HPGDS	O60760
HPS1	Q92902	HPSE	Q9Y251	HPSE2	Q8WWQ2
HRAS	P01112	HRASLS	Q9HDD0	HRASLS5	Q96KN8
HRC	P23327	HRH1	P35367	HRH2	P25021
HRH3	Q9Y5N1	HS2ST1	Q7LGA3	HS3ST3A1	Q9Y663
HS3ST3B1	Q9Y662, Q9Y663	HS3ST4	Q9Y661	HS3ST5	Q8IZT8, Q96QI5
HS6ST1	O60243	HS6ST2	Q96MM7	HS6ST3	Q8IZP7
HSBP1	O75506	HSD11B1L	P28845	HSD11B2	P80365
HSD17B10	Q99714	HSD17B11	Q8NBQ5	HSD17B12	Q53GQ0
HSD17B13	Q7Z5P4	HSD17B14	Q9BPX1	HSD17B3	P37058
HSD17B4	P51659	HSD17B7	P56937	HSF1	Q00613
HSP90AA1	P07900	HSP90AB1	P08238	HSP90B1	P14625
HSPA12A	O43301	HSPA12B	Q96MM6	HSPA14	Q0VDF9
HSPA1B	P0DMV8, P0DMV9	HSPA1L	P34931	HSPA2	P54652
HSPA4	P34932	HSPA5	P11021	HSPA8	P11142
HSPA9	P38646	HSPB1	P04792	HSPB8	Q9UJY1
HSPD1	P10809	HSPE1	P61604	HSPG2	P98160
HSPH1	Q92598	HTATSF1	O43719	HTR1B	P28222
HTR1D	P28221	HTR2A	P28223	HTR2C	P28335
HTR3A	P46098	HTR4	Q13639	HTR5A	P47898
HTRA1	Q92743	HTRA2	O43464	HUS1	O60921
HVCN1	Q96D96	HYAL1	Q12794	HYAL2	Q12891
HYAL3	P38567	HYKK	A2RU49	HYOU1	Q9Y4L1
IARS2	Q9NSE4	ICAM1	P05362	ICAM2	P13598
ICAM5	Q9UMF0	ICE1	Q9Y2F5	ICE2	Q659A1
ICOS	Q9Y6W8-1	ICT1	Q14197	ID1	P41134
ID2	Q02363	ID3	Q02535	ID4	P47928
IDE	P14735	IDH1	O75874	IDH2	P48735, P50213
IDH3B	O43837	IDH3G	P51553	IDI1	Q13907
IER3	P46695	IFI16	Q16666	IFI27	P40305
IFI27L2	P09912	IFI35	P80217	IFI44	Q8TCB0-1
IFI44L	Q53G44	IFI6	P09912	IFIH1	Q9BYX4
IFIT1	P09914	IFIT2	P09913	IFIT3	O14879, P09914
IFITM1	P13164	IFITM2	P13164, Q01629	IFITM3	Q01628
IFNAR1	P17181	IFNAR2	P48551-2	IFNGR1	P15260
IFNGR2	Q13651	IFNLR1	Q8IU57	IFT140	Q96RY7
IFT20	Q8IY31	IFT22	Q9H7X7	IFT27	Q9BW83
IFT43	Q96FT9	IFT52	Q9Y366	IFT74	Q96LB3
IFT88	Q13099	IGF1	P05019	IGF1R	P08069
IGF2	P01344	IGF2BP1	Q9NZI8	IGF2BP3	O00425
IGFBP2	P18065	IGFBP3	P17936	IGFBP4	P22692
IGFBP5	P24593	IGFBP6	P24592	IGFBP7	Q16270
IGHA1	P01876	IGHA2	P01877	IGHG1	P01857, P01861

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
IGHG2	P01859	IGHG4	P01861	IGKC	P01834
IGKV3-20	P01619	IGKV4-1	P06312	IGLC2	P01834, P0DOY2
IGLC3	P0DOY3	IGSF3	Q93033	IK	Q13123
IKBKAP	O95163	IKBKB	O14920	IKBKE	Q14164
IKBKG	Q9Y6K9	IKZF1	Q13422-1	IL10RA	Q13651
IL10RB	Q08334	IL11	P20809	IL12RB1	P42701
IL12RB2	Q99665	IL13RA1	P78552	IL15	P40933
IL15RA	Q13261	IL16	Q14005	IL17RA	Q96F46
IL17RB	Q9NRM6	IL17RC	Q8NAC3	IL17RD	Q8NFM7
IL17RE	Q8NFR9	IL18	Q14116, Q14116-2	IL18R1	Q13478
IL18RAP	O95256	IL1B	P01584	IL1R1	Q9HB29
IL1R2	P27930	IL1RAP	Q9NPH3-1	IL1RAPL1	Q9NZN1
IL1RAPL2	Q9NP60	IL1RN	P18510	IL20RA	Q9UHF4
IL20RB	Q6UXL0	IL21R	Q9HBE5	IL27RA	Q6UWB1
IL2RA	P01589	IL2RB	P14784	IL2RG	P31785
IL31RA	Q8NI17	IL32	P24001	IL33	O95760
IL34	Q6ZMJ4	IL3RA	P26951	IL4R	P24394
IL6	P05231	IL6R	P08887, P08887-2	IL6ST	P40189
IL7	P13232	IL7R	P16871	ILF2	Q12905
ILF3	Q12906	IMMT	Q16891	IMP3	Q9NV31
IMP4	Q96G21	IMPA2	O14732	IMPACT	Q9P2X3
IMPAD1	Q9NX62	IMPDH1	P20839	IMPDH2	P12268
INADL	Q8NI35	ING2	Q9H160	ING3	Q9NXR8
ING4	Q9UNL4	ING5	Q8WYH8	INHBB	P09529
INHBE	P58166	INMT	O95050	INO80	Q9ULG1
INO80C	Q6PI98	INO80D	Q53TQ3	INO80E	Q8NBZ0
INPP1	P49441	INPP4B	O15327	INPP5A	Q14642
INPP5B	P32019	INPP5D	Q92835	INPP5E	Q10713
INPP5F	Q9Y2H2	INPP5J	Q15735	INPPL1	O15357
INSM1	Q01101	INSR	P06213	INTS1	Q8N201
INTS10	Q9NVR2	INTS12	Q96CB8	INTS2	Q9H0H0
INTS4	Q96HW7	INTS5	Q6P9B9	INTS6	Q9UL03
INTS8	Q75QN2	INTS9	Q9NV88	INTU	Q9ULD6
IP6K1	Q92551, Q96PC2	IP6K2	Q9UHH9	IP6K3	Q96PC2
IPMK	Q8NFU5	IPO5	O00410	IPO8	O15397
IPP	P49441	IQCB1	Q15051	IQCE	Q6IPM2
IQGAP1	P46940	IQGAP2	Q13576	IQSEC3	Q9UPP2
IRAK1	P51617	IRAK2	O43187	IRAK3	Q9Y616
IRAK4	Q9NWZ3	IRF1	P10914	IRF2	P14316
IRF4	Q15306	IRF5	Q13568	IRF6	O14896
IRF8	Q02556	IRS1	P35568	IRS2	Q9Y4H2
IRX1	P78414	IRX2	Q9BZI1	ISCA1	Q9BUE6
ISG15	P05161	ISG20	Q96AZ6	ISG20L2	Q9H9L3
ISL1	P61371	ISLR	O14498	ISM1	Q9NSE4
ISPD	A4D126	IST1	P53990	ISYNA1	Q9NPH2
ITCH	Q96J02	ITFG2	Q969R8	ITGA1	P56199
ITGA10	O75578	ITGA11	Q9UKX5	ITGA2	P17301
ITGA3	P26006	ITGA4	P13612	ITGA5	P08648
ITGA6	P23229	ITGA7	Q13683	ITGA8	P53708

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
ITGA9	Q13797	ITGAE	P38570	ITGAL	P20701
ITGAM	P11215	ITGAV	P06756	ITGAX	P20702
ITGB1	P05556	ITGB2	P05107	ITGB3BP	Q13352
ITGB4	P16144	ITGB5	P18084	ITGB7	P26010
ITGB8	P26012	ITGBL1	O95965	ITIH2	P19823
ITIH3	Q06033	ITK	Q08881	ITM2B	Q9Y287
ITPA	Q9BY32	ITPK1	Q13572	ITPKA	P23677
ITPKB	P27987	ITPR1	Q14643	ITPR2	Q14571
ITPR3	Q14573	ITSN1	Q15811	ITSN2	Q9NZM3
IWS1	Q96ST2	JADE2	Q9NQC1	JAG1	P78504
JAG2	Q9Y219	JAK1	P23458	JAK2	O60674
JAK3	P52333	JAM2	P57087	JAM3	Q9BX67
JARID2	Q92833	JCHAIN	P01591	JMJD1C	Q15652
JMJD4	Q9H9V9	JMJD6	Q6NYC1	JMY	Q8N9B5
JOSD1	Q15040	JUN	P05412	JUNB	P17275
JUND	P17535	JUP	P14923	KALRN	O60229
KANK1	Q14678	KANSL1	Q7Z3B3	KANSL2	Q9H9L4
KARS	Q15046	KAT2A	Q92830	KAT2B	Q92831
KAT5	Q92993	KAT6A	Q92794	KAT6B	Q8WYB5
KAT8	Q9H7Z6	KAZN	Q674X7	KBTBD6	Q86V97
KCNA1	Q09470	KCNA4	P22459	KCNAB2	Q13303
KCNB2	Q92953	KCNC1	P48547	KCNC2	Q96PR1
KCNC3	Q14003	KCNC4	Q03721	KCND1	Q9NSA2
KCND2	Q9NZV8	KCND3	Q9UK17	KCNE3	Q9Y6H6
KCNE4	Q8WWG9	KCNF1	Q9H3M0	KCNG1	Q9UIX4
KCNG3	Q8TAE7, Q8TDN1	KCNG4	Q8TDN1	KCNH1	O95259
KCNH2	Q12809	KCNH3	Q9ULD8	KCNH5	Q8NCM2
KCNH7	Q9NS40	KCNH8	Q96L42	KCNIP1	Q9NZI2
KCNIP2	Q9NS61	KCNIP3	Q9Y2W7	KCNIP4	Q6PIL6
KCNJ10	P78508	KCNJ11	Q14654	KCNJ14	Q9UNX9
KCNJ16	Q9NPI9	KCNJ3	P48549	KCNJ6	P48051
KCNJ8	Q15842	KCNJ9	Q92806	KCNK1	O00180, Q9Y2U2
KCNK10	P57789	KCNK12	Q9HB15	KCNK13	Q9HB14
KCNK17	Q96T54	KCNK18	Q7Z418	KCNK2	O95069
KCNK3	O14649	KCNK5	O95279	KCNK6	Q9Y257
KCNK9	Q9NPC2	KCNMA1	Q12791	KCNMB1	Q16558
KCNMB3	Q9NPA1, Q9Y691	KCNMB4	Q86W47	KCNN1	Q92952
KCNN2	Q9H2S1	KCNN4	O15554	KCNQ1	P51787
KCNQ2	O43526	KCNQ3	O43525	KCNQ4	P56696
KCNQ5	Q9NR82	KCNS1	Q96KK3	KCNS2	Q9ULS6
KCNS3	Q9BQ31	KCNV1	Q6PIU1	KCTD1	Q719H9
KCTD10	Q8WZ19	KCTD13	Q8WZ19	KCTD15	Q96SI1
KCTD3	Q9Y597	KCTD7	Q96MP8	KDELRL1	P24390
KDELRL2	P24390, P33947	KDELRL3	O43731	KDM1A	O60341
KDM1B	Q8NB78	KDM2A	Q9Y2K7	KDM2B	Q8NHM5
KDM4A	O75164	KDM4B	O94953	KDM4C	Q9H3R0
KDM5A	P29375	KDM5B	Q9UGL1	KDM5C	P41229, Q9BY66
KDM5D	Q9BY66	KDM6A	O15550	KDM6B	O15054
KDM7A	Q6ZMT4	KDR	P35968	KDSR	Q06136

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
KEAP1	Q14145	KEL	Q9H2C0	KHDRBS1	Q07666
KHDRBS2	Q5VWX1	KHDRBS3	O75525	KHK	P50053
KHSRP	Q92945	KIAA0355	O15063	KIAA1456	Q9P272
KIAA1549	Q9HCM3	KIAA1671	Q9C0C2	KIAA1919	Q5TF39
KIDINS220	Q9ULH0	KIF11	P52732	KIF13A	Q9H1H9
KIF13B	Q9NQT8	KIF15	Q9NS87	KIF16B	Q96L93
KIF17	Q9P2E2	KIF19	Q2TAC6	KIF1A	Q12756
KIF1C	O43896	KIF20B	Q96Q89	KIF21B	O75037
KIF22	Q14807	KIF23	Q02241	KIF24	Q5T7B8
KIF26A	Q9ULI4	KIF26B	Q2KJY2	KIF27	Q86VH2
KIF2A	O00139, Q8N4N8	KIF3A	Q9Y496	KIF3B	O15066
KIF3C	O14782	KIF5A	Q12840	KIF5B	P33176
KIF5C	Q6PKD1	KIF9	Q9HAQ2	KIFAP3	Q92845
KIFC2	Q96AC6	KIFC3	Q9BVG8-2	KIN	O60870
KIRREL	Q96J84	KIRREL3	Q8IZU9	KISS1	Q15726
KISS1R	Q969F8	KIT	P10721	KITLG	P21583-1
KL	Q9UEF7-1, Q9UEF7-2	KLB	Q86Z14	KLC1	Q07866
KLC2	Q6P597, Q9H0B6	KLC4	Q9NSK0	KLF15	Q9UIH9
KLF16	Q9BXX1	KLF4	O43474	KLF5	Q13887
KLF9	Q13886	KLHDC3	Q9BQ90	KLHL11	Q9NVR0
KLHL12	Q53G59	KLHL13	Q9P2J3, Q9P2N7	KLHL2	O95198
KLHL21	Q9UJP4	KLHL3	Q9UH77	KLHL41	O60662
KLHL42	Q9P2K6	KLHL5	Q96PQ7	KLHL9	Q9P2J3
KLK7	P49862	KLKB1	P03952	KLRD1	Q13241
KLRF1	Q9NZS2	KLRG1	Q96E93	KMO	O15229
KMT2A	Q03164	KMT2B	Q9UMN6	KMT2C	Q8NEZ4
KNTC1	P50748	KPNA2	P52292	KPNA4	O00629
KPNA5	O60684	KPNA6	O60684	KPNB1	Q14974
KPTN	Q9Y664	KRBOX4	Q5JUW0	KREMEN1	Q9ULT6
KREMEN2	Q8NCW0	KRR1	Q13601	KRT10	P13645
KRT14	P02533	KRT18	P05783	KRT19	P08727
KRT24	P08727	KRT32	Q14532	KRT8	P13647
KRTAP5-10	Q6L8G5	KSR1	Q8IVT5	KSR2	Q6VAB6, Q8WXI2
KTN1	Q86UP2	KYNU	Q16719	L1CAM	P32004
L2HGDH	Q9H9P8	L3MBTL1	Q9Y468	L3MBTL2	Q969R5
LAGE3	Q14657	LAIR1	Q6GTX8	LAMA1	P25391
LAMA2	P24043	LAMA3	Q16787	LAMA4	Q16363
LAMA5	O15230	LAMB1	P07942	LAMB2	P11047, P55268
LAMC1	P11047	LAMC2	Q13753	LAMC3	Q9Y6N6
LAMP2	P13473	LAMTOR1	Q6IAA8	LAMTOR2	Q9Y2Q5
LAMTOR3	Q9UHA4	LAMTOR4	Q0VGL1	LAMTOR5	O43504
LAP3	Q13867	LARGE	O95461	LARP1	Q6PKG0
LARS2	Q15031	LAS1L	Q9Y4W2	LAT2	Q9UHI5
LATS1	O95835	LATS2	Q9NRM7	LCAT	P04180
LCK	P06239	LCLAT1	Q6UWP7	LCMT1	Q9UIC8
LCMT2	O60294	LCN2	P80188	LCP1	P13796
LCP2	Q13094	LDB1	Q86U70	LDHA	P00338
LDHB	P07195	LDHD	Q86WU2	LDLR	P01130
LDLRAP1	Q5SW96	LEF1	Q9UJU2	LEMD2	Q8NC56

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
LEMD3	Q9Y2U8	LENG1	Q96BZ8	LEO1	Q8WVC0
LEP	P41159	LEPR	P48357-1	LETM1	O95202
LFNG	Q8NES3	LGALS1	P09382	LGALS3	P17931
LGALS3BP	Q08380	LGALS9	O00182	LGI1	O95970
LGI2	Q8N0V4	LGI3	Q8N145	LGI4	Q8N135
LGR4	Q9BXB1	LGR5	O75473	LGR6	Q9HBX8
LHFPL2	Q6ZUX7	LHPP	Q9H008	LHX1	P48742
LHX4	Q969G2	LIFR	P42702	LIG1	P18858
LIG4	P49917	LILRA1	O75019	LILRB1	Q8NHL6
LILRB2	Q8N423	LILRB4	Q8NHJ6	LIMA1	Q8TDZ2
LIMD1	Q9UGP4	LIMD2	Q8TDZ2	LIMK1	P53667
LIMK2	P53671	LIMS1	P48059	LIMS2	Q7Z4I7
LIN37	Q96GY3	LIN52	Q52LA3	LIN54	Q6MZP7
LIN7A	O14910	LIN7B	Q9HAP6	LIN9	Q5TKA1
LINGO1	Q96FE5	LIPA	P18054	LIPC	P11150
LIPT2	A6NK58	LMAN1	P49257	LMAN2	Q12907
LMBRD1	Q9NUN5	LMCD1	Q9NZU5	LMF2	Q9BU23
LMNA	P02545	LMNB1	P20700	LMO1	P25800
LMO2	P25791	LMO3	P25800	LMO7	Q8WWI1
LMOD1	P29536	LNPEP	Q9UIQ6	LNK1	Q8TBB1
LONP1	P36776	LONP2	Q86WA8	LONRF1	Q17RB8
LOR	P23490	LOX	P28300	LOXL1	P58215, Q08397
LOXL2	Q9Y4K0	LOXL3	P58215	LPAR1	Q92633
LPAR2	Q9HBW0	LPAR3	Q9UBY5	LPAR5	Q9H1C0
LPAR6	P43657	LPCAT1	Q8NF37	LPCAT2	Q7L5N7
LPCAT4	Q643R3	LPGAT1	Q92604	LPIN1	Q14693
LPIN2	Q92539	LPIN3	Q9BQK8	LPL	P06858
LPPR2	Q96GM1	LPPR3	Q6T4P5	LPPR5	Q32ZL2
LRAT	O95237	LRFN1	Q9P244	LRFN3	Q9BTN0
LRFN4	Q6PJG9	LRIG1	Q96JA1	LRMP	Q12912
LRP1	Q13901	LRP10	O75096	LRP4	O75096
LRP5	O75197	LRP6	O75581	LRP8	Q14114
LRRC1	Q9BTT6	LRRC14	Q15048	LRRC16A	Q5VZK9
LRRC41	Q15345	LRRC4B	Q9NT99	LRRC52	Q8N7C0
LRRC7	Q96NW7	LRRC8A	Q8IWT6	LRRC8B	Q6P9F7
LRRC8C	Q8TDW0	LRRFIP1	Q32MZ4	LRRK1	Q96J92
LRRK2	Q5S007	LRRN2	O43300	LRRTM2	O43300
LRRTM3	Q86VH5	LRRTM4	O43300, Q86VH4	LRSAM1	Q6UWE0
LSAMP	Q13449	LSM1	O15116	LSM10	Q969L4
LSM11	P83369	LSM2	Q9Y333	LSM3	P62310
LSM4	Q9Y4Z0	LSM5	Q9Y4Y9	LSM7	Q9UK45
LSR	Q86X29	LSS	P48449	LST1	Q9Y6L6
LTB4R	Q15722, Q9NPC1	LTBP1	Q14766	LTBP2	Q14767
LTBP3	Q14767, Q9NS15	LTBP4	Q8N2S1	LTBR	P36941
LTF	P02788	LTN1	O94822	LUC7L3	O95232
LUM	P51884	LY6H	O94772	LY86	O95711
LY96	Q9Y6Y9	LYL1	P12980	LYN	P07948
LYPD1	Q8N2G4	LYPD5	Q6UWN5	LYPD6B	Q8NI32
LYPLA2	O95372	LYRM2	Q9NU23	LYRM7	Q5U5X0

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
LYVE1	Q9Y5Y7	LYZ	P61626	LZTFL1	Q9NQ48
M6PR	P20645	MAD1L1	Q9Y6D9	MAD2L1	Q13257
MAD2L2	Q9UI95	MADD	Q8WXG6-3	MAEA	Q7L5Y9
MAEL	Q96JY0	MAF	O75444	MAF1	Q9H063
MAFB	Q9Y5Q3	MAFG	O15525	MAG	P20916
MAGED1	Q9Y5V3-1, Q9Y5V3-2	MAGED2	Q9UNF1	MAGEF1	Q96MG7
MAGI2	Q86UL8	MAGT1	Q9H0U3	MAL	P58753
MALT1	Q9UDY8	MAML2	Q8IZL2	MAML3	Q96JK9
MAMLD1	Q13495	MAN1A1	P33908	MAN1A2	O60476
MAN1B1	Q9UKM7	MAN1C1	Q9NR34	MAN2A1	Q16706
MAN2A2	P49641	MANBA	O00462	MANEA	Q5SRI9
MANF	P55145	MAOA	P21397	MAOB	P21397
MAP1LC3A	Q9H492	MAP1LC3B	Q9GZQ8	MAP2	P50579
MAP2K2	P36507	MAP2K3	P46734	MAP2K4	P45985
MAP2K5	Q13163	MAP2K6	P52564	MAP2K7	O14733
MAP3K1	Q13233	MAP3K11	Q16584	MAP3K14	Q99558
MAP3K3	Q99759	MAP3K5	Q99683	MAP3K7	O43318
MAP3K8	P41279	MAP4K4	O95819	MAPK1	P28482
MAPK10	P53779	MAPK11	Q15759	MAPK12	P53778
MAPK14	Q16539	MAPK3	P27361, P28482	MAPK4	P31152
MAPK6	Q16659	MAPK7	Q13164	MAPK8	P45983
MAPK9	P45984	MAPKAP1	Q9BPZ7	MAPKAPK2	P49137
MAPKAPK5	Q8IW41	MAPRE1	Q15691	MAPRE3	Q9UPY8
MAPT	P10636	MARCKS	P29966	MARCO	Q9UEW3
MARK3	P27448	MARK4	Q96L34	MARS	P56192
MASP1	P48740	MAST1	Q7Z460	MAT2A	P31153
MAT2B	Q9NZL9	MATK	P42679	MATN1	P21941
MATN3	O15232	MAU2	Q9Y6X3	MB	P02144
MB21D1	Q8N884	MBD1	Q9UIS9	MBD2	Q9UBB5
MBD3	O95983	MBD4	O95243	MBD5	Q9P267
MBOAT1	Q6ZNC8	MBOAT2	Q6ZWT7	MBOAT7	Q96N66
MBP	P13727	MBTPS2	O43462	MC1R	Q01726
MCAM	P43121	MCAT	Q8IVS2	MCCC1	Q96RQ3
MCCC2	Q9HCC0	MCEE	Q96PE7	MCF2	P10911
MCF2L	O15068	MCFD2	Q8NI22	MCHR1	Q99705
MCL1	Q07820	MCM2	P33993, P49736	MCM4	P33991
MCM5	P33992	MCM7	P33993	MCOLN1	Q9GZU1
MCOLN2	Q8IZK6	MCOLN3	Q8TDD5	MCPH1	Q8NEM0
MCRS1	Q96EZ8	MCU	Q8NE86	MDC1	Q14676
MDGA1	Q8NFP4	MDGA2	Q7Z553	MDH1	P40925, P40926
MDH2	P40926	MDK	P21741	MDM2	Q00987
MDM4	O15151	MDP1	P54098	ME1	P48163
ME3	Q16798	MEAF6	Q9HAF1	MECOM	Q03112
MECR	Q9BV79	MED10	Q9BTT4	MED11	Q9P086
MED13	Q9UHV7	MED13L	Q71F56	MED14	O60244
MED16	Q9Y2X0	MED18	Q9BUE0	MED19	A0JLT2
MED20	Q9H944	MED21	Q13503	MED22	Q15528
MED23	Q9ULK4	MED24	O75448	MED25	Q71SY5, Q9NWA0
MED27	Q6P2C8	MED28	Q9H204	MED29	Q9NX70

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
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MEFV	O15553	MEGF11	P07942	MEI1	O43521
MEIS1	O00470	MEN1	O00255	MERTK	Q12866
MET	P08581	METAP2	P50579	METTL1	Q9UBP6
METTL14	Q9HCE5	METTL20	Q8IXQ9	METTL22	Q9BUU2
METTL3	Q86U44	METTL7A	Q9H8H3	MEX3C	Q5U5Q3
MFAP1	P55081	MFAP3	P55082	MFAP4	P55083
MFAP5	Q13361	MFGE8	Q08431	MFN1	Q8IWA4
MFN2	O95140	MFNG	O00587	MFSD2A	Q8NA29
MGA	O43451	MGAM	O43451	MGAT2	Q10469
MGAT3	Q09327	MGAT4A	Q9UM21	MGAT4B	Q9UBM8, Q9UQ53
MGAT4C	Q9UBM8	MGAT5	Q09328	MGAT5B	Q3V5L5
MGLL	Q99685	MGMT	P16455	MGRN1	O60291
MGST1	P10620	MGST2	Q99735	MGST3	O14880
MIB1	Q86YT6	MIB2	Q96AX9	MICA	Q29983
MICAL1	Q8TDZ2	MICB	Q29980	MICU1	Q9BPX6
MICU2	Q8IYU8	MICU3	Q86XE3, Q8IYU8	MID1	Q9NQW6
MID1IP1	Q92748, Q9NPA3	MIEF1	Q9NQG6	MIF	P14174
MINA	Q8IUF8	MINK1	Q8N4C8	MINOS1	Q5TGZ0
MIS12	Q9H081	MIS18BP1	Q6P0N0	MITF	O75030-9
MKKS	Q9NPJ1	MKL1	Q969V6	MKLN1	Q9UL63
MKNK1	Q9BUB5	MKRN1	Q9UHC7	MKRN3	P31947
MKS1	Q9NXB0	MLC1	P60660	MLH3	Q9UHC1
MLKL	Q8NB16	MLLT1	Q03111	MLLT3	P42568
MLPH	Q9BV36	MLST8	Q9BVC4	MLX	Q9UH92
MLXIP	Q9NP71	MLXIPL	Q9NP71	MMAB	Q96EY8
MMADHC	Q9H3L0	MME	P08473	MMP1	P22894
MMP10	P08254, P09238	MMP11	P24347	MMP14	P50281
MMP15	P51511	MMP17	Q9ULZ9	MMP19	Q99542
MMP2	P08253	MMP24	Q9Y5R2	MMP3	P08254
MMP8	P22894	MMRN1	Q13201	MMS19	Q96T76
MNAT1	P51948	MNDA	P41218	MOB1A	Q9H8S9
MOB1B	Q7L9L4	MOBP	Q8NFW9	MOCOS	Q96EN8
MOCS1	Q9NZB8-1, Q9NZB8-5	MOCS3	O95396	MOGS	Q13724
MON1A	Q86VX9	MON1B	Q7L1V2	MORC2	Q9Y6X9
MORF4L1	Q9UBU8	MORF4L2	Q15014	MOSPD2	Q8NHP6
MOV10	Q9HCE1	MOV10L1	Q9BXT6	MPC2	O95563
MPDU1	O75352	MPG	P29372	MPHOSPH10	O00566
MPHOSPH8	Q99549	MPO	P05164	MPP2	Q08050
MPP5	Q8N3R9	MPP6	Q99547	MPP7	Q5T2T1
MPST	P25325	MPZ	P25189	MPZL2	Q86YT9
MR1	Q8N490-2	MRAS	O14807	MRC1	P22897
MRC2	Q9UBG0	MRE11A	P49959	MRGBP	Q9NV56
MRPL10	Q7Z7H8	MRPL11	Q9Y3B7	MRPL12	P52815
MRPL14	Q6P1L8	MRPL15	P49406, Q9P015	MRPL16	Q9NX20
MRPL17	Q9NRX2	MRPL18	Q9H0U6	MRPL19	P49406
MRPL2	Q5T653, Q9BZE1	MRPL20	Q9BYC9	MRPL21	Q7Z2W9
MRPL23	Q16540	MRPL24	Q96A35	MRPL27	Q8IXM3, Q9P0M9

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
MRPL28	Q13084, Q8TCC3	MRPL3	P09001	MRPL30	Q8TCC3
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MRPL36	Q9P0J6	MRPL37	Q9BZE1	MRPL38	Q96DV4
MRPL4	Q9BYD3	MRPL40	Q9NQ50	MRPL41	Q8IXM3
MRPL43	Q8N983	MRPL44	Q9H9J2	MRPL45	Q9BRJ2
MRPL46	Q9H2W6	MRPL47	Q9HD33	MRPL49	Q13405
MRPL50	Q8N5N7	MRPL51	Q4U2R6	MRPL52	Q86TS9
MRPL53	Q96EL3	MRPL54	Q6P161	MRPL55	Q7Z7F7
MRPL57	Q9BQC6	MRPL9	Q9BYD2	MRPS11	P82912
MRPS12	O15235	MRPS14	O60783	MRPS15	P82914
MRPS16	Q9Y3D3	MRPS18B	Q9Y676	MRPS18C	Q9Y3D5
MRPS2	Q9Y399	MRPS21	P82921	MRPS23	Q9Y3D9
MRPS24	Q96EL2	MRPS25	P82663	MRPS26	Q9BYN8
MRPS28	P82673, Q9Y2Q9	MRPS30	Q9NP92	MRPS33	Q9Y291
MRPS34	P82930	MRPS35	P82673, Q9Y2Q9	MRPS36	P82909
MRPS6	P82932	MRPS7	Q9Y2R9	MRS2	Q9HD23
MRV11	Q9Y6F6	MSH3	P20585	MSH4	O15457
MSH6	P52701	MSI1	O43347	MSI2	Q96DH6
MSL1	Q68DK7	MSL2	Q9HCI7	MSL3	Q8N5Y2
MSLN	Q13421	MSMO1	Q15800	MSN	P26038
MSR1	P21757	MSRA	Q9UJ68	MSRB1	Q9NZV6
MSRB2	Q9Y3D2	MSRB3	Q8IXL7	MT-ATP6	P00846
MT-ATP8	P03928	MT-CO1	P00395	MT-CO2	P00403
MT-CO3	P00414	MT-CYB	P00156	MT-ND1	P03886
MT-ND2	P03891	MT-ND3	P03897	MT-ND4	P03905
MT-ND4L	P03901	MT-ND5	P03915	MT-ND6	P03923
MT-RNR2	Q8IVG9	MT1E	P04732	MT1F	P04733
MT1M	P13640, Q8N339	MT1X	P80297	MT2A	P02795
MT3	P25713	MTA1	Q13330	MTA2	O94776
MTA3	Q9BTC8	MTBP	Q96DY7	MTERF1	Q99551
MTERF4	Q7Z6M4	MTHFD1	P11586	MTHFD1L	Q6UB35
MTHFD2	P13995	MTHFD2L	Q9H903	MTHFR	P42898
MTM1	Q13496	MTMR10	Q9NXD2	MTMR12	Q9C0I1
MTMR3	Q96QG7	MTMR4	Q9NYA4	MTMR8	Q96QG7
MTMR9	Q96QG7	MTR	Q99707	MTRF1L	Q9UGC7
MTRR	Q9UBK8	MTX1	Q13505	MUC13	Q9H3R2
MUC19	Q7Z5P9	MUC3A	Q02505	MUL1	Q969V5
MUM1	Q15306	MUS81	Q96NY9	MUT	P22033
MVD	P53602	MVK	Q03426	MVP	Q14764
MX1	P20591	MX2	P20591, P20592	MXD4	Q14582
MXRA8	Q9BRK3	MYBBP1A	Q9BQG0	MYBL1	P10243
MYBPC1	Q00872	MYBPC2	Q14324	MYC	P01106
MYCN	P04198	MYD88	Q99836	MYDGF	Q969H8
MYH10	P35580	MYH11	P35749	MYH14	Q7Z406
MYH2	Q9UKX2	MYH3	P11055	MYH9	P35579
MYL1	P05976	MYL12B	O14950	MYL2	P10916
MYL3	P08590	MYL6	P60660	MYL6B	P14649
MYL9	P24844	MYLIP	Q8WY64	MYLK	Q15746
MYO10	Q9HD67	MYO18B	P35626	MYO19	Q96H55

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MYO9A	B2RTY4	MYO9B	Q13459	MYOCD	Q8IZQ8
MYRIP	Q8NFW9	MYSM1	Q5VVJ2	MYT1	Q99640
MZT2A	Q6P582	MZT2B	Q6NZ67	N4BP1	O75113
N6AMT1	Q9Y5N5	NAA30	Q147X3	NAA35	Q5VZE5
NAA38	Q9BRA0	NAAA	Q02083	NAALAD2	Q9Y3Q0
NAB1	P08865	NAB2	Q15742	NABP1	Q96AH0
NABP2	Q9BQ15	NACA	Q13765	NADK	O95544
NADK2	Q4G0N4	NADSYN1	Q6IA69	NAF1	Q15025
NAGK	Q9UJ70	NAGLU	P54802	NALCN	Q8IZF0
NAMPT	P43490	NANOS3	P60323	NANS	Q9NR45
NAPA	Q9Y2A7	NAPB	Q9H115	NAPEPLD	Q6IQ20
NAPG	Q99747	NAPRT	Q6XQN6	NARF	Q8WVD3
NARFL	Q9H6Q4	NARS	O43776	NAT8L	Q8N9F0
NBAS	A2RRP1	NBN	O60934	NCALD	P61601
NCAN	O14594	NCAPG	Q9BPX3	NCAPG2	Q86XI2
NCAPH2	Q6IBW4	NCBP2	P52298	NCEH1	Q6PIU2
NCF2	P19878	NCF4	Q15080	NCK1	O43639, P16333
NCKAP1	Q9Y2A7	NCKAP1L	P55160	NCL	P19338
NCOA1	Q15788	NCOA2	Q15596	NCOA3	Q9Y6Q9
NCOR2	Q9Y618	NCR3LG1	Q68D85	NCS1	P43080
NCSTN	Q92542	NDC1	Q9BTX1	NDC80	O14777
NDE1	Q9NXR1	NDEL1	Q9GZM8	NDN	Q99608
NDRG1	Q92597	NDST3	O95803, Q9H3R1	NDUFA1	O15239
NDUFA11	Q86Y39	NDUFA12	Q9UI09	NDUFA13	Q9P0J0
NDUFA2	O43678	NDUFA3	O95167	NDUFA4	O00483
NDUFA5	Q16718	NDUFA6	P56556	NDUFA7	O95182
NDUFA8	P51970	NDUFA9	Q16795	NDUFAB1	O14561
NDUFAF1	Q9Y375	NDUFAF2	Q8N183	NDUFAF3	Q9BU61
NDUFAF4	Q9P032	NDUFAF6	Q330K2	NDUFB1	O75438
NDUFB10	O96000	NDUFB11	Q9NX14	NDUFB2	O95178
NDUFB3	O43676	NDUFB4	O95168	NDUFB6	O95139
NDUFB7	P17568	NDUFB8	O95169	NDUFB9	Q9Y6M9
NDUFC1	O43677	NDUFC2	O95298	NDUFS2	O75306
NDUFS3	O75489	NDUFS5	O43920	NDUFS6	O75380
NDUFS7	O75251	NDUFS8	O00217	NDUFV1	P49821
NDUFV2	P19404	NEB	Q15058	NECAP1	Q8NC96
NECAP2	Q9NVZ3	NEDD1	Q8NHV4	NEDD4	P46934, Q96PU5
NEDD8	Q15843	NEFL	P07196	NEGR1	Q7Z3B1
NEIL2	Q969S2	NEK1	Q96PY6	NEK11	Q8NG66
NEK6	Q9HC98	NEK7	Q8TDX7	NEK8	Q8TD19
NEK9	Q8TD19	NELFB	Q8WX92	NELFCD	Q8IXH7
NELFE	P18615	NELL2	Q99435	NEMF	O60524
NES	Q6P1A2	NET1	Q7Z628	NEU1	Q99519
NEURL1	O76050	NEURL1B	A8MQ27	NF1	P21359
NFAM1	Q8NET5	NFATC1	O95644	NFATC2	Q12968, Q13469
NFATC3	Q12968	NFE2L2	Q16236	NFIA	Q12857
NFIB	O00712	NFIC	P08651	NFIX	Q14938

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NFYA	P23511	NGB	Q9NPG2	NGEF	Q8N5V2
NGF	P01138	NGFR	P08138	NGFRAP1	Q00994
NGRN	Q9NPE2	NHLRC2	Q8NBF2	NHLRC3	Q5JS37
NHP2	Q9NX24	NHS	Q6T4R5	NICN1	Q9BSH3
NID1	P14543	NID2	Q14112	NINL	Q9Y2I6
NIP7	Q9Y221	NIPA2	Q8N8Q9	NIPAL1	Q6NVV3
NIPAL2	Q9H841	NIPAL3	Q6P499	NIPBL	Q6KC79
NISCH	Q9Y2I1	NIT2	Q9NQ4	NKAP	Q8N5F7
NKIRAS2	Q9NYR9	NKX2-5	P52952	NLE1	Q6Q0C0
NLGN1	Q8N2Q7	NLGN2	Q8NFZ4	NLGN4X	Q8N0W4
NLGN4Y	Q8NFZ3	NLK	Q9UBE8	NLN	Q9BYT8
NLRC4	Q9NPP4	NLRC5	Q86WI3	NLRP1	Q9C000
NLRP12	P59046	NLRP3	Q96P20	NLRX1	Q86UT6
NMB	Q14956	NMBR	P28336	NME1	P15531
NME2	P22392	NME3	Q13232	NME4	O00746
NME7	Q9Y5B8	NMI	Q13287	NMNAT2	Q9BZQ4
NMNAT3	Q96T66	NMRAL1	Q9HBL8	NMRK1	Q9NWW6
NMT1	P30419	NMT2	O60551	NNMT	P40261
NOB1	Q9ULX3	NOC2L	Q9Y3T9	NOC4L	Q9BVI4
NOD1	Q9Y239	NOD2	Q9HC29	NOL11	Q9H8H0
NOL12	Q9UGY1	NOL4L	Q86VX2	NOL6	Q9H6R4
NOP10	Q9NPE3	NOP2	P46087	NOP56	O00567
NOS1	P29475	NOS2	P35228	NOS3	P29474
NOSIP	Q9Y314	NOSTRIN	Q8IVI9	NOTCH1	P46531
NOTCH2	Q04721	NOTCH3	Q9UM47	NOTCH4	Q99466
NOTUM	Q6P988	NOV	Q9UIW2	NOX4	Q9NPH5
NOX5	Q96PH1	NOXA1	Q86UR1	NPAS2	Q99743
NPB	Q8NG41	NPC1	O15118	NPC1L1	Q9UHC9-2
NPC2	P61916	NPDC1	Q9NQX5	NPEPPS	P55786
NPFFR2	Q9Y5X5	NPHP1	O15259	NPHP4	O75161
NPIPB3	Q92617	NPL	Q9BXD5	NPLOC4	Q8TAT6
NPM1	P06748	NPM2	Q86SE8	NPNT	Q6UXI9
NPR1	P16066	NPRL2	Q8WTW4	NPSR1	Q6W5P4
NPY1R	P25929	NPY5R	Q15761	NQO1	P15559
NR0B1	P51843-1, P51843-2	NR1D1	P20393	NR1D2	Q14995
NR1H3	Q13133	NR2C1	P13056	NR2C2	P49116
NR2C2AP	Q86WQ0	NR2F1	P10589	NR2F2	P24468
NR2F6	P10588	NR3C1	P04150-1, P04150-2, P04150-3, P04150-4, P04150-5, P04150-6, P04150-7, P04150-8, P04150-9	NR3C2	P08235-1, P08235-2, P08235-3, P08235-4
NR4A1	P22736	NR4A2	P43354	NR4A3	Q92570-1, Q92570-2
NR5A2	O00482-1, O00482-2, O00482-3	NR6A1	Q15406-1, Q15406-2, Q15406-3, Q15406-4, Q15406-5	NRAS	P01111
NRBF2	Q96F24-1, Q96F24-2	NRBP1	Q9UHY1	NRCAM	Q92823
NRF1	Q16656	NRG1	Q02297, Q02297-10	NRG2	O14511
NRG3	P56975	NRGN	Q92686	NRIP1	P48552
NRN1	Q9NPD7	NRP2	O60462	NRXN1	P58400, Q9ULB1
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Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
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NSRP1	Q9H0G5	NSUN6	Q8TEA1	NT5C	Q8TCD5
NT5C3B	Q969T7	NT5E	P21589	NT5M	Q8TCD5, Q9NPB1
NTF3	P20783	NTHL1	P78549	NTM	Q9P121
NTN1	O95631	NTN3	O00634	NTN4	Q9HB63
NTNG1	Q9Y2I2	NTNG2	Q96CW9	NTRK1	P04629
NTRK3	Q16288	NTS	P30990	NUAK1	O60285
NUBP1	P53384	NUBP2	Q9Y5Y2	NUBPL	Q8TB37
NUDC	Q9Y266	NUDT1	P36639-1, P36639-2, P36639-3, P36639-4	NUDT10	Q9BW91-1
NUDT11	Q96G61	NUDT13	Q86X67	NUDT14	O95848
NUDT15	Q9NV35	NUDT16	Q96DE0	NUDT2	P50583
NUDT3	O95989	NUDT4	Q96G61, Q9NZJ9	NUDT7	P0C024
NUDT8	Q8WV74	NUF2	Q9BZD4	NUMB	P49757
NUP133	Q8WUM0	NUP153	P49790	NUP205	Q92621
NUP214	P35658, Q96HA1	NUP35	Q8NFH5	NUP37	Q8NFH4
NUP43	Q8NFH3	NUP50	Q9UKX7	NUP54	Q7Z3B4
NUP62	P37198	NUP98	P52948-3, P52948-4, P52948-5	NUPL1	Q9BVL2-1, Q9BVL2-2
NXF1	Q9GZY0, Q9UBU9	NXT1	Q9UKK6	NXT2	Q9UKK6
OAS1	P00973	OAS2	P29728	OAS3	Q9Y6K5
OASL	Q15646	OAZ1	P54368	OAZ2	O95190
OBFC1	Q9H668	OBSL1	Q5VST9	OCA2	Q04671
OCLN	Q16625	OCRL	Q01968	ODC1	P11926
ODF2	Q5BJF6	OFD1	O75665	OGFOD1	Q8N543
OGG1	O15527-4	OGN	P20774	OGT	O15294
OLFM4	Q6UX06	OLR1	P78380	OMA1	Q96E52
OMD	Q99983	OMG	P23515	ONECUT3	O60422
OPCML	Q14982	OPHN1	O60890	OPLAH	O14841
OPRD1	P41143	OPRK1	P41145	OPRL1	P41146
OPRM1	P35372	OPTN	Q96CV9	OR2L13	Q8N349
OR7A5	Q15622	OR8S1	Q8NH09	ORA12	Q96SN7
ORC2	Q13416	ORC4	O43929	ORC5	O43913
ORC6	Q9Y5N6	ORMDL2	Q53FV1	ORMDL3	Q8N138
OS9	Q13438	OSBP	P22059	OSBPL10	Q9BXB5
OSBPL1A	Q9BXW6	OSBPL3	Q9H4L5	OSBPL8	Q9BZF1
OSBPL9	Q96SU4	OSGEP	Q9NPF4	OSM	P13725
OSMR	Q99650	OSR1	Q8TAX0	OST4	P0C6T2
OSTC	Q9NRP0	OSTF1	Q92882	OTOF	Q9HC10
OTOGL	Q3ZCN5	OTUB1	Q96FW1	OTUB2	Q96DC9
OTUD1	Q5VV17	OTUD5	Q96G74	OTUD7A	Q8TE49
OTUD7B	Q6GQQ9	OTULIN	Q96BN8	OVCH1	Q7RTY9
OXA1L	Q15070	OXER1	Q8TDS5	OXSM	Q9NWX1
P2RX1	P51575	P2RX3	P56373	P2RX5	Q93086
P2RX6	O15547	P2RX7	Q99572	P2RY1	P47900
P2RY10	O00398	P2RY11	Q96G91	P2RY12	Q9H244
P2RY13	Q9BPV8	P2RY14	Q15391	P2RY2	P41231
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P4HA1	P13674	P4HA2	O15460	P4HB	P07237

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PAFAH1B1	P43034	PAFAH1B2	P68402	PAFAH1B3	Q15102
PAG1	Q9NWQ8	PAICS	P22234	PAIP1	Q9H074
PAK1	Q13153	PAK3	Q13153	PAK7	Q9P286
PAM16	Q9Y3D7	PAN2	Q9HBH5	PAN3	Q58A45
PANK1	Q8TE04	PANK4	Q9NVE7	PANX1	Q96RD7
PANX2	Q96RD6	PAPD5	Q8NDF8	PAPOLA	P51003
PAPOLG	Q9BWT3	PAPPA	Q13219	PAPPA2	Q9BXP8
PAPSS1	O43252	PAPSS2	O95340	PARD3	Q8TEW0
PARD6A	Q9NPB6	PARD6B	Q9BYG5	PARK2	O60260
PARK7	Q99497	PARL	Q9H300	PARN	O95453
PARP1	P09874	PARP10	Q53GL7	PARP14	Q460N5
PARP2	Q9UGN5	PARP4	Q9UUKK3	PARP6	Q2NL67
PARP8	Q8N3A8	PARP9	Q8IXQ6	PARVA	Q9NVD7
PARVB	Q9HBI1	PAX3	P23760	PAX8	Q06710
PBX1	P40424	PBX2	Q9Y4H4	PC	P11498
PCBD1	P61457	PCBP1	Q15365	PCBP2	Q15366
PCBP4	P57723	PCCA	P05165	PCDH15	Q96QU1
PCDH19	Q8TAB3	PCDH7	O60245	PCGF2	P35227
PCGF5	Q86SE9	PCGF6	Q9BYE7	PCK1	P05771
PCLO	Q9Y6V0	PCM1	Q15154	PCMT1	P22061
PCNA	P12004	PCOLCE	Q15113	PCSK2	P16519
PCSK5	Q92824	PCSK6	P29122	PCSK7	Q16549
PCSK9	Q8NBP7	PCTP	Q9UKL6	PCYOX1L	Q8NBM8
PCYT2	Q99447	PDAP1	Q13442	PDCD11	Q14690
PDCD1LG2	Q9BQ51	PDCD4	Q53EL6	PDCD6IP	Q8WUM4
PDCL	Q13371	PDE10A	Q9Y233	PDE11A	Q9HCR9
PDE1A	P54750	PDE1B	Q01064	PDE1C	Q01064
PDE2A	O00408	PDE3A	Q14432	PDE3B	Q13370
PDE4A	P27815	PDE4B	Q07343-3	PDE4C	Q08493
PDE4D	Q08499-6	PDE5A	O76074	PDE6D	O43924
PDE7A	Q13946	PDE7B	Q9NP56	PDE8A	O60658
PDE8B	O95263	PDE9A	O76083	PDGFA	P04085-1, P04085-2
PDGFB	P01127	PDGFC	Q9NRA1-1	PDGFD	Q9GZP0
PDGFRA	P16234	PDGFRB	P09619	PDHA1	P08559, P29803
PDHB	P11177	PDHX	O00330	PDIA2	Q13087
PDIA3	P30101	PDIA5	Q14554	PKD3	Q15120
PDK4	Q16654	PDLIM5	Q96HC4	PDP2	Q9P2J9
PDPK1	O15530	PDPN	Q86YL7	PDPR	Q8NCN5
PDS5A	Q29RF7	PDS5B	Q9NTI5	PDSS2	Q86YH6
PDZD11	Q5EBL8	PEA15	Q15121	PEAK1	Q9H792
PEAR1	P11047	PEBP1	P30086	PECAM1	P16284
PECR	Q9BY49	PELI1	Q96FA3	PELI2	Q9HAT8
PELO	Q9BRX2	PELP1	Q8IZL8	PER1	O15534
PER2	O15055	PER3	P56645	PERP	Q96FX8
PES1	O00541	PET100	P0DJ07	PEX10	O60683
PEX11B	O96011	PEX14	O75381	PEX16	Q9Y5Y5

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
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PFDN4	Q9NQP4	PFDN5	Q99471	PFDN6	O15212
PFKFB1	P16118	PFKFB2	O60825	PFKFB3	Q16875
PFKFB4	Q16877	PFKM	P08237	PFKP	Q01813
PFN1	P07737	PFN2	P07737, P35080	PGAM1	P18669
PGAM5	Q96HS1-2	PGAP1	Q75T13	PGD	P52209
PGF	P49763	PGGT1B	P53609	PGK1	P00558, P07205
PGLS	O95336	PGM1	P36871	PGM2	P36871, Q96G03
PGM2L1	Q6PCE3	PGP	A6NDG6	PGR	P06401
PGRMC1	O00264	PGS1	Q32NB8	PHACTR2	O75167
PHAX	Q9H814	PHB	P35232	PHB2	Q99623
PHC1	P78364	PHC2	P78364, Q8IXK0	PHC3	Q8NDX5
PHF1	O43189	PHF10	Q8WUB8	PHF19	Q5T6S3
PHF20	Q9BVI0	PHF20L1	A8MW92	PHF21A	Q96BD5
PHF5A	Q7RTV0	PHF8	Q9UPP1	PHGDH	O43175
PHIP	Q8WWQ0	PHKA1	P46020	PHKA2	P46019
PHKB	Q93100	PHKG1	Q16816	PHKG2	P15735
PHLDA1	Q8WV24	PHLPP1	O60346	PHLPP2	O43747
PHYKPL	Q8IUZ5	PI4K2A	Q9BTU6	PI4KA	P42356
PIANP	Q8IYJ0	PIAS1	O75925	PIAS2	O75928-1
PIAS3	Q9Y6X2	PIAS4	Q8N2W9	PICALM	Q13492
PICK1	Q9NRD5	PIEZO1	Q92508	PIGB	Q92521
PIGF	Q07326	PIGH	Q14442	PIGK	Q92643
PIGL	Q9Y2B2	PIGN	O95427	PIGO	Q8TEQ8
PIGQ	Q9BRB3-2	PIGS	Q96S52	PIGT	Q969N2
PIGU	Q9H490	PIGW	Q7Z7B1	PIGX	Q8TBF5
PIGZ	Q86VD9	PIK3AP1	Q6ZUJ8	PIK3C2A	O00443
PIK3C2B	O00750	PIK3C3	Q8NEB9	PIK3CA	P42336
PIK3CB	P42338	PIK3CD	O00329	PIK3CG	P48736
PIK3R1	P27986	PIK3R4	Q99570	PIK3R5	Q8WYR1
PIK3R6	Q5UE93	PIKFYVE	Q9Y2I7	PILRA	Q9UKJ1
PIM1	P11309	PIN1	Q13526	PINK1	Q9BXM7
PIP4K2A	P48426	PIP4K2B	P78356	PIP4K2C	Q8TBX8
PIP5K1A	Q99755	PIP5K1B	O14986	PIP5K1C	O60331
PIPOX	Q9P0Z9	PIR	O00625	PITPNA	Q00169
PITPNM1	O00562	PITRM1	Q5JRX3	PITX2	Q99697
PIWIL2	Q8TC59	PIWIL4	Q7Z3Z4	PJA1	Q8NG27
PKD1	P98161	PKD2	Q13563	PKM	P14618
PKMYT1	Q99640	PKN1	Q16512	PKN2	Q16513
PKN3	Q6P5Z2	PKNOX1	P55347	PKP2	Q99959
PKP4	Q99569	PLA1A	Q53H76	PLA2G12A	Q9BZM1
PLA2G15	Q8NCC3	PLA2G16	P53816	PLA2G2A	P14555, Q9UNK4
PLA2G3	Q9NZ20	PLA2G4A	P47712	PLA2G4C	Q9UP65
PLA2G7	Q13093	PLA2R1	Q13018	PLAC8	Q9NZF1
PLAGL1	Q9UM63	PLAT	P00750	PLAU	P00749
PLAUR	Q03405	PLB1	Q3MJ16	PLBD1	Q6P4A8
PLCB1	Q9NQ66	PLCB2	Q00722	PLCB3	Q01970

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PLCG2	P16885	PLCH1	Q4KWH8	PLCH2	O75038
PLCZ1	Q86YW0	PLD1	Q13393	PLD2	O14939
PLD3	Q8IV08	PLD4	Q96BZ4	PLEC	Q15149
PLEK	P08567	PLEKHA1	Q9HB21	PLEKHA2	Q9HB19
PLEKHA3	Q9HB20	PLEKHA4	Q9H4M7	PLEKHA5	Q9HAU0
PLEKHA6	Q9Y2H5	PLEKHA8	Q96JA3	PLEKHG1	Q9ULL1
PLEKHG2	Q9H7P9	PLEKHG3	A1L390	PLEKHG4	Q58EX7
PLIN1	O60240	PLIN2	Q99541	PLIN3	Q8IUC6
PLIN4	O60664	PLK2	Q9NYY3	PLK3	Q9H4B4
PLN	P26678	PLOD1	Q02809	PLOD2	O00469
PLRG1	O43660	PLS1	Q14651	PLTP	P55058-1, P55058-2
PLXNA1	Q9UIW2	PLXNA2	O75051	PLXNA3	P51805
PLXNA4	Q9HCM2	PLXNB1	O43157	PLXNB3	O43157
PLXNC1	O60486	PLXND1	Q9Y4D7	PM20D1	Q6GTS8
PMEL	P40967	PMEPA1	Q969W9	PMF1	Q6P1K2
PMM1	Q92871	PMM2	O15305	PMP2	A6NFH5
PMP22	Q9NR77	PMPCA	Q10713	PMPCB	O75439
PMS1	P54278	PMVK	Q15126	PNKD	Q8N490-2
PNKP	Q96T60	PNN	Q9H307	PN01	Q9NRX1
PNP	P00491	PNPLA2	Q96AD5	PNPLA3	Q9NST1
PNPLA7	Q6ZV29	PODXL2	Q9NZ53	POFUT1	Q9H488
POFUT2	Q9Y2G5	POGLUT1	Q8NBL1	POLA1	P09884
POLB	P06746	POLD2	P49005	POLD3	Q15054
POLDIP3	Q9BY77	POLE2	P56282	POLE3	Q9NRF9
POLE4	Q9NR33	POLI	Q9UNA4	POLK	Q9UBT6
POLL	P57775	POLN	Q7Z5Q5	POLQ	O75417
POLR1C	O15160	POLR1D	P0DPB5	POLR1E	O15160
POLR2A	P24928	POLR2B	P30876	POLR2C	P19387
POLR2D	O15514	POLR2E	P19388	POLR2F	P61218
POLR2G	P62487	POLR2H	P52434	POLR2I	P36954
POLR2J	P52435	POLR2K	P53803	POLR2L	P62875
POLR3A	O14802	POLR3C	Q9BUI4	POLR3D	P05423
POLR3G	O15318, Q9BT43	POLR3GL	Q9BT43	POLR3H	Q9Y535
POLR3K	Q9Y2Y1	POLRMT	O00411	POM121	Q96HA1
POM121C	A8CG34, Q96HA1	POMGNT1	Q8WZA1	POMK	Q9H5K3
POMP	Q9Y244	POMT1	Q9Y6A1	POMT2	Q9UKY4
PON2	Q15165	PON3	Q15166	POP1	Q9UKB1
POP4	O95707	POP5	Q969H6	POP7	O75817
POR	P16435	PORCN	Q9H237	POT1	P09110
POU2F1	P14859	POU2F2	P09086	POU3F1	Q03052
POU3F2	P20265	POU4F1	Q01851	POU4F2	Q12837
PPA2	Q15181, Q9H2U2	PPAP2A	O14494	PPAP2B	O14495
PPAPDC1A	Q5VZY2	PPAPDC1B	Q8NEB5	PPAPDC2	Q8IY26
PPARA	Q07869	PPARD	Q03181	PPARG	P37231
PPARGC1A	Q9UBK2	PPARGC1B	Q86YN6	PPAT	Q06203
PPCDC	Q96CD2	PPCS	Q9HAB8	PPEF1	O14829
PPFIA1	Q13136	PPFIA3	O75145	PPFIBP1	Q86W92

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
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PPIF	P62937	PPIG	Q13427	PPIL1	Q9Y3C6
PPIL2	Q13356	PPIL3	Q9H2H8	PPIL4	Q8WUA2
PIIP5K1	Q6PFW1	PPL	P23434	PPM1B	O75688
PPM1D	O15297	PPM1E	Q8WY54	PPM1L	Q5SGD2
PPME1	Q9Y570	PPOX	O43612	PPP1CA	P62136
PPP1CB	P62140	PPP1R10	Q96QC0	PPP1R12A	O14974
PPP1R12B	O60237	PPP1R13B	Q96KQ4	PPP1R13L	Q8WUF5
PPP1R15A	O75807	PPP1R16A	P24298	PPP1R1B	Q9UD71
PPP1R3C	Q9UQK1	PPP1R8	Q12972	PPP2CA	P67775
PPP2CB	P62714	PPP2R1A	P30153	PPP2R1B	P30154
PPP2R2A	P63151	PPP2R5A	Q15172	PPP2R5B	Q15173
PPP2R5C	Q13362	PPP2R5D	Q14738	PPP2R5E	Q16537
PPP3CA	Q08209	PPP3CC	P48454	PPP3R1	P63098
PPP4C	P60510	PPP4R2	Q9NY27	PPP5C	P53041
PPP6R1	Q9UPN7	PPP6R3	Q5H9R7	PPRC1	Q5VV67
PPT2	Q9UMR5	PPWD1	Q96BP3	PQBP1	O60828
PQLC2	Q6ZP29	PRC1	O43663	PRCC	Q92733
PRCP	P42785	PRDM1	O75626	PRDM16	Q9HAZ2
PRDM4	Q9UKN5	PRDM8	Q9NQV8	PRDX1	Q06830
PRDX2	P32119	PRDX3	P30048	PRDX4	Q13162
PRDX5	P30044-2	PREB	Q9HCU5	PRELID1	Q9Y255
PRELP	P51888	PREX1	Q8TCU6	PREX2	Q70Z35
PRF1	P14222	PRG4	Q96GM1	PRICKLE1	Q96MT3
PRIM2	P49643	PRKAA1	Q13131	PRKAA2	P54646
PRKAB1	Q9Y478	PRKACA	P17612, P22612	PRKACB	P22694
PRKAG1	P54619	PRKAG2	Q9UGJ0	PRKAR1A	P10644
PRKAR1B	P31321	PRKAR2A	P13861	PRKAR2B	P31323
PRKCA	P17252	PRKCB	P05771	PRKCD	Q05655
PRKCE	Q02156	PRKCG	P05129	PRKCH	P24723
PRKCQ	Q04759	PRKCSH	P14314	PRKD1	Q15139
PRKD2	Q9BZL6	PRKD3	O94806	PRKDC	P78527
PRKG1	Q13976-1	PRKG2	Q13237	PRKRA	O75569
PRKX	P51817	PRLR	P16471	PRMT1	Q99873
PRMT2	Q96LA8	PRMT3	O60678	PRMT5	O14744
PRMT6	Q96LA8	PRMT9	Q86XK2	PRNP	P04156
PROCR	Q9UNN8	PRODH	O43272, Q9UF12	PROS1	P07225
PRPF19	Q9UMS4	PRPF3	O43395	PRPF31	Q8WWY3
PRPF38A	Q8NAV1	PRPF4	O43172	PRPF6	O94906
PRPF8	Q6P2Q9	PRPS1	P60891	PRPS2	P11908
PRR3	Q96QC0	PRR5	P85299	PRSS23	O95084
PRSS3	P35030	PRX	Q9BXM0, Q9BXM0-1	PSAT1	Q9Y617
PSEN2	P49810	PSENEN	Q9NZ42	PSIP1	O75475
PSMA1	P25786	PSMA3	P25788	PSMA4	P25789
PSMA5	P28066	PSMA6	P60900	PSMA7	O14818
PSMB1	P20618	PSMB2	P49721	PSMB3	P49720
PSMB4	P28070	PSMB5	P28074	PSMB6	P28072
PSMB7	Q99436	PSMB8	P28062	PSMB9	P28065

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
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PSMC6	P62333	PSMD1	Q99460	PSMD10	O75832
PSMD11	O00231	PSMD13	Q9UNM6	PSMD2	Q13200
PSMD3	O43242	PSMD4	P55036	PSMD6	Q15008
PSMD7	P51665	PSMD8	P48556	PSMD9	O00233
PSME2	Q9UL46	PSME3	Q06323	PSME4	Q14997
PSMF1	Q92530	PSMG1	O95456	PSMG3	Q9BT73
PSMG4	Q5JS54	PSPN	O60542	PSTPIP1	O43586
PTAFR	P25105	PTCH1	Q13635	PTCH2	Q9Y6C5
PTDSS1	P48651	PTDSS2	Q9BVG9	PTEN	P60484
PTGDR	Q13258	PTGDS	P31025	PTGER2	P35408, P43116
PTGER3	P43115	PTGER4	P35408	PTGES	O14684
PTGES2	Q9H7Z7	PTGFR	P43088	PTGIR	P43119
PTGR1	Q14914	PTGS1	P23219	PTGS2	P35354
PTH1R	Q03431	PTH2R	P49190	PTHLH	P12272
PTK2	Q05397	PTK2B	Q14289	PTK7	P42679
PTPMT1	Q8WUK0	PTPN1	P18031	PTPN11	Q06124
PTPN12	Q05209	PTPN13	Q12923	PTPN14	Q15678
PTPN2	P17706	PTPN20	Q4JDL3	PTPN22	Q9Y2R2
PTPN23	Q9H3S7	PTPN4	P29074	PTPN5	P54829
PTPN6	P29350	PTPN7	P35236	PTPN9	P43378
PTPRB	P23467	PTPRC	P08575	PTPRD	P23468
PTPRJ	Q12913	PTPRK	Q15262	PTPRN2	Q92932
PTPRO	Q92729	PTPRU	Q92729	PTPRZ1	P23471
PTRF	Q6NZI2	PTRH2	Q9Y3E5	PTTG1	O95997
PTX3	P26022	PUF60	Q9UHX1	PUM1	Q14671
PUM2	Q8TB72	PURA	P30520	PUS3	Q9BZE2
PVALB	P20472	PVRL1	Q15223	PVRL2	Q92692
PVRL3	Q9NQS3	PWP2	Q15269	PXDN	Q92626
PXMP2	Q9NR77	PXN	P11678	PXYLP1	Q8TE99
PYCR1	P32322	PYCR2	Q96C36	PYCRL	Q53H96
PYGB	P11216	PYGL	P06737	PYGM	P11217
PYGO1	Q9Y3Y4	PYGO2	Q9BRQ0	PYURF	Q96I23
QARS	P07814, P47897	QDPR	P09417	QKI	Q96PU8
QPRT	Q15274	QRFPR	Q96P65	QTRT1	Q9BXR0
RAB11A	P62491	RAB11B	Q15907	RAB11FIP2	Q7L804
RAB11FIP3	O75154	RAB12	Q6IQ22	RAB13	P51153
RAB14	P61106	RAB15	P59190	RAB1A	P62820
RAB1B	Q9H0U4	RAB20	Q9NX57	RAB21	Q9UL25
RAB22A	Q9UL26	RAB23	Q9ULC3	RAB24	Q969Q5
RAB27A	P51159	RAB27B	O00194	RAB29	O14966
RAB2B	Q8WUD1	RAB30	Q15771	RAB31	Q13636
RAB33A	Q14088	RAB34	Q9BZG1	RAB35	Q15286
RAB36	O95755	RAB37	Q96AX2	RAB38	P57729
RAB39A	Q14964	RAB39B	Q96DA2	RAB3A	P20336
RAB3B	P20337	RAB3GAP1	Q15042	RAB3GAP2	Q9H2M9
RAB3IL1	Q8TBN0	RAB3IP	Q96QF0	RAB40A	Q8WXH6
RAB40C	Q96S21	RAB4B	P61018	RAB5B	P61020

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RAB8B	Q92930	RAB9B	Q9NP90	RABEP1	Q15276
RABEPK	Q7Z6M1	RABGAP1	Q9Y3P9	RABGGTA	Q92696
RABGGTB	P53611	RAC1	P63000	RAC2	P15153
RAC3	P15153	RAD1	Q92889	RAD18	Q9NS91
RAD23A	P54725	RAD51B	O15315	RAD51D	O75771
RAD52	P43351	RAD54L	P46100	RAD9A	Q99638
RAD9B	Q6WBX8	RAE1	P78406	RAG1	P15918
RAI1	Q7Z5J4	RALA	P11233	RALB	P11234
RALGAPA1	Q6GYQ0	RALGAPA2	Q2PPJ7	RALGAPB	Q86X10
RALGDS	Q12967	RAMP1	O60894	RAMP3	O60896
RAN	P62826	RANBP1	P43487	RANBP10	Q6VN20
RANBP2	P49792	RANBP9	Q96S59	RANGAP1	P46060
RANGRF	Q9HD47	RAP1A	P62834	RAP1B	P61224
RAP1GAP	P47736	RAP1GDS1	P52306	RAP2A	P61225
RAPGEF2	Q9Y4G8	RAPGEF3	O95398	RAPGEF4	Q8WZA2
RAPGEF6	Q9Y4G8	RARA	P10276	RARB	P10826
RARG	P13631	RARRES2	Q99969	RARRES3	Q96KN8, Q9UL19
RARS	P54136	RASA1	P20936	RASA2	Q15283
RASA3	Q14644	RASAL1	O95294	RASAL2	Q9UJF2
RASAL3	Q86YV0	RASGEF1A	Q8N9B8	RASGRF1	Q13972
RASGRF2	O14827	RASGRP1	O95267	RASGRP2	Q7LDG7
RASGRP3	Q8IV61	RASGRP4	Q8TDF6	RASSF4	Q8WYP3
RB1	P06400	RB1CC1	Q8TDY2	RBBP4	Q09028
RBBP7	Q16576	RBBP8	Q99708	RBFOX1	Q9NWB1
RBFOX2	O43251	RBFOX3	A6NFN3	RBL1	P28749
RBL2	Q08999	RBM10	P52756, P98175	RBM14	Q96PK6
RBM22	Q9NW64	RBM23	Q14498	RBM26	Q5T8P6
RBM28	Q9NW13	RBM39	Q14498	RBM42	Q9BTD8
RBM4B	P52272	RBM5	P52756	RBM7	Q9Y580
RBM8A	Q9Y5S9	RBMX	P38159	RBP1	P84103
RBP4	P02753	RBPJ	Q06330	RBX1	P62877
RCAN3	Q9UKA8	RCC1	P18754	RCC2	Q9P258
RCE1	Q9Y256	RCL1	Q9Y2P8	RCN1	Q15293
RCOR1	Q9UKL0	RDH10	Q8IZV5	RDH11	Q96NR8
RDH12	Q96NR8	RDH13	Q8NBN7	RDH16	O75452
RDX	P35241	REC8	O95072	RECK	O95980
REEP1	Q9H902	REL	Q00653	RELB	Q01201
RELN	P78509	REM1	P18074	RENBP	P51606
REPIN1	Q9BS31	REPS2	Q8NFBH8	REST	Q13127
RET	P07949	REV1	Q9UBZ9	REV3L	O60673
REXO2	Q9Y3B8	RFC1	P35251	RFC2	P35249, P35250
RFC3	P40937, P40938	RFC4	P35249, P35250	RFK	Q969G6
RFNG	Q9Y644	RFT1	Q9NWF4	RFWD2	Q8NHY2
RGCC	Q9H4X1	RGL1	Q9NZL6	RGL2	O15211
RGL3	Q3MIN7	RGMA	Q96B86	RGMB	Q6NW40
RGN	O43493	RGP1	Q92546	RGS1	Q9UGC6
RGS10	O43665	RGS11	O94810	RGS14	O43566

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
RGS16	O15492	RGS18	Q9NS28	RGS2	Q9UGC6
RGS20	O76081	RGS22	Q8NE09	RGS3	P49796
RGS6	P49758	RGS7	P49802	RGS8	O15492
RGS9	O75916	RGS9BP	Q6ZS82	RHBDF2	Q6PJF5
RHEB	Q15382	RHNO1	Q9BSD3	RHOBTB1	O94844
RHOBTB2	Q9BYZ6	RHOBTB3	O94955	RHOC	P08134
RHOD	O00212	RHOG	P84095	RHOH	Q15669
RHOJ	Q9H4E5	RHOQ	P17081	RHOT1	Q8IXI2
RHOT2	Q8IXI1	RHOU	Q7L0Q8	RHOV	Q96L33
RIC1	Q4ADV7	RICTOR	Q6R327	RIF1	Q5UIP0
RIMS1	Q86UR5	RIN1	Q13671	RIN2	Q8WYP3
RIN3	Q8TB24	RING1	Q06587	RIOK1	Q9BRS2
RIOK2	Q9BVS4	RIOK3	O14730	RIPK1	Q13546
RIPK2	O43353	RIPK3	Q9Y572	RIPPLY2	Q5TAB7
RIT2	Q99578	RLF	P51460	RMI1	Q9H9A7
RMND5A	Q9H871	RMND5B	Q96G75	RNASE1	P07998
RNASE6	Q93091	RNASEK	Q6P5S7	RNASEL	Q05823
RNASET2	O00584	RND2	P52198	RND3	P61587
RNF113A	O15541	RNF115	Q9Y4L5	RNF123	Q5XPI4
RNF125	Q96EQ8	RNF126	Q9BV68	RNF128	Q8TEB7
RNF135	Q8IUD6	RNF138	Q8WVD3	RNF144B	Q7Z419
RNF168	Q8IYW5	RNF181	Q9P0P0	RNF182	Q8N6D2
RNF185	Q96GF1	RNF19A	Q9NV58	RNF19B	Q6ZMZ0
RNF20	Q5VTR2	RNF213	Q63HN8	RNF216	Q9NWF9
RNF217	Q8TC41	RNF219	Q5W0B1	RNF220	Q5VTB9
RNF25	Q96BH1	RNF34	Q969K3	RNF40	O75150
RNF41	Q9H4P4	RNF43	Q68DV7	RNF5	Q96GF1, Q99942
RNF7	Q9UBF6	RNF8	O76064	RNGTT	O60942
RNH1	O60930	RNLS	Q5VYX0	RNMT	O43148
RNMTL1	Q9HC36	RNPC3	Q96LT9	RNPS1	Q15287
RNY1	O00584	ROBO1	Q9Y6N7	ROBO2	Q9HCK4
ROBO3	Q96MS0-1	ROCK1	Q13464	ROCK2	O75116
ROR1	Q01973	ROR2	Q01974	RORA	P35398
RORB	Q92753	RPA3	P35244	RPAP2	Q8IXW5
RPE	Q2QD12, Q96AT9	RPF1	P46934	RPL10	P27635, Q96L21
RPL10A	P62906	RPL11	P62913	RPL12	P30050
RPL13	P26373, P40429	RPL13A	P40429	RPL14	P50914
RPL15	P61313	RPL17	P18621	RPL18	Q07020
RPL18A	Q02543	RPL19	P84098	RPL21	P46778
RPL22	P35268	RPL23	P62829	RPL23A	P62750
RPL24	P83731	RPL26	P61254, Q9UNX3	RPL26L1	Q9UNX3
RPL27	P61353	RPL27A	P46776, P61353	RPL28	P46776, P46779
RPL29	P47914	RPL3	P39023, Q92901	RPL30	P62888
RPL31	P62899	RPL32	P62888, P62910	RPL34	P49207, P62899
RPL35	P42766	RPL35A	P18077, P42766	RPL36	Q9Y3U8
RPL36A	P83881, Q969Q0	RPL36AL	Q969Q0	RPL37	P61927
RPL37A	P18077, P61513	RPL38	P63173	RPL4	P36578
RPL5	P46777	RPL6	Q02878	RPL7	P18124
RPL7A	P18124, P62424	RPL8	P62424, P62917	RPLP0	P05388

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
RPLP1	P05386	RPLP2	P05387	RPN1	P04843
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RPP40	O75818	RPRD1B	Q9NQG5	RPRD2	Q5VT52
RPS10	P46783	RPS11	P62280	RPS12	P25398
RPS13	P62277	RPS14	P62263	RPS15	P62841
RPS15A	P62244	RPS16	P62249	RPS17	P08708
RPS18	P62269	RPS19	P39019	RPS19BP1	Q86WX3
RPS2	P08865, P15880, P46782	RPS20	P60866	RPS21	P62841, P63220
RPS23	P62266	RPS24	P62244, P62847	RPS25	P62851
RPS26	P62854	RPS27	P42677, Q71UM5	RPS27A	P62979, P62987, Q71UM5
RPS27L	Q71UM5	RPS28	P62857	RPS29	P62273
RPS3	P23396	RPS3A	P61247	RPS4X	P62701, Q8TD47
RPS4Y1	P22090, Q8TD47	RPS5	P46782	RPS6	P62753
RPS6KA3	P51812	RPS6KA5	O75582	RPS6KA6	Q9UK32
RPS6KB1	P23443	RPS6KB2	Q9UBS0	RPS7	P62081
RPS8	P62241	RPS9	P46781	RPSA	P08865
RPTOR	Q8N122	RPUSD3	Q6P087	RPUSD4	Q96CM3
RQCD1	Q92600	RRAD	P55042	RRAGA	Q5VZM2, Q7L523
RRAGB	Q5VZM2	RRAGC	Q9HB90	RRAS	P10301
RRAS2	P62070	RRBP1	Q9P2E9	RRM1	P23921
RRM2B	Q7LG56	RRN3	Q9NYV6	RRP1	P05386
RRP36	Q96EU6	RRP7A	Q9Y3A4	RRP8	O43159
RRP9	O43818	RSAD2	Q8WXG1	RSF1	Q16629
RSPO2	Q6UXX9	RSPO3	Q9BXY4	RSPO4	Q2I0M5
RSU1	Q15404	RTCB	Q9Y3I0	RTF1	Q92541
RTKN	Q9BST9	RTN3	O95197	RTN4	Q9NQC3
RTN4R	Q9HBX8	RTN4RL1	Q86UN2	RTN4RL2	Q86UN3
RTP1	P59025	RUFY1	Q96T51	RUNX1	Q01196
RUNX2	Q13950-1, Q13950-2	RUNX3	Q13761	RUVBL1	Q9Y265
RUVBL2	Q9Y230	RWDD1	Q9H446	RWDD2B	Q9Y3V2
RXFP1	Q9HBX9	RXRA	P19793	RXRB	P28702
RXRG	P48443	RYBP	Q8N488	RYK	P34925
RYR1	P21817	RYR2	Q92736	RYR3	Q15413
S100A1	P23297	S100A11	P31949	S100A12	P80511
S100A8	P05109	S100A9	P06702	S100B	P04271
S1PR1	P21453	S1PR3	Q99500	SACM1L	Q9NTJ5
SAE1	Q9UBE0	SAFB	Q15424	SALL1	Q9NSC2
SAMD10	Q96KM6	SAMD8	Q96LT4	SAMHD1	Q9Y3Z3
SAMM50	Q9Y512	SAP18	O00422	SAP30	O75446, Q9HAJ7
SAP30BP	Q9UHR5	SARDH	Q9UL12	SARM1	Q6SZW1-1
SARNP	P82979	SARS	P49591	SARS2	Q9NP81
SART1	O43290	SAT2	Q96QD8	SATB1	Q01826
SAV1	Q9H4B6	SBF1	O95248	SBF2	Q86WG5
S BSPON	Q8IVN8	SCAF1	O14548	SCAI	Q8N9R8
SCAMP1	O15126	SCAP	Q12770	SCARA5	P21757, Q6ZMJ2
SCARB1	Q8WTV0-2	SCARF1	Q14162	SCD	O00767
SCFD1	Q8WVM8	SCG2	O00255, P13521	SCG3	Q8WXD2
SCLT1	Q96NL6	SCMH1	Q96GD3-2	SCN10A	Q9Y5Y9

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SCN11A	Q9UI33	SCN1A	P35498	SCN1B	Q07699
SCN2A	Q99250	SCN2B	O60939	SCN3A	Q99250, Q9NY46
SCN3B	Q9NY72	SCN4A	P35499	SCN4B	Q8IWT1
SCN5A	Q14524	SCN7A	Q01118	SCN8A	Q9UQD0
SCN9A	Q15858	SCNN1A	P37088	SCO2	O43819
SCP2	P22307-1	SCRIB	Q14160	SCUBE1	Q8IWIY4
SCUBE2	Q9NQ36	SCUBE3	Q8IX30	SDC1	P18827
SDC2	P34741	SDC3	O75056	SDC4	P31431
SDCCAG8	Q86SQ7	SDHAF1	A6NFY7	SDHAF2	Q9NX18
SDHAF3	Q9NRP4	SDHAF4	Q5VUM1	SDHC	Q99643
SDK1	Q7Z5N4	SDK2	Q58EX2	SDR16C5	Q8N3Y7
SDSL	Q96GA7	SEC11C	Q9BY50	SEC13	P55735
SEC16A	O15027	SEC22A	Q96IW7	SEC23A	Q15436
SEC24B	O95487	SEC24C	P53992	SEC24D	O94855
SEC31B	Q9NQW1	SEC61A1	P61619	SEC61B	P60468
SEC61G	P60059	SEL1L	Q9UBV2	SELE	P16581
SELL	P14151	SELP	P14151	SELPLG	Q14242
SEMA3A	Q14563	SEMA3E	O15041	SEMA4F	O95754
SEMA5A	Q13591	SEMA6A	Q9H2E6	SENP1	Q9P0U3
SENP3	Q9H4L4	SENP5	Q96HI0	SEPHS2	Q99611
SEPP1	P49908	SEPSECS	Q9HD40	SERINC1	Q9NRX5
SERINC2	Q96SA4	SERINC5	Q86VE9	SERP1	Q9Y6X1
SERPINA1	P01009	SERPINA5	P05154	SERPINB1	P30740
SERPINB6	P35237	SERPIND1	P05546	SERPINE1	P05121
SERPINE2	P07093	SERPING1	P05155	SERPINH1	P50454
SESN1	Q9Y6P5	SESN2	P58004	SESN3	P58005
SET	Q01105	SETD1A	O15047, Q9UPS6	SETD1B	Q9UPS6
SETD2	Q9BYW2	SETD3	Q86TU7	SETD6	Q8TBK2
SETD7	Q8WTS6	SETD9	Q8NE22	SETDB1	Q15047
SETDB2	Q96T68	SF1	Q13285	SF3A1	Q15459
SF3A2	Q15428	SF3A3	Q12874	SF3B1	O75533
SF3B2	Q13435	SF3B3	Q15393	SF3B4	Q15427
SF3B5	Q9BWJ5	SF3B6	Q9Y3B4	SFPQ	P23246
SFRP1	Q8N474	SFRP2	Q96HF1	SFTPD	P35247
SFXN4	Q6P4A7	SGCB	Q16585	SGCD	Q92629
SGCE	O43556	SGCG	Q13326	SGCZ	Q96LD1
SGIP1	O14526, Q9BQI5	SGK1	O00141	SGK2	Q9HBY8
SGK223	Q86YV5	SGK3	Q96BR1	SGMS1	Q86VZ5
SGMS2	Q8NHU3	SGPP1	Q9BX95	SGPP2	Q8IWX5
SGSM2	Q9GZT4	SGTA	O43765	SH2B1	Q9NRF2-2
SH2B2	O14492	SH2B3	Q9UQQ2	SH3BP1	Q9Y3L3
SH3BP4	Q9P0V3	SH3D19	Q5HYK7	SH3GL1	Q99961
SH3GL2	Q99962	SH3GL3	P82987	SH3GLB2	Q9UJU6
SH3KBP1	Q96B97	SH3PXD2A	Q5TCZ1	SH3RF1	Q7Z6J0
SHANK1	Q9Y566	SHANK2	Q9UPX8	SHANK3	Q9BYB0
SHARPIN	Q9H0F6	SHB	Q15464	SHC1	P29353-1, P29353-2, P29353-3
SHC2	P98077	SHC3	Q92529	SHH	Q15465
SHISA5	Q8N114	SHKBP1	Q8TBC3	SHMT1	P34896
SHMT2	P34897	SHOC2	Q9UQ13	SHPRH	Q149N8

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
SHQ1	Q6PI26	SIAH2	O43255	SIGIRR	Q6IA17
SIGLEC1	Q9BZZ2	SIGLEC10	A6NMB1	SIGLEC8	Q96PQ1, Q9NYZ4
SIGLEC9	Q9Y336	SIGMAR1	Q99720	SIPA1	Q96FS4
SIPA1L1	O43166	SIPA1L2	Q96FS4	SIRPA	P78324
SIRPB2	Q9P1W8	SIRT1	Q96EB6	SIRT2	Q8IXJ6
SIRT3	Q9NTG7	SIRT4	Q9Y6E7	SIRT6	Q8N6T7
SIX1	Q15475	SKAP2	O75563	SKI	Q5VTY9
SKIL	P12757	SKIV2L2	P42285	SKP1	P63208
SKP2	Q13309	SLA	Q13239	SLA2	Q9H6Q3
SLAMF6	Q96DU3	SLAMF7	Q9NQ25	SLBP	Q14493
SLC10A2	Q12908	SLC10A6	Q3KNW5	SLC11A1	P49279
SLC11A2	P49281	SLC12A2	P55011	SLC12A3	P55017
SLC12A4	Q9UP95	SLC12A5	Q9H2X9	SLC12A6	Q9UHW9
SLC12A7	Q9Y666	SLC13A3	Q8WWT9	SLC13A4	Q9UKG4
SLC13A5	Q86YT5	SLC14A1	Q13336	SLC14A2	Q15849
SLC15A3	Q8IY34	SLC15A4	Q8N697	SLC16A10	Q8TF71
SLC16A2	P36021	SLC16A3	O15427	SLC16A7	O60669
SLC17A6	Q9P2U8	SLC17A7	Q9P2U7	SLC18A2	Q05940
SLC19A1	P41440	SLC19A2	O60779	SLC19A3	Q9BZV2
SLC1A1	P43005	SLC1A2	P43004	SLC1A3	P43003
SLC1A5	Q15758	SLC1A6	P48664	SLC20A1	Q8WUM9
SLC20A2	Q08357	SLC22A1	O15245	SLC22A15	Q8IZD6
SLC22A16	Q86VW1	SLC22A17	Q8WUG5	SLC22A2	O15244
SLC22A3	O75751	SLC22A5	O76082	SLC22A8	Q8TCC7
SLC23A2	Q9UGH3	SLC24A1	O60721	SLC24A2	Q9UI40
SLC24A3	Q9HC58	SLC24A4	Q8NFF2	SLC24A5	Q71RS6
SLC25A1	P53007	SLC25A11	Q02978	SLC25A12	O75746
SLC25A13	Q9UJS0	SLC25A14	O95258	SLC25A15	Q9Y619
SLC25A16	P16260	SLC25A18	Q9H1K4	SLC25A21	Q9BQT8
SLC25A22	Q9H936	SLC25A26	Q70HW3	SLC25A27	O95847
SLC25A29	Q8N8R3	SLC25A37	Q9NYZ2	SLC25A4	P12235
SLC25A44	Q96H78	SLC25A5	P05141, P12236	SLC25A6	P12236
SLC26A1	Q9H2B4	SLC26A11	Q86WA9	SLC26A2	P50443
SLC26A4	O43511	SLC26A5	Q8TE54	SLC26A6	Q9BXS9
SLC26A7	Q8TE54	SLC27A1	Q6PCB7	SLC27A4	Q6P1M0
SLC27A6	Q9Y2P4	SLC29A2	Q14542	SLC29A3	Q9BZD2
SLC29A4	Q7RTT9	SLC2A1	P11166	SLC2A10	O95528
SLC2A11	Q9BYW1	SLC2A12	Q8TD20	SLC2A13	Q96QE2
SLC2A3	P11169	SLC2A4	P14672	SLC2A5	P22732
SLC2A6	Q9UGQ3	SLC2A9	Q9NRM0	SLC30A1	Q9Y6M5
SLC30A5	Q8TAD4	SLC30A8	Q8IWU4	SLC33A1	O00400
SLC35A1	P78382	SLC35A2	P78381	SLC35A3	Q9Y2D2
SLC35A4	Q96G79	SLC35B2	Q8TB61	SLC35B3	Q9H1N7
SLC35B4	Q969S0	SLC35C1	Q96A29	SLC35D1	Q9NTN3
SLC35D2	Q76EJ3	SLC36A1	Q7Z2H8	SLC36A2	Q495M3
SLC37A1	P57057	SLC37A2	Q8TED4-2	SLC38A1	Q9H2H9
SLC38A2	Q96QD8	SLC38A4	Q969I6	SLC38A9	Q8NBW4
SLC39A1	Q9NP94, Q9NY26	SLC39A14	Q15043	SLC39A3	Q9BRY0
SLC39A4	Q6P5W5	SLC39A6	Q13433	SLC39A7	Q92504

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SLC39A8	Q9C0K1	SLC3A2	P08195	SLC40A1	Q9NP59
SLC41A1	Q8IVJ1	SLC41A2	Q96JW4	SLC43A1	O75387
SLC44A1	Q8WWI5	SLC44A2	Q8IWA5	SLC44A3	Q8N4M1
SLC45A3	Q96JT2	SLC47A1	Q96FL8	SLC4A1	P02730
SLC4A3	P48751	SLC4A4	Q9Y6R1	SLC4A7	Q9Y6M7
SLC4A8	Q2Y0W8	SLC50A1	Q9BRV3	SLC51B	Q86UW2
SLC52A2	Q9HAB3	SLC52A3	Q9NQ40	SLC5A4	Q9NY91
SLC5A6	Q9Y289	SLC5A7	Q9GZV3	SLC6A1	P30531
SLC6A11	P48066	SLC6A12	P48065	SLC6A13	Q9NSD5
SLC6A15	Q9H2J7	SLC6A20	Q9NP91	SLC6A4	P31645
SLC6A6	P31641	SLC6A7	Q99884	SLC6A8	P48029
SLC7A1	P30825	SLC7A11	Q9UPY5	SLC7A2	P52569-1, P52569-2
SLC7A3	Q8WY07	SLC7A5	Q01650	SLC7A6	Q92536
SLC7A7	Q9UM01	SLC7A8	Q9UHI5	SLC8A1	P32418
SLC8A2	Q9UPR5	SLC8A3	P57103	SLC8B1	Q6J4K2
SLC9A1	P19634	SLC9A2	Q9UBY0	SLC9A3	P48764
SLC9A8	Q9Y2E8	SLC9A9	Q8IVB4	SLC9B1	Q4ZJI4
SLC9B2	Q4ZJI4, Q86UD5	SLC9C1	Q4G0N8	SLC9C2	Q5TAH2
SLCO1A2	P46721	SLCO1C1	Q9NYB5	SLCO2A1	Q92959
SLCO2B1	O94956-3	SLCO3A1	Q9UIG8-1	SLIRP	Q9GZT3
SLIT1	O75093	SLIT2	O94813	SLIT3	O75094
SLITRK1	Q96PX8	SLITRK2	Q9H156	SLITRK4	Q8IW52
SLITRK5	O94991	SLK	Q9H2G2	SLMO1	Q96N28
SLPI	P03973	SLU7	O95391	SLX4	Q8IY92
SMAD2	Q15796	SMAD3	P84022	SMAD4	Q13485
SMAD5	Q99717	SMAD6	O43541	SMAD7	O15105
SMAD9	O15198	SMAP2	Q9H0E9	SMARCA2	P51531
SMARCA4	P51532	SMARCB1	Q12824	SMARCC1	Q92922
SMARCC2	Q8TAQ2	SMARCD1	Q96GM5	SMARCD3	Q6STE5
SMARCE1	Q969G3	SMC1A	Q14683	SMC2	O95347
SMC3	Q9UQE7	SMC4	Q9NTJ3	SMC5	Q8IY18
SMDT1	Q9H4I9	SMG1	P62308	SMG5	Q9UPR3
SMG6	Q86US8	SMG8	Q8ND04	SMG9	Q9H0W8
SMIM4	Q8WV10	SMN2	Q16637	SMO	Q9NWM0-3
SMOX	Q9NWM0-3	SMPD3	Q9NY59	SMPD4	Q9NXE4
SMS	P52788	SMUG1	Q53HV7	SMURF1	Q9HCE7
SMURF2	Q9HAU4	SMYD2	Q9NRG4	SMYD3	Q9H7B4, Q9NRG4
SNAI1	O95863	SNAI2	O43623	SNAP23	O00161
SNAP25	P60880	SNAP29	O95721	SNAP91	O60641
SNAPC1	Q16533	SNAPC2	Q13487	SNAPC4	Q5SXM2
SNAPIN	O95295	SNCA	P37840	SNCAIP	Q9Y6H5, Q9Y6H5-2
SNCB	Q16143	SND1	Q7KZF4	SNF8	Q96H20
SNRNP200	O75643	SNRNP25	Q9BV90	SNRNP27	Q8WVK2
SNRNP35	Q16560	SNRNP48	Q6IEG0	SNRNP70	P08621
SNRPA	P09012	SNRPB	P14678	SNRPB2	P08579, P09012
SNRPC	P09234	SNRPD1	P62314, P62316	SNRPD2	P62316
SNRPD3	P62318	SNRPE	P62304	SNRPF	P62306
SNRPG	P62308	SNRPN	P63162	SNTA1	Q13424
SNTB1	Q13884	SNTB2	Q13425	SNTG2	Q9NY99

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
SNU13	P55769	SNW1	Q13573	SNX1	O60749
SNX12	Q8WY07	SNX18	Q96RF0	SNX2	O60749
SNX3	O60493	SNX5	Q9Y5X3	SNX9	Q9Y5X1
SOAT1	P35610	SOAT2	O75908	SOCs2	O14508
SOCs3	O14543	SOCs4	Q8WXH5	SOCs5	O75159
SOCs6	O14544	SOCs7	Q8WXH5	SOD3	P08294
SORBS1	Q9BX66	SORBS3	O60504	SORCS3	Q17R88
SORL1	Q92673	SORT1	Q99523	SOS1	P42766
SOS2	Q07890	SOWAHC	Q53LP3	SOX10	P56693
SOX13	Q9UN79	SOX17	Q9H6I2	SOX4	Q06945
SOX6	P35712	SOX9	P48436	SP1	Q8N907
SP100	P23497	SP3	Q9Y2H2	SPAG9	O60271
SPARC	Q14515	SPARCL1	Q14515	SPATA2	Q9UM82
SPC25	Q15005	SPCS1	Q9Y6A9	SPCS2	Q15005
SPCS3	P61009	SPG7	Q9UQ90	SPHK1	Q9NYA1
SPHK2	Q9NRA0	SPI1	P62826	SPIDR	Q14159
SPINT1	O43278	SPINT2	O43291	SPNS2	Q8IVW8
SPOCK3	Q9BQ16	SPON1	Q9HCB6	SPON2	Q9BUD6
SPOPL	Q6IQ16	SPP1	P49642	SPPL2A	Q8TCT8
SPPL2B	Q8TCT7	SPR	P35270	SPRED1	Q7Z699
SPRED2	Q7Z698	SPRED3	Q2MJR0	SPRN	Q5BIV9
SPRY1	O43609	SPRY2	O43597	SPSB1	Q96BD6
SPSB2	Q99619	SPSB3	Q6PJ21	SPSB4	Q96A44, Q99619
SPTA1	P02549	SPTAN1	Q13813	SPTB	P11277
SPTBN1	Q01082	SPTBN2	O15020	SPTBN4	Q9H254
SPTBN5	Q9NRC6	SPTLC1	O15269	SPTLC3	Q9NUV7
SPTSSA	Q969W0	SPTSSB	Q8NFR3	SQLE	Q14534
SQSTM1	Q13501	SRC	P12931	SRD5A1	P18405
SREBF1	P36956-1	SREBF2	Q12772	SRF	P11831
SRGAP1	Q7Z6B7	SRGAP2	O75044	SRGAP3	O43295
SRGN	P10124	SRI	P30626	SRL	Q9UBK2
SRM	P19623	SRP14	P37108	SRP19	P09132
SRP68	Q9UHB9	SRP72	O76094	SRP9	P49458
SRPK1	Q96SB4	SRPK2	P78362	SRPRB	Q9Y5M8
SRR	Q9GZT4	SRRT	Q9BXP5	SRSF1	Q07955
SRSF11	Q05519	SRSF12	Q8WXF0	SRSF2	Q01130
SRSF5	Q13243	SRSF6	Q13247	SRSF7	Q16629
SRSF8	Q9BRL6	SRSF9	Q13242	SS18	Q15532
SS18L1	O75177	SSB	P05455	SSC5D	A1L4H1
SSNA1	O43805	SSPN	Q14714	SSPO	A2VEC9
SSR2	P43308	SSR3	Q9UNL2	SSR4	P51571
SSRP1	Q08945	SSTR3	P32745	SSU72	Q9NP77
ST13	P50502	ST3GAL1	Q11201	ST3GAL2	Q16842
ST3GAL3	Q11203	ST3GAL4	Q11206	ST3GAL5	Q9UNP4
ST3GAL6	Q9Y274	ST5	P78524	ST6GAL2	Q96JF0
ST6GALNAC1	Q9NSC7	ST6GALNAC2	Q9UJ37	ST6GALNAC3	Q8NDV1
ST6GALNAC5	Q9BVH7	ST8SIA1	Q92185	ST8SIA2	Q92186
ST8SIA3	O43173	ST8SIA4	Q92187	ST8SIA5	O15466
ST8SIA6	P61647	STAB1	Q9NY15	STAB2	Q8WWQ8

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
STAG1	Q8WVM7	STAG2	Q8N3U4	STAG3	Q9UJ98
STAM	O75886, Q92783	STAM2	O75886	STAR	P25092
STARD10	Q9Y365	STARD13	Q9Y3M8	STARD3NL	O95772
STARD5	Q9NSY2	STARD7	Q9NQZ5	STARD8	Q92502
STAT1	P42224	STAT2	P52630	STAT3	P40763
STAT4	Q14765	STAT5A	P42229, P51692	STAT5B	P51692
STAT6	P42226	STEAP3	Q658P3	STEAP4	Q687X5
STIM1	Q13586	STIP1	P31948	STK10	O94804
STK11	Q15831	STK24	Q9Y6E0	STK26	Q9P289
STK3	Q13188	STK38	Q15208	STK4	O75914
STMN1	P16949	STMN2	Q93045	STOM	P27105
STOML2	Q9UJZ1	STON1	Q9Y6Q2	STON2	Q8WXE9
STRA13	A8MT69	STRADA	Q7RTN6	STRADB	Q9COK7
STRAP	Q9Y3F4	STRN	O43815	STS	P08842
STT3B	Q8TCJ2	STUB1	Q9UNE7	STX17	P56962
STX18	Q9P2W9	STX1A	Q16623	STX1B	P61266
STX6	O43752	STXBP1	P61764	STXBP3	O00186
SUCLG1	P53597	SUCLG2	Q96I99	SUFU	Q9UMX1
SUGP1	Q8IWZ8	SUGT1	Q9Y2Z0	SULT1A1	P50225
SULT1A2	P50226	SULT1B1	O75897	SULT1C2	O75897
SULT1C4	O75897	SULT4A1	Q9BR01	SUMF1	Q8NBK3
SUMF2	Q92526	SUMO1	P63165	SUMO2	P55854, P61956
SUMO3	P55854	SUN1	O94901	SUN2	Q9UH99
SUOX	P51687	SUPT16H	Q9Y5B9	SUPT3H	O75486
SUPT4H1	P63272	SUPT5H	O00267	SUPT6H	Q7KZ85
SUPT7L	O94864	SURF1	Q15526	SURF4	O15260
SUV39H1	O43463	SUV39H2	Q9H5I1	SUV420H2	Q86Y97
SUZ12	Q15022	SV2A	Q7L0J3	SVBP	Q8N300
SVIP	Q8NHG7	SWAP70	Q9UH65	SYCP2	Q9BX26
SYDE1	Q6ZW31	SYDE2	Q5VT97	SYF2	O95926
SYK	P43405	SYMPK	Q92797	SYN1	Q13884
SYN2	Q9NY99	SYN3	O14994	SYNE1	Q8NF91
SYNE2	Q8WXH0	SYNGAP1	Q96PV0, Q9UJF2	SYNGR1	O43759
SYNJ1	O43426	SYNJ2	O15056	SYP	P08247
SYT1	P21579, Q8NBV8	SYT10	Q6XYQ8	SYT11	Q9BT88
SYT2	Q8N9I0	SYT4	Q9BT88	SYT5	O00445
SYT7	O43581	SYTL2	Q9HCH5	SYTL4	Q96C24
SYVN1	Q86TM6	SZT2	Q5T011	TAB1	Q15750
TAB2	Q9NYJ8	TAC1	P20366	TAC3	Q9UHF0
TACC3	Q9Y6A5	TACO1	Q9BSH4	TACR1	P25103
TACR3	P29371	TADA2B	Q86TJ2	TADA3	O75528
TAF10	Q12962	TAF11	Q15544	TAF15	Q16514, Q92804
TAF1B	Q53T94	TAF1C	Q15572	TAF1D	Q9H5J8
TAF2	Q6P1X5	TAF3	Q5VWG9	TAF4	O00268, Q92750
TAF4B	Q92750	TAF5	Q15542	TAF5L	O75529
TAF6	P49848	TAF6L	Q9Y6J9	TAF7	Q15545, Q5H9L4
TAF8	Q7Z7C8	TAF9	Q16594	TAF9B	Q16594, Q9HBM6
TAGAP	Q8N103	TAL1	P37837	TALDO1	P37837
TANK	Q92844	TAOK1	Q7L7X3	TAOK3	Q9H2K8

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
TAP1	Q03518	TAPBP	O15533	TARBP2	Q15633
TARS	P26639	TARS2	Q9BW92	TAS1R3	Q7RTX0
TAS2R5	Q9NYW4	TASP1	Q9H6P5	TATDN2	Q93075
TAX1BP1	Q86VP1	TBC1D1	Q86TI0	TBC1D10A	Q9BXI6
TBC1D10B	Q4KMP7	TBC1D13	Q9NVG8	TBC1D14	Q9P2M4
TBC1D15	Q8TC07	TBC1D16	Q8TBP0	TBC1D17	Q9HA65
TBC1D20	Q96BZ9	TBC1D24	Q9ULP9	TBC1D25	Q3MII6
TBC1D4	O60343	TBC1D8B	Q0IIM8	TBCA	O75347
TBCB	P30622	TBCC	Q15814	TBK1	Q9UHD2
TBL1X	O60907	TBL1XR1	Q9BZK7	TBL3	Q12788
TBRG4	Q969Z0	TBX1	O43435	TBX2	Q13207
TBX3	O15119	TBXA2R	P21731, P21731-2, P21731-3	TBXAS1	P24557
TCEA1	P23193	TCEB1	Q15369	TCEB2	Q15370
TCEB3	Q14241	TCF12	Q99081	TCF25	Q9BQ70
TCF3	P15923	TCF4	Q9NQB0	TCF7	P36402
TCF7L1	Q9HCS4	TCF7L2	Q9NQB0	TCHH	Q07283
TCIRG1	Q13488	TCN2	P20062	TCP1	P17987
TCTEX1D1	Q8N7M0	TCTEX1D2	Q8WW35	TDG	Q13569
TDRD1	Q9BXT4	TDRD12	Q587J7	TDRD6	O60522
TDRD9	Q8NDG6	TDRKH	Q9Y2W6	TEAD1	P28347
TEAD2	Q15562	TEAD3	Q99594	TEAD4	Q15561, Q15562
TEC	Q92570-1, Q92570-2	TECR	Q9NZ01	TEK	Q02763
TEP1	P60484	TERF1	P54274	TERF2	Q15554
TERF2IP	Q9NYB0	TES	Q9NZU5	TESK1	Q15569
TET1	Q8NFU7	TET2	Q6N021	TET3	O43151
TEX15	Q9BXT5	TEX2	Q8IWB9	TEX264	Q9Y6I9
TF	P02787	TFAP2A	P05549	TFAP2C	Q92754
TFB1M	Q8WVM0	TFDP1	Q14186	TFDP2	Q14188
TFE3	P19532	TFEB	P19484	TFEC	O14948
TFIP11	Q9UBB9	TFPI	P10646	TFPI2	P48307
TFPT	P0C1Z6	TFRC	P02786	TG	P00488
TGFA	P01135	TGFB1	P01137	TGFB2	P61812
TGFB3	P10600	TGFB1	Q15582	TGFBR1	P36897
TGFBR2	P37173	TGFBR3	Q03167	TGIF1	Q15583
TGS1	Q96RS0	THADA	Q6YHU6	THBD	P07204
THBS1	P07996	THBS2	P35442	THBS4	P35443
THEM4	Q5T1C6	THOC1	Q96FV9	THOC2	Q8NI27
THOC3	Q96J01	THOC5	Q13769	THOC6	Q86W42
THOC7	Q6I9Y2	THOP1	P52888	THRA	P10827
THRB	P10828	THRSP	Q92748	THSD1	Q9NS62
THSD4	Q6ZMP0	THSD7A	Q9UPZ6	THSD7B	Q9C0I4
TIA1	P31483	TIAM1	Q13009	TIAM2	Q8IVF5
TICAM1	Q8IUC6	TIMM10	P62072	TIMM13	Q9Y5L4
TIMM17A	Q99595	TIMM17B	O60830, Q99595	TIMM21	Q9BVV7
TIMM22	Q9Y584	TIMM23	O14925	TIMM23B	O14925
TIMM44	O43615	TIMM50	Q3ZCQ8	TIMM8A	O60220
TIMM8B	Q9Y5J9	TIMM9	Q9Y5J7	TIMP1	P01033
TIMP3	P35625	TINF2	Q9BSI4	TIPIN	Q9BVG5
TJP1	Q07157	TJP2	Q9UDY2	TK2	O00142

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
TKFC	Q3LXA3	TKT	P29401	TLE2	Q04725
TLE3	Q04726-3	TLE4	Q04727	TLL1	O43897
TLL2	Q9Y6L7	TLN1	Q9Y490	TLR1	Q15399, Q9Y2C9
TLR10	Q9BXR5	TLR2	O60603	TLR3	O15455
TLR4	O00206	TLR5	O60602	TLR7	Q9NYK1
TLR8	Q9NR97	TM7SF2	O76062	TMBIM1	Q969X1
TMC1	Q8TDI8	TMC6	Q7Z403	TMED10	P49755
TMED3	Q9Y3Q3	TMED5	Q9Y3A6	TMED9	Q9BVK6
TMEM11	P17152	TMEM115	Q12893	TMEM126A	Q8IUX1, Q9H061
TMEM129	Q14493	TMEM132A	Q24JP5	TMEM132E	Q24JP5
TMEM133	A6NI28	TMEM173	Q86WV6	TMEM177	Q53S58
TMEM186	Q96B77	TMEM216	Q9P0N5	TMEM219	Q86XT9
TMEM223	A0PJW6	TMEM258	P61165	TMEM261	Q96GE9
TMEM55B	Q86T03	TMEM57	Q8N5G2	TMEM63A	O94886
TMEM67	Q5HYA8	TMEM86B	Q8N661	TMEM87A	Q8NBN3
TMEM87B	Q8NBN3	TMF1	P82094	TMIGD3	P0DMS8
TMLHE	Q9NVH6	TMOD1	P28289	TMOD3	Q9NYL9
TMPO	P42167	TMPRSS2	O15393	TMSB4X	P62328
TMX3	Q96JJ7	TNC	P24821	TNFAIP1	Q13829
TNFAIP3	P21580	TNFAIP6	P98066	TNFAIP8	O95379
TNFAIP8L1	Q8WVP5	TNFAIP8L3	Q5GJ75	TNFRSF10A	O00220
TNFRSF10D	Q9UBN6	TNFRSF11A	Q9Y6Q6	TNFRSF11B	O00300
TNFRSF14	Q92956	TNFRSF19	Q9HAV5	TNFRSF1A	P19438
TNFRSF1B	P20333	TNFRSF21	O75509	TNFRSF8	P28908
TNFSF10	P50591	TNFSF12	O43508	TNFSF13	O75888
TNFSF13B	Q9Y275	TNFSF15	O95150	TNFSF4	P23510
TNFSF8	P32971	TNIK	Q9UKE5	TNIP1	Q15025
TNIP2	Q8NFZ5	TNKS	O95271, Q9H2K2	TNKS1BP1	Q9C0C2
TNKS2	P16157	TNNC1	P19429, P63316	TNNC2	P02585
TNNI1	P19237	TNNI2	P48788	TNNI3	P19429
TNNT1	P13805	TNNT2	P45379	TNNT3	P45378
TNRC6A	Q8NDV7	TNRC6B	Q9UPQ9	TNRC6C	Q9HCJ0
TNS3	Q68CZ2	TNXB	P22105	TOLLIP	Q9H0E2
TOM1	Q8IYU2	TOMM20	Q15388	TOMM22	Q9NS69
TOMM40	O96008	TOMM40L	O96008	TOMM5	Q8N4H5
TOMM7	Q9P0U1	TOMM70A	O94826	TOP1	P11387
TOP2B	Q02880	TOR1A	O14656	TOR1AIP1	Q5JTV8
TOR1B	O14657	TOR4A	Q9NXH8	TP53BP1	Q12888
TP53BP2	Q13625	TP53INP1	Q96A56	TP53RK	Q96S44
TP63	Q9H3D4-2	TPCN1	Q9ULQ1	TPCN2	Q8NHX9
TPD52	Q9Y2Q9	TPD52L1	Q16890	TPGS1	Q6ZTW0
TPGS2	Q68CL5	TPH2	Q8IWU9	TPI1	P60174
TPK1	Q9H3S4	TPM2	P07951	TPM3	P06753
TPO	P07202	TPP1	Q96AP0	TPP2	P29144
TPRKB	Q9Y3C4	TPRN	Q4KMQ1	TPRXL	Q17RH7
TPSB2	Q15661	TPST1	O60507	TPTE2	P56180, Q6XPS3
TPX2	Q9ULW0	TRAC	P01848	TRAF1	Q13077
TRAF2	Q12933	TRAF3	Q13114	TRAF3IP1	Q8TDR0
TRAF7	Q6Q0C0	TRAIP	Q9BWF2	TRAM1	Q15629

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
TRAP1	Q12931	TRAPPC1	Q9Y5R8	TRAPPC10	P48553
TRAPPC12	Q8WVT3	TRAPPC13	A5PLN9	TRAPPC2	P0DI81
TRAPPC2L	Q9UL33	TRAPPC3	O43617	TRAPPC4	Q9Y296
TRAPPC5	Q8IUR0	TRAPPC6A	O75865	TRAPPC6B	O75865, Q86SZ2
TRAPPC8	Q9Y2L5	TRAPPC9	Q96Q05	TRAT1	Q6PIZ9
TRDMT1	O14717	TRDN	Q13061	TREM1	Q9NP99
TREM2	Q9NZC2	TRH	P20396	TRIAP1	O43715
TRIB3	Q96RU7	TRIM11	Q96F44	TRIM14	Q14142
TRIM21	P19474	TRIM22	Q8IYM9	TRIM24	O15164
TRIM25	Q14258	TRIM27	P14373	TRIM28	Q13263
TRIM29	Q14134	TRIM3	P53365	TRIM32	Q13049
TRIM33	Q9UPN9	TRIM35	Q9UPQ4	TRIM36	Q9NQ86
TRIM38	O00635	TRIM4	Q9C037	TRIM41	Q8WV44
TRIM45	Q9H8W5	TRIM46	Q7Z4K8	TRIM47	Q8WV44
TRIM5	Q9C035	TRIM56	Q9BRZ2	TRIM69	Q86WT6
TRIM71	Q2Q1W2	TRIM9	Q9C026	TRIO	O75962, Q96PX9
TRIOBP	Q9H2D6-4, Q9H2D6-5	TRIP10	Q15642	TRIP11	Q15643
TRIP12	Q14669	TRMT1	O75648	TRMT10C	Q7L0Y3
TRMT11	Q7Z4G4	TRMT112	Q9UI30	TRMT12	Q53H54
TRMT13	Q9NUP7	TRMT44	Q8IYL2	TRMT5	Q32P41
TRMT6	Q9UJA5	TRMT61A	Q96FX7	TRNT1	Q96Q11
TRPA1	O75762	TRPC1	P48995	TRPC3	Q13507
TRPC4	Q9UBN4	TRPC5	Q9UL62	TRPC6	Q9Y210
TRPC7	O94759	TRPM2	O94759	TRPM3	Q9HCF6
TRPM6	Q9BX84	TRPM7	Q96QT4	TRPM8	Q7Z2W7
TRPV3	Q8NET8	TRPV4	Q9HBA0	TRPV5	Q9NQA5
TRRAP	Q9Y4A5	TRUB2	O95900	TSC2	P49815
TSC22D4	Q99576	TSEN15	Q8WW01	TSEN2	Q8NCE0
TSEN34	Q9BSV6	TSEN54	Q7Z6J9	TSFM	P43897
TSG101	Q99816	TSHR	P16473	TSPAN14	Q8NG11
TSPAN15	O95858	TSPAN33	Q86UF1	TSPAN5	P62079
TSPAN7	P41732	TSPO	P30536	TSPYL2	Q9H2G4
TSR1	Q2NL82	TSR3	Q9UJK0	TSTA3	Q13630
TSTD1	Q8NFU3	TTBK2	Q6IQ55	TTC19	Q6DKK2
TTC21B	Q7Z4L5	TTC30B	Q8N4P2	TTC5	Q8N0Z6
TTC8	Q8TAM2	TTC9	Q9UIM3	TTF1	Q15361
TTLL1	O95922	TTLL11	Q8NHH1	TTLL12	Q14166
TTLL4	Q14679	TTLL5	Q6EMB2	TTLL7	Q6ZT98
TTLL9	Q3SXZ7	TTN	Q8WZ42	TTPA	P49638
TTYH1	Q9BSA4, Q9H313-3	TTYH2	Q9BSA4	TTYH3	Q9C0H2
TUBA1A	Q71U36	TUBA1B	P68363	TUBA1C	Q9BQE3
TUBA4A	P68366	TUBB	P04350	TUBB2A	Q13885, Q9BVA1
TUBB2B	Q13885, Q9BVA1	TUBB4A	P04350	TUBB4B	P04350, P68371
TUBG1	P23258, Q9NRH3	TUBGCP2	Q9BSJ2	TUBGCP3	Q96CW5
TUBGCP4	Q9UGJ1	TUBGCP6	Q96RT7	TUFM	P49411
TULP3	O75386	TUSC3	Q13454	TUT1	Q9H6E5
TWF2	Q6IBS0	TWIST1	Q15672	TWIST2	Q8WVJ9
TXK	P42681	TXN	P10599	TXN2	Q99757
TXNDC11	Q6PKC3	TXNIP	Q9H3M7	TXNL4A	P83876

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
TXNRD1	Q16881	TXNRD2	Q9NNW7	TYMP	P19971
TYRO3	Q06418	TYROBP	O43914	TYRP1	P17643
TYW1	Q9NV66	TYW3	Q6IPR3	U2AF1L4	Q8WU68
U2AF2	P26368	U2SURP	O15042	UACA	Q9BZF9
UAP1	Q16222	UBA1	P22314	UBA52	P62979, P62987
UBA6	A0AVT1	UBA7	P22314, P41226	UBAC1	Q9BSL1
UBB	P62979, P62987	UBC	P62987	UBE2A	P49459
UBE2D2	P62837	UBE2D3	P61077	UBE2D4	Q9Y2X8
UBE2E1	P51965	UBE2E2	Q96LR5	UBE2E3	Q969T4
UBE2F	Q969M7	UBE2G1	P62253	UBE2G2	P60604
UBE2I	P63279	UBE2J1	Q9Y385	UBE2J2	Q8N2K1-1
UBE2K	P61086	UBE2L3	P68036	UBE2L6	O14933
UBE2M	P61081	UBE2N	P61088	UBE2O	Q9C0C9
UBE2Q1	Q7Z7E8	UBE2Q2	Q8WVN8	UBE2S	Q16763
UBE2T	Q9NPD8	UBE2V1	Q13404	UBE2Z	Q9H832
UBE3A	Q05086	UBE3C	Q15386	UBE3D	Q7Z6J8
UBE4A	Q14139	UBIAD1	Q9Y5Z9	UBL4A	P11441
UBL5	Q9BZL1	UBN1	Q9NPG3	UBOX5	O94941
UBQLN1	Q9UMX0	UBQLN2	Q9UHD9	UBR1	Q8IWV7
UBR2	Q8IWV7, Q8IWV8	UBTF	P17480	UBXN1	Q04323
UBXN11	Q5T124	UCHL1	P09936	UCHL3	P15374
UCHL5	Q9Y5K5	UCK1	Q9HA47	UCKL1	Q9NWZ5
UCP2	P55851, P55916	UFD1L	Q92890	UFL1	O94874
UGCG	Q16739	UGGT1	Q9NYU2	UGGT2	Q9NYU1
UGT2B7	O75795, P16662	UGT8	Q16880	UHMK1	Q8TAS1
UHRF1BP1L	A0JNW5	UHRF2	Q96PU4	UIMC1	Q96RL1
ULBP1	Q9BZM5	ULBP2	Q9BZM5	ULBP3	Q9BZM4
ULK1	O75385	ULK3	Q6PHR2	UNC119B	A6NIH7
UNC13D	Q70J99	UNC5A	Q6ZN44	UNC5B	Q8IZJ1
UNC5C	O95185	UNC5D	Q6UXZ4	UNC80	Q8N2C7
UNC93B1	Q9H1C4	UNG	P13051-1	UNK	Q9H9P5
UPB1	Q9UBR1	UPF1	Q92900	UPF2	Q9HAU5
UPF3A	Q9H1J1	UPF3B	Q9BZI7	UPP1	Q16831
UPP2	O95045	UQCC2	Q9BRT2	UQCC3	Q6UW78
UQCR10	Q9UDW1	UQCR11	O14957	UQCRB	P14927
UQCRC1	P31930	UQCRC2	P22695	UQCRCFS1	P47985
UQCRH	P07919	UQCRQ	O14949	URM1	Q9BTM9
UROD	P06132	UROS	P10746	USE1	Q9NZ43
USF1	Q8NFM3	USF2	Q15853	USH1G	Q495M9
USMG5	Q96IX5	USO1	O60763	USP1	O94782
USP11	P51784	USP12	O75317, Q86T82	USP13	Q92995
USP14	P54578	USP15	Q9Y4E8	USP16	Q9Y5T5
USP18	Q9UMW8	USP19	O94966	USP2	O75604
USP20	Q9Y2K6	USP21	Q9UK80	USP22	Q9UPT9
USP24	Q9UPU5	USP25	Q9UHP3	USP28	Q96RU2
USP3	Q9Y6I4	USP34	Q70CQ2	USP37	Q86T82
USP39	Q53GS9	USP4	Q13107	USP44	Q70CQ1, Q9H0E7
USP45	Q70EL2	USP47	Q96K76	USP48	Q86UV5
USP5	P45974	USP6NL	Q92738	USP8	P40818

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UTP3	Q9NQZ2	UTRN	P46939	UTS2	O95399
UTS2B	O95399, Q765I0	UTY	O14607	UVRAG	Q9P2Y5
UVSSA	Q2YD98	UXS1	Q8NBZ7	UXT	Q9UBK9
VAC14	Q08AM6	VAMP1	P23763	VAMP2	P63027
VAMP3	Q15836	VAMP4	O75379	VAMP7	P51809
VAMP8	Q9BV40	VANGL1	Q8TAA9	VANGL2	Q9ULK5
VARS	P26640	VARS2	P26640	VASH1	Q7L8A9
VASH2	Q86V25	VASP	Q2MJR0	VAT1	Q99536
VAV1	P15498	VAV2	P52735	VBP1	P61758
VCAM1	P19320	VCAN	P13611	VCL	P18206
VCP	P55072	VCPIP1	Q96JH7	VDAC1	P21796
VDAC2	P45880	VDAC3	Q9Y277	VDR	P11473
VEGFA	P15692, P49765	VEGFB	P49765	VEGFC	P49767
VENTX	O95231	VGF	O15240	VHL	P40337
VIM	P08670	VIMP	Q9BQE4	VIPR1	P32241
VIPR2	P41587	VIT	O75030	VKORC1	Q9BQB6
VNN1	O95497	VNN2	O95497	VPS11	Q9H270
VPS16	Q9H269	VPS18	Q9P253	VPS25	Q9BRG1
VPS28	Q9UK41	VPS29	Q9UBQ0	VPS33A	Q96AX1
VPS33B	Q9H267	VPS35	Q96QK1	VPS36	Q86VN1
VPS37A	Q8NEZ2	VPS37D	Q86XT2	VPS39	Q96JC1
VPS41	P49754	VPS4A	Q9UN37	VPS51	Q9UID3
VPS54	Q9P1Q0	VPS72	Q15906	VRK1	Q99986
VRK2	Q86Y07	VRK3	Q8IV63	VSIG10	Q9NYS7
VTI1A	Q96AJ9	VTI1B	Q9UEU0	VWA1	Q6PCB0
VWF	P04275	WAC	Q9BTA9	WAPAL	Q7Z5K2
WARS	P23381	WARS2	Q9UGM6	WAS	P42768
WASF1	Q92558	WASF2	Q9Y6W5	WASL	O00401
WBP1	P39656	WBP11	Q9Y2W2	WBSCR17	Q6IS24
WBSCR22	O43709	WDR1	O75083	WDR11	Q8WWQ0
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WDR24	Q96S15	WDR3	Q9UNX4	WDR33	Q9C0J8
WDR34	Q96EX3	WDR35	Q9P2L0	WDR36	Q8NI36
WDR4	P57081	WDR43	Q15061	WDR45	Q9Y484
WDR45B	Q5MNZ6	WDR46	O15213	WDR5	P61964
WDR59	Q6PJI9	WDR6	Q9NNW5	WDR60	Q8WVS4
WDR61	Q9GZS3	WDR70	Q9NW82	WDR77	Q9BQA1
WDR82	Q6UXN9	WDR83	Q9BRX9	WDR91	A4D1P6
WDTC1	Q8N5D0	WFDC2	Q14508	WFS1	O76024
WHAMM	Q8TF30	WHSC1	O96028	WIPF1	O43516
WIPI2	Q9Y4P8	WLS	Q5T9L3	WNK1	Q9H4A3
WNK2	Q9Y3S1	WNK3	Q9BYP7	WNK4	Q96J92
WNT10B	O00744	WNT11	O96014	WNT2B	Q93097
WNT3	P56703	WNT4	O14905, P56705	WNT5A	P41221
WNT5B	Q9H1J7	WNT6	Q9Y6F9	WNT7A	O00755
WNT7B	P56706	WNT9A	O14904	WNT9B	O14905
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XAB2	Q9HCS7	XAF1	Q6GPH4	XBP1	P17861-2
XDH	Q06278	XIAP	P98170	XIRP2	A4UGR9
XPC	Q01831	XPNPEP1	O43895	XPNPEP2	O43895
XPO1	O14980	XPO5	Q9HAV4	XPOT	O43592
XRCC1	P18887	XRCC3	O43542	XRCC4	Q13426
XRCC5	P13010	XRCC6	P12956	XRN1	Q8IZH2
XYLB	O75191	XYLT1	Q86Y38	XYLT2	Q9H1B5
YAF2	Q8IY57	YAP1	P46937	YARS	P54577
YBX1	P67809	YES1	P07947	YIF1A	O95070
YKT6	O15498	YOD1	Q5VVQ6	YRDC	Q86U90
YTHDC2	Q9H6S0	YWHAB	P31946	YWHAE	P62258
YWHAG	P61981	YWHAH	Q04917	YWHAQ	P27348
YWHAZ	P63104	ZAP70	P43403	ZBTB16	Q05516
ZBTB17	Q13105	ZBTB33	Q86T24	ZC3H12A	Q5D1E8
ZC3H15	Q8WU90	ZC3H18	Q86VM9	ZC3H6	Q8N5P1
ZC3H8	Q8N5P1	ZC3HAV1	Q7Z2W4	ZC3HC1	Q86WB0
ZCCHC11	Q5TAX3	ZCCHC6	Q5VYS8	ZCCHC7	Q8N3Z6
ZCCHC8	Q6NZY4	ZCRB1	Q8TBF4	ZDHHC18	P31947
ZDHHC2	Q9UIJ5	ZDHHC20	Q5W0Z9	ZDHHC21	Q8IVQ6
ZDHHC5	Q9C0B5	ZDHHC9	Q9Y397	ZEB1	P37275
ZEB2	O60315	ZFAND6	Q6FIF0	ZFC3H1	O60293
ZFHX3	Q15911	ZFP1	Q9BS31	ZFP2	Q6ZN57, Q9NQZ8
ZFP28	Q8NHY6	ZFP36	P26651	ZFP36L1	Q07352
ZFP36L2	P47974	ZFP69	Q49AA0	ZFP69B	Q9UJL9
ZFP90	Q8TF47	ZFPM1	Q8IX07	ZFPM2	Q8WW38
ZFYVE16	Q7Z3T8	ZFYVE9	O95405-1	ZIC1	Q15915
ZIC2	O95409	ZIK1	Q3SY52, Q9BWM5	ZKSCAN1	P17029
ZKSCAN4	Q969J2	ZKSCAN5	Q9Y2L8	ZKSCAN7	Q6ZNG0, Q9P0L1
ZMAT2	Q96NC0	ZMYM2	Q9UBW7	ZNF100	Q8IYN0
ZNF114	Q8NC26	ZNF124	Q15973	ZNF138	P52744
ZNF14	P17017	ZNF141	Q15928	ZNF154	A6NNF4, Q13106
ZNF155	Q12901	ZNF169	Q14929	ZNF175	Q9Y473
ZNF195	O14628	ZNF197	O14709	ZNF200	P98182
ZNF202	O95125	ZNF205	O95201	ZNF208	O43345
ZNF211	Q13398	ZNF212	Q6NUN9, Q9UDV6	ZNF213	O14771
ZNF215	Q8NE65, Q9UL58	ZNF217	O75362	ZNF224	Q9NZL3
ZNF225	P18146	ZNF235	Q14590, Q86WZ6	ZNF248	Q8NDW4
ZNF25	P17030	ZNF250	P15622	ZNF254	O75437
ZNF257	Q9Y2Q1	ZNF26	P17031	ZNF260	Q8TAQ5
ZNF263	O14978	ZNF264	O43296	ZNF266	Q14584
ZNF267	Q14586	ZNF268	Q96BR6	ZNF273	Q14593
ZNF282	Q9UDV7	ZNF287	Q9HBT7	ZNF3	P17036, Q6AW86
ZNF30	P17039	ZNF300	Q96RE9	ZNF302	Q9NR11
ZNF304	Q9GZX5	ZNF311	Q5JNZ3	ZNF320	A2RRD8
ZNF324	O75467	ZNF324B	Q6AW86	ZNF331	Q9NQX6
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ZNF33B	Q06730	ZNF34	Q8IZ26	ZNF345	Q8NA42

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ZNF394	Q53GI3	ZNF416	Q9BWM5	ZNF420	Q8TAQ5
ZNF425	Q6IV72	ZNF429	Q86V71	ZNF43	P17038
ZNF430	Q9H8G1	ZNF431	Q8TF32	ZNF433	Q8N7K0
ZNF436	Q9C0F3	ZNF445	P59923	ZNF446	Q9NWS9
ZNF460	Q14592	ZNF467	Q7Z7K2	ZNF471	Q9BX82
ZNF473	Q8WTR7	ZNF483	Q8TF39	ZNF484	Q5JVG2, Q8N8J6
ZNF486	Q96H40	ZNF490	Q9ULM2	ZNF493	Q6ZR52
ZNF496	Q96IT1	ZNF506	Q5JVG8	ZNF512B	Q96KM6
ZNF514	Q96K75	ZNF517	Q6ZMY9	ZNF519	Q8TB69
ZNF521	Q96K83	ZNF530	Q6P9A1	ZNF534	Q76KX8
ZNF544	Q6NX49	ZNF546	Q86UE3	ZNF549	Q6P9A3
ZNF552	Q9H707	ZNF554	Q86TJ5	ZNF555	Q96BR6
ZNF558	Q96NG5	ZNF561	Q8N587	ZNF562	Q6V9R5
ZNF563	Q8TA94	ZNF566	Q969W8	ZNF568	Q3ZCX4
ZNF570	Q96NI8	ZNF571	Q7Z3V5	ZNF573	Q86YE8
ZNF577	Q9BSK1	ZNF584	Q8IVC4	ZNF586	Q9NXT0
ZNF587	Q96SQ5	ZNF589	Q86UQ0	ZNF595	Q8IYB9
ZNF596	Q8TC21	ZNF598	Q86UK7	ZNF599	Q08ER8, Q96NL3
ZNF600	Q6ZNG1	ZNF606	Q8WXB4	ZNF611	Q8N823
ZNF615	Q8N8J6	ZNF619	Q8N2I2	ZNF620	Q6ZNG0
ZNF627	Q7L945	ZNF628	P51786	ZNF638	Q14966
ZNF641	Q96N77	ZNF648	Q96GC6	ZNF664	Q8N3J9
ZNF667	Q5HYK9	ZNF668	Q96K58	ZNF669	Q96BR6
ZNF671	Q8TAW3	ZNF676	Q8N7Q3	ZNF677	Q86XU0
ZNF678	Q5SXM1	ZNF680	Q8NEM1	ZNF682	Q95780
ZNF684	Q5T5D7	ZNF688	P0C7X2	ZNF689	Q96CS4
ZNF691	Q5VV52	ZNF692	Q96SR6	ZNF696	Q9H7X3
ZNF697	Q5TEC3	ZNF699	Q32M78	ZNF701	Q9NV72
ZNF703	Q9H7S9	ZNF704	Q6ZNC4	ZNF706	Q9Y5V0
ZNF707	Q96C28	ZNF71	Q9NQZ8	ZNF711	Q9Y462
ZNF714	Q96N38	ZNF717	Q9BY31	ZNF718	Q3SXZ3
ZNF721	Q8TF20	ZNF726	A6NNF4	ZNF727	A8MUV8
ZNF730	Q6ZMV8	ZNF732	B4DXR9	ZNF735	P0CB33
ZNF736	B4DX44	ZNF738	Q8NE65	ZNF74	Q16587
ZNF740	Q8NDX6	ZNF746	Q6NUN9	ZNF749	O43361
ZNF75D	P51815	ZNF764	Q96H86	ZNF767P	Q75MW2
ZNF77	Q15935	ZNF771	Q7L3S4	ZNF775	Q96BV0
ZNF776	Q68DI1	ZNF777	Q9ULD5	ZNF778	Q96MU6
ZNF782	Q6ZMW2	ZNF785	A8K8V0	ZNF786	Q8N393
ZNF790	Q6PG37	ZNF804B	A4D1E1	ZNF805	O43296
ZNF830	Q96NB3	ZNF846	P17019	ZNF850	Q96PQ6
ZNF853	Q6IV72	ZNF883	P52742	ZNF99	A8MXY4
ZNRD1	Q9P1U0	ZNRF1	Q8ND25	ZNRF2	Q8NHG8
ZNRF3	Q9ULT6	ZP3	P21754	ZSCAN22	Q68DY9
ZSCAN32	Q9NX65	ZSWIM8	A7E2V4	ZWILCH	Q9H900
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ABCB4	ENSG00000005471	ABCC1	ENSG00000103222	ABCC3	ENSG00000108846		
ABCF2	ENSG00000033050	ABCG1	ENSG00000160179	ABCG2	ENSG00000118777		
ACACB	ENSG00000076555	ACLY	ENSG00000131473	ACOX1	ENSG00000161533		
ACSL1	ENSG00000151726	ACSS3	ENSG00000111058	ACTA2	ENST00000224784		
ACTB	ENST00000331789	ADAP1	ENSG00000105963	ADD1	ENSG00000087274		
ADIPOQ	ENSG00000181092	AGPAT1	ENSG00000204310	AGT	ENSG00000135744		
AIFM2	ENSG00000042286	AIP	ENSG00000110711	AKT1	ENSG00000142208		
ALAS1	ENSG00000023330	ALDH1A3	ENST00000329841	ALK	ENSG00000171094		
ALOX5	ENSG00000012779	AMER1	ENSG00000184675	AMH	ENSG00000104899		
AMY2B	ENST00000361355	ANGPTL4	ENSG00000167772	ANPEP	ENST00000300060		
ANXA1	ENST00000257497	ANXA3	ENST00000264908	ANXA4	ENST00000394295		
ANXA9	ENST00000368947	APAF1	ENSG00000120868	APC2	ENSG00000234906		
APOC1	ENSG00000130208	APOD	ENSG00000189058	APOE	ENSG00000130203		
APP	ENSG00000142192	AQP1	ENST00000311813	ARC	ENSG00000198576		
AREG	ENST00000395748	ARF1	ENSG00000143761	ARFGAP1	ENSG00000101199		
ARG1	ENSG00000118520	ARID3A	ENSG00000116017	ARL4C	ENSG00000188042, ENST00000339728		
ARNTL	ENSG00000133794	ASAH1	ENSG00000104763	ASNS	ENSG00000070669		
ATF3	ENSG00000162772	ATF4	ENSG00000128272	ATF5	ENSG00000169136, ENST00000423777		
ATM	ENSG00000149311, ENST00000278616	ATP5B	ENSG00000110955	ATP6AP2	ENSG00000182220		
ATP6V0A1	ENSG00000033627	ATP6V0B	ENSG00000117410	ATP6V0C	ENSG00000185883		
ATP6V0D1	ENSG00000159720	ATP6V0E1	ENSG00000113732	ATP6V0E2	ENSG00000171130		
ATP6V1A	ENSG00000114573	ATP6V1B2	ENSG00000147416	ATP6V1D	ENSG00000100554		
ATP6V1E1	ENSG00000131100	ATP6V1F	ENSG00000128524	ATP6V1G1	ENSG00000136888		
ATP6V1H	ENSG00000047249	ATR	ENSG00000175054, ENST00000350721	B2M	ENSG00000166710		
BACH1	ENSG00000156273	BAX	ENSG00000087088	BBC3	ENSG00000105327		
BCL2	ENSG00000171791	BCL2L1	ENSG00000171552	BCL2L11	ENSG00000153094		
BCL6	ENSG00000113916	BDNF	ENSG00000176697	BDP1	ENSG00000072135		
BHLHE40	ENSG00000134107	BHLHE41	ENSG00000123095	BID	ENSG00000015475		
BIRC7	ENSG00000101197	BMP4	ENSG00000125378	BNIP3L	ENSG00000104765		
BOC	ENSG00000144857	BOLA2B	ENSG00000183336	BRCA1	ENSG00000012048		
BRWD1	ENSG00000185658	BST2	ENSG00000130303	BTG1	ENSG00000133639		
BTG2	ENSG00000159388	C10orf2	ENSG00000107815	CA1	ENSG00000133742		
CA2	ENST00000285379	CALR	ENSG00000179218	CASP1	ENSG00000137752		
CASP10	ENSG00000003400	CAT	ENSG00000121691	CAV1	ENSG00000105974		
CBFA2T2	ENSG00000078699	CBX5	ENSG00000094916	CCL2	ENSG00000108691		
CCL3	ENSG00000277632	CCL4	ENSG00000275302	CCL5	ENSG00000271503		
CCNA1	ENSG00000133101	CCNA2	ENSG00000145386	CCND1	ENSG00000110092		
CCNE1	ENSG00000105173	CCNL1	ENSG00000073111	CCR1	ENSG00000163823		
CCR2	ENSG00000121807	CCR5	ENSG00000160791	CD163	ENSG00000177575		
CD24	ENST00000606017	CD274	ENST00000381573.3	CD36	ENSG00000135218		
CD44	ENST00000428726	CD74	ENST00000009530	CD80	ENSG00000121594		
CD86	ENSG00000114013	CDC25A	ENSG00000164045	CDC42	ENSG00000070831		

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CDK2	ENSG00000123374	CDK5	ENSG00000164885	CDK5R1	ENSG00000176749
CDKN1A	ENSG00000124762	CDKN1B	ENSG00000111276	CDKN2A	ENSG00000147889, ENST00000579755
CDKN2B	ENSG00000147883	CEACAM1	ENSG00000079385	CEBPA	ENSG00000245848
CEBPB	ENSG00000172216	CEBPD	ENSG00000221869	CEL	ENST00000372080
CENPF	ENSG00000117724	CES1	ENSG00000125148	CFL1	ENSG00000172757
CHAC1	ENSG00000128965	CHD1	ENSG00000153922, ENST00000284049.7	CHEK1	ENSG00000149554, ENST00000438015
CHL1	ENSG00000013573	CIDEA	ENSG00000176194	CIDEC	ENSG00000187288
CIITA	ENSG00000179583	CITED2	ENSG00000164442	CLDN5	ENSG00000184113
CLOCK	ENSG00000134852	CNN2	ENSG00000064666	CNOT6L	ENST00000264903
COL17A1	ENST00000648076	COL1A1	ENSG00000108821	COL1A2	ENSG00000164692
COL4A6	ENSG00000197565	COX7A1	ENSG00000161281	CPEB1	ENST00000563519
CPS1	ENSG00000021826	CPT1A	ENSG00000110090	CPT2	ENSG00000157184
CR1	ENSG00000203710	CREB1	ENSG00000118260	CRY1	ENSG00000008405
CSF1	ENSG00000184371	CSF1R	ENSG00000182578	CSF3R	ENSG00000119535
CSPG4	ENST00000308508	CSRP1	ENSG00000159176	CTDSP2	ENSG00000175215
CTGF	ENSG00000118523	CTSD	ENSG00000117984	CUL7	ENSG00000044090
CXCL10	ENSG00000169245	CXCL12	ENSG00000107562	CXCL2	ENSG000000081041
CXCL8	ENSG00000169429	CXCR4	ENST00000241393	CXXC1	ENSG00000154832
CXXC5	ENSG00000171604	CYCS	ENSG00000172115	CYP1A1	ENSG00000140465
CYP4A11	ENSG00000187048	CYP51A1	ENSG00000001630	DBP	ENSG00000105516
DCT	ENSG00000080166	DCTN1	ENSG00000204843	DDB2	ENSG00000134574
DDIT3	ENSG00000175197, ENST00000346473	DDIT4	ENSG00000168209	DDX11	ENSG00000013573
DEDD2	ENSG00000160570	DFNA5	ENSG00000105928	DGAT2	ENSG00000062282
DHCR7	ENSG00000172893	DHH	ENSG00000139549	DICER1	ENSG00000100697
DIO2	ENSG00000211448, ENST00000438257	DLL1	ENSG00000198719	DLL3	ENSG00000090932
DNAJB11	ENSG00000090520	DNAJB6	ENSG00000105993	DNAJC3	ENSG00000102580
DNMT1	ENSG00000130816, ENST00000340748	DPPA4	ENSG00000121570	DRG1	ENSG00000104419
EBF1	ENSG00000164330	EBI3	ENSG00000105246	EDEM1	ENSG00000134109
EDIL3	ENSG00000164176	EED	ENSG00000074266	EEP1	ENSG00000122547
EGF	ENSG00000138798	EGFR	ENST00000275493	EGR1	ENSG00000120738
EGR2	ENSG00000122877	EGR4	ENSG00000135625	EIF2AK2	ENSG00000055332
ELL2	ENST00000237853	ELOVL2	ENSG00000012660	ELOVL5	ENSG00000012660
ELOVL6	ENSG00000170522	ENPP2	ENST00000427067	EPAS1	ENSG00000116016
EPCAM	ENST00000263735	EPHA4	ENSG00000116106	ERBB2	ENST00000269571
ERBB4	ENSG00000178568	ERICH5	ENST00000318528	ESR1	ENST00000206249
ESRRA	ENSG00000173153	EXTL1	ENSG00000158008	EXTL2	ENSG00000162694
EXTL3	ENSG00000012232	EYA1	ENSG00000104313	F13A1	ENSG00000124491
F3	ENSG00000117525	FABP4	ENSG00000170323	FABP6	ENSG00000170231
FABP7	ENSG00000164434	FAN1	ENSG00000075618	FANCC	ENSG00000158169
FANCD2	ENSG00000144554, ENST00000419585	FAS	ENSG00000026103	FASN	ENSG00000169710
FBXO32	ENSG00000156804	FCGR1A	ENSG00000150337	FCGR1B	ENSG00000198019
FDFT1	ENSG00000079459	FDPS	ENSG00000160752	FGF2	ENSG00000138685

Input	Ensembl Id	Input	Ensembl Id	Input	Ensembl Id
FGF9	ENSG00000102678	FHL2	ENSG00000115641	FKBP14	ENSG00000106080
FKBP4	ENSG00000004478	FKBP5	ENSG00000096060	FLT4	ENSG00000037280
FN1	ENSG00000115414	FOS	ENSG00000170345	FOXD3	ENSG00000187140
FOXO1	ENSG00000150907	FOXO3	ENSG00000118689	FPR1	ENSG00000171051
FSCN1	ENSG00000075618	FTH1	ENST00000273550	FTL	ENST00000331825
G0S2	ENSG00000123689	G6PD	ENSG00000160211	GABPA	ENSG00000154727
GADD45A	ENSG00000116717	GAMT	ENSG00000130005	GATA6	ENST00000269216
GBP1	ENSG00000117228	GBP2	ENSG00000162645	GBP3	ENSG00000117226
GBP4	ENSG00000162654	GBP5	ENSG00000154451	GCK	ENSG00000106633
GCLC	ENSG00000001084	GCLM	ENSG00000023909	GDNF	ENSG00000168621
GEM	ENSG00000164949	GFAP	ENSG00000131095	GGPS1	ENSG00000152904
GIP	ENSG00000159224	GJB1	ENSG00000169562, ENST00000361726.6	GLI1	ENSG00000111087
GLI2	ENSG00000074047	GLIPR1	ENSG00000139278	GLS2	ENSG00000135423
GMPR	ENSG00000137198	GOSR2	ENSG00000108433	GPAM	ENSG00000119927
GPRC5A	ENST00000014914	GPRIN1	ENSG00000169258	GRB10	ENSG00000106070
GREB1	ENSG00000196208, ENST00000381486.6	GREM1	ENSG00000166923	GRIA2	ENSG00000120251
GRIN2A	ENSG00000183454	GRIN2B	ENSG00000273079	GSDMD	ENSG00000104518
GSK3A	ENSG00000105723	GXYLT2	ENSG00000172986	H19	ENST00000252900
HBB	ENSG00000244734	HBD	ENSG00000223609	HBG2	ENSG00000196565
HDGF	ENSG00000143321	HERPUD1	ENSG00000051108	HES1	ENST00000232424
HES7	ENSG00000179111	HEY1	ENSG00000164683	HEY2	ENSG00000135547
HEYL	ENSG00000163909	HGF	ENSG00000019991	HHEX	ENST00000282728
HHIP	ENSG00000164161	HIF1A	ENSG00000100644	HLA-A	ENSG00000206503
HLA-B	ENSG00000234745	HLA-C	ENSG00000204525	HLA-DPA1	ENSG00000231389
HLA-DPB1	ENSG00000223865	HLA-DQA1	ENSG00000196735	HLA-DQA2	ENSG00000237541
HLA-DQB1	ENSG00000179344	HLA-DQB2	ENSG00000232629	HLA-DRA	ENSG00000204287
HLA-DRB1	ENSG00000196126	HLA-DRB5	ENSG00000198502	HLA-E	ENSG00000204592
HLA-F	ENSG00000204642	HLA-H	ENSG00000206341	HMGCR	ENSG00000113161
HMGCS1	ENSG00000112972	HMOX1	ENSG00000100292	HNFB1	ENST00000610754
HNF4G	ENSG00000164749	HNRNPA2B1	ENSG00000122566	HNRNPDL	ENSG00000152795
HOXA2	ENSG00000105996, ENST00000222718	HOXA3	ENSG00000105997, ENST00000317201	HOXB2	ENSG00000173917, ENST00000330070
HOXB3	ENSG00000120093, ENST00000311626	HOXC4	ENSG00000198353, ENST00000303406	HOXD1	ENSG00000128645, ENST00000331462
HOXD3	ENSG00000128652, ENST00000249440	HSP90AA1	ENSG00000080824	HSP90B1	ENSG00000166598
HSPA1B	ENSG00000204388, ENSG00000212866, ENSG00000224501, ENSG00000231555, ENSG00000232804	HSPA1L	ENSG00000206383, ENSG00000226704, ENSG00000234258, ENSG00000236251	HSPA5	ENSG00000044574
HSPA8	ENSG00000109971	HSPA9	ENSG00000113013	HSPB1	ENSG00000106211
HSPD1	ENSG00000144381	HSPE1	ENSG00000115541	HSPH1	ENSG00000120694
HTRA2	ENSG00000115317	HYOU1	ENSG00000149428	ICAM1	ENSG00000090339
ICOS	ENSG00000163600	ID1	ENSG00000125968	ID2	ENSG00000115738
ID3	ENSG00000117318	ID4	ENSG00000172201	IDH1	ENSG00000138413
IDI1	ENSG00000067064	IFI27	ENSG00000165949	IFI35	ENSG00000068079
IFI6	ENSG00000126709	IFIT1	ENSG00000185745	IFIT2	ENSG00000119922
IFIT3	ENSG00000119917	IFITM1	ENSG00000185885	IFITM2	ENSG00000185201

Input	Ensembl Id	Input	Ensembl Id	Input	Ensembl Id
IFITM3	ENSG00000142089	IGF2	ENST00000337883, ENST00000381406	IGFBP3	ENSG00000146674
IGFBP7	ENSG00000163453	IGHG1	ENSG00000211896	IGHG4	ENSG00000211892
IL10RA	ENSG00000110324	IL12RB2	ENSG00000081985	IL18	ENSG00000150782
IL1B	ENSG00000125538, ENST00000263341	IL1R1	ENSG00000115594	IL1R2	ENSG00000115590
IL1RN	ENSG00000136689	IL2RA	ENSG00000134460	IL2RG	ENSG00000147168
IL4R	ENSG00000077238	IL6	ENSG00000136244	IL6R	ENSG00000160712
INSM1	ENSG00000173404	IP6K2	ENSG00000068745	IQSEC3	ENSG00000120645
IRAK1	ENSG00000184216	IRF1	ENSG00000125347	IRF2	ENSG00000168310
IRF4	ENSG00000137265	IRF5	ENSG00000128604	IRF6	ENSG00000117595
IRF8	ENSG00000140968	IRX1	ENSG00000170549	IRX2	ENSG00000170561
ISG15	ENSG00000187608	ISG20	ENSG00000172183	ISL1	ENSG00000016082
ITCH	ENSG00000078747	ITGA2	ENSG00000164171	ITGA4	ENSG00000115232
ITGA5	ENSG00000161638	ITGA6	ENST00000442250	ITGAL	ENSG00000005844
ITGAM	ENSG00000169896	ITGAX	ENSG00000140678	ITGB1	ENST00000302278
ITGB2	ENSG00000160255	ITGBL1	ENSG00000198542	JAG1	ENSG00000101384
JUNB	ENSG00000171223, ENST00000302754	KDEL3	ENSG00000100196	KDM1B	ENST00000297792
KDM4B	ENST00000381759	KDM4C	ENST00000381306	KDM5A	ENST00000399788
KDM5B	ENST00000235790	KDM6B	ENSG00000132510	KEAP1	ENSG00000079999
KIT	ENST00000288135	KLF15	ENSG00000163884	KLF4	ENSG00000136826
KLF5	ENSG00000102554	KLHDC3	ENSG00000124702	KPNA2	ENST00000330459.7
KRT14	ENST00000167586	KRT18	ENST00000388835	KRT19	ENST00000361566
KRT8	ENST00000293308	LAMA5	ENSG00000130702	LCN2	ENSG00000148346
LCP1	ENSG00000136167	LDHB	ENST00000350669	LEP	ENSG00000174697
LFNG	ENSG00000106003	LGALS3	ENSG00000131981	LHX1	ENSG00000273706
LIFR	ENSG00000113594	LIG1	ENSG00000105486	LMNA	ENSG00000160789
LMNB1	ENSG00000113368	LONP1	ENSG00000196365	LPIN1	ENSG00000134324
LPL	ENSG00000175445	LRIG1	ENST00000273261	LSS	ENSG00000160285
LYZ	ENST00000261267	MAG	ENSG00000105695	MAOA	ENSG00000189221
MAP4K4	ENSG00000071054	MAPK4	ENST00000400384	MAPK6	ENSG00000069956
MAPKAPK5	ENSG00000089022	MCL1	ENSG00000143384	MCM2	ENSG00000073111
MCM5	ENSG00000100297	MDC1	ENSG00000137337, ENST00000376406	MDM2	ENSG00000135679
ME1	ENSG00000065833	MECOM	ENSG00000085276	MED12L	ENST00000273432
MED13	ENST00000397786	MEF2C	ENSG00000081189	MET	ENSG00000105976
MFGE8	ENST00000268150	MGLL	ENSG00000074416	MID1	ENSG00000101871
MIF	ENSG00000104899	MINK1	ENSG00000141503	MITF	ENSG00000187098
MME	ENST00000360490	MMP1	ENSG00000196611	MMP10	ENSG00000166670
MMP2	ENSG00000087245	MMP3	ENSG00000149968	MOBP	ENSG00000168314
MPZ	ENSG00000158887	MRPL18	ENSG00000112110	MSN	ENSG00000147065
MT-ATP6	ENST00000361899	MT-ATP8	ENST00000361851	MT-CO1	ENST00000361624
MT-CO2	ENST00000361739	MT-CO3	ENST00000362079	MT-CYB	ENST00000361789
MT-ND1	ENST00000361390	MT-ND2	ENST00000361453	MT-ND3	ENST00000361227
MT-ND4	ENST00000361381	MT-ND4L	ENST00000361335	MT-ND5	ENST00000361567
MT-ND6	ENST00000361681	MT-RNR1	ENST00000389680	MT-RNR2	ENST00000387347
MT2A	ENSG00000125148	MTERF1	ENSG00000127989	MUM1	ENSG00000137265
MVD	ENSG00000167508	MVK	ENSG00000110921	MX1	ENSG00000157601
MX2	ENSG00000183486	MXD4	ENSG00000123933	MYC	ENST00000613283
MYCN	ENSG00000134323	MYDGF	ENSG00000074842	MYL9	ENSG00000101335

Input	Ensembl Id	Input	Ensembl Id	Input	Ensembl Id
MYLIP	ENSG00000007944	MYOCD	ENSG00000141052	NAB2	ENSG00000166886
NAMPT	ENSG00000105835	NANOS3	ENSG00000187556	NDN	ENSG00000182636
NDRG1	ENSG00000104419	NFE2L2	ENSG00000116044	NKX2-5	ENSG00000183072
NLK	ENST00000407008	NLRC4	ENSG00000091106	NLRP3	ENSG00000162711
NOP2	ENSG00000111641	NOS2	ENSG00000007171	NOS3	ENSG00000187556
NOTCH1	ENSG00000148400	NOTCH2	ENSG00000134250, ENST00000256646	NOTCH3	ENSG00000074181, ENST00000263388
NOTCH4	ENSG00000206312, ENST00000383264	NOX4	ENSG00000086991	NPAS2	ENSG00000170485
NPNT	ENSG00000168743	NQO1	ENSG00000181019	NR1D1	ENSG00000126368
NR1H3	ENSG00000025434, ENST00000441012	NR4A3	ENSG00000119508	NR5A2	ENSG00000116833
NRF1	ENSG00000106459	OAS1	ENSG00000089127	OAS2	ENSG00000111335
OAS3	ENSG00000111331	OASL	ENSG00000135114	OCLN	ENSG00000197822
ONECUT3	ENSG00000205922	OPRD1	ENSG00000116329	OPRK1	ENSG00000082556
OPRM1	ENSG00000112038	OR2L13	ENSG00000196071	OR7A5	ENSG00000188269
OR8S1	ENSG00000284723	OSM	ENSG00000099985	OSR1	ENSG00000143867
PAIP2	ENST00000265192	PAX3	ENSG00000135903	PAX8	ENSG00000125618
PBX1	ENSG00000185630	PCBP4	ENSG00000090097	PCDH19	ENSG00000165194
PCK1	ENSG00000124253	PCNA	ENSG00000132646	PDCD4	ENSG00000150593
PDGFA	ENSG00000197461	PDIA2	ENST00000219406	PDIA5	ENSG00000065485
PDK4	ENSG00000004799	PDPN	ENSG00000162493	PER1	ENSG00000179094
PER2	ENSG00000132326	PER3	ENSG00000049246	PERP	ENSG00000112378
PGD	ENSG00000142657	PGR	ENST00000325455	PIK3R1	ENSG00000145675
PIM1	ENSG00000137193	PINK1	ENSG00000158828	PITPNA	ENSG00000174238
PITX2	ENSG00000164093	PLAC8	ENSG00000145287	PLAGL1	ENSG00000118495
PLIN1	ENSG00000166819	PLIN2	ENSG00000147872	PLIN4	ENSG00000167676
PLK2	ENSG00000145632	PLK3	ENSG00000173846	PLTP	ENSG00000100979
PLXNA4	ENSG00000221866	PLXND1	ENSG00000004399	PMEL	ENSG00000185664
PMP22	ENSG00000109099	PMVK	ENSG00000163344	PNPLA2	ENSG00000177666
POLA1	ENSG00000101868	POLR2D	ENST00000272645	POLRMT	ENSG00000099821
POU2F1	ENSG00000143190	POU3F1	ENSG00000185668	POU3F2	ENST00000328345
PPARA	ENSG00000186951	PPARG	ENSG00000132170	PPARGC1A	ENSG00000109819
PPIA	ENSG00000196262	PPP1R15A	ENSG00000087074, ENST00000200453	PPP2R5B	ENSG00000068971
PRDM1	ENSG00000057657	PRDM16	ENSG00000142611	PRDX1	ENSG00000117450
PREB	ENSG00000138073	PRF1	ENSG00000180644	PRKCB	ENSG00000166501
PRKCQ	ENSG00000065675	PRSS3	ENST00000379405	PRX	ENSG00000105227
PSMB8	ENSG00000204264	PSME2	ENSG00000100911	PTAFR	ENSG00000169403
PTCH1	ENSG00000185920	PTCH2	ENSG00000117425	PTEN	ENST00000371953
PTGDS	ENSG00000107317	PTGS2	ENSG00000073756	PTPN1	ENSG00000196396
PTPN11	ENSG00000179295	PTPN12	ENSG00000127947	PTPN13	ENSG00000163629
PTPN14	ENSG00000152104	PTPN2	ENSG00000175354	PTPN20	ENSG00000204179
PTPN23	ENSG00000076201	PTPN4	ENSG00000088179	PTPN5	ENSG00000110786
PTPN6	ENSG00000111679	PTPN7	ENSG00000143851	PTPN9	ENSG00000169410
PVALB	ENSG00000100362	PXDN	ENSG00000130508	PXN	ENSG00000089159
RAB27A	ENSG00000069974	RABGGTA	ENSG00000100949	RAD51D	ENSG00000185379, ENST00000345365
RAG1	ENSG00000166349	RALA	ENSG00000006451	RAP1B	ENSG00000127314
RBBP4	ENSG00000162521	RBBP8	ENSG00000101773	RBFOX1	ENSG00000078328
RBFOX3	ENSG00000167281	RBL1	ENSG00000080839	RBL2	ENSG00000103479

Input	Ensembl Id	Input	Ensembl Id	Input	Ensembl Id
RET	ENSG00000165731	RGCC	ENSG00000102760	RGL1	ENSG00000143344
RHOH	ENSG00000116574	RIPPLY2	ENSG00000203877	RNASE1	ENST00000397967
RNASEL	ENSG00000135828	RNU4-1	ENSG00000200795	RNU5A-1	ENSG00000199568
ROBO1	ENSG00000169855	ROBO2	ENSG00000185008	ROBO3	ENSG00000154134
RORA	ENSG00000069667	RPLP0	ENSG00000089157	RPS2	ENST00000343262
RRAD	ENSG00000166592	RRM2B	ENSG00000048392	RSAD2	ENSG00000134321
RSP03	ENSG00000146374	RUNX1	ENSG00000159216, ENST00000344691	RUNX2	ENSG00000124813
S100B	ENSG00000160307	S1PR1	ENSG00000170989	SALL1	ENSG00000103449
SAMHD1	ENSG00000101347	SCD	ENSG00000099194	SCO2	ENSG00000130489
SCP2	ENSG00000175215	SDC4	ENST00000372733	SERP1	ENSG00000120742
SERPINE1	ENSG00000106366	SERPINH1	ENSG00000149257	SESN1	ENSG00000080546
SESN2	ENSG00000130766	SESN3	ENSG00000149212	SFTPD	ENSG00000133661
SGK1	ENST00000237305	SHC1	ENSG00000160691	SHH	ENSG00000164690
SIRT1	ENSG00000096717	SIRT3	ENSG00000142082	SIX1	ENSG00000126778
SLC27A1	ENSG00000130304	SLC2A3	ENSG00000059804	SLC2A4	ENSG00000181856
SLC4A4	ENST00000264485	SLC7A11	ENSG00000151012	SLIT2	ENSG00000145147
SLPI	ENST00000338380	SMAD6	ENSG00000137834	SMAD7	ENSG00000101665
SMURF1	ENSG00000198742	SNAI1	ENSG00000124216	SNAI2	ENSG0000019549
SNCB	ENSG00000074317	SOCS2	ENSG00000120833	SOCS3	ENSG00000184557
SOCS4	ENSG00000180008	SOCS7	ENSG00000180008	SOD3	ENSG00000109610
SOX17	ENST00000297316	SOX9	ENST00000245479	SP100	ENSG00000067066
SPARC	ENSG00000113140	SPI1	ENSG00000066336	SPP1	ENSG00000118785
SQLE	ENSG00000104549	SQSTM1	ENSG00000161011	SREBF1	ENSG00000072310
SRF	ENSG00000112658	SRPRB	ENSG00000144867	STAT1	ENSG00000115415
STAT3	ENSG00000168610	STAT5A	ENSG00000126561	STEAP3	ENSG00000115107
STK11	ENSG00000118046	STMN1	ENSG00000117632	STRA13	ENSG00000134107
STT3B	ENSG00000163527	SUZ12	ENSG00000178691	SYT10	ENSG00000110975
SYVN1	ENSG00000162298	TALDO1	ENSG00000177156	TATDN2	ENSG00000157014
TBX2	ENSG00000121068	TCF4	ENSG00000196628	TCP1	ENSG00000120438
TEC	ENSG00000119508	TEP1	ENSG00000171862	TET2	ENST00000265149
TFAP2A	ENST00000379613	TFAP2C	ENST00000201031	TFB1M	ENSG00000029639
TFPI2	ENST00000222543	TFRC	ENST00000360110	TGFA	ENSG00000163235
TGFB1	ENSG00000105329	TGFBR3	ENSG00000069702, ENST00000212355	THBS1	ENSG00000137801
THRSP	ENSG00000151365	TIAM2	ENSG00000146426	TIMP1	ENSG00000102265
TJP1	ENSG00000104067	TKT	ENSG00000163931	TLN1	ENSG00000137076
TM7SF2	ENSG00000149809	TMEM97	ENST00000226230	TNFRSF10A	ENSG00000104689
TNFRSF10D	ENSG00000173530	TNFRSF1A	ENSG00000067182	TNFRSF1B	ENSG00000028137
TNFRSF21	ENSG00000146072	TNIK	ENSG00000154310	TP53INP1	ENSG00000164938
TP63	ENST00000354600	TPP1	ENSG00000166340	TRIAP1	ENSG00000170855
TRIB3	ENSG00000101255	TRIM14	ENSG00000106785	TRIM21	ENSG00000132109
TRIM22	ENSG00000132274	TRIM25	ENSG00000121060	TRIM29	ENSG00000137699
TRIM3	ENSG00000110171	TRIM35	ENSG00000104228	TRIM38	ENSG00000112343
TRIM45	ENSG00000134253	TRIM46	ENSG00000163462	TRIM5	ENSG00000132256
TRPC3	ENSG00000138741	TSC22D1	ENSG00000102804	TSPYL2	ENSG00000184205
TUBB4B	ENST00000340384	TWIST1	ENSG00000122691	TXN	ENSG00000136810
TXNRD1	ENSG00000198431	TYRP1	ENSG00000107165	UBB	ENSG00000170315
UXT	ENSG00000126756	VCAM1	ENSG00000162692	VEGFA	ENSG00000112715
VGF	ENSG00000128564	VIM	ENST00000544301	WFDC2	ENSG00000101443

Input	Ensembl Id	Input	Ensembl Id	Input	Ensembl Id
WFS1	ENSG00000109501	WNT11	ENSG000000085741	WNT4	ENSG00000162552
WNT9B	ENSG00000158955	WWC1	ENSG00000113645	XAF1	ENSG00000132530
XBP1	ENSG00000100219	YAP1	ENST00000282441	YIF1A	ENSG00000174851
ZAP70	ENSG00000115085	ZBTB17	ENSG00000116809	ZCCHC11	ENSG00000134744
ZCCHC6	ENSG00000083223	ZEB1	ENSG00000148516	ZEB2	ENSG00000169554
ZIC1	ENSG00000152977	ZIC2	ENSG00000043355	ZNF225	ENSG00000120738
ZNF385A	ENSG00000161642	ZNF521	ENSG00000198795		

Input	NCBI_NUCLEOTIDE Id
SNORD3A	NR_006880

Interactors (8512)

Input	UniProt Id	Interacts with	Input	UniProt Id	Interacts with
A2M	P01023	Q8NBL1	A2ML1	A8K2U0	P01160
AAED1	Q7RTV5	Q9NRD5	AAGAB	Q6PD74	O75843, Q12846
AAK1	Q2M2I8	Q9NQ11	AAMDC	Q9H7C9	P54252
AAMP	C9JG97	O14503	AANAT	Q29495	P63104
AAR2	Q9Y312	Q9BUQ8	AARS	P49588	P13569
AARSD1	Q9BTE6-2	P63172	AASDHPPT	Q9NRN7	Q9UQK1
AASS	Q9UDR5	P0DTC5	AATK	Q6ZMQ8-2	Q15078
ABCA1	O95477	P02647	ABCB1	P08183	P17931
ABCB10	Q9NRK6	Q71RS6	ABCC1	P33527	P17931
ABCC2	Q92887	P19838	ABCC4	O15439	P17931
ABCD1	P33897	P46059	ABCD3	P28288	P40855
ABCD4	O14678	P23945	ABCF3	A0A0S2Z645	P61289
ABCG2	Q9UNQ0	Q9UNQ0	ABCG4	Q9H172	O76024
ABHD10	Q9NUJ1	Q15637, Q96NC0	ABHD11	Q8NFV4-4	Q15915
ABHD14A	Q9BUJ0	Q9UNA3	ABHD15	Q6UXT9	P21673
ABHD16A	O95870	P05177	ABHD17C	Q6PCB6	O14832
ABHD3	Q8WU67	P04201	ABHD4	Q8TB40	Q6ZPD8
ABI1	Q8IZP0	Q07889	ABI2	Q9NYB9-2	Q13177
ABI3	Q9P2A4	O43639	ABL2	P42684	P19174
ABLIM2	Q6H8Q1-8	P14136	ABLIM3	O94929-2	Q13573
ABT1	Q9ULW3	P06748	ABTB1	Q969K4	P29692
ACAA1	P09110	O00628	ACAA2	P42765	P0DTC3
ACACB	O00763	P50747	ACAD10	Q6JQN1	Q99677
ACADL	P28330	P22735	ACADS	P16219	Q9NU23
ACAN	P16112	P05067	ACAP1	Q15027	P17252
ACAT1	P35610	O75326	ACBD6	Q9BR61	P12931
ACCS	Q96QU6	P14927	ACD	Q96AP0	Q92900
ACHE	P22303	P06733	ACIN1	Q9UKV3	P09651
ACKR3	P25106	P49407	ACLY	P53396	P13569
ACO1	P21399	P15056	ACO2	Q99798	Q8IWL3
ACOT11	Q8WXI4-2	Q9NQT4	ACOT12	Q8WYK0	O15357
ACOT7	O00154	P13569	ACOT8	O14734	P04601
ACOT9	Q9Y305	P13671	ACOX2	Q99424	P63172
ACP1	P24666	P60709	ACP2	P11117	P21453
ACPP	A0ABF7PH85	Q9H1K1	ACSF2	Q96CM8	Q6ZPD8
ACSF3	Q4G176	P14373	ACSL4	O60488	P13569

Input	UniProt Id	Interacts with	Input	UniProt Id	Interacts with
ACSL5	Q9ULC5	O00559	ACSS3	Q9H6R3	Q96CV9
ACTA1	P68133	Q16665	ACTA2	P62736	P15498
ACTB	P60709	P23528	ACTG1	P63261	P23528
ACTL6A	O96019	P39748	ACTL6B	O94805	Q9Y265
ACTN1	P12814	P12931	ACTN2	P35609	Q92934
ACTN4	O43707	Q07157	ACTR10	Q9NZ32	Q8N5V2
ACTR1A	P61163	Q14203	ACTR1B	P42025	Q8N5V2
ACTR2	P61160	P13569	ACTR3C	Q9C0K3	Q9H668
ACTR5	Q9H9F9	O96019	ACTR8	Q9H981	Q9Y5K5
ACVR1	Q04771	Q9NPF0	ACVR1B	P36896	P62942
ACVR2A	P27037	Q14160	ACVR2B	Q13705	P27037
ACY3	Q96HD9	O75928	ADAL	Q6DHSV7-2	P08670
ADAM10	O14672	P48509	ADAM12	O43184-2	O95633
ADAM17	P78536	P05556	ADAM22	Q9P0K1-3, Q9P0K1	P31946, P63104
ADAM23	Q9UKQ2	P00749	ADAM28	Q9UKQ2	P00749
ADAM32	Q8TC27	P08648	ADAM33	Q08AM2	Q16585
ADAM9	Q13443	Q99962	ADAMTS1	Q9UHI8	O00233
ADAMTS13	Q76LX8	P04275	ADAMTS18	Q8TE60	Q6Y288
ADAMTS4	O75173	O00462	ADAMTSL3	P82987	Q7Z3S9
ADAMTSL4	Q6UY14-3	P09681	ADAMTSL5	Q6ZMM2-2	Q7Z3S9
ADAP1	O75689	Q08379	ADARB1	P78563-4	P19525
ADAT2	Q7Z6V5	Q96GM5	ADCK3	Q8NI60	Q9H2K0
ADCY3	O60266	P17931	ADCY4	Q8NFM4	P37231
ADCY9	O60503	P15514	ADCYAP1	P18509	Q9Y5J6
ADCYAP1R1	P41586-3	P18509	ADD1	P35611	Q13153
ADD2	P35612	P35611	ADGRA1	Q86SQ6	Q12959
ADGRA2	Q96PE1	Q12959	ADGRA3	Q8IWK6	P34925
ADGRB3	O60242	O60883	ADGRE2	Q9UHX3	A1L3X0
ADGRE5	P48960	P08174	ADGRF5	Q8IZF2	Q9NPB3
ADGRG5	Q8IZF4	P17302	ADGRL2	O95490	P21926
ADGRL3	Q9HAR2	Q9NZU0	ADI1	Q9BV57	Q9UKG1
ADIPOQ	Q15848	P49327	ADIPOR1	Q96A54	P25103
ADIPOR2	Q86V24	Q9P0P0	ADIRF	Q15847	Q14116
ADORA2B	P29275	P29274	ADRA1A	P35348	P61073
ADRA1B	P35368	P61073	ADRA1D	P25100	P46939
ADRB2	P07550	Q99685	ADRBK1	P25098	Q13635
ADRBK2	P35626	P25098	ADRM1	Q16186	P60896
ADSL	P30566	P13569	AEBP2	Q6ZN18-2	Q9H2G4
AEN	Q8WTP8-2, Q8WTP8	Q13685	AES	Q08117-2	O00267
AFF1	P51825	P04608	AFF3	P51826	P50750
AFF4	Q9UHB7	P55199	AFG3L2	Q9Y4W6	P42858
AGAP2	Q99490-2	P42229	AGAP3	Q96P47	P61981
AGFG1	P52594	P51809	AGFG2	O95081	Q9NQZ5
AGO2	Q9UKV8	P04156	AGO3	Q9H9G7	O15397, Q8NDV7
AGPAT1	Q99943	Q8TDQ0	AGPAT4	Q9NRZ5	Q96BA8
AGPAT6	Q86UL3	O60443	AGPAT9	Q53EU6	Q9H2J7
AGRN	O00468	O15530	AGT	P01019	P11021
AGTR1	P30556	Q03135	AGTRAP	Q6RW13-2	Q96E11
AHCTF1	Q8WYP5	P06748	AHCY	P9WGV3	P10145

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AHCYL2	Q96HN2	P31152	AHDC1	Q5TGY3	Q13185, P45973
AHI1	Q8N157	P50570	AHNAK	Q09666	P18827
AHNAK2	Q8IVF2-3	Q15836	AHR	P35869	Q9Y618
AHSA1	O95433	Q13451	AIF1	P55008	P06396
AIFM1	O95831	P62937	AIG1	Q9NVV5	Q96PM5
AIM1	Q96GD4	P06493	AIM2	O14862	P29466
AIMP1	Q12904-2	Q96CV9	AIMP2	Q13155	P03508
AIP	O00170	Q92985	AJUBA	Q96IF1	Q13526
AK1	P00568	P13569	AK3	Q9UIJ7	P02649
AK4	P27144	Q8WVK2	AK8	Q96MA6	P26441
AKAP1	Q92667	P13861	AKAP11	Q9UKA4	P51530
AKAP12	Q02952	P13861	AKAP13	Q12802	P31946, P63104
AKAP17A	Q02040	P61981	AKAP7	Q9P0M2	P17612
AKAP8L	Q9ULX6	P60484	AKAP9	Q99996-3	P06396
AKIP1	Q9NQ31	P06396	AKIRIN2	Q53H80	P28066, P49721, P49720
AKR1A1	P14550	O76024	AKR1B10	O60218	Q13085
AKR1C1	Q04828	P42330	AKR1C2	P52895	P42330
AKR7A2	O43488	P54252	AKT1	P31749	P54278
AKT1S1	Q96B36	P31946, P63104	AKT3	Q9Y243	P07900
ALAS1	P13196	Q16822	ALAS2	P22557	Q8N9N5
ALDH16A1	Q8IZ83	Q9Y315	ALDH1A1	P00352	P14136
ALDH1B1	P30837	Q9UMX1	ALDH2	P05091	P02647
ALDH3B1	P43353	Q13231	ALDH6A1	Q02252	Q9H2A2
ALDH8A1	Q9H2A2	Q02252	ALDOC	P09972	P13569
ALG1	Q9BT22	P13569	ALG10	P50076	Q96BA8
ALG10B	Q5I7T1	P06576	ALG13	Q9NP73-4	Q01974
ALG3	Q92685	Q9H015	ALG5	Q9Y673	P0DTC3
ALG6	Q9Y672	Q9BVK2	ALG8	Q9BVK2	Q96BA8
ALK	Q9UM73	P46109	ALKBH2	Q6NS38	Q08379
ALKBH4	Q9NXW9	P42568	ALMS1	Q8TCU4	Q8IWZ6
ALOX15B	O15296	P19474	ALOX5	P09917	Q9P0N9
ALOX5AP	P20292	O15552	ALOXE3	Q9BYJ1	Q8NFP9
ALS2CR11	Q53TS8	P48742	ALS2CR12	Q96Q35-2	Q9UI12
ALYREF	Q86V81	Q92900	AMBRA1	Q9C0C7, Q9C0C7-3	O75385
AMD1	P17707	P18509	AMDHD2	Q9Y303-2	P55212
AMER1	Q5JTC6	O14965	AMER3	Q8N944	P25054
AMICA1	Q86YT9	Q9NPF0	AMMECR1	Q9Y4X0	P14373
AMMECR1L	Q6DCA0	P49354	AMOT	Q4VCS5	Q8N3R9
AMOTL1	Q8IY63	Q8N3R9	AMPD2	Q01433	P18827
AMPD3	Q01432	P08238	AMPH	P49418	Q05193
ANAPC1	Q9H1A4	Q9UM11, Q12834	ANAPC10	Q9UM13	Q12834
ANAPC11	Q9NYG5-2	P54646	ANAPC13	Q9BS18	O76024
ANAPC15	P60006	P51955	ANAPC2	Q9UJX6	Q9UM11, Q12834
ANAPC4	Q9UJX5	P51955	ANAPC7	Q9UJX3	Q12834
ANGEL1	Q9UNK9	P06730	ANGPT1	Q15389	Q02763, Q15389
ANGPT2	O15123, O15123-1	Q02763	ANGPTL2	Q9UKU9	P11021
ANGPTL4	Q9BY76	Q14790	ANGPTL7	O43827	P17987
ANK3	Q12955	Q15796	ANKEF1	Q9NU02	P07237
ANKHD1	Q8IWZ3-3, Q8IWZ3	P61981	ANKIB1	Q9P2G1	P22736
ANKMY2	Q8IV38	Q15185	ANKRD10	Q9NXR5-2	Q15915

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ANKRD13C	Q8N6S4	P80370	ANKRD13D	Q6ZTN6-2	P04792
ANKRD17	O75179	P61981	ANKRD2	Q9GZV1	Q16659
ANKRD27	Q96NW4	P25963	ANKRD28	O15084	Q14694
ANKRD29	Q8N6D5	P50747	ANKRD36	A1A5B0	Q08379
ANKRD36B	Q8N2N9-4	P55040	ANKRD44	Q8N8A2	P46531
ANKRD46	Q86W74-2	Q969X2	ANKRD50	Q9ULJ7	Q3SXY8
ANKRD54	Q6NXT1	Q06187	ANKS1A	Q49AR9	P50458
ANKS1B	Q7Z6G8-3	P61586	ANO1	Q5XXA6	P13569
ANO4	Q32M45	Q96S44	ANO5	Q75V66	P43119
ANO6	Q4KMQ2	P80370	ANOS1	P23352	P11362
ANP32A	P39687	O15169	ANP32B	Q92688	P04792
ANP32E	Q9BT0	P52294	ANTXR1	Q9H6X2	Q16625
ANTXR2	P58335	P13569	ANXA1	P04083	Q13546
ANXA3	P12429	P13569	ANXA4	P09525	P55212
ANXA5	P08758	Q96CV9	ANXA9	O76027	O60437
AOC3	Q16853	Q5SRI9	AP1B1	Q10567-3	Q5SW96
AP1G2	O75843	Q96PK6	AP1M1	Q9BXS5	P18848
AP1S1	P61966-2	P37173	AP1S2	P56377	O75843
AP2A1	O95782	P29353	AP2A2	O94973	P29353
AP2B1	P63010	P12236	AP2M1	Q96CW1	P29353
AP2S1	P53680	Q96D71, Q2M2I8, P42566	AP3B1	O00203	P0DTC4
AP3D1	O14617	Q96EV8	AP3M1	Q9Y2T2	Q9UBH0
AP4M1	O00189	P49354	AP5Z1	O43299	P21462
APAF1	O14727	P99999	APBA1	Q02410	O14936
APBA2	O35431	P05067-4	APBA3	O96018	Q8N9R8
APBB1	O00213	P05067-4	APBB2	Q92870-2	Q969S8
APBB3	O95704	Q9BYZ6	APC2	Q9UJX6	Q9UM11, Q12834
APCDD1	Q8J025	O75197	APEH	P13798	P13798
APEX1	P27695	Q09472	APH1A	Q96BI3	P06241
APH1B	Q8WW43	P05067	APIP	Q96GX9	P63208
APITD1	Q8N2Z9	P42858	APLF	Q8IW19	P20585
APLN	APLN	P06396	APLP1	P51693	P05067-4
APLP2	Q06481-5	Q13190	APMAP	Q9HDC9	P13569
APOB	P04114	P01130	APOBEC3C	Q9NRW3	P05771
APOBEC3G	Q9HC16	P69723	APOC1	P02654	Q5XKP0
APOD	P05090	Q96A29	APOE	P02649	P02748
APOL2	Q9BQE5	Q96PP8	APOL3	O95236-2	Q16585
APOL6	Q9BWW8	Q9UHA4	APOOL	Q6UXV4	P46059
APP	P05067	Q92664	APPBP2	Q92624	O60942
APPL1	Q9UKG1	O14908	APPL2	Q8NEU8	O14908
APRT	P07741	Q6DKK2	AQP1	P29972	P08151
AQP3	Q92482	P01375	AQP7	O14520	O60240
AQP9	O14520	O60240	AQR	O60306	Q13573
AR	P10275	Q15648	ARAP1	Q96P48	O43639
ARAP3	Q8R5G7	P12931	ARC	O60936	Q14790
ARCN1	P48444	P13569	AREL1	O15033	P46109
ARF1	P84080	Q9NZ52	ARF3	P61204	P05067
ARF4	P18085	Q96PM5	ARF5	P84085	O94955
ARFGAP1	Q9NP61	P17302	ARFGAP2	Q8N6H7	P05155
ARFGAP3	Q9NP61	P17302	ARFGEF1	Q9Y6D6	Q9NRM0

Input	UniProt Id	Interacts with	Input	UniProt Id	Interacts with
ARFGEF2	Q9Y6D5	P05919	ARFGEF3	Q5TH69	P15907
ARFIP1	P53367	O60906	ARFIP2	P53365	Q92730
ARG1	P05089	P03372	ARGLU1	Q9NWB6	Q6NYC1
ARHGAP1	Q07960	P61586	ARHGAP10	Q5T5U3	P29692
ARHGAP12	Q8IWW6	P60709	ARHGAP15	Q53QZ3	P63000
ARHGAP17	Q68EM7	Q4VCS5-2	ARHGAP19	Q14CB8	P61981
ARHGAP24	Q8N264	P21333	ARHGAP26	Q9UNA1	A1A4S6
ARHGAP29	Q52LW3	P32249	ARHGAP31	Q2M1Z3	Q9Y490
ARHGAP32	A7KAX9	P50458	ARHGAP33	O14559	P06241
ARHGAP42	A6N128	P46108	ARHGAP44	Q17R89	P49662
ARHGAP5	Q13017-1	Q8IZD9	ARHGAP9	Q9BRR9	P62993
ARHGDIA	P52565	P61586	ARHGDIB	P52566	P01100
ARHGDIG	Q99819	Q9Y6A5	ARHGEF1	Q92888	P61586
ARHGEF10	O15013	P11142	ARHGEF10L	Q9HCE6	Q8NHY2
ARHGEF11	O15085	O43157	ARHGEF12	Q9NZN5	Q9UM54
ARHGEF16	Q5VV41	P31946, P63104	ARHGEF17	Q96PE2	P31946
ARHGEF26	Q96DR7	P31327	ARHGEF28	Q8N1W1-4	P27797
ARHGEF3	Q9NR81	P14373	ARHGEF4	Q9NR80	P63104
ARHGEF6	Q15052	Q13177	ARHGEF7	O55043, Q14155	Q13177
ARID1B	Q8NFD5	P51531	ARID2	Q68CP9	P51532
ARID3A	Q99856	P23025	ARID3B	Q8IVW6	P48431
ARID4A	P29374	Q13547	ARID4B	Q4LE39	Q92769
ARID5A	Q03989	P40937	ARID5B	Q14865	Q92769
ARIH1	Q9Y4X5	Q9BYZ6	ARL1	P40616	P13569
ARL10	Q8N8L6	P21453	ARL13B	Q3SXY8	P48509
ARL15	Q9NXU5	O43772	ARL16	Q0P5N6	O43924
ARL2	P36404	Q9BTW9	ARL2BP	Q9Y2Y0	P36404
ARL3	P36405	Q14191	ARL4A	P40617	Q08379
ARL4C	P56559	P52306	ARL5B	Q96KC2	Q9Y5M8
ARL6	Q9H0F7	O43290	ARL6IP1	Q15041	Q00059
ARL6IP4	Q66PJ3-3, A0A0C4DG62	Q9H4P4	ARL6IP6	Q8N6S5	O15498
ARL8A	Q96BM9	Q96EN8	ARL8B	Q9NVJ2	P51149
ARMC10	Q8N2F6-2	Q96GM5	ARMC5	Q96C12	Q8IVS8
ARMC6	Q6NXE6	O43819	ARMC7	Q9H6L4	P08670
ARMC8	Q8IUR7	Q99081	ARMCX1	Q9P291	Q92917, Q9UBB9
ARMCX3	Q9UH62	P04004	ARMCX5	Q6P1M9	Q15185
ARNT	P27540	Q16665, Q99814	ARNT2	Q9HBZ2	Q16665
ARNTL	O00327-8	O14977	ARNTL2	Q8WYA1-3	Q99814
ARPC1A	Q92747	Q9UJU6	ARPC1B	O15143	O00401
ARPC2	O15144	Q16236	ARPC3	O15145	P48552
ARPC4	P59998	O00401	ARPC5	O15511	P00533
ARPC5L	Q9BPX5	P13569	ARRB2	P32121	Q13233
ARRDC1	Q8N5I2	Q96PU5	ARRDC2	Q8TBH0	Q9UGI0
ARRDC3	Q96B67	P51668	ARRDC4	Q8NCT1	Q96PU5
ARSA	O43681	P10451	ARSB	P15848	P01185
ARSG	Q96EG1	Q8NBK3	ARSI	Q5FYB1	P11021
ARSJ	Q8N5N6	Q13077	ARSK	Q6UWY0	Q03405
ARV1	Q9H2C2	P01730	ARVCF	O00192	Q96RT1
ASAH1	Q13510, Q13510-2	Q13285	ASAP1	Q9ULH1	P12931

Input	UniProt Id	Interacts with	Input	UniProt Id	Interacts with
ASAP2	O43150	P19174	ASAP3	Q8TDY4	Q86UT5
ASB1	Q9Y576	P01222	ASB12	Q8WXXK4-2	Q96BZ8
ASB13	Q8WXXK3	Q93034	ASB15	Q8WXXK1	P55040
ASB2	Q96Q27-2	Q93034	ASB3	Q9Y575-3	P04792
ASB4	Q9Y574-2	Q96CV9	ASB6	Q9NWX5	O60551
ASB7	Q9H672	Q93034, Q9UBF6	ASB8	Q9H765	Q15369
ASB9	Q96DX5	Q15369, Q15370	ASCC1	Q8N9N2-2	Q13114
ASCC2	Q9H1I8	Q99633	ASCC3	Q8N3C0	P63244
ASF1A	Q9Y294	P49736	ASGR1	P07306	Q96BA8
ASH1L	Q9NR48	Q09028	ASH2L	Q9UBL3	Q09472
ASIC1	P78348	Q9NRD5	ASIC2	Q16515	Q9NRD5
ASIC4	Q96FT7	Q9HD20	ASMTL	O95671	Q96CV9
ASNA1	O43681	P10451	ASNSD1	Q9NWL6	Q14160
ASPA	P45381	O75925	ASPG	Q86U10	P57764
ASPH	Q12797-6	Q9HBE4	ASPHD2	Q6ICH7	O75063
ASPM	Q8IZT6	P11802	ASRGL1	Q7L266	Q13287
ASS1	P00966	Q9Y4Z2	ASTN1	O14525-2	Q14145
ASXL2	Q76L83	O43791	ATAD1	Q8NBU5	P04201
ATAD2	Q6PL18	Q9Y6Q9	ATAD3A	Q9NVI7	Q14197
ATAD3B	Q5T9A4	P21802	ATAD5	Q96QE3	P40937, P35249
ATAT1	Q5SQI0	P31946, P63104	ATCAY	Q86WG3	Q9UNE7
ATE1	O95260	P01185	ATF1	P18846	P18846
ATF2	P15336	P19367	ATF3	P18847	P18847
ATF4	P18848	P18847	ATF5	Q9Y2D1	P49715
ATF6	P18850	Q8NHY2	ATF6B	Q99941	Q8TC07
ATF7IP	Q6VMQ6	Q13485	ATG10	Q9H0Y0	Q9H1Y0, O94817
ATG101	Q9BSB4	O75385	ATG12	O94817	Q9H1Y0, Q9H0Y0, Q9NT62, Q676U5
ATG13	O75143	O75385	ATG14	Q6ZNE5	P41208
ATG16L2	Q8NAA4	Q9H1Y0	ATG2B	Q96BY7	Q13309
ATG4B	Q9Y4P1	Q9GZQ8	ATG4C	Q96DT6	O00401
ATG5	Q9H1Y0	Q9GZQ8	ATG7	O95352	P29692
ATG9A	Q7Z3C6	Q96QE2	ATG9B	Q674R7	Q9Y496
ATIC	P31939	P00533	ATL2	Q8NHH9	P63104
ATL3	Q6DD88	Q92911	ATM	Q13315	Q9NZQ7
ATMIN	O43313	Q13315	ATN1	Q86V38	Q13371
ATOX1	O00244	P35670	ATP11B	A0A0C4DG94	A6N115
ATP13A1	Q9HD20	Q96FT7	ATP13A2	Q9NQ11	Q2M2I8
ATP13A3	Q9H7F0	Q9NRZ7	ATP1A1	P05023	P12931
ATP1A3	P13637	P21453	ATP1A4	Q13733	P20309
ATP1B1	P05026	P21926	ATP2A2	P16615	P0DTC4
ATP2A3	Q93084-2	P37173	ATP2B1	P20020	Q9Y639
ATP2B2	P23634, Q01814	P01258	ATP2B4	P23634	P01258
ATP5A1	P25705	P63104	ATP5B	P06576	Q9H244
ATP5C1	P36542	Q96H96	ATP5D	P30049	P46199
ATP5E	P21306	P00747	ATP5F1	P24539	Q9H244
ATP5G1	P05496	P21217	ATP5G3	P48201	P06576
ATP5H	O75947	P06576	ATP5I	P56385	P06576
ATP5J	P18859	Q9H2H9	ATP5J2	P56134	P01106
ATP5L	O75964	P13569	ATP5O	P48047	P05919

Input	UniProt Id	Interacts with	Input	UniProt Id	Interacts with
ATP6AP1	Q15904	P27449, P61421	ATP6AP2	O75787	Q15904
ATP6V0A1	Q93050	Q15904	ATP6V0A2	Q9Y487	Q15904
ATP6V0B	Q99437	P22732	ATP6V0C	P27449	Q15904
ATP6V0D1	P61421	Q15904	ATP6V0E1	O15342	P05090
ATP6V1A	P38606	Q9UII2	ATP6V1B2	P21281	Q9UII2
ATP6V1D	Q9Y5K8	Q9UII2	ATP6V1E1	P36543	Q9UII2
ATP6V1F	Q16864	Q9UII2	ATP6V1G1	O75348	P0C6X7
ATP6V1G2	O95670	P21281, P36543	ATP6V1H	Q9UII2	P42772
ATP8A1	Q9Y2Q0	Q9NP91	ATP9A	O75110	P03372
ATPAF1	Q5TC12	P06576	ATPAF2	Q8N5M1	P49247
ATPIF1	Q9UII2	Q96QG7	ATR	Q13535	Q9NZQ7
ATRX	P46100	Q9UER7	ATXN1	P54253	Q14012
ATXN10	Q9UBB4	Q16611	ATXN2	Q99700, Q99700-5	P11940
ATXN2L	Q8WWM7	P11940	ATXN3	P54252	Q969S8
ATXN7	O15265	Q8N2S1	ATXN7L1	Q9ULK2-2	P06396
ATXN7L2	Q5T6C5	P43897	ATXN7L3	Q14CW9	O75528, Q92830, Q96ES7, Q92831
AUP1	Q9Y679	P35372	AURKAIP1	Q9NWT8	Q8TB40
AURKB	Q96GD4	P06493	AUTS2	Q8WXX7	Q09472
AVPI1	Q5T686	Q9NPB3	AXL	P30530	P19174
AZGP1	P25311	P04156	B2M	P61769	P59596
B3GALNT1	O75752	Q9BYJ1	B3GALNT2	Q8NCR0	Q9H2A9
B3GALT2	O43825	O94822	B3GALT5	Q9Y2C3	P81534
B3GALT6	Q96L58	P40855	B3GAT3	O94766	O43752
B3GLCT	Q6Y288	P13671	B4GALT3	O60512	P27824
B4GALT4	O60513	P11021	B4GALT5	O43286	P16410
B4GALT6	Q9UBX8	P01185	B4GALT7	Q9UBV7	P53611
B4GAT1	O43505	P21453	B9D2	Q9BPU9	P47897
BABAM1	Q9NWX8	P38398	BACE1	P56817	P10997
BACE2	Q9Y5Z0	P10997	BACH1	P97302, O14867	O15525
BACH2	Q9BYV9	O15525	BAD	Q92934	O43521
BAG1	Q99933	P11142	BAG2	O95816	Q9ULW0
BAG3	O95817	Q9BRX2	BAG4	O95429	P0DTC4
BAG5	Q9UL15	P11142	BAG6	P46379-2	P10451
BAHCC1	Q9P281	Q9H160	BAHD1	Q8TBE0	Q00403
BAIAP2L1	Q9UHR4	Q9UQB8	BAK1	Q16611	P55957
BANF1	O75531	O75531	BANK1	Q8NDB2	P07948
BANP	Q8N9N5-2	Q96E11	BAP1	Q99496	P08183
BARD1	Q99728	Q05048	BASP1	P80723	P10636-8
BATF3	Q9NR55	P18847	BAX	Q07812	P55957
BAZ1A	Q9NRL2	P61981	BAZ2B	Q9UIF8	Q96GX9
BBC3	Q9BXH1	Q07817	BBOX1	O75936	Q9UHD2
BBS10	Q8TAM1	Q8IWX6	BBS2	Q9BXC9	Q9NWB1
BBS9	Q3SYG4	Q8IWX6, Q96RK4, Q8N3I7, Q8NFI9, Q8TAM2, Q9BXC9	BBX	Q8WY36	Q13547
BCAM	P50895	P21926	BCAN	Q96GW7	P05067
BCAP29	Q9UHQ4	P51572	BCAP31	P51572	Q14790
BCAR1	P56945	Q68CZ2	BCAR3	O75815	O95995
BCAS3	Q9H6U6	P08670	BCAS4	Q8TDM0-3	Q9C005
BCAT1	P54687-4	P06396	BCAT2	O15382	O75380

Input	UniProt Id	Interacts with	Input	UniProt Id	Interacts with
BCCIP	Q9P287	P06748	BCDIN3D	Q7Z5W3	Q9NPC8
BCKDHB	P21953	P06746	BCKDK	O14874	O95395
BCL11A	Q9H165-2	O43639	BCL11B	Q9C0K0	P46531
BCL2	P10415	Q9C000	BCL2L1	Q07817	Q9C000
BCL2L11	O43521	Q92934	BCL2L13	Q9BXK5	P78382
BCL3	P20749	O95999	BCL6	P41182	P56524
BCL6B	A8KA13	P41182	BCL7A	Q4VC05	Q9H8M2
BCL7B	Q9BQE9	Q969G3, Q96GM5, Q12824	BCL7C	Q8WUZ0	Q9H8M2
BCOR	Q6W2J9	Q13363	BCORL1	Q5H9F3	Q13363
BCS1L	Q9Y276	O14786	BDH1	Q02338	P46059
BDNF	P23560-2	P18428	BDP1	Q99952	P00533
BEGAIN	Q9BUH8	P28749	BEND3	Q5T5X7	Q13547
BEND5	Q7L4P6	Q96GM5	BEND7	Q8N7W2-2	P11309
BET1	O15155	O95249, Q13190, O15498	BET1L	Q9NYM9	Q9P2W9
BEX1	Q9HBH7	Q9NRD5	BEX2	Q9BXY8	P19883
BEX5	Q5H9J7	P06396	BFAR	Q9NZS9	P51668
BHLHE23	Q8NDY6	P68400, P67870	BHLHE40	O14503	Q16613
BHLHE41	Q99PV5, Q9C0J9	Q16665	BHMT2	Q9H2M3	Q15813
BICD1	Q96G01	P49841	BID	P70444, P55957, P70444-PRO_0000223236	Q07812
BIK	Q13323	O15554	BIN1	O00499	Q13496
BIN3	Q9NQY0	Q9BRT9	BIRC2	Q13490	Q13546, Q9Y572
BIRC3	Q13489	Q13546, Q9Y572	BIRC6	Q9NR09	P53350
BIRC7	Q96CA5-1	P62837	BLCAP	P62952	O95237
BLMH	Q13867	P04271	BLNK	Q8WV28	Q06187
BLOC1S1	P78537	Q9NUP1, Q9UL45, O14964, Q96EV8, Q6QNY0	BLOC1S2	Q6QNY1	A1L4H1
BLOC1S4	Q9NUP1	Q9UL45, Q96EV8, P78537	BLVRA	P53004	Q9H244
BLVRB	P30043	Q8WVK2	BLZF1	Q9H2G9	P36941
BMF	Q96LC9	Q07817	BMI1	P35226	P17096
BMP2K	Q9NSY1	Q8WY64	BMP4	P12644	P27797
BMP5	P22003	P68871	BMP7	P18075	Q9BUH8
BMPR1A	P36894	P12644	BMS1	Q14692	P06748
BMX	P51813	P40763	BNIP1	Q12981	Q9Y5M8
BNIP2	Q12982	Q8TB40	BNIP3L	O60238	Q9NQ11
BOC	Q9BWV1	Q14623	BOD1	Q96IK1-2	P78333
BOK	Q9UMX3	P16284	BOLA2B	Q9H3K6	P19320
BOLA3	Q53S33	Q86SX6	BOP1	Q14137	P06748
BORA	Q6PGQ7	P51530	BPGM	P07738	Q9BXXK1
BPNT1	O95861	Q8NE62	BPTF	Q12830	Q09028
BRAF	P15056	Q12967	BRAT1	Q6PJG6	Q13315
BRCA1	P38398	Q9ULW0, O75330	BRCA2	P51587	Q86YC2
BRD2	P25440	Q13761	BRD3	Q15059	P01106
BRD4	O60885	P0DTC4	BRD8	Q9H0E9	O96019
BRD9	Q9H8M2	Q96GM5	BRE	Q9NXR7	P38398
BRF2	Q9HAW0	P20226	BRICD5	Q6PL45	P98160
BRINP1	O60477	P09651	BRINP2	Q9C0B6	P27824
BRINP3	Q76B58	O75197	BRIP1	Q9BX63	P54278
BRIX1	Q8TDN6	P06748	BRK1	Q8WUW1	P26641

Input	UniProt Id	Interacts with	Input	UniProt Id	Interacts with
BRMS1	Q9HCU9	P56524	BROX	Q5VW32	P23193
BRPF1	P55201	P62805	BRPF3	Q9ULD4-2	P07550
BRSK2	Q8IWQ3	Q92905	BRWD1	Q9NSI6-4	P07550
BSCL2	J3KQ12	O75063	BSG	P35613	P20020
BSPRY	Q6P6S3	P63104	BST2	Q10589	Q10589
BTAF1	O14981	P46059	BTBD2	Q9BX70	Q16401
BTBD6	Q96KE9	Q9UER7	BTBD9	Q96Q07-2	P55212
BTBD	P43251	P01008	BTF3	P20290-2	O60551
BTF3L4	Q96K17	Q9NRG4	BTG1	P62324	Q13158
BTG2	P78543	P10275	BTG3	Q14201	Q01094
BTK	Q06187	Q06187	BTN2A1	Q7KYR7	P03886
BTN2A2	Q8WVV5	P25788	BTN3A1	O00481	P04155
BTN3A2	P78410	Q96CV9	BTNL9	Q6UXG8-3	P01375
BUB1	O43683	P51530	BUB3	O43684	Q12834
BUD13	Q9BRD0	Q9ULW0	BUD31	P41223	O95926
BYSL	Q13895	Q13573	BZW2	Q9Y6E2	Q7Z434
C10orf107	Q8IVU9	P42858	C10orf12	Q96JN0, Q96JN0-2	Q13363
C10orf2	Q96RR1	Q86VM9	C10orf35	Q96D05	O43914
C10orf54	Q9H7M9	P27449	C11orf1	Q9H5F2	P31273
C11orf24	Q96F05	Q96BA8	C11orf30	Q7Z589-5	P35638
C11orf31	Q8IZQ5	Q9Y4X5	C11orf54	Q9H0W9-4	P04792
C11orf57	Q6ZUT1	Q9NRZ7	C11orf65	Q8NCR3	P12830
C11orf68	Q9H3H3	Q15056	C11orf84	Q9BUA3	O75928-2
C11orf87	Q6NUJ2	Q7Z3S9	C12orf10	Q9HB07	Q9BRX2
C12orf45	Q8N5I9	Q9Y2X3	C12orf49	Q9H741	P40855
C12orf57	Q99622	P31946	C14orf1	Q9UKR5	O00501
C14orf119	Q9NWQ9	P61587	C14orf142	Q9BXV9	Q96S44
C14orf159	Q7Z3D6	P25788	C14orf166	Q9Y224	P13569
C14orf2	P56378-2	P02654	C14orf37	Q86TY3	Q9NPF0
C15orf26	Q6P656	P41182	C15orf27	Q2M3C6	O60341
C15orf39	Q6ZRI6	Q13363	C15orf41	Q9Y2V0	Q9Y294
C15orf57	Q9BV29	Q9NUX5	C15orf59	Q2T9L4	Q96EZ8
C16orf45	Q96MC5	Q8WZA2	C16orf58	Q96GQ5	Q6ZPD8
C16orf59	Q7L2K0	Q7Z3S9	C17orf49	Q8IXM2	O76024
C17orf59	Q96GS4	Q96NL6	C17orf62	Q9BQA9	Q9H2K0
C17orf80	Q9BSJ5	Q8N2W9	C17orf85	Q53F19	Q86U42
C17orf89	A1L188	Q5TEU4	C17orf96	A6NHQ4	Q15022
C17orf97	Q6ZQX7-4	P08684	C18orf25	Q96B23	P63165
C18orf54	Q8IYD9	Q53GS7	C18orf8	Q96DM3	P07196
C19orf25	Q9UFG5	Q9H244	C19orf38	A8MVS5	O60443
C19orf43	Q9BQ61	O75533	C19orf47	Q8N9M1-2	P60900
C19orf52	Q9BSF4	P16050	C19orf53	Q9UNZ5	P10636-8
C19orf54	Q5BKX5-3	Q9NQ94	C19orf60	Q96EN9	P51665, O00487
C19orf66	Q9NUL5-3	P04792	C19orf68	Q86XI8	Q13185
C19orf70	Q5XKP0	P55056, P02654	C1GALT1	Q9NS00	P26842
C1QB	P02746	Q70CQ3	C1QBP	Q07021	Q8N726
C1QC	P02747	P02745	C1QL4	Q86Z23	Q9UBD6
C1QTNF1	Q9BXJ1	P02751	C1QTNF2	Q9BXJ5	P33992
C1R	P00736	P05155	C1S	P09871	P05155
C1orf109	Q9NX04	Q9H4P4	C1orf112	Q9NSG2	Q6PIW4

Input	UniProt Id	Interacts with	Input	UniProt Id	Interacts with
C1orf122	Q6ZSJ8	Q16576	C1orf198	Q9H425	Q9Y4D1
C1orf216	Q8TAB5	P17302	C1orf226	A1L170	P53350
C1orf27	Q5SWX8	Q96FT7	C1orf35	Q9BU76	Q96CW1
C1orf50	Q9BV19	P23526	C1orf52	Q8N6N3	Q13257
C1orf54	Q8WWF1	Q14703	C1orf74	Q96LT6	Q9GZV5
C1orf94	Q6P1W5	Q9NWB1	C2	P06681	Q15125
C20orf195	Q9BVV2	P38936	C20orf196	Q8IYI0	P04792
C20orf24	Q9BUV8	O60704	C20orf27	Q9GZN8	P62136
C21orf2	O43822	O00244	C21orf59	P57076	O00628
C22orf39	Q6P5X5	Q9NPB3	C2CD2L	O14523	P23945
C2CD5	Q86YS7	P80370	C2orf15	Q8WU43	Q15109
C2orf40	Q9H1Z8	O76024	C2orf42	Q9NWW7	P13196
C2orf44	Q9H6R7	P53350	C2orf68	Q2NKX9	Q86US8
C2orf72	A6NCS6	O15431	C3	P01024	Q16659
C3AR1	Q16581	Q9UBV7	C3orf14	Q9HBI5	Q96ST8
C3orf33	Q6P1S2	P40855	C3orf38	Q5JPI3	P11142
C3orf52	Q5BVD1	Q15109	C3orf62	Q6ZUJ4	Q09472
C4A	P0C0L4	P46059	C4orf22	Q6V702	Q16236
C4orf29	Q0P651	Q15116	C4orf3	Q8WVX3-2	Q9UBD6
C4orf32	Q8N8J7	Q96FE5	C4orf45	Q96LM5	P41182
C4orf46	Q504U0	Q9NRD5	C5	P01031	P13671, P02748, P10643
C5AR1	P21730	P18031	C5AR2	Q9P296	P09601
C5orf22	Q49AR2	P13725	C5orf28	Q0VDI3	Q5QGT7
C5orf45	Q6NTE8	P20618	C5orf51	A6NDU8	Q6PJG9
C5orf64	Q2M2E5	P16104	C6	P13671	Q6Y288
C6orf118	Q5T5N4	Q96ES7	C6orf120	Q7Z4R8	Q9Y5M8
C6orf226	Q5I0X4	O00628, P09110, O14832	C6orf47	O95873	P29371
C7	P10643	P01031	C7orf43	Q8WVR3	P22735
C7orf50	Q9BRJ6	P06748	C7orf55	Q96HJ9	Q9NUJ1
C7orf60	Q1RMZ1	Q9UJK0	C8orf33	Q9H7E9	P78563
C8orf46	Q8TAG6	Q9NPB3	C8orf48	Q96LL4	Q96CV9
C9orf106	Q8NAJ2	P25788, Q9BT73	C9orf114	Q5T280	P0DTC9
C9orf142	Q9BUH6	Q14191	C9orf16	Q9BUW7	Q70UQ0
C9orf24	Q8NCR6	Q9NPB3	C9orf78	Q9NZ63	O75152
C9orf85	Q96MD7	P46736, Q9NWW8	C9orf89	Q96LW7	O95999
C9orf9	Q96E40	O00471	CA10	Q9NS85	O95456, Q969U7, Q5JS54, Q9BT73
CA11	Q9NS71	Q9Y6X1	CA14	Q9ULX7	Q96BI1
CA5B	Q9Y2D0	P11182	CA8	P35219	P02768
CAB39	Q9Y376	Q15831	CAB39L	Q9H9S4	Q15831
CABIN1	Q9Y6J0	Q96JC9	CABLES1	Q8TDN4	P04637
CABLES2	Q9BTV7	P80370	CABYR	O75952-5, O75952-3	P49841
CACFD1	Q9UGQ2	Q96T60	CACNA1A	O00555	Q8N2S1
CACNA1C	Q13936	Q9NZU7	CACNB1	Q02641	P55040
CACNB2	Q08289	O14744	CACNB3	P54287, P54284	P55040
CACNB4	O00305-2	P21579	CACNG3	O60359	P53350
CACNG5	Q9UF02	Q8N9R8	CACTIN	Q8WUQ7	Q14562
CACUL1	Q86Y37	P03372	CACYBP	Q9HB71	Q13451
CAD	O76075	O00273	CADM1	Q9BY67	P29074
CADM4	Q8NFFZ8	Q9UMR7	CADPS	A2RRN7	Q96CV9

Input	UniProt Id	Interacts with	Input	UniProt Id	Interacts with
CALB1	P05937	Q9Y6K9	CALCA	P01258	P01258
CALCOCO2	Q13137	P30626	CALD1	Q05682	Q13813
CALHM1	Q8IU99	P05067	CALM1	P62161	P52888
CALM2	P62161	P52888	CALM3	P62161	P52888
CALR	P27797	Q9ULW0	CALU	O43852	Q05513
CAMK1D	Q8IU85	Q14012	CAMK1G	Q96NX5	O76024
CAMK2A	Q9UQM7	Q13224	CAMK2B	Q13554-3	P26640
CAMK2D	Q13557	Q96PM5	CAMKK2	Q96RR4	Q16539
CAMKMT	Q7Z624	P08123	CAMKV	Q8NCB2	P98160
CAMLG	P49069	Q8TC07	CAMSAP1	Q5T5Y3	P63104
CAMSAP2	Q08AD1	P29692	CAMSAP3	Q9P1Y5	P31946
CAMTA2	O94983-5	Q9Y316	CANT1	Q8WVQ1	Q96JA1
CAP1	Q01518	P60709	CAPG	P40121	P11142
CAPN1	P07384	Q92934	CAPN10	Q9HC96	P10636
CAPN13	Q6MZZ7	Q9UPT8	CAPN2	P17655	P07237
CAPN6	Q9Y6Q1	Q6PJG9	CAPN7	Q9Y6W3	Q9HBZ2
CAPNS1	P04632	P01375	CAPRIN1	Q14444	P62280
CAPRIN2	Q6IMN6	O75197	CAPS	Q9ULU8-4	O00560
CAPZA2	P47755	Q14203	CAPZB	P47756	P09651
CARD11	Q9BXL7	Q9UDY8	CARD8	Q9Y2G2	Q9Y2G2
CARD9	Q9H257	O95999	CARF	Q9NXV6	O94913, Q9H0D6
CARHSP1	Q9Y2V2	Q13077	CARKD	Q8IW45	Q7Z3S9
CARM1	Q86X55	P17676	CARNMT1	Q8N4J0	Q13547
CARS	P49589	P26641	CARTPT	Q16568	P13688
CASC3	O15234	Q92900	CASK	O14936	Q13009
CASKIN1	Q8VHK2, Q8WXD9	O14936	CASKIN2	Q8WXE0	O43639
CASP1	P29466	Q9C000	CASP10	Q92851	Q14790
CASP12	Q920D5	O95786	CASP3	P42574	O60551
CASP4	P49662	P51878	CASP7	P55210	Q13177
CASP8	Q14790-2, Q14790	O15519-1	CASP8AP2	Q9UKL3	Q14790
CASQ2	O14958	P62736	CASS4	Q9NQ75-2	Q08379
CAST	O15446	O15160, O95602, P19388, P61218	CAT	P04040	Q06124
CAV1	P33724	P59632	CAV2	P51636	Q03135
CBARP	Q8N350-4	O43639	CBFA2T2	O43439	P17987
CBFA2T3	O75081	O43439	CBFB	Q13951	Q13761
CBLB	Q13191	Q13882	CBLN4	Q9NTU7	Q86XF0
CBR3	O75828	O00560	CBS	P35520	P25786
CBX1	P83916	Q8N9R8	CBX2	Q14781-2	Q00403
CBX3	Q13185	Q9UI95	CBX4	O00257	Q16665
CBX5	P45973	Q14191	CBX7	Q8VDS3	Q9NZI8
CBX8	Q9HC52	P11686	CBY1	Q9Y3M2	P63104
CC2D1A	Q6P1N0	Q9H444	CCAR2	Q8N163	P01106
CCDC102A	Q96A19	O15392	CCDC102B	Q68D86	P01189
CCDC106	Q9BWC9	P40429	CCDC107	Q8WV48	P01375
CCDC112	Q8NEF3-2	Q9Y2T1	CCDC113	Q9H0I3	P36941
CCDC115	Q96NT0	O15342	CCDC120	Q96HB5	P21802
CCDC124	Q96CT7	P06748	CCDC127	Q96BQ5	P41208
CCDC134	Q9H6E4	O75478	CCDC136	Q96JN2-2	P56524
CCDC137	Q6PK04	P78563	CCDC138	Q96M89-2	Q16512

Input	UniProt Id	Interacts with	Input	UniProt Id	Interacts with
CCDC14	Q49A88-3	P06396	CCDC141	Q6ZP82-1	P06396
CCDC146	Q8IYE0, Q8IYE0-2	O75496	CCDC148	Q8NFR7	P21673
CCDC150	Q8NCX0-3	P28074	CCDC151	A5D8V7	P06396
CCDC152	Q4G0S7	O95363	CCDC155	Q8N6L0	P16234
CCDC158	Q5M9N0-2	P07550	CCDC167	Q9P0B6	P25103
CCDC172	P0C7W6	Q9H1K1	CCDC183	G5E9W6	Q969G3
CCDC184	Q52MB2	Q13114	CCDC22	O60826	Q13614
CCDC24	Q8N4L8	Q9ULX6	CCDC26	CCDC26	P07237
CCDC28A	Q7Z6N9	P24592	CCDC28B	Q9BUN5-3	P62993
CCDC33	Q8N5R6	P84022	CCDC34	Q96HJ3-2	P61981
CCDC36	Q8IYA8	O94955	CCDC40	Q4G0X9	Q9HBG6
CCDC43	Q96MW1, Q86WV7	Q96T60	CCDC51	Q96ER9	Q8WVK2
CCDC57	Q2TAC2-2	P04732	CCDC59	Q9P031	P98164
CCDC6	Q16204	Q13164	CCDC68	Q9H2F9	Q9Y3P9
CCDC71	Q8IV32	Q13547	CCDC74B	Q96LY2-2	Q5T8I9
CCDC8	Q9H0W5	Q16539	CCDC84	Q86UT8	Q14691
CCDC85B	Q15834	Q9NX63	CCDC85C	A6NKD9	P01730
CCDC86	Q9H6F5	P06748	CCDC88B	A6NC98	P20700
CCDC88C	CCDC88C	O76024	CCDC89	Q8N998	P43897
CCDC90B	Q9GZT6	P61981	CCDC92	Q53HC0	Q9UPV0
CCDC94	Q9BW85	P52747	CCDC97	Q96F63	P18846
CCK	P06307	Q8WUX9	CCL13	Q99616	P48061
CCL2	CCL2	Q04206	CCL26	Q9Y258	P48061
CCL28	Q9NRJ3	Q9UNE7	CCL3	P10147	P10147
CCL4	P13236	Q3SXY8	CCL4L2	Q8NHW4	Q3SXY8
CCL5	P13501	P13500	CCNA1	P78396	P38936
CCNA2	P20248	Q13415	CCNB1IP1	Q9NPC3	Q93034
CCND1	P24385	Q13761	CCND2	P30279	P11802
CCND3	P30281	P06493	CCNDBP1	O95273	Q96T88
CCNE1	P24864	P61024	CCNF	P41002	O43303
CCNH	P51946	Q15669	CCNI	Q14094	Q00535
CCNJ	Q5T5M9	P24941	CCNJL	Q8IV13	O00443
CCNL1	P49736	P03433	CCNL2	Q96S94	P68400
CCNT2	O60583	Q9UHB7	CCNY	Q8ND76	P31946
CCR1	P32246	P04156	CCR10	O00590	Q9UKG4
CCR5	P51681	P01730	CCSER2	Q9H7U1	P61981
CCT2	P78371	Q13371	CCT3	P49368	Q13371
CCT4	P50991	P06493	CCT5	P48643	Q13371
CCT6A	P40227	Q8NCC3	CCT7	Q99832	Q13371
CCT8	P50990	Q13882	CCZ1	P86791	P51149
CD109	Q6YHK3	P23946	CD14	EBI-11177705, EBI-11177518	O43474
CD164	Q04900	P61073	CD2	P08921	P06239
CD200	P41217	Q8TD46	CD200R1	Q8TD46-4	P11686
CD209	Q9NNX6-10	P54652	CD22	P20273	P19174
CD226	Q15762	P21359	CD24	P25063	P07948
CD244	Q9BZW8	Q06124	CD247	P20963	P43403, Q9Y2R2
CD248	Q9HCU0	P08887	CD274	Q9NZQ7	Q13315
CD28	P10747	P27986	CD2AP	Q9Y5K6	Q8WUM4
CD2BP2	O95400	Q92530	CD300A	Q9UGN4	A8K4G0

Input	UniProt Id	Interacts with	Input	UniProt Id	Interacts with
CD300C	Q08708	A8K4G0	CD300LB	A8K4G0	Q496F6, A8K4G0
CD300LF	Q8TDQ1	A8K4G0	CD302	Q8IX05	Q3SXY8
CD320	Q9NPF0	P20062	CD33	P20138	Q06124
CD36	P16671	Q9UKG9	CD37	P11049	P08034
CD3E	P07766	P43403	CD3EAP	O15446	O15160, O95602, P19388, P61218
CD4	P01730	P01730	CD40	P25942	O15397
CD44	P16070	P10451	CD46	P15529	P78504
CD47	Q08722	P78324, Q9P1W8, O43914	CD48	P09326	Q9BZW8
CD5	P06127	P27986	CD52	Q6IBD0	Q9BRI3
CD53	P19397	P78382	CD55	P08174	P48960
CD58	P19256	P17931	CD59	P13987	P13569
CD6	P30203	P17931	CD63	P08962	P01033
CD68	P34810	Q15058	CD69	Q07108	P21453
CD72	P21854	Q96BA8	CD74	P04233	P61073
CD79B	P40259	P15056	CD80	P33681	P08138
CD82	P27701	P01033	CD83	Q01151	P40818
CD84	Q9UIB8	Q9UIB8	CD86	P42081	P16410
CD8A	P01732	P06493	CD9	P21926	P48509
CD93	Q9NPY3	Q9BQ95	CD96	P40200	P15151
CDA	P32320	Q96QG7	CDC14A	Q9UNH5	P61981
CDC14B	O60729	P43246	CDC25A	P30304	O96017
CDC26	Q99JP4, Q8NHZ8	P51955	CDC27	P30260	P51955
CDC34	P49427	Q13616	CDC37	Q16543	Q06187
CDC40	O60508	O95926, Q14562	CDC42	P60953	P52565
CDC42BPA	Q5VT25	P63104	CDC42BPP	Q9Y5S2	Q07157
CDC42EP1	Q00587-2	P21673	CDC42EP2	O14613	Q9NRD5
CDC42EP3	Q9UKI2	Q5S007	CDC42EP4	Q9H3Q1	Q16181
CDC42SE2	Q9NRR3	P60953	CDC7	O00311	P14373
CDC73	Q6P1J9	O00267	CDCA3	Q99618	Q13077
CDCA4	Q9BXL8	Q96LA8	CDCA7L	Q96GN5	P46087
CDCP1	Q9H5V8	P12931	CDH1	P12830	Q68CZ2
CDH10	Q9Y6N8	P35222	CDH13	P55290	P05556
CDH2	P19022	O60716	CDH3	P22223	P35222
CDH5	P33151	Q14118	CDH8	P55286	P35221, O60716
CDHR3	Q6ZTQ4	Q969S2	CDIP1	Q9H305	P0CG47, P0CG48
CDIPT	O14735	Q8N9R8	CDK1	P06493	Q99741
CDK10	Q15131	P11142	CDK11B	P21127	Q13451
CDK12	Q9NYV4	P61981	CDK13	Q14004	P11802
CDK14	O94921	P38936	CDK15	Q96Q40	Q13451
CDK16	Q00536	P31946, P63104	CDK17	Q00537	P06493
CDK18	Q07002	P31946, P63104	CDK19	Q9BWU1	Q15648
CDK2	P24941	Q09472	CDK20	Q8IZL9	Q9UNE7
CDK2AP2	O75956	Q92769	CDK4	P11802	P42772
CDK5	Q03114	Q16620	CDK5R1	Q15078	Q9ULW0
CDK5RAP2	Q96SN8	P17612	CDK5RAP3	Q96JB5	Q9NZQ7
CDK6	Q00534	P42772	CDK8	P49336	Q15648
CDK9	P50750	P04608	CDKAL1	Q5VV42	P06239
CDKL1	Q00532	Q96EB6	CDKL5	O76039	Q08379
CDKN1A	P38936	Q99741	CDKN1B	P46527	P06493

Input	UniProt Id	Interacts with	Input	UniProt Id	Interacts with
CDKN1C	P49918	P11802	CDKN2A	Q8N726	P08047, P04198
CDKN2AIP	Q9NXV6	O94913, Q9H0D6	CDKN2AIPNL	Q96HQ2	Q9H0D6
CDKN2B	P42772	Q9UI12	CDKN2C	P42773	P01185
CDKN2D	P55273	P22736	CDON	O35158	Q13635
CDPF1	Q6NVV7	Q15915	CDR2	Q01850	P51530
CDR2L	Q86X02	Q969Z0	CDS1	O96017	O96017
CDYL	Q9Y232	Q92769	CDYL2	Q8N8U2	O14744
CEACAM1	P13688	P13688	CEACAM8	P31997	P13688
CEBPA	P49715	P18848	CEBPB	P17676	P17096
CEBPD	P49716	Q9BXW9	CEBPG	P53567	P18847
CEBPZ	Q03701	P49841	CECR5	Q9BXW7	P10809
CELF5	Q8N6W0	O15234	CELSR2	Q9HCU4	Q92838
CEND1	Q8N111	P00533	CENPB	P07199	P67870
CENPC	Q03188	P06748	CENPF	P49454	Q96QG7
CENPJ	Q9HC77	P31946	CENPL	P38265	P14136
CENPO	Q9BU64	Q13352	CENPP	Q6IPU0	P49247
CENPQ	Q7L2Z9	P15336	CENPT	Q96BT3	Q13352
CENPV	Q7Z7K6	P61981	CEP112	Q8N8E3	P31946, P63104
CEP126	Q9P2H0	Q99728	CEP131	Q9UPN4	P31946
CEP135	Q66GS9	Q07157	CEP152	O94986	Q13568
CEP162	Q5TB80	Q9NPJ6	CEP164	Q9UPV0	Q6IQ55
CEP170B	Q9Y4F5	P31946, P63104	CEP19	Q96LK0	Q9H2G4
CEP192	Q8TEP8	O14965	CEP290	O15078	Q8IW35, O43303
CEP295	Q9C0D2	Q9NV70	CEP350	Q5VT06	P35221
CEP44	Q9C0F1	O43242	CEP57L1	Q8IYX8-2	Q13895
CEP63	Q96MT8-3, Q96MT8	Q8TC07	CEP70	Q8NHQ1	P43897
CEP83	Q9Y592-2	P18848	CEP85	Q6P2H3-3	O00560
CEP85L	Q5SZL2-5	O00233	CEP89	Q96ST8	P31946
CEP95	Q96GE4	P36941	CEP97	Q8IW35	O43303
CERCAM	Q5T4B2	Q9BZL1	CERK	Q8TCT0	O75496
CERKL	Q49MI3	P35813	CERS2	Q96G23	O15245
CERS5	Q8N5B7	Q9H244	CERS6	Q6ZMG9	Q9H244
CES1	P02795	P09429	CES2	P23141-3	O95231
GETN2	Q9R1K9, P41208	Q01831	CFAP20	Q9Y6A4	Q9Y2Y0
CFAP36	Q96G28	P36404	CFAP46	Q8IYW2	Q9UJK0
CFH	P08603	P01024	CFI	P05156- PRO_0000027569, P05156- PRO_0000027570, P05156	P08603
CFL1	P23528	P53667	CFL2	Q549N0, Q9Y281	P23528
CFLAR	O15519, O15519-1	Q14790	CGNL1	Q0VF96	P31946
CHAF1A	Q13111	Q9UIS9	CHAMP1	Q96JM3	Q9UI95
CHAT	P28329-3	Q8WUX9	CHCHD1	Q96BP2	P04618
CHCHD10	Q8WYQ3	Q9BRQ6, Q16891	CHCHD2	Q9Y6H1	P07237
CHCHD3	Q9NX63	Q96QF0	CHCHD4	Q8N4Q1	O43768
CHCHD5	Q9BSY4	P53701	CHD1	O14646-2	O76024
CHD1L	Q86WJ1	Q9NTX7	CHD3	Q12873	Q99728
CHD4	Q14839	P01106	CHD5	Q8TDI0	Q13547
CHD8	Q9HCK8	P06748	CHD9	Q3L8U1-3	Q07869

Input	UniProt Id	Interacts with	Input	UniProt Id	Interacts with
CHEK1	O14757	Q06609	CHEK2	O96017	O96017
CHERP	Q8IWX8	P21802	CHFR	Q96EP1-2, Q96EP1	P51668
CHGA	P26339	P00441	CHGB	P16014	P00441
CHI3L1	P36222	P17931	CHIC2	Q9UKJ5	O96015
CHID1	Q9BWS9	Q15848	CHL1	O00533	O00533
CHM	P24386	Q92696	CHMP1A	Q9HD42	P40818
CHMP2A	O43633	Q9Y3E7	CHMP2B	Q9UQN3	Q9Y3E7
CHMP4B	Q9H444	Q96FZ7, Q8WUX9, Q9Y3E7	CHMP5	Q9NZZ3	Q9Y3E7
CHMP6	Q96FZ7	Q9H444	CHMP7	Q8WUX9	P06396
CHN1	P15882	O43639	CHN2	P52757	O43639
CHORDC1	Q9UHD1	P53041	CHPF	Q8IZ52-2	O60260
CHPF2	Q9P2E5	P0DTC8	CHPT1	Q8WUD6	Q9Y2B1
CHRD	Q9H2X0	P84022	CHRNA3	P32297	Q8NC56
CHRNA4	P43681	Q16585	CHRNA5	P30532	P59665
CHRNA7	P36544	Q13573	CHRNA9	Q9UGM1	O14757
CHRNB1	P11230	Q5T9L3	CHRNB3	Q05901	Q8NC56
CHST14	Q8NCH0	P27824	CHST3	Q7LGC8	P01185
CHST7	Q9NS84	P40855	CHST8	Q9H2A9	Q9Y3A6
CHST9	Q7L1S5	O75197	CHSY1	Q86X52	Q03405
CHSY3	Q70JA7	Q03405	CHTF18	Q8WVB6	P40937, P35249
CHTF8	P0CG13	P40937, Q9BVC3, P35249	CHTOP	Q9Y3Y2	P13569
CIAO1	O76071	P51530	CIAPIN1	Q6FI81	Q9H3K6
CIART	Q8N365	O75553	CIC	Q96RK0	P31946, P63104
CIDEB	Q9UHD4	P23434	CIITA	P33076	Q96EB6
CINP	Q9BW66	Q9H3L0	CIR1	Q86X95	Q07955, Q01130
CIRBP	Q14011	Q92973	CIRH1A	Q969X6	P50552
CISD2	Q8N5K1	P36941	CIT	O14578	O14907
CITED1	Q99966	P05412	CITED2	Q99967	Q09472
CIZ1	Q9ULV3	Q9P0V3	CKAP4	Q07065	O75326
CKAP5	Q14008	Q13573	CKM	P06732	P28065
CKMT1A	P12532	P17980	CKMT1B	P12532	P17980
CKS1B	P61024	P06493	CLASP1	Q7Z460	P31946, P63104
CLCN4	P51793	Q8WTS1	CLDN1	Q9NY35	Q9ULC8
CLDN11	O75508	P26715	CLDN15	Q96FX9	Q8IY26
CLDN19	Q8N6F1-2	Q8N6U8	CLDN5	O00501	Q9Y5Q5
CLDN6	P56747	P45983	CLDN9	O95484	Q92935
CLEC10A	Q8IUN9	P01375	CLEC11A	Q9Y240	Q14703
CLEC14A	Q86T13	Q01453	CLEC16A	Q2KHT3	Q9H4P4
CLEC1A	Q8NC01	Q16623	CLEC2B	Q92478	Q86TM6
CLEC2D	Q9UHP7-3	P27449	CLEC4A	Q9UMR7	P13591
CLEC5A	Q9NY25	Q9BRQ6	CLEC7A	Q9BXN2	Q01638
CLGN	O14967	Q15904	CLHC1	Q8NHS4	Q96ES7
CLIC1	O00299	O75832	CLIC3	O95833	Q9NUX5
CLIC4	Q9Y696	O15247	CLINT1	Q14677	O00443
CLIP2	Q9UDT6	P10636-8	CLIP3	Q96DZ5	P04792
CLIP4	Q8N3C7	Q9NPC8	CLK4	Q9HAZ1	P52294
CLMP	Q9H6B4	Q15058	CLOCK	O15516	O00327
CLP1	Q92989	Q9UPV0	CLPB	Q9H078	P46199
CLPP	Q16740	P20618	CLPTM1	O96005	P31213

Input	UniProt Id	Interacts with	Input	UniProt Id	Interacts with
CLPTM1L	Q96KA5	Q9H1C4	CLSTN1	O94985-2	Q13573
CLSTN3	Q8IUW6	P02649	CLTA	P09496	Q9Y6A5
CLTB	P09497	O00443, P42566	CLTCL1	P53675	P13569
CLU	P10909	P30101	CLUH	O75153	Q96PE7
CLUHP3	Q96NS8	P06396	CMAS	Q8NFW8	P48059
CMKLR1	Q99788	Q9BUV8	CMSS1	Q9BQ75	P59595
CMTM3	Q96MX0	Q3SXY8	CMTM5	Q96DZ9-2	Q96E11
CMTM6	Q9NX76	Q9NZQ7	CMTM7	Q96FZ5	Q15109
CMTM8	Q8IZV2	P00533	CMYA5	Q8N3K9	O75923
CNDP1	Q96KN2	Q5SRI9	CNDP2	Q96KP4	P10636-8
CNGA3	Q16281	Q86XL3	CNIH1	O95406	Q6IN84
CNIH2	Q6PI25	Q96BA8	CNIH3	Q8TBE1	O15554
CNKSR1	Q969H4	P51587	CNKSR2	Q8WXI2	P00533
CNKSR3	Q6P9H4	P42772	CNN1	P51911	Q8N0S2
CNN2	Q99439	P11142	CNN3	Q15417	P11142
CNNM1	Q9NRU3	Q12974	CNNM2	Q9H8M5	Q12974
CNOT11	Q9UKZ1	Q9BXI6	CNOT2	Q9NZN8	P28070
CNOT3	O75175	P27797	CNOT4	O95628-2	Q9UBH0
CNOT6L	Q96LI5	P47756	CNOT7	Q60809	P06730
CNOT8	Q9UFF9	P62324	CNP	P09543	Q12824
CNPPD1	Q9BV87	Q00535	CNPY2	Q9Y2B0	Q8WY64
CNPY3	Q9BT09	Q96JA1	CNPY4	Q8N129	Q9BRQ8
CNR1	P21554	P14416	CNRIP1	Q96F85	O95070
CNST	Q6PJW8	Q9NUX5	CNTF	P26441	P42702
CNTFR	P26992	P26441, O75462	CNTN2	Q02246	P00533
CNTNAP1	P78357	P05919	CNTNAP2	Q9UHC6	Q12860
CNTNAP3	Q9BZ76	P01222	CNTNAP4	Q9C0A0-1	Q13009
CNTROB	Q8N137	Q2TBE0	COA3	Q9Y2R0	P00395
COA6	Q5JTJ3	Q9NPB3	COA7	Q96BR5	P29279
COASY	Q13057	P13569	COBL	O75128	P13196
COBLL1	Q53SF7	P31946	COCH	O43405	P08185
COG1	Q8WTW3	Q9NZQ7	COG2	Q14746	Q9NZQ7
COG3	Q96JB2	Q9NZQ7	COG4	Q9H9E3	Q9Y639
COG5	Q9UP83	Q9Y639	COG6	Q9Y2V7	O95999
COG7	P83436	Q9NZQ7	COG8	Q96MW5	Q9NZQ7
COIL	P38432	Q96HC4	COL12A1	Q99715	Q6ISS4
COL13A1	Q5TAT6	Q6ISS4	COL14A1	Q05707	P32119
COL17A1	Q9UMD9	Q9Y3C6	COL18A1	P39060- PRO_0000005794	P19338
COL1A1	P02452	P30101	COL1A2	P08123	Q7Z624
COL25A1	Q8NE08	P50454	COL26A1	Q96A83-2	Q8WVX9
COL3A1	P02461-1	P09486	COL4A1	P02462	P14780
COL4A2	P08572	P14780	COL4A3BP	Q9Y5P4	O95197
COL4A5	P29400-2	Q8TB40	COL4A6	Q14031	P05067
COL5A1	P20908	P14780	COL5A2	P05997	Q6ISS4
COL6A1	P12109	P14780	COL6A2	P12110	P14780
COL6A3	P12111	Q6ISS4	COL8A1	P27658	O43559
COL8A2	Q4VAQ0	Q00059	COL9A3	Q14050	P00533
COLEC10	Q9Y6Z7	Q9UBS3	COLEC12	Q5KU26	P17931
COLGALT1	Q8NBJS	P02452	COLGALT2	Q8IYK4	P02452

Input	UniProt Id	Interacts with	Input	UniProt Id	Interacts with
COLQ	Q9Y215	P22303	COMMD1	Q8N668	P35670
COMMD10	Q9Y6G5	P01100	COMMD2	Q86X83	Q96PC5
COMMD3	Q9UBI1	Q96PC5	COMMD4	Q9H0A8	Q14186, O75461
COMMD6	Q7Z4G1	Q14186, O75461	COMP	P49747	O00244
COMT	P21964	P30518	COMTD1	Q86VU5	Q14012
COPB1	P53618	P13569	COPB2	P35606	P19838
COPE	O14579	P13569	COPG2	Q9UBF2	P13569
COPRS	Q9NQ92	P15880, Q9BRS2	COPS2	P61201	Q09472
COPS3	Q9UNS2	Q16143	COPS4	Q9BT78	P61081
COPS5	Q92905	Q15843	COPS6	Q7L5N1	O94844
COPS7A	Q9UBW8	O94844	COPS7B	Q9H9Q2	P61081
COPS8	Q99627	Q15831	COQ5	Q5HYK3	Q13304
COQ9	O75208	Q14162	CORIN	Q9Y5Q5	O00501
CORO1A	P31146	P26640	CORO1B	Q9BR76	P13569
CORO1C	Q9ULV4	P13569	CORO2A	Q92828	Q8N695
CORO6	Q6QEF8	Q9UL45	COTL1	Q14019	P09917
COX20	Q5RI15	Q9BZP6	COX4I1	P13073	P00395, P00403
COX5A	P20674	P00403	COX5B	P10606	P15954
COX6A1	P12074	P42858	COX6B1	P14854	P00403
COX6C	P09669	P00403	COX7A2	P14406	P00403
COX7A2L	O14548	O14949, P14927	COX7C	P15954	P00403
COX8A	P10176	P06748	CPA3	Q9UI42	P01160
CPA6	Q8N4T0	Q9UPY3	CPD	O75976	P0DTC3
CPEB1	Q9BZB8	Q92797	CPLX1	O14810	P63027
CPLX2	Q6PUV4	P43897	CPNE1	Q99829	P13569
CPNE2	Q96FN4	Q8TBC4	CPNE5	Q9HCH3	O60216
CPNE7	Q9UBL6-2	Q9UGI0	CPNE8	Q86YQ8-2	P04271
CPS1	P31327	P63104	CPSF1	Q10570	O95639
CPSF2	Q9P2I0	Q92797, Q9C0J8, Q6UN15	CPSF3	Q9UKF6	O95639
CPSF3L	Q5TA45	O95947	CPSF4	O95639-3, O95639	Q92797, Q9C0J8, Q6UN15, Q10570
CPSF6	Q16630	O95639	CPT2	P23786	Q9POJ0
CPVL	Q9H3G5	P47756	CRABP1	P29762	P13569
CRABP2	P29373	P30281	CRACR2A	Q9BSW2	P35638
CRACR2B	Q8N4Y2-3	Q9UBH0	CRADD	P78560	Q9Y5W5
CRAMP1L	Q96RY5	Q9Y4X5	CRB1	P82279	Q8N3R9
CRBN	Q96SW2	P62877	CRCP	O75575	Q99081
CREB1	P16220	P18846	CREB3	O43889	O15554
CREB3L2	Q70SY1	Q96BA8	CREB5	Q02930-3	P19883
CREBBP	Q92793	Q13568	CREBL2	O60519	P35638
CREG1	O75629	O96006	CRELD1	Q96HD1-2	P19404
CRELD2	Q6UXH1-2	P12830	CREM	EBI-10048213	P08047
CRHBP	P24387	O95067	CRHR1	P34998	P06850
CRIP2	P52943	P82979	CRK	P46108-1, P46108	Q13905
CRKL	P46109	Q13905	CRLF1	O75462-PRO_0000011039, O75462	P26992, Q9UBD9
CRLF3	Q8IUI8	O14832	CRMP1	Q14194	P08670
CROT	Q9UKG9	P16671	CRTAM	O95727-1, O95727	Q9BY67
CRTAP	O75718	O95076	CRTC1	Q6UUV9	P31946, P63104
CRTC3	Q6UUV7	P31946, P63104	CRY1	Q16526	P63208, Q13616

Input	UniProt Id	Interacts with	Input	UniProt Id	Interacts with
CRYAB	P02511	P12830	CRYBB3	P26998	P05814
CRYL1	Q9Y2S2	O00443	CRYZL1	O95825	Q9HCM2
CSAD	Q9Y600	Q9NRF2-2	CSAG2	Q9Y5P2	Q96EB6
CSAG3	Q9Y5P2	Q96EB6	CSDE1	O75534	Q9Y3F4
CSF1	P09603	P07333	CSF1R	P07333	P29323
CSF2RB	P32927	O60674	CSF3R	Q99062	P40763
CSGALNACT1	Q8TDX6	Q9Y6M5	CSGALNACT2	Q8N6G5	Q8NBL1
CSK	P41240	Q68CZ2	CSMD1	Q96PZ7	P84022
CSNK1D	P48730	O75330	CSNK1E	P49674	O75695
CSNK1G1	Q9HCP0	P78368	CSNK1G2	P78368	Q16539
CSNK1G3	Q9Y6M4	P78368	CSNK2A1	P68400	P46100
CSNK2A2	P19784	P46100	CSPG4	Q6UVK1	P17931
CSPP1	Q1MSJ5-1	Q08379	CSRNP1	Q96S65	Q99633
CSRNP2	Q9H175	P62136	CSRP1	P21291	Q16790
CSRP2	Q16527	Q7LC44	CSRP2BP	Q9H8E8	P14373
CST3	P01034	P05067	CST7	O76096	P07711
CSTA	P01040	P04792	CSTB	P04080	P13569
CSTF2	P33240	Q99728	CTAGE5	Q96PC5	P18846
CTBP1	Q13363	P48552	CTBP2	P56545	Q8IVP5
CTBS	Q01459	Q01201	CTDNEP1	O95476	Q9Y5M8
CTDP1	Q9Y5B0	O94822	CTDSP1	Q9GZU7	P02686
CTDSP2	O14595	Q9HCG8	CTDSPL2	Q05D32	P52292
CTGF	P29279	P02751	CTNNA1	P35221	P25963
CTNNA2	P26232	P35222	CTNNA3	Q9UI47-2	O75582
CTNNAL1	Q9UBT7	P46939	CTNNBIP1	Q9NSA3	Q9BRT9
CTNNBL1	Q8WYA6	Q9UKN5	CTPS1	P17812	P01106
CTR9	Q6PD62	O00267	CTSA	P10619	Q9UM11
CTSB	P07858	P40692	CTSD	P07339	Q8NB7J
CTSF	Q9UBX1	P59665	CTSG	P08311	O00462
CTSH	P09668	P07711	CTSK	P43235	P07711
CTSO	P43235	P07711	CTSS	P25774	P14316
CTSZ	Q9UBR2	Q7Z3S9, P0DPK4	CTTN	Q14247	P32121
CTTNBP2	Q8WZ74	O43815	CTU1	Q7Z7A3	P50552
CTXN3	Q4LDR2	Q9H2K0	CUBN	O60494	P27352
CUEDC2	Q9H467	Q9UM11	CUL3	Q13618	Q9BYZ6
CUL5	Q93034	Q9Y314	CUL7	Q14999	Q16539
CUTA	O60888	P22736	CUX1	P39880	P06400
CWC15	Q9P013	Q96EB6	CWC22	Q9HCG8	P61981
CWC27	Q6UX04	P0DTC4	CWF19L2	Q2TBE0	Q13573
CX3CL1	P78423	P05556	CXADR	P97792-1	P19793
CXCL10	P02778	P48061	CXCL12	P48061	P78556, P13501, O15444, Q99616
CXCL14	O95715	P48061	CXCL16	CXCL16	Q3SXY8
CXCL2	P19875	P48061	CXCL8	P10145	P25025
CXCL9	Q07325	Q9Y3A6	CXCR4	P61073	O15554
CXXC1	Q9P0U4	Q09472	CXXC4	Q9EQC9	O14640
CXorf40A	Q8TE69	P02795	CXorf56	Q9H5V9	Q9Y3E7
CXorf57	Q6NSI4	Q13451	CYB561	P49447	P09601
CYB561A3	Q8NBI2	Q9NUQ2	CYB561D1	Q8N8Q1	Q05329
CYB5A	P00167	O76024	CYB5B	O43169	Q8WWI5

Input	UniProt Id	Interacts with	Input	UniProt Id	Interacts with
CYB5D2	Q8WUJ1	P17152	CYB5R2	Q6BCY4	P62333
CYB5R3	P00387	P01375	CYBRD1	Q53TN4	P09601
CYC1	P08574	P40855	CYCS	P99999	Q96EB6
CYFIP1	Q7L576	Q9UQB8	CYFIP2	Q96F07	Q9UQB8
CYHR1	P0DTL6	P18827	CYLD	Q9NQC7	O95786
CYP1A1	P04798	O60493	CYP4F12	Q9HCS2	Q9NPL8
CYP51A1	Q16850	O15431	CYP7B1	O75881	P0CG47
CYR61	O00622	Q04721	CYSLTR1	Q9Y271	Q9NS75
CYSLTR2	Q9NS75	Q14160	CYSRT1	A8MQ03	P15514
CYTH1	Q15438	Q9UNE7	CYTH2	Q99418	P29274
CYTH3	O43739-2	P53365	CYTH4	Q9UIA0	P20618
CYTIP	O60759	Q96EZ8	DAB1	O75553-5, O75553	Q13526
DAB2	P98078, P98082	Q9UM54	DACH1	Q9UI36	Q13485
DAD1	P61803	P23945	DAG1	Q14118	Q14118
DAGLB	Q8NCG7	Q92989	DALRD3	Q5D0E6-2	O76024
DAP3	P51398	Q14197	DAPK1	P53355	P14618-1
DAPK3	O43293	Q9BQ90	DAPL1	A0PJW8	Q15915
DAPP1	Q9UN19	Q8WY64	DARS	P14868	P05412
DAXX	Q9UER7	P46100	DAZAP1	Q96EP5	P13569
DAZAP2	Q15038	Q9HAU4	DBF4	Q9UBU7	P14136
DBN1	Q16643	P60709	DBNDD1	Q9H9R9	P48730
DBNL	Q9UJU6	P06396	DBP	Q10586	P35638
DBT	P11182	Q96E11	DCAF10	Q5QP82	Q13356
DCAF11	Q8TEB1	P11182	DCAF12	Q5T6F0	P98170
DCAF13	Q9NV06	P04271	DCAF15	Q66K64	O15431
DCAF4	Q8WV16	Q9Y4Z2	DCAF7	P61962	Q13233
CAKD	Q8WVC6	Q9Y2C2	DCBLD1	Q8N8Z6	P32971
DCBLD2	Q96PD2	P46109	DCC	P43146	O95631
DCLK1	O15075	O43602	DCLK2	Q8N568	P62258
DCLK3	Q9C098	Q00535	DCLRE1B	Q9H816	Q96EB6
DCLRE1C	Q96SD1	P78527	DCN	P07585	O75063
DCP1A	Q9NPI6	Q92900	DCP2	Q8IU60	Q92900
DCT	P40126	P27824	DCTN1	Q14203	P30622
DCTN2	Q13561	Q96GM5	DCTN3	O75935	Q9Y244
DCTN5	Q9BTE1	Q14203	DCTN6	O00399	Q9Y5N6
DCTPP1	Q9H773	P25786	DCUN1D1	Q96GG9	P61081
DCX	O43602	Q9BSE5	DCXR	Q7Z4W1	Q7Z4W1
DDA1	Q9BW61	Q9C0C7	DDAH1	O94760	O75832
DDAH2	O95865	P48431	DDB1	Q16531	Q8NHY2
DDB2	Q92466	Q9NYZ3	DDC	P20711	P10275
DDI2	Q5TDH0-2	P14136	DDIT3	P35638	P17676
DDIT4	Q9NX09	Q9Y297	DDIT4L	Q96D03	P55957
DDN	O94850	Q96B97	DDOST	P39656	Q14165
DDR1	Q08345	Q06124	DDR2	Q16832	Q16288
DDRKG1	Q96HY6	P55017	DDX1	Q92499-1	P04618
DDX10	Q13206	P06748	DDX17	Q92841	Q13882
DDX19A	Q9NUU7	P13569	DDX19B	Q9UMR2	P37173
DDX20	Q9UHI6	P01106	DDX21	Q9NR30	P40429
DDX23	Q9BUQ8	Q13573	DDX24	Q9GZR7	P01106
DDX25	Q9UHL0	Q9UBB9	DDX27	Q96GQ7	P06748

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DDX28	Q9NUL7	Q9NYY8	DDX31	Q9H8H2	P20061
DDX39A	O00148	P21673	DDX39B	Q13838	O00560
DDX41	Q9UJV9	Q13573	DDX42	Q86XP3	O43143
DDX49	Q9Y6V7	P04591	DDX5	P17844	Q09472
DDX51	Q8N8A6	P06748	DDX52	Q9Y2R4	P06748
DDX54	Q8TDD1	P06748	DDX56	Q9NY93	P62753
DDX58	O95786	P59596	DDX6	P26196	Q92900
DEAF1	O75398	P49841	DECRI	Q16698	O95997
DEDD	O75618	Q13217	DEF6	Q9H4E7	Q8WZA2
DEF8	Q6ZN54-2	Q14957	DEK	P35659	Q9UER7
DENND1A	Q8TEH3	P31946, P63104	DENND2A	Q9ULE3-2	Q96B97
DENND2C	Q68D51	P61981	DENND2D	Q9H6A0	P55199
DENND4A	Q7Z401	P31946, P63104	DENND4C	Q5VZ89	P31946, P63104
DENND6A	Q8IWF6	P37173	DENR	O43583	P25963
DEPDC1B	Q8WUY9	P61981	DEPDC5	O75140-2	Q6W0C5
DEPTOR	Q8TB45	Q13077	DERA	Q9Y315	Q9Y3C6
DERL1	Q9BUN8	Q8TB40	DERL2	Q9GZP9	Q9H1C4
DES	P31001, P17661	Q13496	DESI1	Q6ICB0	P0CG47, P62987, P62979, P0CG48
DESI2	Q9BSY9	Q99541	DEXI	O95424	P17252
DFFA	O00273	O76075	DFNA5	O60443	Q86UL3
DFNB31	Q9P202	Q9UMD9	DGAT1	O75907	P61073
DGCR14	Q96DF8	Q96GM5	DGCR2	P98153	Q8TC07
DGCR6	Q14129	Q9ULX6	DGCR6L	Q9BY27	Q15742
DGCR8	Q8WYQ5	P46087	DGKA	P23743	P12931
DGKD	Q16760	O75923	DGKE	P52429	Q9NQC8
DGKQ	P52824	Q8TDQ0	DGKZ	Q13574-2	Q08999, P28749, P06400
DHCR24	Q15392	P62736	DHCR7	Q9UBM7	P13569
DHH	O43323	Q4KMG0	DHPS	P49366	Q13895
DHRS1	Q96LJ7	Q12982	DHRS11	Q6UWP2-2	Q92796
DHRS13	Q6UX07	P32971	DHRS2	Q13268	P18827
DHRS3	O75911	Q9P296	DHRS4	Q9BTZ2	P06865
DHRS7	Q9Y394	Q9Y639	DHRS9	Q9BPW9	Q8TF71
DHRSX	Q8N5I4	O96006	DHX16	O60231	Q92917, Q92620
DHX30	Q7L2E3	Q99558	DHX35	Q9H5Z1	Q9Y4Z2
DHX37	Q8IY37	P06748	DHX40	Q8IX18	P14373
DHX57	Q6P158	Q9Y4Z2	DHX58	Q96C10	Q9UKV8, Q9UPY3
DHX8	Q14562	Q9Y6X9	DHX9	Q08211	Q86U42
DIABLO	Q9NR28	Q6ZPD8	DIAPH3	Q9NSV4	P21709
DICER1	Q9UPY3, Q9UPY3-1	Q9UKV8	DIP2A	Q14689	P19883
DIP2B	Q9P265	Q8TDQ0	DIRAS1	O95057	P52306
DISC1	Q9NRI5	P13639	DISP1	Q96F81	O76024
DIXDC1	Q155Q3	O14641	DKC1	O60832	Q13247
DLC1	P63167	O75330	DLEU1	O43261	Q92993
DLG1	Q12959	Q9NTX7	DLG2	Q15700	O60469
DLG3	Q62936, Q92796	Q13224	DLG4	P78352	P27037
DLGAP1	O14490, P97836-5	P78352	DLGAP2	Q9P1A6-3	O76024
DLGAP3	O95886	Q08379	DLGAP4	Q9Y2H0-1	Q8N3R9
DLK1	P80370	Q8WVC6	DLST	P36957	Q9ULW0

Input	UniProt Id	Interacts with	Input	UniProt Id	Interacts with
DMAP1	Q9NPF5	P20700	DMC1	Q14565	Q06609
DMD	P11532	P46939	DMKN	Q6E0U4	P11021
DMRT2	Q9Y5R5	Q9Y5V3	DMTN	Q08495	P31946, P63104
DMWD	G5E9A7	Q8WUX9	DMXL1	Q9Y485	P21281
DMXL2	Q8TDJ6	O75695	DNA2	P51530	Q9ULX3
DNAAF2	Q9NVR5	Q8NAV1	DNAJA1	P31689	Q9GZX7
DNAJA2	O60884	P51530	DNAJA3	Q96EY1-2, Q96EY1	P08581
DNAJA4	Q8WW22-2	Q8N9N5	DNAJB11	Q9UBS4	P11021
DNAJB14	Q8TBM8	P13569	DNAJB5	O75953	P25685
DNAJB6	O75190-2	Q9BRX5	DNAJC1	Q96KC8	P27449
DNAJC10	Q8IXB1	Q9UBX5	DNAJC11	Q9NVH1	Q16891, Q9NX63
DNAJC12	Q9UKB3	P17752	DNAJC13	O75165	Q16625
DNAJC14	Q6Y2X3	P82979	DNAJC16	Q9Y2G8	Q6UX41
DNAJC18	Q9H819	P23945	DNAJC19	Q96DA6	P46059
DNAJC28	Q9NX36	P06576	DNAJC3	Q27968	P19525
DNAJC30	Q96LL9	P21802	DNAJC4	Q9NNZ3	O76024
DNAJC5	Q9H3Z4	Q92692	DNAJC6	O75061	P10636-8
DNAJC7	Q99615	Q9BQ52	DNAJC8	O75937	P02545
DNAL4	O96015	O96015	DNALI1	O14645	Q8WUX9
DNASE1L1	P49184	P12830	DNASE1L2	Q92874	P11021
DNASE2	O00115	P27824	DND1	Q8IYX4	Q96JH7
DNLZ	Q5SXM8	P38646	DNM1	Q05193	Q05193
DNM1L	O00429	P62937	DNM3	Q9UQ16	Q05193
DNMBP	Q6XZF7	P50570	DNMT1	P26358	P48552
DNMT3B	Q9UBC3	Q9Y6K1	DNPH1	O43598	P17252
DNTTIP1	Q9H147	Q92769	DNTTIP2	Q5QJE6	Q6NZI2
DOC2A	Q14183	Q7Z4L5	DOCK1	Q14185	P46109
DOCK10	Q96BY6	P42357	DOCK11	Q5JSL3	P31946, P63104
DOCK2	Q92608-2	P10636	DOCK4	Q8N1I0	P62993
DOCK5	Q9H7D0	P46109	DOCK7	Q96N67	Q9UM54
DOCK8	Q8NF50	Q99836	DOCK9	Q9BZ29	P84022, Q15796
DOHH	Q9BU89	Q9NQZ5	DOK2	O60496	Q9NUX5
DOK3	Q7L591-3	Q9NQ94	DOK4	Q8TEW6	Q06710
DOK6	Q6PKX4	Q16288	DOLK	Q9UPQ8	Q9NZQ7
DOT1L	Q8TEK3	P42568, Q03111	DPAGT1	Q9H3H5	P42858
DPCD	Q9BVM2	Q92696	DPF1	Q92782	Q8NFD5, P51531
DPF2	Q92785	Q92925, Q8NFD5, P51531, O96019, O94805, O75177, Q15532	DPF3	Q92784-1, Q92784, Q92784-2	Q8NFD5, P51531, O96019
DPH2	Q9BQC3	P11142	DPH6	Q7L8W6	P43119
DPM1	O60762	P01106	DPM3	Q9P2X0-2	O75396
DPP3	Q9NY33	Q14145	DPP4	P27487	P27487
DPP8	Q6V1X1	P14678	DPPA4	Q7L190	Q9BYJ1
DPY19L1	Q2PZI1	O75326	DPY19L3	Q6ZPD9	P13671
DPY30	Q9C005	P17980, O00487	DPYSL2	Q16555	Q8NFD5
DPYSL4	O14531	P19174	DPYSL5	Q9BPU6	P06396
DRAM1	Q8N682	Q9NPL8	DRAP1	Q14919	Q00403
DRAXIN	Q8NBI3	O95631	DRD2	P14416	P14416
DRG1	Q92597	P61981	DRICH1	Q6PGQ1	P49674, P48730
DSC2	Q02487	Q14126	DSC3	Q02487	Q14126

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DSCC1	Q9BVC3	P00740	DSE	Q9UL01	Q9NRD5
DSP	P15924	P07332	DST	Q03001	Q96CV9
DSTN	P60981	P60709	DTNA	Q9Y4J8-9	P40763
DTNB	O60941-5	Q9BUH8	DTNBP1	Q96EV8	A1L4H1
DTWD1	Q8N5C7	O15431	DTWD2	Q8NBA8	P07437
DTX1	Q86Y01	Q96J02	DTX2	Q86UW9	P51668
DTX3	Q8N9I9	P51668	DUS3L	Q96G46	Q96CV9
DUSP1	P28562	Q16539	DUSP10	Q9Y6W6	P60709
DUSP12	Q9UNI6	P55010	DUSP15	Q9H1R2	P19174
DUSP16	Q9BY84	Q16539	DUSP19	Q8WTR2	P39748
DUSP23	Q9BVJ7	Q8TBC4	DUSP27	Q68J44	P28074
DUSP4	Q13115	Q13077	DUSP6	Q16828	P28482
DUSP7	Q16829	Q16539	DUSP8	Q13202	P45983
DUSP9	Q99956	Q16539	DVL1	O14640	Q9Y4D1
DVL3	Q92997	P23025	DXO	O77932	P08670
DYDC2	Q96IM9	O95602	DYM	Q7RTS9	P21453
DYNC1I1	O14576-2	Q13501, Q9GZQ8	DYNC1I2	Q13409	P43034
DYNC1LI1	Q9Y6G9	P43034	DYNLL1	P63167	O75330
DYNLL2	Q96FJ2	O75330	DYNLRB1	Q9NP97	P10636-8
DYNLT1	P63172	Q9BRX2	DYNLT3	P51808	P43034
DYRK1A	Q13627	Q09472	DYRK1B	Q9Y463	Q09472
DYSF	O75923	Q96CV9	DZIP1	Q86YF9	P61981
DZIP1L	Q8IYY4	P61981	E2F4	Q16254	P11802
E2F6	O75461	Q92769, P51532	E4F1	Q66K89	P67809
EAF1	Q96JC9	P55199	EAF2	Q96CJ1	Q16665
EAPP	Q56P03	Q13895	EBF2	Q9HAK2	Q13077
EBF3	Q9H4W6-2	Q9HAK2	EBNA1BP2	Q99848	P19525
EBP	Q15125	Q9H244	ECD	O95905	Q9Y265
ECE1	P42892	P17931	ECEL1	O95672	Q13217
ECH1	Q13011	P26641	ECHDC2	Q86YB7	O75380
ECHS1	P30084	P11182	ECI2	O75521	Q16659
ECM1	Q16610	Q9NQ94	ECSIT	Q9BQ95	Q9H244
ECT2	Q9H8V3	P12830	ECT2L	Q008S8	P63208
EDA	Q92838	P17480	EDA2R	Q9HAV5	Q13114
EDAR	Q9UNE0	Q92838	EDARADD	Q8WWZ3	P63279
EDC3	Q96F86	P11309	EDC4	Q6P2E9	Q8IZH2
EDEM1	Q92611	Q9UMX1	EDEM2	Q9BV94	Q8TAS1
EDEM3	Q2HXL6	P30101	EDF1	O60869	P37231
EDNRA	P25101	P32121, P49407	EDRF1	Q3B7T1	O14832
EEA1	Q15075	P13569	EED	O75530	Q9Y6K1
EEF1A1	P68104	Q9NYA1	EEF1A2	Q05639	P24557
EEF1B2	P24534	P26641, P68104	EEF1DP3	Q658K8	P07550
EEF1E1	O43324	O00560	EEF1G	P26641, P26641-2	P68104, P29692, P24534
EEF2	P13639	Q16539	EEF2K	O00418	Q9H4G0
EEF2KMT	Q96G04	Q9NRD5	EEPD1	Q7L9B9	P18827
EFCAB14	O75071	P68400	EFCAB6	Q5THR3-2	Q08379
EFEMP1	Q12805, A0A0S2Z3V1	Q12933	EFEMP2	O95967	Q9UBX5, P15502
EFHC2	Q5JST6	O94955	EFHD1	Q9BUP0	P03372
EFNA2	O43921	P54764	EFNA3	P52797	P31749

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EFNA5	P52803	Q9UBD6	EFNB1	P98172	O15397
EFNB3	Q15768	P54764	EFS	O43281-2	Q13526
EFTUD2	Q15029	P01106	EGF	P01133	P00533
EGFL7	Q9UHF1	Q9UBX5	EGFL8	Q99944	P08311
EGFR	P00533	P01135, P01133	EGLN1	Q9GZT9	Q16665, Q99814
EGLN2	Q96KS0	Q16665	EGLN3	Q9H6Z9	P14618-1
EGR1	P18146	Q8N726	EGR2	A0A0S2Z3U4	Q08379
EHBP1	Q8NDI1	Q9NWB7	EHBP1L1	Q8N3D4	Q96ES7
EHD1	Q9H4M9	P13569	EHD2	Q9NZN4	Q86UL3
EHD3	Q9NZN3	Q9H1K0	EHD4	Q9H223	Q9H1K0
EHHADH	Q08426	Q16585	EHMT1	Q9H9B1	Q92769
EHMT2	Q96KQ7	Q92769	EI24	O14681	Q12982
EIF1AD	Q8N9N8	Q9H668, Q2NKJ3	EIF1AY	O14602	P47813
EIF1B	O60739	P62249	EIF2A	P05198	P19525
EIF2AK1	Q9BQI3	P11142	EIF2AK2	P19525	P55265
EIF2AK3	Q9NZJ5	P11021	EIF2AK4	Q9P2K8	P36873
EIF2B1	Q14232	P07550	EIF2B2	P49770	P0CG47, P62987
EIF2B4	Q9UI10	P01019	EIF2B5	Q13144	P21579
EIF2D	P41214	P17980	EIF2S2	P20042	P01138
EIF3A	Q14152	P47813	EIF3B	P55884	Q04637, P06730
EIF3C	Q99613	P47813	EIF3D	O15371	P21673
EIF3F	O00303	Q9Y5J6	EIF3G	O75821	Q9BRX2
EIF3I	Q13347	Q04637	EIF3J	O75822	Q04637
EIF3K	Q9UBQ5	P30281	EIF3L	Q9Y262	P13569
EIF3M	Q7L2H7	O00232	EIF4A1	P60842	Q04637, Q15056, P06730
EIF4A2	Q14240	Q04637, P06730	EIF4A3	P38919	Q13573
EIF4E2	O60573	Q99814	EIF4EBP1	Q13541	P06730
EIF4EBP2	Q13542	P06730	EIF4ENIF1	Q9NRA8	P06730
EIF4G1	Q04637	P06730	EIF4G2	P78344	P06748
EIF4G3	O43432-1, O43432	P06730	EIF4H	Q15056-2	Q9NRD5
EIF5	P55010	P47813	EIF5A	P63241	O00560
EIF5A2	Q9GZV4	O00560	EIF5B	O60841	O76024
EIF6	P56537	P19525	ELAVL1	Q15717	Q9NZI8
ELAVL2	Q12926	O00425, Q9NZI8	ELAVL3	Q14576	Q13627
ELAVL4	P26378-2	Q00994	ELF1	P32519	P08047
ELF2	Q15723	Q01196	ELK1	P19419	P63279
ELK3	P41970	O00712, Q12857	ELL	P55199	Q9UHB7, Q96JC9
ELL2	O00472	P07550	ELMO1	Q92556	Q8IZD9
ELMO2	Q96JJ3	Q8IZD9	ELMSAN1	Q6PJG2	Q16539
ELN	P15502	Q9UBX5	ELOVL2	Q9NYP7	P09601
ELOVL4	Q9GZR5	P09601	ELOVL5	Q9NYP7	P09601
ELOVL6	Q9H5J4	P09601	ELOVL7	A1L3X0	Q96EV8
ELP3	Q9H9T3	P11142	EMB	Q6PCB8	P17931
EMC10	Q5UCC4	Q5J8M3	EMC2	Q15006	O43868
EMC3	Q9P0I2	P27824	EMC4	Q5J8M3	P27824
EMC6	Q9BV81	Q3KNW5	EMC7	Q9NPA0	P27824
EMC8	O43402	P27824	EMC9	Q9Y3B6	Q5J8M3
EMD	P50402	Q99962	EME2	A4GXA9	Q96NY9
EMILIN1	Q9Y6C2	Q8NBN7	EMILIN2	Q9BXX0	Q99435
EMILIN3	Q9NT22	P09238	EML1	O00423	Q9H9L3

Input	UniProt Id	Interacts with	Input	UniProt Id	Interacts with
EML3	Q32P44	P63208	EML4	Q9HC35	P63208
EMP1	P54849	Q9Y5Q5	EMP3	P54852	A1L3X0
ENC1	O14682	Q13501	ENDOD1	O94919	Q9NRZ7
ENDOG	Q14249	P11142	ENG	P17813	P08648
ENKD1	Q9H0I2	P52747	ENKUR	Q8TC29	O43464
ENO1	P06733	P03496	ENO2	P09104	P06733
ENOPH1	Q9UHY7	Q9H4P4	ENOX1	Q8TC92	O00560
ENPP1	P22413	Q9UNQ0	ENPP4	Q9Y6X5	P17931
ENPP6	Q6UWR7	O15228	ENTHD2	Q96N21	O95848
ENTPD1	P49961	Q96S59	ENTPD2	Q9Y5L3	Q96CV9
ENTPD3	O75355-2	Q3SXY8	ENTPD5	O75356	P42858
ENTPD6	O75354	P27824	EOGT	Q5NDL2	Q03405
EP300	Q09472	Q13761	EP400	Q96L91	O96019
EPAS1	Q99814	P40818	EPB41	P11171	P61981
EPB41L1	Q9H4G0	Q9UHB7	EPB41L2	O43491	P42568, Q03111
EPB41L3	Q9Y2J2	P49841	EPB41L5	Q9HCM4	Q8N3R9
EPC1	Q9H2F5	O96019	EPC2	Q52LR7	P52294
EPCAM	P16422	P12830	EPDR1	Q9UM22	P01222
EPGN	Q6UW88-2	Q9H2K0	EPHA10	Q5JZY3-3	Q8N9N5
EPHA2	P29317	Q06124	EPHA3	P29320	Q13485
EPHA4	P54764	P05177	EPHA6	Q9UF33	P04070
EPHA7	Q15375	Q16288	EPHA8	P29322	P11021
EPHB1	P54762	Q14451	EPHB2	P54763	Q13009
EPHB4	P54760	P01588	EPHB6	O15197	Q13451
EPHX1	P07099	Q13077	EPM2A	O95278, Q9WUA5	Q9UQK1, Q6VVB1
EPN1	Q9Y6I3	P42566	EPN2	O95208-2	P52888
EPN3	I6L9I8	O76024	EPRS	P07814	Q8IWL3
EPS15	P42566	Q9P0V3	EPS8	Q12929	P49419
EPS8L1	Q8TE68	Q9UQB8	EPSTI1	Q96J88-3	P17252
ERAL1	O75616	P08123	ERAP1	Q9NZ08	P21462
ERBB2	P04626	P12931	ERBB2IP	Q96RT1	P63104
ERBB3	P21860	Q14451	ERBB4	Q15303	P23528
ERC1	Q8IUD2	P01222	ERC2	O15083	P10242
ERCC1	P07992-3, P07992	P23025	ERCC2	P18074	P15056
ERCC3	P19447	P28715	ERCC5	P28715	P24522
ERF	P50548	Q9UGI0	ERG	P11308	O00712, Q12857
ERGIC2	Q96RQ1	P15907	ERGIC3	Q9Y282	Q02094
ERH	P84090	Q9Y5B0	ERI1	Q8IV48	Q14493
ERICH1-AS1	Q9P1A6-3	O76024	ERICH5	Q6P6B1	Q14145
ERLIN1	O75477	Q86TM6	ERMP1	Q7Z2K6	Q9UBD6
ERN1	O75460-2	P38936	ERO1A	Q96HE7	P30101
ERO1B	Q86YB8	P0DTC8	ERP29	P30040	Q15848
ERP44	Q9BS26	P43251	ERVK3-1	Q96A10	O95429
ESF1	Q9H501	P06748	ESM1	Q9NQ30	O76024
ESR1	P03372	P40429	ESR2	Q92731	P29279
ESRP1	Q6NXG1-3	Q96CV9	ESRRA	P11474	Q9UBK2
ESRRB	Q61539	Q16665	ESRRG	P62508, P62508-3	Q9UBK2
ESYT2	A0FGR8	P61981	ETFA	P13804	Q8IWL3
ETNK2	Q9NVF9	Q16623	ETS1	P14921-3	Q12778
ETV1	P50549	Q8NHY2	ETV4	P43268-3	Q13485

Input	UniProt Id	Interacts with	Input	UniProt Id	Interacts with
ETV5	P41161	Q8NHY2	ETV6	P41212	O75197
ETV7	Q9Y603	O00712, Q12857	EVA1B	Q9NVM1	Q8WVC6
EVA1C	P58658	O14786	EVI2A	P22794	P27449
EVI2B	P34910-2	P22732	EVI5	O60447	P53350
EVI5L	Q96CN4	O95995	EVL	Q9UI08	Q13813
EVPL	Q92817	Q9Y324	EWSR1	Q01844	Q09472
EXOC1	Q9NV70	P18848	EXOC3-AS1	Q8N2X6	Q12933
EXOC4	Q96A65	P04601	EXOC5	O00471	Q9ULZ3
EXOC6	Q8TAG9	Q9Y639	EXOC6B	Q9Y2D4	O60645, Q8TAG9, Q8IYI6, O00471, Q9UPT5, Q96A65, Q9NV70, Q96KP1
EXOC7	Q9UPT5-1, Q9UPT5	A1L4H1	EXOSC1	Q9Y3B2	P52747
EXOSC2	Q13868	Q99547, Q9Y2L1, Q13901, P42285, Q01780, Q9NQT4	EXOSC3	Q9NQT5	P06748, Q92841
EXOSC4	Q9NPD3	Q99547, Q9Y2L1, Q13901, P42285, Q01780, Q9NQT4	EXOSC5	Q9NQT4	Q8WY64
EXOSC9	Q06265	P63279	EXT1	Q16394	Q03405
EXTL2	Q9UBQ6	P14136	EXTL3	O43909	P61073
EYA1	Q99502	Q15475, Q9NPC8	EYA2	O00167-2	Q13526
EYA3	Q99504	O76024	EZH1	Q92800	P15056
EZR	P15311	P52565	F10	Q9BQ89	P49674
F12	P00748	P11142	F13A1	P00488	P19388
F2R	P25116	Q9UNN8	F3	P13726	P08709
F5	P12259	P78380	F8	P00451-PRO_0000002968	P00740
F8A1	P23610	P42858	F8A3	P23610	P42858
FA2H	Q7L5A8	Q96BA8	FAAH	Q7L5A8	Q96BA8
FAAP100	Q0VG06	P99999	FAAP20	Q6NZ36	Q9UBZ9
FABP3	P05413	P05556	FABP4	P15090	P08670
FABP5	Q01469	P13569	FABP7	O15540	O76024
FADD	Q13158	Q13546	FADS2	Q8NFF5-2	P29373
FADS6	Q8N9I5	P49638	FAF1	Q9UNN5	P11142
FAH	P16930	P00533	FAHD1	Q6P587-2	P04792
FAHD2A	Q96GK7	P11182	FAM102A	Q5T9C2-3	P56945
FAM103A1	Q9BTL3	O43148	FAM105A	Q9NUU6	O43464
FAM107A	O95990-4, O95990	Q12933	FAM109A	Q8N4B1-4	O43639
FAM110A	Q9BQ89	P49674	FAM110B	Q8TC76	P61981
FAM111A	Q96PZ2	P04637	FAM114A1	Q8IWE2	Q9UBN6
FAM117A	Q9C073	P31946	FAM118A	Q9NWS6	O00244
FAM120A	Q9NZB2	Q92900	FAM120B	Q96EK7	Q12933
FAM120C	Q9NX05	P07910	FAM122B	Q7Z309	P31946, P63104
FAM122C	Q6P4D5-2	P27797	FAM124A	Q86V42	Q96QF0
FAM126A	Q9BYI3, A0A0S2Z6M9	P14373	FAM126B	Q8IXS8	P14373
FAM127A	A6ZKI3	P42772	FAM127C	Q17RB0	O00560
FAM129A	Q9BZQ8	P06748	FAM129B	Q96TA1	Q14145
FAM131B	Q86XD5	Q99741	FAM133A	Q8N9E0	Q9NRZ7
FAM134A	Q8NC44	Q9GZQ8	FAM134B	Q9H6L5-1	Q9GZQ8
FAM135B	Q49AJ0-4	P04792	FAM136A	Q96C01	O00443
FAM13A	O94988	P31946, P63104	FAM150B	Q6UX46	Q9UM73
FAM156A	Q8NDB6	P21673	FAM160B1	Q5W0V3	Q13304

Input	UniProt Id	Interacts with	Input	UniProt Id	Interacts with
FAM161A	Q3B820	Q9BUH8	FAM161B	Q96MY7	O95995
FAM163A	Q96GL9	P08670, P06396	FAM167A	Q96KS9	P46939
FAM168B	A1KXE4-2	Q01974	FAM171A2	A8MVW0	Q16288
FAM172A	Q8WUF8	P63208	FAM173A	Q9BQD7	O14964
FAM174A	Q8TBP5	Q6UWP7	FAM175A	Q6UWZ7	P53611
FAM184A	Q8NB25	P06396	FAM185A	Q8N0U4	P19404
FAM188A	Q9H8M7	P78549	FAM189A2	Q15884	Q9UM54
FAM189B	P81408	Q96PU5	FAM192A	Q9GZU8	Q99436, P28066, P20618, P28065, P61289, P49720, P25786
FAM195A	Q9BUT9	O15234	FAM196A	Q6ZSG2	O15111
FAM199X	Q6PEV8	P49674, P48730	FAM19A1	Q7Z5A9	P53611
FAM19A4	Q96LR4	P08648	FAM206A	Q9NX38	Q7L7X3
FAM208A	Q9UK61	Q99549	FAM208B	Q5VWN6	P40937
FAM20C	Q8IXL6	P10451	FAM210A	Q96ND0	Q9Y366
FAM210B	Q96KR6	P09601	FAM212A	Q96EL1	P02545
FAM213A	Q9BRX8	Q9UNA3	FAM213B	Q8TBF2	O43776
FAM214A	Q32MH5	P14373	FAM214B	Q7L5A3	Q12933
FAM217B	Q9NTX9	Q8WY64	FAM218A	Q96MZ4	P25788
FAM219A	Q8IW50	O43924	FAM219B	Q5XKK7	Q9NRD5
FAM21A	Q641Q2	P51530	FAM21C	Q9Y4E1	P47756
FAM222A	Q5U5X8	Q9UBE8	FAM222B	Q8WU58	Q9ULX6
FAM24B	Q8N5W8	A8MZ59	FAM26E	Q8N5C1	Q00534
FAM32A	Q9Y421	Q9HCG8	FAM35A	Q86V20-2	Q12888
FAM3A	Q9BU27	Q9BXN2	FAM3B	P58499	O75197
FAM3C	Q92520	Q9UNA3	FAM46A	Q96IP4	P43405
FAM46B	Q96A09	P06396	FAM53C	Q9NYF3	P53350
FAM60A	Q9NP50	Q9Y6N5	FAM65C	Q96MK2	P61586
FAM69A	Q5T7M9	Q9UNA3	FAM69B	Q5VUD6	P21673
FAM73B	Q7L4E1	P11142	FAM76B	Q5HYJ3-3	O76024
FAM78A	Q5JUQ0	P42858	FAM78B	Q5VT40	Q9UI95
FAM83F	Q8NEG4	P48729	FAM83H	Q6ZRV2	O43639
FAM84B	Q96KN1	Q06609	FAM86C1	Q9NVL1-2	P21673
FAM89B	Q9QUI1	P12755	FAM91A1	Q658Y4	P08173
FAM92A1	A1XBS5-3, A1XBS5	Q9Y3M2	FAM96B	Q9Y3D0	P08670, P06396
FAM98A	Q8NCA5	P59595	FAM98C	Q17RN3	Q00403
FAN1	Q16658	P54278	FANCB	Q8NB91	P99999
FANCC	Q00597	Q13485	FANCD2	Q9BXW9	O00255
FANCE	Q9HB96	P31749	FANCF	Q9NPI8	P42858
FANCG	O15287	P08684	FANCL	Q9NW38	Q13485
FANCM	Q8IYD8	P49736	FANK1	Q8TC84	P04792
FAR1	Q8WVX9	P13569	FAR2	Q96K12	P40855
FARP1	Q9Y4F1	P18827	FARP2	O94887	P49674
FARS2	O95363	Q14353	FARSA	Q9Y285	P10636-8
FARSB	Q9NSD9	P10636-8	FAS	P25445-1, P25445	Q14790
FASN	P49327	Q16665	FASTKD5	Q7L8L6	P31152
FAT1	Q14517	P23946	FAU	P62861	P06748
FAXC	Q5TGI0	Q9H668, Q2NKJ3	FAXDC2	Q96IV6	Q3SXY8
FBL	P22087	P05455	FBLIM1	Q8WUP2	P13569
FBLN1	P23142-4	Q96FE5	FBLN2	P98095	P14543

Input	UniProt Id	Interacts with	Input	UniProt Id	Interacts with
FBLN5	Q9UBX5	Q8N2S1	FBN1	P35555	P15502
FBN2	P35556	P04608	FBP1	P09467	Q16665, Q99814
FBR5	Q9HAH7	Q9BYE7	FBRSL1	Q9HCM7	P68400, P67870
FBXL12	Q9NXK8	P63208, Q13616	FBXL14	Q8N1E6	P35813
FBXL15	Q9H469	Q9HAU4	FBXL16	Q8N461	Q16665
FBXL17	Q9UF56	O14867	FBXL18	Q96ME1	P63208, Q13616
FBXL19	Q6PCT2-2	Q6ZPD8	FBXL20	Q96IG2	P63208
FBXL21	Q8BFZ4	P63208, Q13616	FBXL3	Q9UKT7	Q9UJ68
FBXL4	Q9UKA2	P17931	FBXL5	Q9UKA1	P63208, Q13616
FBXL6	Q8N531	P63208	FBXO10	Q9UK96	O43924
FBXO11	Q86XK2	P62877, P63208, Q13616	FBXO16	Q8IX29	P41208
FBXO17	Q96EF6	O60551	FBXO18	Q8NFZ0	P63208
FBXO2	Q9UK22	Q8N2S1	FBXO21	O94952	P62877, P63208, Q13616
FBXO25	Q8TCJ0	P60709	FBXO28	Q9NVF7	Q13177
FBXO30	Q8TB52	Q9UM54	FBXO31	Q5XUX0	Q13315
FBXO33	Q7Z6M2	P04004	FBXO34	Q9NWN3	Q9UM54
FBXO4	Q9UKT5	P17707	FBXO42	Q6P3S6	Q06330
FBXO44	Q9H4M3-2, Q9H4M3	P63208, Q13616	FBXO45	P0C2W1	P63208
FBXO9	Q9UK97-2, Q9UK97	P63208	FBXW4	P57775	Q13371
FBXW5	Q969U6	P63208, Q13616	FBXW7	Q969H0	P46531
FBXW8	Q8N3Y1	P01106	FBXW9	Q5XUX1	P63208, Q13616
FCER1G	P30273	P43405	FCGBP	Q9Y6R7	Q07654
FCGR1A	P12314	Q86VB7	FCGR2B	P31994	P19174
FCGRT	P55899	P09958	FCHO1	O14526	Q13882
FCHO2	Q0JRZ9	P98082	FCHSD2	O94868-3	P06396
FCN1	O00602	P18433	FDFT1	P37268	Q3SXY8
FDPS	P14324	Q14997	FECH	P22830	P16035
FEM1A	Q9BSK4	P09110	FEM1B	Q9UK73	P49593
FEM1C	Q96JP0	Q93034	FEN1	P39748	Q14191
FER	P16591	Q7LC44	FERMT2	Q96AC1	P35222
FERMT3	Q86UX7-2	Q99633	FES	P07332	P15311
FEZ1	Q9Y250	P30307	FEZ2	Q9UHY8	Q9BXM7
FGD4	Q96M96	P17252	FGD5	Q6ZNL6	O43463
FGD6	Q6ZV73	P31946, P63104	FGF1	P05230	P98160
FGF10	O15520	P21802	FGF11	Q92914	Q96HN2
FGF13	Q92913	Q96EB6	FGF14	Q92915-2	P49638
FGF17	O60258	Q9UGY1	FGF2	P09038	P21802
FGFR1	P11362-2	Q9BRX2	FGFR1OP	O95684	Q96NL6
FGFR1OP2	Q9NVK5	Q13043	FGFR2	P21802	O15520
FGFR3	P22607	Q8N5V2	FGFRL1	Q8N441	P34741
FGL2	Q14314	Q2M1P5	FGR	P09769	P00533
FH	P07954	P11309	FHIT	P49789	P35222
FHL1	Q13642	P04156	FHL2	Q14192	Q9UBZ9
FHL3	Q13643	Q96E11	FHL5	Q5TD97	O96015
FHOD1	Q9Y613	P19474	FHOD3	Q2V2M9, Q2V2M9-4	Q13501
FIBP	O43427	P80370	FIGF	O43915	Q15836
FIGN	Q5HY92	O14964	FIP1L1	Q6UN15	O95639
FIS1	Q9Y3D6	Q14790	FITM2	Q8N6M3	Q9H244

Input	UniProt Id	Interacts with	Input	UniProt Id	Interacts with
FIZ1	Q96SL8	Q04721	FKBP10	Q96AY3	P02452
FKBP14	Q9NWM8	P01222	FKBP1A	Q0VDC6	Q8WUX9
FKBP2	P26885	P42858	FKBP3	Q00688	Q00987
FKBP4	Q02790	Q9UKV8	FKBP5	Q13451	O43318
FKBP8	Q14318	Q13371	FKBP9	O95302	Q14118
FKBPL	Q9UIM3	P49674, P48730	FKRP	Q9H9S5	Q9Y2B1
FKTN	O75072	Q9Y2B1	FLAD1	Q8NFF5	Q8NFF5
FLI1	Q01543	P84022	FLII	Q13045	P13569
FLNA	P21333-2	Q9BRX2	FLNB	O75369	Q13233
FLNC	Q14315	Q92629	FLOT1	P41440	Q03135
FLOT2	Q14254	P13569	FLT1	P17948	P46109
FLT3	P36888	Q06124	FLT4	P35916	P35968
FLVCR1	Q9Y5Y0	P31785	FLYWCH1	Q4VC44	P50458
FLYWCH2	Q96CP2	P11142	FMN2	Q9NZ56	Q12974
FMNL1	O95466	O75044	FMNL3	Q6ZPF4	O75044
FMO1	Q01740	Q3KNW5	FMO3	P31513	Q8IUN9
FMOD	P13605	P08603	FMR1	Q06787	Q04637
FN1	P02751	P29279	FN3KRP	Q9HA64	P02788
FNBP1	Q8R511	P52198	FNBP1L	Q5T0N5	Q9Y4D1
FNBP4	Q8N3X1	O75400	FNDC3A	Q9Y2H6	O43511
FNDC3B	Q53EP0	Q9NPD5	FNDC4	Q9H6D8	Q03135
FNTA	P49354	Q12974	FNTB	P49356	P01185
FOS	P01100	Q8NHY2	FOSB	P53539	Q9BZS1
FOSL1	P15407	P26640	FOSL2	P15408	Q8NFP9
FOXC1	Q12948	P08651	FOXC2	Q99958	Q99958
FOXD2	O60548	Q9Y5V3	FOXD3	Q9UJU5	Q01860
FOXD4	Q12950	P10515	FOXD4L1	Q9NU39	P61289
FOXK2	Q01167	O43278	FOXN2	P32314	Q9Y297
FOXN3	O00409	Q13573	FOXO1	Q12778	P06493
FOXO3	O43524	P28749, P06400	FOXO4	P98177	P98177
FOXP1	Q9H334	Q9BZS1	FOXP2	Q75MZ5	P49247
FOXP4	Q8IVH2	Q9BZS1	FOXS1	O43638	P67775
FPR1	P21462	Q92538	FRA10AC1	Q70Z53	Q13123
FRAS1	Q86XX4	P08311	FRAT1	Q92837	P49841
FRAT2	O75474	P49841	FRG1	Q14331	Q9HCG8
FRK	P42685	P41240	FRMD1	Q8N878	P02788
FRMD3	A2A2Y4	Q9Y6X1	FRMD4A	Q9P2Q2	P61981
FRMD5	Q7Z6J6	Q92823	FRMD6	Q96NE9, Q96NE9-2	P31946, P17252, P63104
FRMPD1	Q5SYB0	Q14160	FRMPD4	Q14CM0	Q14160
FRS2	Q8WU20	Q06124	FRS3	O43559	Q96FE5
FRYL	O94915	P46531	FSCN1	Q16658	P54278
FSD1	Q9BTV5	P30740	FST	P19883	P10599
FSTL1	Q12841	P08571	FSTL5	Q8N475	O75554
FTCD	O95954	Q8N3R9	FTH1	P02794	Q9UER7
FTL	P02792	Q9UJK0	FTO	Q9C0B1-2	Q92989
FTSJ1	Q9UET6	Q9Y5P4-2	FTSJ2	Q9UI43	P36776, P10809
FTSJ3	Q8IY81	P19525	FUBP3	Q96I24	P09651
FUNDC1	Q8IVP5	Q9GZQ8	FUNDC2	Q9BWH2	Q8IVP5
FUS	P35637	Q15149	FUT8	Q9BYC5	P26572
FUT9	Q9Y231	Q9NWM8	FXN	Q16595	O75925

Input	UniProt Id	Interacts with	Input	UniProt Id	Interacts with
FXR1	P51114	Q13416	FXR2	P51116	Q04637
FXYD1	O00168	Q92597	FXYD3	Q14802-3	O15554
FXYD6	Q9H0Q3	Q9UBD6	FYB	O15117	O43639
FYCO1	Q9BQS8	Q9GZQ8	FYN	P06241	P12931
FZD10	Q9ULW2	O14786	FZD2	Q14332	P43119
FZD4	Q9ULV1	P42858	FZD7	O75084	P41221
FZD8	Q9H461	P56704	FZR1	Q9UM11	Q9HAW4
G0S2	P27469	Q07817	G2E3	Q7L622	O76024
G3BP1	Q13283	Q04637	G3BP2	Q9UN86-2, Q9UN86	Q14694
G6PD	P11413	Q8IXJ6	GAB1	Q13480	P46109
GABARAP	O95166	O75385	GABARAPL2	P60520	Q13685
GABBR1	Q9UBS5	P46459	GABBR2	O75899	P46459
GABPA	Q06546	P08047	GABPB1	Q06547	Q13573
GABPB2	Q06547	Q13573	GABRA2	P47869	P00533
GABRD	O14764	P18850	GABRE	P78334	O60427
GABRG1	Q8N1C3	Q96CV9	GABRG2	P18507	P27824
GADD45A	P24522	P28715	GADD45B	O75293	O14733
GADD45G	O95257	P68104	GADD45GIP1	Q8TAE8	Q14197
GAK	O14976	Q9NQ11	GAL	P22466	P09238
GAL3ST1	Q99999	Q7Z7B1	GALK2	Q01415	P13987
GALNT10	Q86SR1	Q07817	GALNT12	Q8IXK2	P02790
GALNT15	Q8N3T1	Q9BRI3	GALNT16	Q8N428	O15397
GALNT18	Q6P9A2	P01185	GALNT2	Q10471	Q9HC29
GALNT6	Q8NCL4	O75063	GALNT7	Q86SF2	Q8IV16
GALR1	P47211	P08908	GAMT	Q14353	O95363
GAN	Q9H2C0	Q99426	GANAB	Q14697	P01106
GAP43	P17677	Q16822	GAPDH	P04406	P35228
GAPVD1	Q14C86	P48729	GAREM	Q9H706	P29353
GAREML	Q75VX8	P62993	GARS	P41250	Q8WXH5
GART	P22102	P13569	GAS2L1	Q99501	Q96B97
GAS2L3	Q86XJ1	Q96GD4	GAS7	O60861-1	O95070
GAS8	O95995	P43897	GATA2	P23769	O00712, P08651, Q12857
GATAD1	Q8WUU5	Q13547	GATAD2B	Q8WXI9	Q92769
GATC	O43716	O14964	GATM	P50440	Q9P1W8
GBAS	O75323	Q05513	GBP1	P32455	P32455
GBP2	P32456	P32455	GBP3	Q8WXF7	P04156
GBP4	Q96PP9	P32455	GBP5	Q96PP8	P32455
GCA	P28676	P28070	GCAT	O75600	O75380
GCC1	Q96CN9	P08151	GCC2	Q8IWJ2	P18509
GCFC2	P16383	O43143, O95926, Q9BUQ8, O94906, Q9UBB9	GCH1	P30793	P63104
GCK	P35557	Q14397	GCLC	P48506	Q6IPR3
GCLM	P48507	P10746	GCNT1	Q02742	P62987
GCSH	P23434, Q53XL7	Q13287	GDA	P16260	Q15116
GDAP2	Q9NXN4	Q9Y3M2	GDE1	Q9NZC3	Q14973
GDF9	O60383	P04183	GDI1	P31150	P51149
GDI2	P50395	P02751	GDNF	P39905- PRO_0000034005	Q92673
GDPD5	Q8WTR4	Q9Y6N7	GEM	P55040	Q9BUH8

Input	UniProt Id	Interacts with	Input	UniProt Id	Interacts with
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GEMIN8	Q9NWZ8	Q9H244	GET4	Q7L5D6	Q9HB29
GFAP	P14136	P52747	GFER	P55789	P11802
GFM1	Q96RP9	P03508	GFOD1	Q9NXC2	P04792
GFPT2	O94808	Q9NUX5	GFRA1	P56159-2, Q62997	Q92673
GFRA3	O60609	Q5T4W7	GGA1	Q9UJY5	Q9NZ52
GGA2	Q9UJY4	Q9NZ52	GGA3	Q9NZ52	Q6VY07
GGCX	P38435	P0DTC5	GGH	Q92820	P27824
GGN	Q86UU5	O43639	GGPS1	O95749	O00244
GGT1	P19440-3	P17252	GGT5	P36269	Q03426
GGT7	Q9UJ14	Q5SRI9	GHITM	Q9H3K2	P0DTC3
GHR	P10912	P01241	GID8	Q9NWU2	O60313
GIGYF2	Q6Y7W6	P31946, P63104	GIMAP2	Q9UG22	P42858
GIMAP5	Q96F15	Q3SXY8	GINM1	Q9NU53	Q14118
GINS2	Q9Y248	O96017	GINS3	Q9BRX5	P49736
GIP	P09681	P48546	GIPC1	O14908	Q9P0V3
GIPC2	Q8TF65	Q9P0V3	GIPR	P48546	P01275
GIT1	Q9Y2X7	Q13177	GIT2	Q14161	Q13177
GJA1	P17302, P08050	Q07157	GJA4	P35212	P35372
GJA5	P36382	Q9UKG4	GJB1	P08034	P02787
GJB2	P29033	Q3SXY8	GJB6	O95452	O75396
GJC1	Q8N144, P28229	Q07157	GK	P32189	P22736
GLB1	P16278	Q9UM11	GLB1L2	Q8IW92	P19526
GLCC1	Q86VQ1	P31946	GLCE	O94923	P40855
GLDC	P23378	Q9BQB4	GLG1	Q92896	O15530
GLI1	P08151	Q13526, Q96J02	GLI2	P10070	P84022, Q15796
GLI3	P10071	Q9UMX1	GLI4	P10075	Q04721
GLIPR1	P48060	Q9BYJ1	GLIPR1L2	Q4G1C9-2	P08670
GLIS2	Q9BZE0	O75496	GLIS3	Q8NEA6-2	Q96RU7
GLMN	Q92990	P15907	GLO1	Q04760	P13569
GLP1R	P43220	Q9HD20	GLRB	P48167	Q6UX41
GLRX2	Q9NS18	Q7Z4W1	GLRX3	O76003	P22301
GLRX5	Q86SX6	Q8IWL3	GLS	O94925	Q9NXA8
GLS2	Q9UI32	Q6GPH4	GLT8D1	Q68CQ7	P48960
GLT8D2	Q9H1C3	P19526	GLTP	Q9NZD2	Q9NQZ5
GLTSCR1	Q9NZM4	Q96GM5	GLTSCR1L	Q6AI39	Q96GM5
GLTSCR2	Q9NZM5	P19525	GLUD1	P00367	Q13418
GLUL	P15104	Q9BZE9	GLYAT	Q6IB77, Q6IB77-2	Q7Z3S9, P0DPK4
GLYR1	Q49A26-4	Q96CV9	GM2A	P17900	P62736
GMCL1	Q96IK5	O43148	GMDS	O60547	O60547
GMFB	P60983	Q86X55	GMPPA	Q96IJ6	Q92917
GMPPB	Q9Y5P6	O15160	GMPPB	Q9Y5P6	O15160
GMPS	P49915	Q93009	GMPPB	Q9Y5P6	O15160
GNA13	Q14344	Q92888	GMPPB	Q9Y5P6	O15160
GNA15	P30679	P11142	GMPPB	Q9Y5P6	O15160
GNAI3	P08754	P13569	GMPPB	Q9Y5P6	O15160
GNAQ	P50148	P49407	GMPPB	Q9Y5P6	O15160
GNAZ	P19086	P54764	GMPPB	Q9Y5P6	O15160
GNB1L	Q9BYB4	P53611	GMPPB	Q9Y5P6	O15160
			GNB1	P62873	Q13371
			GNB2	P62879	Q13371

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GNB2L1	P63244	Q86UK7	GNB4	Q9HAV0	P18827
GNB5	Q96F32	P61981	GNE	Q9Y223	Q969Y2
GNG10	P50151	Q13371	GNG12	Q9UBI6	Q14192
GNG2	P59768	Q99835	GNG3	P63215	Q13371
GNG4	P50150	Q13371	GNL1	P36915	Q9UG63
GNL2	Q13823	P06748	GNL3	Q9BVP2	Q9UKV8
GNL3L	Q9NVN8	Q00987	GNPDA1	P46926	Q9Y303
GNPDA2	Q8TDQ7	Q9Y303	GNS	P15586	Q8NBK3
GOLGA2	Q08379	Q96T21	GOLGA4	Q13439	Q96ES7
GOLGA7	Q7Z5G4	Q8TDQ0	GOLGA7B	Q2TAP0	O14964
GOLIM4	O00461	Q9Y2C2	GOLM1	Q8NBJ4	P11686
GOLPH3	Q9H4A6	Q9UJK0	GOLPH3L	Q9H4A5	P21453
GOLT1A	Q6ZVE7	Q9NRZ7	GON4L	Q3T8J9	P16104
GOPC	Q9HD26	Q6UWP7	GORAB	Q5T7V8	Q96PM5
GORASP1	Q9BQQ3	P30622	GORASP2	Q9H8Y8	P11926
GOSR2	O14653	Q13190	GOT1	P17174	P15056
GOT2	P00505	P13569	GPAA1	O43292	Q12974
GPAM	Q9HCL2	P12236	GPANK1	O95872	Q13190
GPATCH11	Q8N954	Q15696	GPATCH2	Q9NW75-2	Q9C005
GPATCH8	Q9UKJ3	Q9H4G0	GPBP1	Q86WP2	P20618
GPBP1L1	Q9HC44	P61981	GPC1	P35052	O15530
GPC3	Q8IYG2	O76024	GPC4	O75487	O15530
GPC5	P78333	Q13485	GPC6	Q9Y625	P32249
GPD1L	Q8N335	P13501	GPD2	P43304	Q8WTR4
GPHN	Q9NQX3-2	P25786	GPIHBP1	Q8IV16	Q14703
GPKOW	Q92917	P23193	GPM6A	P51674	Q09328
GPM6B	P35803	P31645	GPN1	Q9HCN4	O15160, O95602, P19388, P61218, P62875
GPN2	Q9H9Y4	P62875	GPNMB	Q14956-1, Q14956	P00533
GPR155	Q7Z3F1	P25090	GPR156	Q8NFN8	Q9ULK5
GPR160	Q9UJ42	P17252	GPR161	Q8N6U8	P17612
GPR17	Q13304	Q7Z5K2	GPR183	P32249	O14786
GPR3	P46089	P05067-4	GPR35	Q9HC97	P17931
GPR37	O15354	P29274	GPR37L1	O60883	Q14162
GPR39	O43194	P08908	GPR50	Q13585	P04731, P02795
GPR61	Q9BZJ8	Q9UKG4	GPR89B	P0CG08	O76024
GPRASP2	Q96D09	P08311	GPRC5A	Q8NFJ5	P13569
GPRC5B	Q9NZH0	Q13190	GPRC5C	Q9NQ84	Q9H015
GPRIN1	Q7Z2K8	P21453	GPRIN2	O60269	P31947
GPS1	Q13098	Q9BYZ6	GPS2	Q13227	P18848
GPSM1	Q86YR5-3	O14503	GPX1	P07203	P13569
GPX7	Q96SL4	P19525	GPX8	Q8TED1	Q92730
GRAMD1A	Q96CP6	P68400	GRAMD1B	Q3KR37	P68400
GRAMD1C	Q8IYS0	P48509	GRAMD2	Q8IUY3	Q12982
GRAMD3	Q96HH9	Q12981	GRAMD4	Q6IC98	Q96QG7
GRAP	Q13588	P19474	GRAP2	O75791, O89100	P40818
GRASP	Q7Z6J2	Q9BYF1	GRB10	Q13322-4	P12931
GRB14	O88900	P11362	GRB2	P62993	P13639
GREM1	O60565	P35968	GRIA1	P42261	Q9UIH9
GRIA2	P42262	P46459	GRIA3	P42263	Q9UBS3

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GRIA4	P48058-2	Q8WTS1	GRID1	Q9ULK0	P68871
GRIK2	Q8IY40	P02649	GRIN1	Q05586-1	Q13224
GRIN2A	Q12879	P78352	GRIN2B	Q13224	Q14160
GRIP2	Q9C0E4	Q15223	GRIPAP1	Q4V328	Q9UER7
GRK5	P34947	P25963	GRK6	P43250-2, P43250	P05067
GRM1	Q13255	P52294	GRM2	Q14416	Q9H2H9
GRM3	Q14832	O43791	GRN	P28799	P98160
GRPEL1	Q9HAV7	P07902	GRPEL2	Q8TAA5	P11182
GRXCR1	A8MXD5	O00560	GSDMD	P57764	P29466
GSE1	Q14687	Q92769	GSG1	Q2KHT4	Q12933
GSK3A	P49840	P49841	GSN	P06396	Q8N5V2
GSPT1	P15170	Q92900	GSPT2	Q8IYD1	Q92900
GSTA4	O15217	O95995	GSTCD	Q8NEC7	P40763
GSTK1	Q9Y2Q3	O00142	GSTM3	P21266	Q14493
GSTM5	P46439	Q92917	GSTO2	Q9H4Y5	Q92922
GSTP1	P09211	Q12933	GSTZ1	O43708	Q12933
GTF2A2	P52657	P20226	GTF2E1	P29083	Q9Y6K9
GTF2E2	P29084	P43897	GTF2F1	P35269	Q9Y5B0
GTF2F2	P13984	P19388	GTF2H1	P32780	Q01831
GTF2H2	Q13888	P50613	GTF2H4	Q92759	Q96JC9
GTF2H5	Q6ZYL4	P50613	GTF3A	Q92664	P05067
GTF3C1	Q12789	P06748	GTF3C4	Q9UKN8	P53350
GTF3C5	Q9Y5Q8	P53350	GTPBP1	O00178	P50552
GTPBP10	A4D1E9	P05388	GTPBP2	Q9BX10	P20618
GTPBP3	Q969Y2	Q969Y2	GTPBP4	Q9BZE4	P19525
GTPBP8	Q8N3Z3	P10809	GUCD1	Q96NT3-2	Q13163
GUCY1A2	P33402	Q14160	GUK1	Q16774	Q6DKK2
GUSBP1	Q15486	P07550	GXYLT1	Q4G148	P27824
GXYLT2	A0PJZ3	Q9Y4Z2	GYPC	P04921	Q3KNW5
GYS1	P13807	Q04446, Q9UQK1	H1F0	P07305	O15315
H1FX	Q92522	P06748	H2AFX	P16104	Q01831, P41208
H2AFY	O75367	Q01831	H2AFY2	Q9P0M6	Q01831
H2AFZ	P0C0S5	Q01831	H3F3A	P84243	P46100
H3F3B	P84243	P46100	HAAO	P46952	Q9NUX5
HACD2	Q6Y1H2	P25103	HACD3	Q9P035	P0DTC4
HACE1	Q8IYU2	Q96CV9	HADH	P40939	O00468
HADHA	P40939	O00468	HADHB	P55084	O15530
HAP1	P27695	Q09472	HAPLN2	Q9GZV7	P14373
HAPLN3	Q96S86	P21217	HARS	P12081	P12081
HARS2	P49590	P12081	HAT1	O14929	P04004
HAUS2	Q9NVX0	Q13426	HAUS4	Q8BFT2	Q9BUH8
HAUS6	Q7Z4H7	O60341	HAVCR1	Q96D42	P0DTC2
HAX1	O00165	Q9BRX2	HBA1	P69905	P11142
HBA2	P69905	P11142	HBB	P68871	P11142
HBD	P02042	P32971	HBP1	Q8IV16	Q14703
HBQ1	P09105	P02100, P02042, P68871, P69892	HBS1L	Q9Y450	Q9BRX2
HCCS	P53701	Q96E11	HCFC1	P51610	P08047
HCFC2	Q9Y5Z7	P11142	HCK	P08631-2	Q8N5V2
HCLS1	P14317	P07948	HCN1	O60741	P01024

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HCRTTR1	O43613	P35414	HCRTTR2	O43614	P62937
HCST	Q9UBK5	Q8NAT1	HDAC1	Q13547	Q9UJW3
HDAC10	Q969S8	Q96CV9	HDAC11	Q96DB2	Q9UPY3
HDAC2	Q92769	P53350	HDAC3	O15379	P00558
HDAC4	P56524	Q13761	HDAC5	Q9UQL6	Q13761
HDAC6	Q9UBN7	Q99836	HDAC7	Q9UKV0	Q06413
HDAC9	Q9UKV0	Q06413	HDDC3	Q8N4P3	O00471
HDGF	P51858	P17844	HDGFRP2	Q7Z4V5	P32121, P49407
HDHD3	Q9BSH5	O75593	HDX	Q7Z353	Q9UMX1
HEATR1	Q9H583	O14975	HEATR3	Q7Z4Q2	P08173
HEATR5A	Q86XA9	O75688	HECTD1	Q9ULT8	O75496
HECTD3	Q5T447	Q6NZI2	HEG1	Q9ULI3	P17931
HELB	Q8NG08	P19388	HELLS	Q9NRZ9	P01106
HELQ	Q8TDG4	O43502	HELZ	P42694	Q8IWL3
HEMK1	Q9Y5R4	P28070	HENMT1	Q5T8I9	Q9Y496
HEPH	Q9BQS7-3	O76003	HERC1	Q15751	O95395
HERC3	Q15034	Q9UNU6	HERC5	Q9UII4	P15880
HERPUD1	Q15011	Q86TM6	HERPUD2	Q9BSE4	P30825
HES1	Q14469	Q96EB6	HES4	Q9HCC6	Q00403
HES6	Q96HZ4	Q14469	HES7	Q9BYE0	P47897
HEXB	P07686	P06865	HEXIM2	Q96MH2	P11309
HEY1	Q9Y5J3	Q14469	HEY2	Q9UBP5	Q96EB6
HFE	Q30201	O14672	HGF	P14210, P14210-6, Q08048	P08581
HGFAC	Q04756	O43278	HGH1	Q9BTY7	P08238
HGS	O14964	Q9Y5J6	HHAT	Q5VTY9	O00560
HHATL	Q9HCP6	P15151	HHEX	Q03014	P06730
HHIP	Q96QV1	Q15465	HIATL1	Q5SR56	Q9NRZ7
HIBADH	P31937	P63027	HIC1	Q14526	Q13363
HIC2	Q96JB3	Q13363	HID1	Q8IV36	Q9BZL1
HIF1A	Q16665	P40818	HIF1AN	Q9NWT6	Q16665
HIF3A	Q9Y2N7-4	Q16665	HIGD1C	A8MV81	P02654
HIGD2A	Q9BW72	P40855	HINFP	Q9BQA5	P04637
HINT1	P49773	P13569	HIP1R	O75146	Q9NZQ7
HIPK1	Q86Z02	P04637	HIRA	P54198	Q9NPG3, Q9Y294
HIRIP3	Q9BW71	P68400, P67870	HIST1H1B	P16401	P19338
HIST1H1D	P16402	P17096	HIST1H1E	P10412	P05455
HIST1H2AE	P04908	Q16695	HIST1H2AG	P0C0S8	O95760
HIST1H2BD	P58876	Q01831	HIST1H2BE	P62807	Q96CW1
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HIST1H2BJ	P06899	P10636-8	HIST1H2BK	O60814	Q12824
HIST1H2BL	Q99880	Q96CW1	HIST1H2BO	P23527	Q01831
HIST1H3A	P68431	Q01831	HIST1H3B	P68431	Q01831
HIST1H3D	P68431	Q01831	HIST1H4A	P62805	Q01831
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HIST2H2AC	Q16777	Q08945	HIST2H2BE	Q16778	Q15843

Input	UniProt Id	Interacts with	Input	UniProt Id	Interacts with
HIST3H2A	Q7L7L0	P19404	HIST4H4	P62805	Q01831
HIVEP1	P15822	P84022	HIVEP2	P31629	P63208
HIVEP3	Q5T1R4	P61981	HK1	P19367	P12931
HK2	P52789	P19367	HK3	P52790	P19367
HKDC1	Q2TB90	P19367	HLA-A	P04439	Q16288
HLA-B	P01889	P61769	HLA-C	P10321	P01222
HLA-DMA	P28067	P02686	HLA-DPA1	P20036	Q14118
HLA-DPB1	A0A1C3PI11, P04440	Q96CV9	HLA-DQA1	P01909	O43612
HLA-DRA	P01903	Q9NZJ5	HLA-DRB1	P01911	Q13370
HLA-DRB5	Q30154	O14757	HLA-E	P13747	Q09328
HLA-F	P30511	P61769	HLA-H	P01893	P11142
HLCS	P50747	Q9NUX5	HLF	Q16534	P35638
HM13	Q8TCT9	P04578	HMBX1	Q6NT76	Q13573
HMBS	P08397	Q9NRD5	HMCES	Q96FZ2	P11142
HMGAI	P17096	P17676	HMGAI2	P52927	P06400
HMGGB2	P26583	P48729	HMGGB3	O15347	O00560
HMGCR	P04035	Q9H1C4	HMGCS1	Q01581	P18827
HMGNS	P82970	Q8WVK2	HMHAI	Q92619	P63000
HMOX1	P09601	Q9NUX5	HMOX2	P30519	P0DTC3
HN1	Q9UK76	P06307	HN1L	Q96RY5	Q9Y4X5
HNF1B	P35680	P20823	HNRNPA0	Q13151	P09651
HNRNPA1	P09651	Q8IZH2	HNRNPA1L2	Q32P51	P09651
HNRNPA2B1	P22626	Q9NZI8	HNRNPA3	P51991	Q9NZI8
HNRNPAB	Q99729	Q9NZI8	HNRNPC	P07910	Q9Y6M1
HNRNPD	Q14103-4	Q04637	HNRNPDL	O14979	Q12906
HNRNPH2	P55795	O43347	HNRNPH3	P31942	P09651
HNRNPK	P61978	P12931	HNRNPL	P14866	P09651
HNRNPLL	Q8WVV9	P61978	HNRNPM	P52272	Q99558
HNRNPR	O43390	P67809	HNRNPU	Q00839	P09429
HNRNPUL1	Q9BUJ2	Q9NPI1	HNRNPUL2	Q1KMD3	P67809
HOMER1	Q86YM7	O75928-2	HOMER3	Q9NSC5	P50458
HOMEZ	Q8IX15-3	O43639	HOOK1	Q9UJC3	P53611
HOOK2	Q96ED9-2	Q9NPB3	HOOK3	Q86VS8	Q9NYA4
HOPX	Q9BPY8	Q99081	HOXA7	P31268	O00560
HOXA9	EBI-1801917	Q03164	HOXB13	Q92826	P17252
HOXB7	P09629	P78527	HOXC4	P09017	Q9Y3P9
HOXD10	P28358	O75925	HOXD3	P31249	Q96GM5
HOXD8	P13378	Q5SW96	HP	P00738	P02647
HPCAL1	P37235	Q99685	HPCAL4	Q9UM19	P13726
HPGDS	O60760	Q13370	HPS1	Q92902	Q9C0C7
HR	O43593	P10745	HRAS	P01112	P21359
HRASLS	Q9HDD0	Q15915	HRASLS5	Q96KN8	Q96JH7
HS1BP3	Q53T59	P06396	HS2ST1	Q7LGA3	P53794
HS6ST1	O60243	Q9Y2A9	HS6ST2	Q96MM7	P27824
HSBP1	O75506	O75385	HSBP1L1	C9JCN9	Q9UBN6
HSD17B10	Q99714	P13639	HSD17B11	Q8NBQ5	Q13190
HSD17B12	Q53GQ0	P06396	HSD17B13	Q7Z5P4	O95237
HSD17B14	Q9BPX1	P29372	HSD17B4	P51659	P13569
HSD17B7	P56937	Q05940	HSDL1	Q3SXM5	Q00169

Input	UniProt Id	Interacts with	Input	UniProt Id	Interacts with
HSF1	Q00613	P49137	HSF2	Q03933	P11142
HSF2BP	O75031	P01160	HSF4	Q9ULV5-2	P56693
HSP90AA1	P07900	P12931	HSP90AB1	P08238	Q13546
HSP90AB2P	Q58FF8	P13569	HSP90B1	P14625	P11021
HSPA12A	O43301	Q9Y639	HSPA12B	Q96MM6	Q12933
HSPA14	Q0VDF9	Q8IWL3	HSPA1B	P08107	P27695
HSPA1L	P34931	Q13233	HSPA2	P54652	Q8WUX9
HSPA4	P34932	Q53H54	HSPA5	P11021	P01222
HSPA8	P11142	Q9UNU6	HSPA9	P38646	Q99650
HSPB1	P04792	Q9BRX2	HSPB6	O14558	P04792
HSPB7	Q9UBY9	Q8N0S2	HSPB8	Q9UJY1	P49247
HSPBAP1	Q96EW2-2	P06396	HSPBP1	Q9NZL4	P32519
HSPD1	P10809	Q16822	HSPE1	P61604	P13569
HSPG2	P98160- PRO_0000391621, P98160- PRO_0000391622, P98160	P35968	HSPH1	Q92598	P11142
HTATIP2	Q9BUP3-3	Q8TB40	HTATSF1	O43719	Q9Y580
HTR2C	P28335	Q8TC07	HTR3A	P46098	Q16585
HTR4	P97288	P12931	HTR5A	Q86XA9	O75688
HTRA1	Q92743	P78504	HTRA2	O43464	Q8WXH5
HTRA3	P83110-2, P83110	P60709	HUNK	P57058	P23528, P53667
HUS1	O60921	Q03405	HYAL2	Q12891	P21217
HYAL3	O43820	O75553	HYI	Q5T013	Q05519
HYLS1	Q96M11	Q08379	HYOU1	Q9Y4L1	P11021
HYPK	Q9NX55	P40222	IARS2	Q9NSE4	Q8IWL3
IBTK	Q9P2D0	Q06187	ICAM1	P05362	P15514
ICAM2	P13598	Q9UI95	ICAM5	Q9UMF0	Q9BRX5
ICE1	Q9Y2F5	P55199	ICE2	Q659A1	P55199
ICOS	Q9Y6W8	O75144	ICT1	Q14197	Q00059
ID1	P41134	O14936	ID2	Q02363	Q96HC4
ID3	Q02535	P51149	ID4	P47928	Q9Y4Z2
IDE	P14735-1	P10997	IDH1	O75874	P0DP23
IDH2	P48735	P46059	IDH3B	O43837	P13569
IDH3G	P51553	P10809	IER2	Q9BTL4	Q53GS7
IER3	P46695	Q9UKG4	IER3IP1	Q9Y5U9	P08034
IER5	Q5VY09	Q00613	IFFO2	Q5TF58	P21802
IFI16	Q16666	P38398	IFI27	P40305-2	P22736
IFI35	P80217-2	Q8N2W9	IFIH1	Q9BYX4	P05161
IFIT1	P09914	P09913, O14879	IFIT2	Q8IZ03	O00308
IFIT3	O14879	Q92905	IFITM2	Q6IBB0	Q14973
IFITM3	Q01628	P43490	IFNAR1	P17181	Q13332
IFNAR2	P48551	Q92793	IFNGR1	P15260	Q9NPF0
IFNGR2	P38484	P13861	IFNLR1	Q8IU57	P23458
IFRD2	Q12894	Q9Y6K5	IFT140	Q96RY7	O75386
IFT20	Q8IY31-3	Q13286	IFT22	Q9H7X7	Q3SXY8

Input	UniProt Id	Interacts with	Input	UniProt Id	Interacts with
IFT27	Q9BW83	Q9NQC8, Q8TDR0, A0AVF1, Q9Y547, Q8IY31, Q9NWB7, Q9Y366, Q9H7X7, Q8WYA0, Q96AJ1, Q9P2H3, Q9UG01, Q96LB3, Q13099	IFT43	Q96FT9-2	P17252
IFT52	Q9Y366	Q3SXY8	IFT74	Q96LB3	Q3SXY8
IFT88	Q13099	Q969G3	IGBP1	P78318	P04792
IGDCC4	Q8TDY8	P02788	IGF1	P05019	P08833
IGF1R	P08069	Q9NZN5	IGF2	P01344	Q95749
IGF2BP1	Q9NZI8	P11940	IGF2BP3	O00425	Q9NZI8
IGFBP2	P18065	P59665	IGFBP3	P17936	Q86XT9
IGFBP4	P22692	P05019	IGFBP5	P24593	Q8TB40
IGFBP6	P24592	P53611	IGFBPL1	Q8WX77	P43146
IGFN1	Q86VF2-5	P14373	IGHA1	P01876	Q00403
IGHA2	P01877	O94844	IGHMBP2	P38935	Q12849
IGLL5	B9A064	P30838	IGSF3	O75054	P21926
IGSF6	O95976	Q9HD47	IGSF8	Q969P0	Q969Z0
IGSF9	Q9P2J2-2	O43169	IK	Q13123	Q00403
IKBKAP	O95163	Q15306	IKBKB	O14920	Q9ULZ3
IKBKE	Q14164	P51532	IKBKG	Q9Y6K9	Q13546
IKZF1	Q13422-7	Q99728	IKZF3	Q9UKT9	Q13882
IKZF4	Q9H2S9	P48730	IKZF5	Q9H5V7	P61024
IL10RA	Q13651	P22301	IL10RB	Q08334	P22301
IL12RB1	P42701	P61916	IL12RB2	Q99665	Q14765
IL13RA1	P78552	P24394, P35225, P05112	IL15	P40933	Q13261
IL15RA	Q13261	P40933	IL16	Q14005-2	P52747
IL17RA	Q96F46	Q9NRM6	IL17RB	Q9NRM6	Q96F46
IL17RD	Q8NFM7	Q9NYV6	IL18	Q14116	Q13478
IL18R1	Q13478	Q14116	IL18RAP	O95256	O76024
IL1R1	P14778	Q9Y4K3	IL1R2	P27930	P01583
IL1RAP	Q9NPH3	P01584	IL1RAPL1	Q9NZN1	Q13332
IL1RAPL2	Q9NP60	Q13332	IL1RN	P18510	P14778
IL20RA	Q9UHF4	Q99741	IL20RB	Q6UXL0	Q96HA1
IL27RA	Q6UWB1	P08240	IL2RA	EBI-1566868	P17096
IL2RB	P14784	P60568	IL2RG	P31785	Q9NPF0
IL32	P24001-2, C6GKH1	Q05655	IL34	Q6ZMJ4-1	P07333
IL3RA	P26951-1	P08700	IL4R	P24394	Q99640
IL6	P05231	P05231	IL6R	P08887-PRO_0000450730, P08887	P05231
IL6ST	P40189	P05231	IL7	P13232	P16871
IL7R	P16871	P27449	ILF2	Q12905	Q92900
ILF3	Q12906	Q92900	ILVBL	A1L0T0	O43592
IMMT	Q16891	Q9NVH1, Q9NX63, Q6UXV4	IMP3	O00425	Q9NZI8
IMP4	Q8TCT7	O75063	IMPA2	O14732	Q8TBC4
IMPACT	Q9P2X3	P20618	IMPDH1	P20839	Q8WVC6
IMPDH2	P12268	Q8WWZ1	INA	Q16352	P08670
INADL	Q8NI35	Q8N3R9	INF2	Q27J81	P50548
ING1	Q9UK53	P15559	ING2	Q9H160	Q92769
ING3	Q9NXR8	P52294	ING4	Q9UNL4	Q00994

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ING5	Q8WYH8	Q00994	INHBE	P58166	Q9Y2C2
INO80	Q9ULG1	O96019	INO80C	Q6PI98	P50454
INO80D	Q53TQ3	Q9Y265	INO80E	Q8NBZ0	O96019
INPP1	A0A0S2Z3X1	P20393	INPP5A	Q14642	P53611
INPP5D	Q92835, Q92835-2	O75525	INPP5E	Q10713	P01160
INPP5F	Q9Y2H2	P31946, P63104	INPP5J	Q15735	P61289
INPPL1	O15357	P26641	INSC	Q1MX18	Q96GM5
INSR	P06213	Q9Y4H2	INSRR	P14616	P08238
INTS1	Q8N201	P03372	INTS10	Q9NVR2	P67775
INTS12	Q96CB8	P61073	INTS2	Q9H0H0	O43639
INTS4	Q96HW7	Q15116	INTS5	Q6P9B9	P08173
INTS6	Q9UL03	P22891	INTS8	Q75QN2	P67775
INTS9	Q9NV88	P52292	IP6K1	Q92551	P21802
IP6K2	Q9UHH9	P04637	IP6K3	Q96PC2	Q13114
IPO11	Q9UI26	Q9NRM0	IPO13	O94829	P47813
IPO4	Q8TEX9	P01106	IPO5	O00410	Q05513
IPO8	O15397	Q9UKV8	IPO9	Q96P70	P08173
IPP	Q9Y573	P28482	IQCB1	Q15051	P36941
IQCC	Q4KMZ1	Q14145	IQCE	Q6IPM2	P25788
IQGAP1	P46940	Q16539	IQGAP2	Q13576	P01106
IQSEC2	Q5JU85	P0DP23	IQUB	Q8NA54	Q15669
IRAK1	P51617	Q92985	IRAK2	O43187	Q5D1E8
IRAK3	Q9Y616	Q9Y4K3	IRAK4	Q9NWZ3	Q15109
IRF1	P15314	Q13114	IRF2	P14316	P51610
IRF2BP1	Q8IU81	Q14160	IRF2BP2	Q7Z5L9	Q8N6T7
IRF2BPL	Q9H1B7	P14316	IRF4	Q15306	Q9H6S0
IRF5	Q13568	Q13568	IRF8	Q02556	Q96CV9
IRGQ	Q8WZA9	Q9H2D1	IRS1	P35568	P11940
IRS2	Q9Y4H2	Q96EB6	ISCA1	Q9BUE6	Q96RU8
ISG15	P05161	P41226	ISG20L2	Q9H9L3	P06748
ISL1	P61371	Q9Y4X5	ISL2	Q96A47	Q9Y4X5
ISLR	O14498	Q99741	ISLR2	Q6UXK2, Q5RKR3	P41159
IST1	P53990, P53990-3	P13798	ISYNA1	Q9NPH2	Q9NUX5
ITCH	Q96J02	P08151	ITFG1	Q8TB96	P11686
ITFG2	Q969R8	Q5T2D3	ITGA1	P56199	P17931
ITGA2	P17301	P05556	ITGA3	P26006	P48509
ITGA4	P13612	P02751	ITGA5	P08648	P02751
ITGA6	P23229	P48509	ITGA7	Q13683	Q14192
ITGA8	P53708	Q6UX41	ITGAL	P20701	P08575
ITGAM	P11215	Q3SXY8	ITGAV	P06756	P02751
ITGB1	P05556	P48509	ITGB2	P05107	P00519
ITGB3BP	Q13352	Q969Y2	ITGB4	P16144	P48509
ITGB5	P18084	P04004	ITGB7	P26010, P26010-1	P21333
ITGB8	P26012	O43914	ITIH2	P19823	O15520
ITK	Q08881	P48023	ITM2A	O43736	P05067-4
ITM2B	Q9Y287	Q86XT9	ITM2C	Q9NQX7	P13569
ITPA	Q9BY32	P56705	ITPKB	P27987	Q8N9R8
ITPR1	P11881	P10415	ITPR2	Q14571	Q9H244
ITPR3	Q14573	P40763	ITPRIP	Q8IWB1	Q96FT7
ITPRIPL1	Q6GPH6-2	Q12933	ITSN1	Q15811	P60880

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JADE2	Q9NQC1-2	P01100	JAG1	P78504	P46531
JAGN1	Q8N5M9	Q13190	JAK1	P23458	O60674
JAK2	O60674	O60674	JAK3	P52333	Q13485
JAKMIP1	Q96N16	P20618	JAM2	P57087	Q9BX67
JAM3	Q9BX67	Q9BX67, P57087	JARID2	Q92833	Q09028
JAZF1	Q86VZ6	Q9H2F5	JCHAIN	P01591	O94844
JDP2	Q8WYK2	P18848	JMJD1C	Q15652	P10275
JMJD6	Q6NYC1	O60885	JMY	Q9QXM1	Q09472
JOSD1	Q15040	Q13077	JPH3	Q8WXH2	Q9NR63
JPH4	Q96JJ6	Q9ULK5	JRK	O75564-2	Q96MT3
JSRP1	Q96MG2	O43561-2	JUN	P05412	P29692
JUNB	P17275	P18847	JUND	P17535	P18847
JUP	P14923	P01160	KANK1	Q14678	Q9UQB8
KANK2	Q63ZY3	Q15669	KANK4	Q5T7N3	P41240
KANSL1	Q7Z3B3	Q13077	KANSL2	Q9H9L4	P61964
KARS	Q15046	P13569	KAT2A	Q92830	Q01094
KAT2B	Q92831	Q16665	KAT5	Q92993	P11926
KAT6A	Q92794	P61964	KAT8	Q9H7Z6	P04637
KATNA1	O75449	Q9Y4X5	KATNAL2	Q8IYT4	P03372
KATNB1	Q9BVA0	Q06187	KATNBL1	Q9H079	Q96GM5
KAZN	Q674X7-2	P25786	KBTBD4	Q9NVX7-2, Q9NVX7	P11142, P04792
KBTBD6	Q86V97	Q6NZI2	KCNA1	Q09470, P10499	P78352
KCNA4	P22459	P21673	KCNAB2	Q13303	P05067
KCNC4	Q03721	P56817	KCND2	Q9NZV8	P11142
KCNE3	Q9Y6H6	P40818	KCNF1	Q9H3M0	P35372
KCNH1	O95259	Q7L8L6	KCNH3	Q9ULD8	P04578
KCNIP1	Q9NZI2-2	P40189	KCNIP3	Q9Y2W7	P40189
KCNIP4	Q6PIL6	P08123	KCNJ10	P78508	Q8IUQ4
KCNJ11	Q14654	Q8WTR4	KCNJ14	Q99712	P63252
KCNJ6	P48051	O75063	KCNK1	O00180	P63165
KCNK2	Q13303	P05067	KCNK3	O14649	P31946
KCNK5	O95279	P29033	KCNMA1	Q08460	P46109
KCNN4	O15554	P61073	KCNS2	Q9ULS6	P0DPK4
KCNS3	Q9BQ31	Q13370	KCNT2	Q6UVM3	Q14160
KCTD1	Q719H9	P25786	KCTD12	Q96CX2	Q9UJK0
KCTD13	Q8WZ19	P04792	KCTD15	Q96SI1, Q96SI1-2	P11142, P04792
KCTD16	Q68DU8	P05067-4	KCTD17	Q8N5Z5	P11686
KCTD18	Q6PI47	Q07954	KCTD21	Q4G0X4	Q9UBS4
KCTD3	Q9Y597	P26045	KCTD4	Q8WVF5	Q9UER7
KCTD5	Q9NXV2	Q9UBU9	KCTD7	Q96MP8-2	P25786
KCTD9	Q7L273	O14786	KDELC1	Q6UW63	P0DTC8
KDELC2	Q7Z4H8	P0DTC8	KDELR1	P24390	Q92796
KDM1A	O60341	P46531	KDM1B	Q8NB78	Q92560
KDM2A	Q9Y2K7	P63208	KDM2B	Q8NHM5	P63208
KDM4A	O75164	Q15831	KDM5A	P29375	P06400
KDM5C	P41229	Q13127	KDM5D	Q9BY66-3	P10275
KDM6A	O15550	O43474	KDM6B	Q5NCY0	Q9BVI0
KDR	P35968	P98160	KDSR	Q06136	P04792

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KEAP1	Q14145	Q86YC2	KEL	P23276	P01031
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KHDRBS3	O75525	P40938	KHK	P50053-2	P50458
KHNYN	O15037	Q15843	KHSRP	Q92945	P05412
KIAA0020	Q15397	P40429	KIAA0040	A0A384DVV8	O76024
KIAA0196	Q12768	P52907	KIAA0226	Q92622	O95995
KIAA0226L	Q9H714	Q9P2Y5	KIAA0319L	Q8IZA0	Q8IUQ4
KIAA0355	O15063	P31946, P63104	KIAA0368	Q5VYK3	Q9NRM0
KIAA0408	Q6ZU52	O43639	KIAA0430	Q9Y4F3	P61964
KIAA0556	O60303	Q13099	KIAA0586	Q9BVV6	Q8IW35, O43303
KIAA0825	Q8IV33	Q96CV9	KIAA0930	Q6ICG6	Q14191
KIAA1033	Q2M389	O75396	KIAA1147	A4D1U4	Q9Y337
KIAA1161	Q6NSJ0	P11021	KIAA1191	Q96A73	Q9NUX5
KIAA1211	Q6ZU35	P47756	KIAA1211L	Q6NV74	P47756
KIAA1217	Q5T5P2-6	Q13882	KIAA1324	Q6UXG2-3	P04792
KIAA1328	Q86T90	Q96QG7	KIAA1429	Q69YN4	P31946, P63104
KIAA1467	A2RU67	O43913	KIAA1468	Q9P260	P21709
KIAA1522	Q9P206	Q9UQB8	KIAA1524	Q8TCG1	Q9NZQ7
KIAA1551	Q9HCM1	Q9NV70	KIAA1644	Q3SXP7	P29033
KIAA1671	Q9BY89	P31946	KIAA1683	A0JP07	P27797
KIAA1715	Q9C0E8-2	O95070	KIAA1919	Q5TF39	Q7L8W6
KIAA2013	Q8IYS2	Q15904	KIDINS220	Q9ULH0	P21802
KIF13B	Q9NQ78	P31946, P63104	KIF16B	Q96L93-6	Q9BRT9
KIF1A	Q12756	O76024	KIF1BP	Q96EK5	Q13451
KIF1C	O43896	P31946, P63104	KIF20B	Q96Q89	P52294
KIF21B	O75037	P61981	KIF22	Q14807	Q8TC07
KIF23	Q02241	P31946, P63104	KIF24	Q5T7B8	P51955
KIF27	Q86VH2	Q9P0N9	KIF2A	O00139	P61981
KIF3A	Q9Y496	Q5T819	KIF3B	O15066	P62195
KIF3C	O14782	Q9Y496	KIF5A	Q12840	P31946, P63104
KIF5B	P33176	O14977	KIF5C	O60282	P31946, P63104
KIF9	Q9HAQ2	P08670	KIFAP3	Q92845	Q13177
KIFC3	Q9BVG8-5	Q99728	KIN	O60870	Q9BUU2
KIRREL	Q96J84	Q07157	KIRREL3	Q8IZU9	P15311
KIRREL3-AS3	Q8N7Y1	P25788	KISS1R	Q969F8	P67775
KIT	P10721	Q14451	KITLG	P21583	P10721
KIZ	Q2M2Z5	Q96NL6	KL	O35082	Q9GZV9
KLC1	Q07866-2	Q9Y6A5	KLC2	Q9H0B6	O14977
KLC4	Q9NSK0	P31946, P63104	KLF10	Q13118	P08047
KLF11	O14901	Q8WUX9	KLF12	Q9Y4X4	Q13363
KLF13	Q9Y2Y9	P43694	KLF15	Q9UIH9	P12931
KLF16	Q9BXX1	Q12857	KLF2	Q9Y5W3	Q9HCE7
KLF3	P57682	O00712, Q12857	KLF4	O43474	Q13363
KLF5	Q13887	P0CG48	KLF6	Q99612	P02545
KLF8	O95600	Q12857	KLHDC2	Q9Y2U9	Q9H015
KLHDC3	Q9BQ90	Q15843	KLHDC8A	Q8IYD2	Q13371
KLHDC9	Q8NEP7	O76003	KLHL11	Q9NVR0	Q9NRD5
KLHL12	Q53G59	O95050	KLHL13	Q9P2N7	P01282
KLHL14	Q9P2G3	Q9UF56	KLHL15	Q96M94	P21854
KLHL17	Q6TDP4	Q9UF56	KLHL18	O94889	Q92905

Input	UniProt Id	Interacts with	Input	UniProt Id	Interacts with
KLHL2	O95198	Q9UF56	KLHL21	Q9UJP4	P23025
KLHL23	Q8NBE8	P29279	KLHL24	Q6TFL4	O43639
KLHL26	Q53HC5	Q15915	KLHL28	Q9NXS3	Q9UF56
KLHL3	Q9UH77	P06396	KLHL32	Q96NJ5	P27361
KLHL33	A6NCF5	P10809	KLHL35	Q6PF15	O43639
KLHL36	Q8N4N3	P27695	KLHL41	O60662	Q96PM5
KLHL42	Q9P2K6	P61081	KLHL5	Q96PQ7	Q9NYA1
KLHL6	Q8WZ60	Q13485	KLHL7	Q8IXQ5	P34741
KLHL9	Q9P2J3	Q9UF56	KLKB1	P03952	O00233
KLRD1	Q13241	P26715	KLRG1	Q96E93	Q3SXY8
KLRG2	A4D1S0	Q9NT99	KMO	O15229-2	P42858
KMT2A	Q03164	O00255	KMT2B	Q9UMN6	O00255
KMT2C	Q8NEZ4	P61964	KMT2E	Q8IZD2	P04637
KNOP1	Q1ED39	P06748	KNSTRN	Q9Y448	Q96ES7
KNTC1	P50748	P46059	KPNA2	P52292	Q96C86
KPNA4	O00629	Q9BQB4	KPNA5	O15131	Q96C86
KPNA6	O60684	P03466	KPNB1	Q14974	Q13724
KPTN	Q9Y664	Q9NYZ3	KRBOX4	Q5JUW0	Q13263
KREMEN2	Q8NCW0	P14138	KRI1	Q8N9T8	P40429
KRIT1	O00522	Q15185	KRR1	Q13601	Q9UER7
KRT10	P13645	P13569	KRT14	P02533	Q15149
KRT18	P05783	Q9NUX5	KRT19	P08727	Q9P0N9
KRT24	Q2M2I5	Q96ES7	KRT8	P05787	P08670
KSR1	Q8IVT5, Q61097	P31946, P63104	KSR2	Q6VAB6	Q93008
KTN1	Q86UP2	P51665	KXD1	Q9BQD3	Q6NZI2
L1CAM	P32004	O00533	L1TD1	Q5T7N2	Q13263
L2HGDH	Q9H9P8	O15431	L3HYPDH	Q96EM0	P46939
L3MBTL1	Q9Y468	P33992, P49736	L3MBTL2	Q969R5	Q92769
L3MBTL3	Q96JM7-2	O94955	L3MBTL4	Q8NA19	Q9Y2I6
LACE1	Q8WV93	Q8N9N5	LACTB2	Q53H82	Q8IUQ4
LAGE3	Q14657	P28065	LAIR1	Q6GTX8	P29350
LAMA1	P25391	P17931	LAMA3	Q16787	P13569
LAMA4	Q16363	P04637	LAMA5	O15230	P59665
LAMB1	P07942	P17931	LAMB2	P55268	Q03405
LAMC1	P11047	P17931	LAMC2	Q13753	Q70UQ0
LAMP2	P13473-2	Q8WUX9	LAMTOR1	Q6IAA8	P08670, P06396
LAMTOR2	Q9Y2Q5	P61421	LAMTOR3	Q9UHA4	Q9Y2Q5, Q0VGL1, Q6IAA8, O43504, Q8NBW4
LAMTOR4	Q0VGL1	O15392	LAMTOR5	O43504	P68104
LANCL2	Q9NS86	Q9Y2W6	LAP3	P28838	Q13185, P45973
LAPTM4B	Q86VI4	P21802	LAPTM5	Q13571	Q96PU5
LARGE	O95461	Q6VN20	LARP1	Q6PKG0	P59595
LARP1B	Q659C4-6	Q3MII6	LARP4	Q71RC2	P06748
LARP4B	Q92615	P11940	LARP7	Q4G0J3	P23193
LARS2	Q15031	O95861	LAS1L	Q9Y4W2	P06748
LASP1	Q14847	Q9H2D1	LAT2	Q9UHI5	O60906
LATS1	O95835	P06493	LATS2	Q9NRM7	O95863
LAYN	Q6UX15-2	P24593	LBX1	P52954	P46940
LCA5	Q86VQ0	Q9UMX1	LCAT	P04180	P02647

Input	UniProt Id	Interacts with	Input	UniProt Id	Interacts with
LCK	P06239	P43403, Q9Y2R2	LCLAT1	Q6UWP7	Q9NRD5
LCN2	P80188	Q969S2	LCOR	Q96JN0, Q96JN0-2	Q13363
LCORL	Q8N3X6	Q13363	LCP1	P13796	Q8NG68
LCP2	Q13094	P19174	LDB1	Q86U70, Q86U70-2	P50458
LDB2	O43679	Q96GM5	LDB3	P40107	Q96BA8
LDHA	P00338	P00338	LDHB	P07195	P00338
LDLR	P01130	Q8NBP7	LDLRAD4	O15165	Q96PU5
LDLRAP1	D3ZAR1	P01130	LDOC1	O95751	P48552
LDOC1L	Q6ICC9	P35638	LEF1	Q9UJU2	Q16236
LEMD1	Q68G75	Q15125	LEMD2	Q8NC56	P80370
LEMD3	Q9Y2U8	P16410	LENG1	Q96BZ8	Q9BUH8
LEO1	Q8WVC0	P23193	LEPR	P48357	O43613
LEPROTL1	O95214	Q3SXY8	LETM1	O95202	P25788
LETMD1	Q6P1Q0	P40855	LGALS1	P09382	P78504
LGALS3	P17931	Q58EX2	LGALS3BP	Q08380	P02751
LGALS8	O00214	P13798	LGALS9	O00182	P51790
LGALSL	Q3ZCW2	P50458	LHFPL1	Q86WI0	O75396
LHFPL2	Q6ZUX7	Q14162	LHFPL4	Q7Z7J7	P06576
LHX4	Q969G2	Q9Y5N6	LIFR	P42702	P26441
LIG1	Q96JA1	P22681	LIG4	P49917	P48729
LILRB1	Q8NHL6	P41240	LILRB4	Q8NHJ6	P48960
LIMA1	Q9UHB6	P31946, P63104	LIMCH1	Q9UPQ0-1	Q96JA1
LIMD1	Q9UGP4	Q5K4L6	LIMD2	Q9BT23	P60900
LIMK1	P53667	Q13177	LIMK2	P53671	P53667
LIMS1	P48059	Q9Y3P9	LIMS2	Q7Z4I7-5	Q96PM5
LIN28B	Q6ZN17	Q92900	LIN37	Q96GY3	P10243
LIN52	Q52LA3	P10243	LIN54	Q6MZP7	P10243
LIN7A	O14910	Q8N3R9	LIN7B	Q9HAP6	Q8N3R9
LIN9	Q5TKA1	P10243	LINC-PINT	A0A455ZAR2	Q08050
LINC00493	A0A096LP01	P06493	LINGO1	Q96FE5	P08138
LINGO2	Q7L985	Q9HAU4	LITAF	Q99732	P21802
LLGL1	Q15334	Q05513	LLGL2	Q6P1M3	Q05513
LLPH	Q9BRT6	P62249	LMAN1	P49257	P04578
LMAN2	Q12907	P13569	LMBRD2	Q68DH5	P25103
LMNA	P02545-1, P02545-2, P02545	P20700	LMNB1	P20700	P02545-1
LMNB2	Q03252	P20700	LMO1	P25800	P52747
LMO2	P25791-3	Q13541	LMO3	Q8TAP4-4	Q9BRX2
LMO4	P61968	Q9BRX2	LMO7	Q8WWI1-3	Q96JA1
LMTK3	Q96Q04	P03372	LNPEP	Q9UIQ6	P17931
LNx1	Q8TBB1	Q15056	LNx2	Q8N448	O95050
LONP1	P36776	Q4G176	LONP2	Q86WA8	Q9P0P0
LONRF1	Q17RB8	Q9UQK1	LONRF2	Q1L5Z9	P99999
LONRF3	Q496Y0	P35232	LOR	P23490	O75752
LOX	P28300	Q9UBX5, P15502	LOXL1	Q08397	P06276
LOXL2	Q9Y4K0	Q03405	LPAL2	Q16609	P06307
LPAR1	P61793	Q9NZN5	LPAR2	Q9HBW0	P17931
LPAR3	Q9UBY5	Q3SXY8	LPAR6	P43657	P11532
LPCAT1	Q8NF37	Q13418	LPCAT2	Q7L5N7	Q7Z5P4
LPCAT4	Q643R3	P08173	LPGAT1	Q92604	P22732

Input	UniProt Id	Interacts with	Input	UniProt Id	Interacts with
LPIN1	Q14693	P12931	LPIN3	Q9BQK8	Q86US8
LPL	P06858	O00233	LPP	Q93052	P05120
LPPR2	Q96GM1	Q9BVC3	LPXN	O60711	Q13177
LRAT	O95237	Q7Z5P4	LRBA	P50851	P16410
LRCH1	Q9Y2L9	Q9Y2Z0	LRCH2	Q5VUJ6	Q92905
LRCH3	Q96II8	Q9UM54	LRFN1	Q9P244	P23468, Q13332
LRFN3	Q9BTN0	P13385	LRFN4	Q6PJG9	P23468, Q13332
LRFN5	Q96NI6	P23468, Q13332	LRIF1	Q5T3J3	Q96A70
LRIG1	Q96JA1	P22681	LRIG2	O94898	Q96JA1
LRIG3	Q6UXM1	Q96JA1	LRMP	Q12912	O94901
LRP1	Q07954	P12236	LRP10	O75096	Q9BQB4
LRP1B	Q9NZR2	P05067	LRP2BP	Q9P2M1	P38936
LRP4	O75096	Q9BQB4	LRP5	O75197	Q9BQB4
LRP6	O75581	P98082	LRP8	Q14114	P35968
LRPAP1	P30533	P98155	LRRC1	Q9BTT6	P13569
LRRC14	Q15048	Q92905	LRRC16A	Q5VZK9	Q00613
LRRC20	Q8TCA0	P60709	LRRC25	Q8N386	P21854
LRRC27	Q9C0I9	P09601	LRRC32	Q14392	O60906
LRRC36	Q1X8D7	Q9UMX1	LRRC41	Q15345	Q93034, Q9UBF6
LRRC45	Q96CN5	P13196	LRRC46	Q96FV0	Q9UM11
LRRC47	Q8N1G4	P10636-8	LRRC4C	Q9HCJ2	O75197
LRRC56	Q8IYG6	Q9HD47	LRRC58	Q96CX6	Q93034
LRRC59	Q96AG4	P67809	LRRC6	Q86X45	O14757
LRRC61	Q9BV99	O14964	LRRC73	Q5JTD7	Q9NRD5
LRRC8A	Q8IWT6	O43914	LRRC8B	Q6P9F7	Q13370
LRRC8C	Q8TDW0	P25090	LRRFIP1	Q32MZ4	Q96LB3
LRRFIP2	Q9Y608	P13569	LRRIQ3	A6PVS8	O43639
LRRK1	Q38SD2	P29353	LRRK2	Q5S007	P13639
LRRN2	O75325	P98170	LRRN4	O75427	A1L3X0
LRRTM2	O43300	P01375	LRRTM3	Q86VH5	O95470
LRRTM4	Q86VH4	O14786	LRSAM1	Q6UWE0	P61088
LRWD1	Q9UFC0	Q9Y5N6	LSAMP	Q13449	Q14154
LSM1	O15116	P28074	LSM10	Q969L4	P62306
LSM11	P83369	P62306, P62304	LSM12	Q3MHD2	P10636-8
LSM14A	Q8ND56	P09651	LSM14B	Q9BX40-2	P14136
LSM2	Q9Y333	Q08117	LSM3	P62310	P67809
LSM4	Q9Y4Z0	Q9BRS2	LSM5	Q9Y4Y9	Q9H4B4
LSM7	Q9UK45	P62316, P62306, P14678, P62318	LSMEM2	Q8N112	P21854
LSR	Q86X29	Q96FT7	LSS	P48449	O95070
LST1	Q9Y6L6	Q9NPD5	LTBP1	Q14766	P09238
LTBP2	Q14767	Q9UBX5	LTBP3	Q14767	Q9UBX5
LTBP4	Q8N2S1	Q9UBX5	LTBR	P36941	Q9NPF0
LTF	P02788	Q9HA64	LTN1	O94822	Q9Y5B0
LUC7L3	O95232	P61981	LUM	P51884	P61586
LURAP1	Q96LR2	Q8NFX5	LURAP1L	Q8IV03	O00560
LUZP1	Q86V48	P31946	LVRN	Q6Q4G3-4	O76024
LXN	Q9BS40	Q9UQK1	LY96	Q9Y6Y9	O00206
LYN	P07948	P16671	LYNX1	P0DP58-2	P06396
LYPD1	Q8N2G4	P41221	LYPD5	Q6UWN5	Q01628

Input	UniProt Id	Interacts with	Input	UniProt Id	Interacts with
LYPLA2	O95372	Q9Y496	LYRM2	Q9NU23	O14561
LYRM5	Q6IPR1	P38117	LYRM7	Q5U5X0	Q9H1K1
LYSMD1	Q96S90	Q99081	LYVE1	Q9Y5Y7	Q9BRI3
LYZ	P61626	P61626	LZTFL1	Q9NQ48	P20618, P25786
LZTR1	Q8N653	Q13485	LZTS1	Q9Y250	P30307
LZTS2	Q9BRK4	Q9Y316	LZTS3	O60299	Q13627
M6PR	P20645	Q9NZ52	MAATS1	Q7Z4T9	Q9C005
MAD1L1	Q9Y6D9	O00267	MAD2L1	Q13257	O43511
MAD2L1BP	Q15013	O43511	MAD2L2	Q9UI95	Q9UBZ9
MADD	Q8WVG6	Q9NUX5	MAEA	Q7L5Y9	P01275
MAF	O75444	P18848	MAF1	Q9H063	Q9UJK0
MAFB	Q9Y5Q3	O14867	MAFG	O15525	O15525
MAGEA2B	P43356	P15954	MAGED1	Q9Y5V3	P56693
MAGED2	Q9UNF1	P52333	MAGED4	Q96JG8	Q15185
MAGEE1	Q9HCI5, Q9UBF1	Q13263	MAGEE2	Q8TD90	Q9Y4X5
MAGEH1	Q9H213	P17252	MAGEL2	Q9UJ55	Q9UMX1
MAGI1	Q6P9H4	P42772	MAGI2	Q86UL8	P01135
MAGI3	Q5TCQ9, Q9EQJ9	P01135	MAGT1	Q9H0U3	Q8TCJ2
MAK	P20794	Q9UM11	MAL	P01732	P06493
MAL2	Q969L2	Q969Z0	MALL	Q13021	Q13324-2
MALT1	Q9UDY8	Q14790	MAML3	Q96JK9	P46531
MAN1A1	P33908	P09238	MAN1A2	O60476	Q14627
MAN1C1	Q9NR34	O95363	MAN2A1	Q16706	Q03405
MANBA	O00462	P08311	MANBAL	Q9NQG1	Q12893
MAOB	P27338	P06396	MAP1A	P78559	Q9GZQ8
MAP1LC3A	Q9H492	Q8IVP5	MAP1LC3B	Q9GZQ8	O75385
MAP1S	Q66K74	O14543	MAP2	P11137	P62993
MAP2K2	P36507	P11142	MAP2K3	P46734	Q16539
MAP2K4	P45985	Q13233	MAP2K5	Q13163	Q13164
MAP2K6	P52564	Q16539	MAP2K7	O14733	O43318
MAP3K1	Q13233	P32121	MAP3K11	Q16584	P27986
MAP3K12	Q12852	P00533	MAP3K14	Q99558	O15111, O14920
MAP3K2	Q9Y2U5	Q13163	MAP3K3	Q99759	Q13163
MAP3K4	Q9Y6R4	Q06187	MAP3K5	Q99683	P25445
MAP3K7	O43318	O15111, O14920	MAP3K7CL	P57077-4	P22415
MAP3K8	P41279	P19838, Q8NFFZ5	MAP3K9	P80192	P17252
MAP4K2	Q12851	Q9NUX5	MAP4K3	Q924I2	Q99962
MAP4K4	O95819	Q9H0R5	MAP4K5	Q9Y4K4	P62993
MAP7	Q14244	P61981	MAP7D1	Q3KQU3	P01106
MAP7D3	Q8IWC1	P61981	MAPK1	P28482	O75330
MAPK10	P53779	P49407	MAPK11	Q15759	P49137
MAPK12	P53778	Q14160	MAPK14	Q16539	P13639
MAPK3	P27361	P19419	MAPK4	P31152	Q96HN2
MAPK6	Q16659	P02768	MAPK7	Q13164	Q13163
MAPK8	P45983-4	Q9BY84	MAPK8IP1	Q9UQF2	O43318
MAPK8IP2	Q13387-4	P05556	MAPK8IP3	Q9UPT6	Q5SW96
MAPK9	P45984	P45983	MAPKAP1	Q9BPZ7	P31749
MAPKAPK2	P49137	Q16539	MAPKAPK5	Q8IW41	Q15109
MAPRE1	Q15691	P30622	MAPRE3	Q9UPY8	P10636
MAPT	P10636-8	P13639	MARCKS	P29966	P13569

Input	UniProt Id	Interacts with	Input	UniProt Id	Interacts with
MARCKSL1	P49006	O60551	MARK2	Q7KZI7	P46939
MARK3	P27448	P11309	MARK4	Q96L34	Q15831
MARS	P56192	P01106	MARVELD2	Q8N4S9	Q8NG68
MASP1	P48740-2	P17927	MAST1	Q9Y2H9	O14734
MAT2A	P31153	Q9BRX2	MAT2B	Q9NZL9	O60733
MATK	P42679	P41240	MATN2	O00339	Q96ES7
MATN3	O15232	P02787	MAU2	Q9Y6X3	P45973
MAZ	P56270	P43146	MB21D1	Q8N884	P09874
MB21D2	Q8IYB1	Q96QF0	MBD1	Q9UIS9	Q92769
MBD2	Q9UBB5	P26358	MBD3	O95983-2	Q9Y5J6
MBD4	O95243-2	P11245	MBNL1	Q86VM6	Q08117
MBNL3	Q9NUK0	O75553	MBOAT7	Q96N66	Q86WV6
MBP	P13727	Q12946	MBTD1	Q05BQ5	Q9H2F5
MCAM	P43121	Q12860	MCAT	Q8IVS2	Q00839
MCC	P23508	P51955	MCCC1	Q96RQ3	P49411
MCCC2	Q9HCC0	P13569	MCEE	Q96PE7	P11182
MCF2L	Q6PDM6	P61586	MCFD2	Q8NI22	P78382
MCL1	P97287, Q07820	Q9BXH1	MCM2	P49736	P03433
MCM4	P33991	P25205, P49736	MCM5	P33992	P25205, P49736
MCM7	P33993	P33992, P25205, P49736	MCM9	Q9NXL9	P52701, P20585
MCOLN1	Q9GZU1	Q9GZU1	MCOLN2	Q8IZK6	Q9GZU1
MCOLN3	Q8TDD5	Q86XT9	MCPH1	Q8NEM0	Q01094
MCRS1	Q96EZ8	Q9UER7	MCU	Q8NE86-1, Q8NE86	Q9H4I9, Q9BPX6
MCUR1	Q96AQ8	P23786	MDC1	Q14676	Q06609
MDFI	Q99750	P43897	MDH1	P40925	P48163
MDH2	P40926	P13569	MDK	P21741	P42568, Q9UHB7
MDM2	Q00987	Q15648	MDM4	O15151	P06400
MDN1	Q9NU22	Q9Y639	MDP1	P54098	P00480
ME1	P48163	P48163	MEA1	Q16626	Q96CW1
MEAF6	Q9HAF1	P40937	MECOM	Q03112	Q13363
MED10	Q9BTT4	Q15648	MED11	Q9P086	Q15648
MED12L	Q86YW9	P61024	MED13	Q9UHV7	Q15648
MED13L	Q71F56	Q15648	MED14	O60244	Q15648
MED16	Q9Y2X0	Q15648	MED18	Q9BUE0	Q15648
MED19	A0JLT2-2	P50747	MED20	Q9H944	Q15648
MED21	Q13503	Q15648	MED22	Q15528-2	O76024
MED23	Q9ULK4	Q15648	MED24	O75448	Q15648
MED25	Q9NWA0, Q71SY5	Q15648	MED27	Q6P2C8	Q15648
MED28	Q9H204	Q15648	MED29	Q9NX70	Q15648
MED31	Q9Y3C7	Q15648	MED4	Q9NPJ6	Q15648
MED8	Q96G25	Q15648	MED9	Q9NWA0	Q15648
MEDAG	Q5VYS4	Q9H0E2	MEF2A	Q02078-6	Q13164
MEF2C	Q06413	P56524	MEFV	O15553	Q9ULZ3
MEGF6	O75095	P00533	MEGF8	Q7Z7M0	P22692
MEI4	A8MW99	Q96QG7	MEIS1	O00470	P40424
MEIS3	Q99687-3	Q96BZ8	MEN1	O00255-2, O00255	P35579
MEOX1	P50221	Q969S2	MEOX2	Q6FHY5	P01236
MESDC1	Q9H1K6	Q9NRD5	MEST	Q5EB52	P13569
MET	P08581	O43157	METAP2	P50579	Q93034
METTTL13	Q8N6R0	Q12933	METTTL15	A6NJ78	P46059

Input	UniProt Id	Interacts with	Input	UniProt Id	Interacts with
METTL16	Q86W50	P53701	METTL17	Q9H7H0	Q9UBX5
METTL20	Q8IXQ9	P10809	METTL22	Q9BUU2	O60870
METTL3	Q86U44	P31689	METTL6	Q8TCB7	Q12948
METTL7A	Q9H8H3	P06396	METTL9	Q9H1A3	P27824
MEX3B	Q6ZN04	P61077	MEX3C	Q5U5Q3	Q93009
MFAP1	P55081	Q96CW1	MFAP3	P55082	P43119
MFAP4	P55083	Q16625	MFGE8	Q08431	P0DTC8
MFN1	Q8IWA4	Q16611	MFN2	O95140, Q80U63	Q16611
MFSD10	Q14728	P37173	MFSD11	O43934	P15151
MFSD3	Q96ES6	Q8N4V1	MFSD4	Q8N468	O43592
MFSD5	Q6N075	P25103	MFSD6	Q6ZSS7	P29033
MFSD8	Q8NHS3	Q9H244	MFSD9	Q8NBP5	P35414
MGA	Q8IWI9	P17980	MGAM	O43451	Q96BA8
MGAT2	Q10469	P40855	MGAT4B	Q9UQ53	Q96CV9
MGAT4C	Q9UBM8	O75197	MGAT5	Q09328	Q03405
MGLL	Q99685	P07550	MGRN1	O60291	P14138
MGST1	P10620	P01133	MGST2	Q99735	Q16585
MGST3	O14880	Q00059	MIB1	Q86YT6	Q9UHD2
MIB2	Q96AX9	Q9UHD2	MICA	Q29983	P17931
MICAL2	O94851	P14136	MICAL3	Q7RTP6, Q7RTP6-1	Q02241
MICALCL	O94851	P14136	MICALL1	Q8N3F8	P31946
MICALL2	Q8IY33	P12814	MICB	Q29980	P17302
MICU1	Q9BPX6	Q9H4I9, Q9BPX6, Q8NE86	MICU2	Q8IYU8	Q9H4I9, Q9BPX6, Q8NE86
MID1	P29372	P04156	MID1IP1	Q9NPA3	P16473
MID2	Q9UJV3-2	P51668	MIDN	Q504T8	P22736
MIEF1	Q9NQG6	Q9UBN6	MIEF2	Q96C03-3	O95070
MIEN1	Q9BRT3	O95231	MIER1	Q8N108-16	Q15109
MIER2	Q8N344	Q92769	MIER3	Q7Z3K6	Q92769
MIF	P14174	Q92905	MIF4GD	A9UHW6	Q14493
MIIP	Q5JXC2	P06396	MILR1	Q7Z6M3	Q92538
MINA	Q8IUF8	P06748	MINK1	Q8N4C8	O43639
MINOS1	Q5TGZ0	P55056	MIPEP	Q99797	Q9BQP7
MIPOL1	Q8TD10	Q8WY64	MIS12	Q9H081, Q9CY25	Q13185, P45973
MIS18BP1	Q6P0N0	P56539	MITF	O75030	P61981
MKI67	P46013	P53667	MKKS	Q9NPJ1	P26641
MKL1	Q969V6	P11831	MKL2	Q9ULH7	P11831
MKLN1	Q9UL63	Q13685	MKNK1	Q9BUB5	Q04637
MKRN1	Q9UHC7	P51668	MKRN3	Q13064	P51668
MKS1	Q9NXB0	O95816	MKX	Q8IYA7	Q9BZK7
MLC1	Q15049	Q00403	MLF2	Q15773	Q9UNE7
MLH3	Q12083	O14641	MLK4	Q5TCX8	O15111
MLKL	Q8NB16	Q9Y572	MLLT1	Q03111	P04608
MLLT10	P55197	P29692	MLLT11	Q13015	P04271
MLLT3	P42568	Q9UHB7, Q9JYC9, Q03111	MLPH	Q9BV36	P51159
MLST8	A0A0A0MR05	P99999	MLX	Q9UH92	Q08117
MLXIP	Q9HAP2	P61981	MMAB	Q96EY8	P11182
MMADHC	Q9H3L0	Q9P0N9	MMD	Q15546	Q3SXY8
MMD2	Q8IY49-2	Q96BA8	MME	P08473	A0PJW6
MMP1	EBI-9996975	P17676	MMP10	P09238	P01033

Input	UniProt Id	Interacts with	Input	UniProt Id	Interacts with
MMP14	P50281	P09958	MMP2	P08253	Q8NBP7
MMP28	Q9H239	P09958	MMP3	P08254	P09238
MMRN1	Q13201	P02649	MMS19	Q96T76	P51530
MMS22L	Q6ZRQ5	Q08945	MNAT1	P51948	P28715
MNDA	P41218	P08581	MNS1	Q8NEH6	P30281
MNT	Q99583	O43639	MOB1A	Q9H8S9	Q13188, Q13043
MOB1B	Q7L9L4	Q13188, Q13043	MOB3B	Q86TA1	Q13188
MOB4	Q9QYW3	Q05193	MOBP	Q13875-3	O00560
MOCOS	Q96EN8	Q9NVD7	MOCOS1	Q9NZB8	Q9BQ90
MOCOS3	O95396	O95076	MOGS	Q13724	P56182
MON2	Q7Z3U7	P21453	MORC2	Q9Y6X9	Q14562
MORC3	Q14149	Q9NWF9	MORC4	Q8TE76	P40763
MORF4L1	Q9UBU8-2, Q9UBU8	Q86YC2	MORF4L2	Q15014	Q86YC2
MORN4	Q8NDC4	O75330	MOSPD2	Q8NHP6	P48023
MOSPD3	O75425	Q3SXY8	MOV10	Q9HCE1	Q92900
MOXD1	Q6UVY6	O76003	MPC2	O95563	Q5TGZ0
MPDZ	O75970	Q8N3R9	MPG	P29372	P04156
MPHOSPH10	O00566	P06748	MPHOSPH8	Q99549	Q15047
MPHOSPH9	Q99550	P61981	MPLKIP	Q8TAP9	Q9ULX6
MPND	Q8N594	P04792	MPO	P05164	Q9UNE7
MPP2	Q08050	Q13416	MPP3	Q13368	Q9HAP6
MPP5	Q8N3R9, Q8N3R9-1	P59637	MPP6	Q99547	Q13901, P42285, Q01780, Q9NQT4
MPP7	Q13112	Q04637	MPPED1	O15442	Q9NWZ5
MPPED2	Q15777	Q9Y244	MPST	P25325	O96006
MPZL1	O95297	Q06124	MR1	Q95460-1	P61769
MRAP	Q8TCY5, Q8TCY5-1	Q01718	MRAP2	Q96G30	Q01718
MRC1	Q61830	P0DTC2	MRC2	Q9UBG0	P17931
MRE11A	P49959	P04637	MRFAP1	Q9Y605	Q9H2K0
MRFAP1L1	Q96HT8	Q969G3	MRGBP	Q9NV56	O96019
MRPL10	Q7Z7H8	Q14197	MRPL11	Q9Y3B7	Q13371
MRPL12	P52815	P43897	MRPL15	P49406	P40429
MRPL16	Q9NX20	P67809	MRPL17	Q9NRX2	P67809
MRPL18	Q9H0U6	Q14197	MRPL19	P49406	P40429
MRPL2	Q9BZE1, Q5T653	Q14197	MRPL20	Q9BYC9	P40429
MRPL21	Q7Z2W9	Q9BQ52	MRPL23	Q16540	P67809
MRPL24	Q96A35	P62241, P62753	MRPL27	Q9P0M9, Q8IXM3	Q14197
MRPL28	Q13084	Q14197	MRPL3	P09001	Q14197
MRPL30	Q8TCC3	O00425	MRPL32	Q9BYC8	P62241
MRPL34	Q9BQ48	P62241	MRPL35	Q9NZE8	Q9HD23
MRPL37	Q9BZE1	Q14197	MRPL38	Q96DV4	Q9NPA3
MRPL4	Q9BYD3	Q14197	MRPL40	Q9NQ50	P67809
MRPL41	Q8IXM3	Q14197	MRPL43	Q8N983-3	P17252
MRPL44	Q9H9J2	P08670	MRPL45	Q9BRJ2	Q6NZI2
MRPL46	Q9H2W6	Q9BQP7	MRPL47	Q9HD33	Q14197
MRPL49	Q13405	P67809	MRPL50	Q8N5N7	Q14197
MRPL51	Q4U2R6	Q14197	MRPL52	Q86TS9	Q14197
MRPL53	Q96EL3	Q15370	MRPL57	Q9BQC6	P62241
MRPL9	Q9BYD2	P21673	MRPS11	P82912	Q00839

Input	UniProt Id	Interacts with	Input	UniProt Id	Interacts with
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MRPS15	P82914	P62753	MRPS16	Q9Y3D3	Q9H9Z2
MRPS18B	Q9Y676	P06400	MRPS18C	Q9Y3D5	Q9UPR3
MRPS2	Q9Y399	Q00839	MRPS21	P82921	P84103
MRPS23	Q9Y3D9	O15160	MRPS24	Q96EL2	P36776, P10809
MRPS25	P82663	P62753	MRPS26	Q9BYN8	P48431
MRPS28	Q9Y2Q9	P67809	MRPS30	Q9NP92	Q14197
MRPS33	Q9Y291	Q00839	MRPS34	P82930	O00425
MRPS35	Q9Y2Q9	P67809	MRPS6	P82932	P42229
MRPS7	Q9Y2R9	P67809	MRS2	Q9HD23	Q8N5V2
MRV11	Q9Y6F6-3	O76024	MS4A14	Q96JA4	Q8N661
MS4A4A	Q96JQ5	O15342	MS4A7	Q9GZW8	P21854
MSANTD3	Q96H12	P06396	MSANTD4	Q8NCY6	P63211
MSH3	P20585	P43246	MSH4	O15457	O43196
MSH6	P52701	P43246	MSI1	O43347	O43347
MSI2	Q96DH6	Q9Y5J6	MSL1	Q68DK7	O75496
MSL2	Q9HCI7	P61981	MSL3	Q8N5Y2	O76024
MSMO1	Q15800	Q6NT55	MSN	P26038	P05919
MSR1	P21757	P27449	MSRB1	Q9NZV6	P14373
MSRB2	Q9Y3D2	P14136	MSRB3	Q8IXL7	Q9UKN5
MSS51	Q4VC12	P55040	MSTO1	Q9BUK6	Q13304
MT-ATP8	P03928	P06576	MT-CO1	P00395	P25103
MT-CO2	P00403	P15954	MT-CO3	P00414	P37840
MT-CYB	P00156	O14949, P14927	MT-ND1	P03886	P43490
MT-ND2	P03891	O75380	MT-ND4	P03905	P19404, O75380, O75489, O43920
MT-ND5	P03915	P25103	MT-ND6	P03923	O75489
MT-RNR2	Q8IVG9	P17936	MT1F	P04733	P46087
MT2A	P02795	P09429	MTA1	Q13330	Q16665
MTA2	O94776	Q92769	MTA3	Q9BTC8	Q92769
MTBP	Q96DY7	P49336	MTCH1	Q9NZJ7	Q92611
MTCH2	Q9Y6C9	P56705	MTDH	Q86UE4	Q9UKV8
MTERF4	Q7Z6M4	Q6ZPD8	MTFP1	Q9UDX5	Q96CV9
MTFR1	Q15390	Q96PC5	MTG1	Q9BT17	Q9NRD5
MTHFD1	P11586	P13569	MTHFD1L	Q6UB35	Q96GW9
MTHFD2	P13995	P49137	MTHFD2L	Q9H903	Q5T8I9
MTHFR	P42898	A0PJW6	MTM1	Q13496	Q9C0I1
MTMR10	Q9NXD2	Q13614	MTMR12	Q9C0I1	Q13614, Q13496
MTMR3	Q13615	P31946	MTMR4	Q9NYA4	P84022, Q15796
MTMR8	Q96QG7	Q13614	MTMR9	Q96QG7	Q13614
MTR	Q99707	Q9Y4U1	MTSS1L	Q765P7	P60709
MTUS1	Q9ULD2-3	Q13190	MTX1	Q13505	Q05940
MUL1	Q969V5	O60443	MUM1	Q9H6H2	P25788
MURC	Q5BKX8	Q13895	MUS81	Q96NY9	P39748
MUT	P22033	P42858	MVK	Q03426	Q9NUX5
MVP	Q14764	P13569	MX1	P20591	Q13507
MX2	P20592	Q96KQ7	MX11	P50539-3	Q7Z3S9, P0DPK4
MXRA8	Q9BRK3	Q9NPB3	MYADM	Q96S97	Q9Y624
MYBBP1A	Q9BQG0	P01106	MYC	P01106	P17480
MYCBP2	O75592	P63208	MYCN	P04198	Q8N726

Input	UniProt Id	Interacts with	Input	UniProt Id	Interacts with
MYD88	Q99836	Q15109	MYDGF	Q969H8	Q14554
MYEOV2	Q8WXC6	Q92905	MYH1	P12882	Q16665
MYH10	P35580	P61981	MYH11	P35749	P13569
MYH14	Q7Z406	O15111	MYH2	Q9UKX2	Q16665
MYH3	P11055	Q16665	MYH7	P12883	Q15475
MYH7B	A7E2Y1-2	Q16134	MYH9	P35579	O00255
MYL1	P02602	P04792	MYL12B	O14950	P60709
MYL3	P08590	P35579	MYL6	P60660	O43809
MYL6B	P14649	P35579	MYL9	P24844	P35579
MYLIP	Q8WY64	Q9NQT4	MYLK3	Q32MK0	P53778
MYLK4	Q86YV6	P08238	MYO10	Q9HD67	P43146
MYO15B	Q96JP2	O43474	MYO1C	Q12965	P50570
MYO1D	O94832	P43251	MYO1E	Q12965	P50570
MYO1G	B0I1T2	P13569	MYO3B	Q8WXR4	P08238
MYO5B	Q9ULV0	P40855	MYO5C	Q9NQX4	O75496
MYO6	Q9UM54	P98082	MYO9A	B2RTY4	Q8TC07
MYOC	Q99972	P09486	MYOCD	Q8IZQ8	P11831
MYOM1	P52179-2	Q9UBB9	MYOM2	P54296	O75923
MYOT	A0A0S2Z4Y0	P12814	MYOZ1	Q9NP98	P12814
MYOZ2	Q9NPC6	P12814	MYOZ3	Q8TDC0	Q13882
MYPOP	Q86VE0	P21673	MYRIP	Q8NFW9	P51159
MYT1	Q99640	P06493	MZT2A	Q6P582	O00571
MZT2B	Q6NZ67	Q9Y6L6	N4BP1	O75113	Q96J02
N4BP2L2	Q92802	Q9Y3C6	N4BP3	O15049	P40937
NAA10	P41227	Q96FF9	NAA15	Q9BXJ9	Q8WVK2
NAA16	Q6N069-4	P14136	NAA20	P61599	P28908
NAA40	Q86UY6	Q01831	NAA50	Q9GZZ1	Q8WVK2
NAA60	Q9H7X0	P55212	NAALAD2	Q9Y3Q0	O76024
NAB2	Q15742	P40937	NABP1	Q96AH0	Q5VWX1
NABP2	Q9BQ15	Q06609	NACA	Q13765	P05067
NACAD	O15069	P42858	NACC1	Q96RE7	Q99633
NADK	O95544	P31946, P63104	NADSYN1	Q6IA69	Q9UNE7
NAF1	Q15025	Q00403	NAGK	Q9UJ70	P33992
NAGLU	P54802	P27824	NAMPT	P43490	P43490
NANOS3	P60323-2	P20618	NAP1L2	Q9ULW6	P42858, Q96HR3
NAP1L3	Q99457	P14136	NAP1L4	Q99733	P04608
NAP1L5	Q96NT1	P18848	NAPA	P54920	O95249, Q13190, O15498
NAPB	Q9H115	Q13190	NAPEPLD	Q6IQ20	Q6IQ20
NAPG	Q99747	P46459	NARF	Q8WVD3	P07550
NARFL	Q9H6Q4	P27540	NARS	O43776	Q8WVK2
NASP	P49321, P49321-2	Q9Y294	NAT14	Q8WUY8	P40855
NAT8L	Q8N9F0	P40855	NATD1	Q8N6N6	Q12933
NAV2	Q8IVL1	P63104	NBAS	A2RRP1	P13569
NBEAL1	Q6ZS30	Q9Y639	NBL1	P41271-2	P04271
NBN	O60934	Q13315	NCALD	P61601	P23352
NCAPG	Q9BPX3	P13569	NCAPH2	Q6IBW4	P61073
NCBP2	P52298	Q92900	NCEH1	Q6PIU2	P27824
NCF2	P19878	P63000	NCK1	P16333	P35968
NCKAP1	Q9Y2A7	Q9UQB8	NCKAP5	O14513	P25054

Input	UniProt Id	Interacts with	Input	UniProt Id	Interacts with
NCKAP5L	Q9HCH0	Q9NV70	NCL	P19338	P19338
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NCOA2	Q15596	O00482-1	NCOA3	Q9Y6Q9	Q09472
NCOA4	Q13772	O95714	NCOA5	Q9HCD5	Q16629
NCOA7	Q8NI08	Q9GZQ8	NCOR2	Q9Y618	Q13573, Q06330
NCR3LG1	Q68D85	P47813	NCS1	P62166	Q9UKN5
NCSTN	Q92542	P27824	NDC1	Q9BTX1	P61073
NDC80	O14777	Q9H211	NDE1	Q9NXR1, Q9CZA6	P43034
NDEL1	Q9GZM8	P23025	NDN	P25233	P08138
NDP	Q00604	Q9ULV1	NDRG1	Q92597	P61981
NDRG2	Q9UN36	P03372	NDRG4	Q9ULP0-2	Q9H2K0
NDUFA1	O15239	O75380	NDUFA11	Q86Y39	O75489
NDUFA12	Q9UI09	P19404, O75380, O75489, P03915, O43920, P56181, O43181	NDUFA13	Q9P0J0	P23786
NDUFA2	O43678	P19404, O75380, Q9UI09, O75489, P49821, O43920, P56181, O43181	NDUFA3	O95167	P01019
NDUFA4	O00483	Q9UI09	NDUFA5	Q16718	P61981
NDUFA6	P56556	Q8IWL3	NDUFA7	O95182	P49638
NDUFA8	P51970	P19404, O75380, P03886, O95139, Q9UI09, O75489, P03915, P03905, O43181	NDUFA9	Q16795	P19404, O75380, O95139, Q9UI09, O75489, P49821, P03915, O43920, P56181, O43181
NDUFAB1	O14561	P26441	NDUFAB1	Q9Y375	Q16625
NDUFAB2	Q8N183	O75396	NDUFAB3	Q9BU61	Q9NPB3
NDUFAB4	Q9P032	O75489, Q9NPL8, O43920, Q9BU61	NDUFB1	O75438	P19404, O75380, O75489
NDUFB10	O96000	P05455	NDUFB11	Q9NX14	P08034
NDUFB2	Q9Y6T4	O00311	NDUFB3	O43676	P19404, O75380, Q9UI09, O75489
NDUFB4	O95168	P19404, O75380, O75489, O43920	NDUFB6	O95139	P19526
NDUFB7	P17568	Q5TAB7	NDUFB8	O95169	P25103
NDUFB9	Q9Y6M9	P19404, O75380, Q9UI09, O75489, O43920	NDUFC2	O95298	P19404, O75380, Q9UI09, Q9NPL8, Q9P032, P49821, P03905, P56181, O43181
NDUFS2	O75306	Q99650	NDUFS3	O75489	P17152
NDUFS5	O43920	P03886, O95139, Q9UI09, O75489, Q9P032, P03915, P03905, P19404, O75380, Q9NPL8, P49821, P56181, O43181	NDUFS6	O75380	P25103
NDUFS7	O75251	Q8NE62	NDUFS8	O00217	Q8IWL3
NDUFV1	P49821	P20618, Q16401	NDUFV2	P19404	Q8IWL3
NEB	P20929	Q07889	NEBL	O76041	P14136
NECAB1	Q8N987	O00244	NECAB2	Q7Z6G3-2	Q00994
NECAP1	Q8NC96	Q96CW1	NECAP2	Q9NVZ3	P42566
NEDD1	Q8NHV4	Q8IW35, O43303	NEDD4	P46934-4	Q9HAU4
NEDD8	Q15843	P68104	NEDD9	Q14511	Q8IZD9
NEFH	P12036	P05067	NEFL	P07196	Q9BRX2
NEFM	P07197	P06400	NEGR1	Q7Z3B1	P61916
NEIL2	Q969S2	P14373	NEK1	Q96PY6	P04004

Input	UniProt Id	Interacts with	Input	UniProt Id	Interacts with
NEK11	Q8NG66	P51955	NEK6	Q9HC98-4	P12830
NEK7	Q8TDX7	Q9ULW0	NEK8	Q8TD19	P53350
NEK9	Q8TD19	P53350	NELFB	Q8WX92	P38398
NELFCD	Q8IXH7	Q8WX92	NELFE	P18615	Q8WX92
NELL2	Q99435	Q6IPR3	NEMP1	O14524-2	Q14162
NENF	Q9UMX5	Q14554	NES	P48681	P08670
NET1	Q7Z628	O43639	NEU1	Q99519	Q96BA8
NEURL4	Q96JN8	O43303	NEXN	Q0ZGT2	P60709
NF1	P21359	P31946	NFAT5	O94916	P17844, Q92841
NFATC1	O95644	P18846	NFATC2	Q13469-2	P05412
NFATC2IP	Q8NCF5-2	O96015	NFATC3	Q12968	Q9UMX1
NFATC4	Q14934	P0CG48	NFE2L1	Q14494	O15525
NFE2L2	Q16236	O15525	NFE2L3	Q9Y4A8	O15525
NFIA	Q12857	P50458	NFIB	O00712	P50458
NFIC	P08651	O00712	NFIX	Q14938	O00712
NFKB1	P19838- PRO_0000030311	O00255	NFKB2	Q00653	P41279, P19838
NFKBIA	P25963	Q15813	NFKBIB	Q15653	P19838
NFKBID	Q8NI38	P01189	NFKBIE	O00221	P19838
NFU1	Q9UMS0	Q9UBB9	NFXL1	Q6ZNB6	Q96FT7
NFYA	P23708	P08047	NGB	Q9NPG2	P14635
NGEF	Q8N5V2	P06396	NGF	P01138- PRO_0000019600, P01138	P08138
NGFR	P07174, P08138	P01138	NGFRAP1	Q00994	Q00994
NGRN	Q9NPE2	O00425	NHLH1	Q5T203	P11309
NHLH2	Q02577	Q99081	NHLRC2	Q8NBF2-2	P04792
NHLRC3	Q5JS37	O00253	NHP2	Q9NX24	O60832, Q9NPE3
NHS	Q6T4R5	O43639	NHSL1	Q5SYE7	O43639
NHSL2	Q5HYW2	P53350	NICN1	Q9BSH3	P0CG47
NID2	Q8IV28	P10745	NIF3L1	Q9GZT8	P42772
NIFK	Q9BYG3	P06748, Q9NS56	NIN	Q8N4C6	P49841
NINJ1	Q92982	Q16585	NINJ2	Q9NZG7	P17302
NINL	Q9Y2I6	Q7L590	NIP7	Q9Y221	P46087
NIPAL1	Q6NVV3	Q09328	NIPAL3	Q6P499	P27449
NIPBL	Q6KC79	P45973	NIPSNAP1	Q9BPW8	Q13501
NIPSNAP3A	Q9UFN0	P10809	NISCH	Q9Y2I1	P03496
NKAP	Q8N5F7	P61981	NKD1	Q969G9	P67775
NKILA	EBI-10826665	P25963	NKRF	O15226	P06748
NKTR	P30414	P61981	NKX2-5	P52952	P43694
NLE1	Q9NVX2	Q07020	NLGN2	Q8NFZ4	Q8N0Z6
NLK	Q9UBE8	O43318	NLN	Q9BYT8	Q5SW96
NLRC4	Q9NPP4	P29466, Q9NPP4	NLRC5	Q86WI3	O15111, O14920
NLRP1	Q9C000	P29466	NLRP12	P59046	P08631
NLRP2	Q9NX02	P16104	NLRP3	Q96P20	Q9ULZ3
NLRX1	Q86UT6	P49411	NMB	Q14956-1, Q14956	P00533
NME1	P15531	P10911	NME2	P19804	Q05193
NME3	Q13232	Q8IWZ6, Q3SYG4, Q9BXC9	NME4	O00746	Q9NPB3
NME7	Q9Y5B8	P25786	NMI	Q13287	P23434
NMRAL1	Q9HBL8	Q96CV9	NMT1	P30419	P04601

Input	UniProt Id	Interacts with	Input	UniProt Id	Interacts with
NMT2	O60551	P01137	NOA1	Q8NC60	P84103
NOB1	Q9ULX3	Q9NXX1	NOC2L	Q9Y3T9	P40429
NOC3L	Q8WTT2	P19525	NOC4L	Q9BVI4	Q9UER7
NOD1	Q9Y239	P14136	NOD2	Q9HC29	P29466
NOL10	Q9BSC4	P06748	NOL11	Q9H8H0	Q13901
NOL12	Q9UGY1	P40429	NOL3	O60936	Q14790
NOL4	O94818-2	P07237	NOL4L	Q96MY1	Q13363
NOL6	Q9H6R4-4	Q8IUQ4	NOL7	Q9UMY1	Q9H8H0
NOLC1	Q14978	P32121, P49407	NOMO1	Q15155	P25090
NOMO3	P69849	P25090	NONO	Q15233	O43913
NOP10	Q9NPE3	Q9NPB3	NOP16	Q9Y3C1	P06748
NOP2	P46087	P04733	NOP56	O00567	Q9Y4Z2
NOP9	Q86U38	O43914	NOS1	P29476	P23634
NOS1AP	O75052	P53350	NOS2	P60321	Q14192
NOS3	P29474	Q12891	NOSIP	Q9Y314	Q93034
NOSTRIN	Q8IVI9	Q05193	NOTCH1	P46531	P78504
NOTCH2	Q04721	Q8N2S1	NOTCH3	Q9UM47	P53611
NOTCH4	Q99466	P05067	NOV	P48745	O43559
NOX4	Q9NPH5	O00206	NOXA1	Q86UR1-2	Q8WY64
NPAS1	Q99742	P11142	NPAS2	Q99743	O43639
NPAS3	Q8IXF0	Q9HBZ2	NPAT	Q14207	Q09028
NPB	Q8NG41	Q9BTV6	NPC1	O15118	Q03135
NPC2	P61916	P00751	NPDC1	Q9NQX5	Q9BW85
NPEPPS	P55786	P13569	NPHP1	C9J082	P01019
NPHP4	O75161	O95816	NPIPB3	Q92617	P59634
NPLOC4	Q8TAT6	O95786	NPM1	P06748	P40429
NPM2	Q86SE8, Q86SE8-2	P06748	NPM3	O75607	P40429
NPTX1	Q15818	P0DTC8	NQO1	P15559	P15559
NR0B1	P51843	Q13285	NR1D1	P20393	O75376
NR1D2	Q6NSM0	Q8WY64	NR1H3	Q13133	Q15788
NR2C1	P13056-2	P10276	NR2C2	Q6PHZ7	P06396
NR2C2AP	Q86WQ0	P42858	NR2F1	P10589	Q00403
NR2F2	P24468	P43119	NR2F6	P10588	P24468
NR3C1	P06536	Q15788	NR3C2	P08235	Q9UBK2
NR4A1	P22736	Q9Y4D7	NR4A2	P43354	P05067
NR4A3	Q92570	Q13526	NR5A2	O00482-1	Q15596
NRAS	P01111	P41440	NRBF2	Q96F24	P07902
NRBP1	Q9UHY1	P51530	NRCAM	Q92823	Q12860
NRD1	O43847	P04637	NRDE2	Q9H7Z3	Q13352
NRF1	Q14494	O15525	NRG1	Q02297-10	P21854
NRIP1	P48552	Q9Y6K1	NRIP2	Q9BQI9	Q8N9N5
NRIP3	Q9NQ35	Q969U7	NRM	Q8IXM6	P04000
NRN1	Q9NPD7	Q86XT9	NRP2	Q99435	Q6IPR3
NRSN2	Q9GZP1	Q8N6M3	NRXN1	Q63373	O14936
NSA2	P40078	O14641	NSD1	Q96L73	P17096
NSF	P46459	P59635	NSFL1C	Q9UNZ2	P55072
NSL1	Q96IY1	P37173	NSMAF	Q92636	Q9NPI6
NSMCE2	Q96MF7	P06396	NSMF	Q6X4W1-6	P14136
NSRP1	Q9H0G5	Q8WUA2	NSUN3	Q9H649	Q6PJG9
NT5C	Q8TCD5	P52306	NT5E	P21589	P17302

Input	UniProt Id	Interacts with	Input	UniProt Id	Interacts with
NTAN1	Q96AB6	Q13546	NTF3	P20783	P23560
NTM	Q9P121-3	Q96P65	NTMT1	Q9BV86	P18754
NTN1	O95631	P43146	NTN4	Q9HB63	P15056
NTPCR	Q9BSD7	P13569	NTRK1	P04629	O75385
NTRK3	Q16288	Q16620	NTS	P20068- PRO_0000019532	P63211
NUAK1	O60285	Q93008	NUBP1	P53384	Q9Y5Y2
NUBP2	Q9Y5Y2	Q8TB37	NUBPL	Q8TB37	Q9Y5Y2
NUCKS1	Q9H1E3	Q12824	NUDC	Q9Y266	Q13451
NUDCD1	Q96RS6	O43143, Q92620, Q14562	NUDCD3	Q8IVD9	P43034
NUDT1	P36639-2	Q53H54	NUDT14	O95848	O95848
NUDT16	Q96DE0	P27930	NUDT16L1	Q9BRJ7	Q9NPB3
NUDT2	P50583	Q14566	NUDT22	Q9BRQ3	P52747
NUDT4	Q9NZJ9	P04792	NUDT6	P53370	P46199
NUF2	Q9BZD4	Q9Y3P9	NUFIP2	Q7Z417	Q9UBX5
NUMB	Q9QZS3-2	P08151	NUMBL	Q9Y6R0	Q9NWB1
NUP133	Q8WUM0	P01106	NUP153	P49790	Q9UPY3
NUP205	Q92621	P13569	NUP214	P35658	Q9UBU9
NUP35	Q8NFH5	Q13526	NUP37	Q8NFH4	Q8WYP5
NUP43	Q8NFH3	P23025	NUP50	Q9UKX7	O43303
NUP54	Q7Z3B4	Q9Y6A5	NUP62	P37198	Q6W0C5
NUP62CL	Q9H1M0	O14641	NUP98	P52948	Q9UBU9
NUPL1	Q9BVL2	Q8WUX9	NUPR1	O60356	O15534
NUTF2	P61970	Q9NX63	NXF1	Q9UBU9	P46060
NXPE3	Q969Y0	Q6IN84	NXPH3	O95157	Q9UMX1
NXPH4	O95158	Q8IV16	NXT1	Q9UKK6	Q9Y3Y2
NXT2	Q9NPJ8	Q9UBU9	NYAP1	Q6PFX7	P27986
OAF	Q86UD1	P56559	OARD1	Q9Y530	P09874
OAS1	P00973-2, P00973	P12814	OAS3	Q9Y6K5	Q7Z434
OASL	Q15646	P67809	OAZ1	P54368	O14977
OAZ2	O95190	O14977	OBFC1	Q9H668	Q9UBD5, Q13416
OBSL1	O75147	Q9UBX5	OCLN	Q16625	P18827
OCRL	Q01968	Q9NYZ3	ODC1	P11926	Q92993
ODF2	Q5BJF6	P53350	ODF2L	Q96GC1	P42858
OFD1	O75665	Q4AC94	OGFOD1	Q8N543	P62266
OGFOD2	Q6N063-2	P55212	OGFRL1	Q5TC84	P11802
OGN	P20774	Q9BT09	OGT	O15294-3	Q8N9R8
OLFM1	Q99784	Q5SRI9	OLFM2	O95897	O14641
OLFM4	Q6UX06	Q86XF0	OLFML2A	Q68BL7	Q15848
OLFML2B	Q68BL8	Q13217	OLR1	P78380	P17987
OMA1	Q96E52	P46059	ONECUT3	O60422	O43639
OPCML	Q14982	P27824	OPRD1	P41143	Q9UKG4
OPRK1	P41145	P35414	OPRM1	P35372	Q5T9L3
OPTN	Q96CV9	P52888	ORC2	Q13416	Q13415
ORC4	O43929	Q13415	ORC5	O43913	Q13415
ORC6	Q9Y5N6	O43929, Q9UBD5, Q13416	ORMDL2	Q53FV1	Q92911
ORMDL3	Q8N138	Q06124	OS9	Q13438	O00468
OSBP	P22059	P05120	OSBP2	Q969R2-2	Q9NRD5
OSBPL1A	Q9BXW6	P51149	OSBPL3	Q9H4L5	P31946, P63104
OSBPL8	Q9BZF1	O15530	OSBPL9	Q96SU4	Q92685

Input	UniProt Id	Interacts with	Input	UniProt Id	Interacts with
OSGEP	Q9NPF4	Q96S44	OSGIN1	Q9UJX0	Q99081
OSM	P13725	P40189	OSMR	Q99650	P50458
OSR1	O95747	Q8WVK2	OSR2	Q8N2R0	P25788
OST4	P0C6T2	A6NI15	OSTC	Q9NRP0	P80370
OSTF1	Q92882	O75695	OTOGL	Q3ZCN5	P42858
OTUB1	Q96FW1	P51668	OTUB2	Q9BWI9	O76024
OTUD1	Q5VV17	O43353	OTUD4	Q01804	P04004
OTUD7A	Q8TE49	Q13485	OTUD7B	Q6GQQ9	Q8NFZ5
OTULIN	Q96BN8	Q12981	OXA1L	Q2M1J6	P17252
OXER1	Q8TDS5	P35680	OXLD1	Q5BKU9	Q4G176
OXNAD1	Q96HP4	P10809	OXR1	Q8N573-5	Q14145
OXSM	Q9NWU1	Q9NU23	P2RX1	P51575	P37268
P2RX7	Q99572	P14373	P2RY10	O00398	Q13315
P2RY12	Q9H244	P53794	P2RY6	Q15077	O76024
P2RY8	Q86VZ1	Q92538	P3H1	Q32P28	Q70UQ0
P3H2	Q8IVL5	O95076	P3H4	Q92791	O95076
P4HA1	P13674	Q6UN15	P4HA2	O15460	P39060
P4HB	P07237	O43521	PA2G4	Q9UQ80	P06400
PABPC1	P11940	Q92900	PABPC4	Q13310	P59595
PABPC4L	P0CB38	P16104	PABPN1	Q86U42	P11940
PACRGL	Q8N7B6-2	P19404	PACS2	Q86VP3	P63104
PACSIN1	Q9BY11	Q9NUX5	PACSIN2	Q9UNF0	P50570
PACSIN3	Q9UKS6	Q07889	PADI4	Q9UM07	Q96RU7
PAF1	Q8N7H5	P23193	PAFAH1B1	P43034	Q14203
PAFAH1B2	P68402	P43034	PAFAH1B3	Q15102	Q8WUX9
PAG1	Q9NWX8	P41240	PAICS	P22234	Q969R5
PAIP1	Q9H074, Q9H074-3	P11940	PAIP2	Q9BPZ3	P11940
PAIP2B	Q9ULR5	O15234	PAK1	Q13153	Q13177
PAK1IP1	Q9NWT1	P62753	PAK3	O75914	P63000
PAK7	Q9P286	P53350	PALD1	Q9ULE6	Q92985
PALLD	Q8WX93	P13569	PALM	O75781	P53611
PALM2	Q9Y2D5	Q96CV9	PALM3	A6NDB9	Q6IA17
PALMD	Q9NP74	O76024	PAM	P19021	P08185
PAN2	Q9HBH5	P25103	PAN3	Q58A45	Q8NDV7
PANX1	Q96RD7	O43914	PANX2	Q495U3	P06396
PAPD5	Q8NDF8	P06748	PAPSS1	O43252	Q8WVK2
PAPSS2	O95340	Q96CV9	PAQR5	Q9NXX6	P26951
PAQR6	Q6TCH4	Q96BA8	PAQR7	Q86WK9	P40763
PAQR8	Q8TEZ7	Q08426	PARD3	Q8TEW0	P59637
PARD3B	Q8TEW8	P0DTC3	PARD6A	Q9NPB6	Q05513
PARD6B	Q9BYG5	Q05513	PARK2	O60260-5	O94844
PARK7	Q99497	Q13158	PARN	O95453	Q99728
PARP1	P09874	Q9NTX7	PARP11	Q9NR21-5	P07550
PARP12	Q9H0J9	P48507	PARP14	Q460N5	O43914
PARP2	Q9UGN5	Q9NTX7	PARP8	Q8N3A8	P61981
PARVA	Q9NVD7	Q96EN8	PARVB	Q9HBI1	Q15404, Q9NVD7, P48059
PARVG	Q9HBI0	Q15404, P48059	PASK	Q96RG2	P08311
PATZ1	Q9HBE1	P40429	PAWR	Q96IZ0, Q62627	P11021
PAX3	P24610	Q08999, P28749	PAX8	Q06710	O00712, Q12857

Input	UniProt Id	Interacts with	Input	UniProt Id	Interacts with
PAX9	P55771	O43474	PAXBP1	Q9Y5B6	Q9BUQ8
PBDC1	Q9BVG4	P68104	PBX1	P40424	Q12948
PBX2	P40425	Q9UHB7	PBX3	Q96AL5	O60906
PBX4	Q9BYU1	Q00403	PBXIP1	Q96AQ6	O15554
PC	P11498	P48163	PCBD1	P61457	P35680
PCBD2	Q9H0N5	P35680	PCBP1	Q15365	P08581
PCBP2	Q15366	P11940	PCBP4	P57723	P11142
PCDH10	Q9P2E7	P43119	PCDH7	O60245	P34925
PCDHB11	Q9Y5F2	Q99741	PCDHB7	Q9Y5E2	O60906
PCED1A	Q9H1Q7	Q7Z3S9	PCED1B	Q96HM7	Q7Z3S9
PCGF1	Q9BSM1	P63208	PCGF2	P35227	Q96GA3
PCGF3	Q3KNV8	P68400, P67870	PCGF5	Q86SE9	Q12933
PCGF6	Q9BYE7	Q8TEY7	PCK1	P35558	Q16822
PCM1	Q15154	Q8IW35, O43303	PCMT1	P22061, P22061-2	P49247
PCMTD1	Q96MG8	P62263	PCNA	P12004	P52701, P20585
PCNP	Q8WW12	O75533	PCOLCE	Q15113	P07996
PCSK1N	Q9UHG2	P09238	PCSK4	Q6UW60-2	P42858
PCSK5	Q92824	Q03405	PCSK6	P29122	P0DTC8
PCSK7	Q16549	P50454	PCSK9	Q8NBP7, Q8NBP7-1	P01130
PCYOX1L	Q8NBM8	P27824	PCYT2	Q99447	P25786
PDAP1	Q13442	Q96FA3	PDCD10	Q9BUL8	Q8WVK2
PDCD11	Q14690	P19838	PDCD1LG2	Q9BQ51	Q9NZQ7
PDCD2	Q16342	O60260	PDCD2L	Q9BRP1	P15880
PDCD4	Q53EL6	P62277	PDCD5	O14737	Q9NRD5
PDCD6IP	Q8WUM4	P12931	PDCL	Q13371	O00628
PDCL3	Q9H2J4	Q13371	PDE1A	P54750-4	P42858
PDE1B	Q01064	Q9NUX5	PDE2A	O00408	O00170
PDE3B	Q13370	O95398	PDE4A	P27815	O75344
PDE4C	Q08493-3, Q08493-2	P30626	PDE4D	Q08499-2	Q9HBX9
PDE4DIP	Q5VU43	Q16891, Q9NX63	PDE6D	O43924	P36404
PDE7A	Q13946	P31946	PDE9A	O76083-2	O00560
PDGFA	P04085	P16234	PDGFB	P01127	P16234
PDGFC	Q9NRA1	P16234	PDGFD	Q9GZP0	O95714
PDGFRA	P16234	P46109	PDGFRB	P09619	P40818
PDGFRL	Q15198	Q9UBW7	PDHA1	P08559	Q16891
PDHB	P11177	P10515	PDHX	O00330	P46199
PDIA3	P30101	P27797	PDIA4	P13667	P13569
PDIA5	Q14554	Q969H8	PDIK1L	Q8N165	Q13451
PDK3	Q15120	P10515	PDK4	Q16654	Q15119
PDLIM1	O00151	P43487	PDLIM4	P50479	P12814
PDLIM5	Q96HC4	P11142	PDPK1	O15530	O00468
PDPR	Q8NCN5	O75771	PDRG1	Q9NUG6	Q9UBK9
PDS5A	Q29RF7	Q7Z5K2	PDS5B	Q9NTI5	Q7Z5K2
PDSS2	Q86YH6	P14324	PDZD11	Q5EBL8	Q04656
PDZD2	O15018	P59637	PDZD4	Q17RL8	Q9UJW3
PDZD7	Q9H5P4	P60484	PDZD8	Q8NEN9	P23945
PDZRN3	Q9UPQ7	Q15418	PDZRN4	Q6ZMN7	Q15418
PEA15	Q15121	P28482	PEAK1	Q9H792	P46109
PEBP1	P30086	P16050	PEBP4	Q96S96	P31749

Input	UniProt Id	Interacts with	Input	UniProt Id	Interacts with
PECAM1	P16284	P12931	PECR	Q9BY49	Q5T8I9
PEG10	Q86TG7-2	Q9NQT4	PEG3	Q9GZU2	P05067
PELI1	Q96FA3	P02545, O14653	PELI2	Q9HAT8	Q12891
PELO	Q9BRX2	P04792	PELP1	Q8IZL8	P19838
PER1	O15534	P07237	PER2	O15055	P49674, P48730
PER3	P56645	O96017	PERP	Q96FX8	P04792
PES1	O00541	P06748	PEX11B	O96011	P40855
PEX14	O75381	P40855	PEX16	Q9Y5Y5	Q96CV9
PEX19	P40855	Q9NZQ7	PEX2	P28328	P40855
PEX26	Q7Z412	Q969S8	PEX3	P56589	Q9Y5Y5, P40855
PEX5	P50542	O14734	PEX7	O00628	Q13371
PFDN1	O60925	Q13164	PFDN2	Q9UHV9	Q13164
PFDN4	Q9NQP4	Q13164	PFDN5	Q99471	Q13164
PFDN6	O15212	Q13164	PFKFB1	P16118	P35557
PFKFB2	O60825	P31946, P63104	PFKFB3	Q16875	P31946, P63104
PFKFB4	Q16877	P31946, P63104	PFKM	P08237	Q8IUQ4
PFKP	Q01813	P31152	PFN1	P07737	P60709
PFN2	P35080	P60709	PGAM1	P18669	P13569
PGAM5	Q96HS1	Q9Y572	PGBD1	Q96JS3	P22736
PGD	P52209	P13569	PGF	P49763	P17948
PGGT1B	P53609	Q9NQ87	PGK1	P00558, P00558-1	O15379
PGM2	Q96G03	P04792	PGPEP1	Q9NXJ5	P68871
PGR	P06401-1	Q13569	PGRMC1	O00264	P08684
PHACTR2	O75167	P60709	PHACTR3	Q96KR7	Q9BZL1
PHACTR4	Q8IZ21	P31946, P63104	PHAX	Q9H814	Q13188
PHB	P35232	P40763	PHB2	Q99623	P35232
PHC1	P78364	P11142	PHC2	Q8IXK0	P08047
PHC3	Q8NDX5	Q13363	PHF1	O43189	P04637
PHF10	Q8WUB8	Q92925, O96019	PHF11	Q9UIL8	O00560
PHF12	Q96QT6	Q13547	PHF13	Q86YI8	Q9UMR3
PHF14	O94880	P16104	PHF19	Q5T6S3	Q14012
PHF20	Q9BVI0	P62805	PHF20L1	A8MW92	P62805
PHF21A	Q96BD5	Q92769	PHF21B	Q96EK2	Q13547
PHF23	Q9BUL5	P99999	PHF24	Q9UPV7	P07204
PHF3	Q92576	P61981	PHF5A	Q7RTV0	O76024
PHF7	Q9BWX1	P62837	PHF8	Q9UPP1	P46531
PHGDH	O43175	P13569	PHKB	Q93100	Q8WUJ3
PHKG2	P15735	P0DP23	PHLDA1	A2BDE7	O76024
PHLDB1	Q86UU1	P63104	PHLDB2	Q86SQ0	P31946, P63104
PHLPP1	O60346	Q8TAF3, O94782	PHLPP2	Q6ZVD8	Q8TAF3, O94782
PHPT1	Q9NRX4	P0DTC3	PHYHD1	Q5SRE7	Q9NUX5
PHYHIP	Q92561	Q9NS87	PHYKPL	Q8IUZ5	Q9NUX5
PI16	Q6UXB8	P27449	PI4K2A	Q9BTU6	Q96J02
PI4KA	P42356	P13569	PIAS1	O75925	Q16143
PIAS2	O75928-2	Q13114	PIAS3	Q9Y6X2	Q13485
PIAS4	Q8N2W9	O60551	PIBF1	Q4G0R1	Q9UER7
PICALM	Q13492-2, Q13492	Q9NYZ3	PICK1	Q9NRD5	P23975
PID1	Q7Z2X4	Q8N9N5	PIEZO1	Q92508	P61073
PIGB	Q92521	P19526	PIGF	Q07326	Q9NPL8
PIGH	Q14442	O95256	PIGN	O95427	P43119

Input	UniProt Id	Interacts with	Input	UniProt Id	Interacts with
PIGO	Q8TEQ8	Q9BRI3	PIGQ	Q9BRB3	Q9P0P0
PIGS	Q96S52	P13569	PIGT	Q969N2	Q8NBL1
PIGW	Q7Z7B1	Q9H8X2	PIH1D1	Q9NWS0	Q96GM5
PIH1D3	Q9NQM4	P54646	PIK3AP1	Q6ZUJ8-3, Q6ZUJ8	O43639
PIK3C2A	O00443	P31946, P63104	PIK3C2B	O00750	P31946, P63104
PIK3C3	Q8NEB9	P61981	PIK3CA	P42336	P03496
PIK3CB	P42338	Q13485	PIK3CD	O00329, O00329-2	P27986
PIK3CG	P48736	Q13370	PIK3IP1	Q96FE7	P78382
PIK3R1	P27986	O15455	PIK3R4	Q99570	P31946
PIK3R6	Q5UE93	Q13370	PILRA	Q9UKJ1	P29350
PIM1	P11309-2	Q9UNQ0	PIN1	Q13526	P08151
PIN4	Q9Y237	P00533	PINK1	Q9BXM7	Q8WXH5
PINX1	Q96BK5	P06748	PIP4K2A	P48426	P55199
PIP4K2B	P78356	Q15831	PIP4K2C	Q8TBX8	Q15831
PIP5K1A	Q99755	P01116	PIP5K1C	O60331	P53611
PIPOX	Q9P0Z9	Q9NUX5	PIR	O00625	P02675
PITPNA	Q00169	P31930	PITPNC1	Q9UKF7-2	O60906
PITRM1	Q5JRX3	P0DTC5	PITX2	Q99697-3	Q12948
PIWIL4	Q7Z3Z4	Q9Y2W6	PJA1	Q8NG27	Q9Y5V3
PKD1	Q15139	P02795	PKD2	Q13563	P21453
PKDCC	Q504Y2	Q9H5K3	PKIA	P61925	P17612
PKIB	Q9C010-2	P17612	PKIG	Q9Y2B9	P17612
PKM	P14618-1	Q16665	PKMYT1	Q99640	P06493
PKN1	Q16512	P26641	PKN2	Q16513	O76075
PKN3	Q6P5Z2	A1A4S6	PKNOX1	P55347	P40424
PKNOX2	Q96KN3	P48552	PKP2	Q99959	P03431
PKP4	Q99569	Q8NI35	PLA1A	EBI-3954042	P78527
PLA2G12A	Q9BZM1	O76024	PLA2G4C	Q9UP65	P42858
PLAA	Q9Y263	Q9Y5M8	PLAC8	Q9UHV8	P0CG47
PLAGL2	Q9UPG8	O94955	PLAT	P00750	Q04721
PLAU	P00749	Q03405	PLAUR	Q03405	Q09328
PLB1	Q6P1J6-2	P14136	PLBD1	Q6P4A8	P15498
PLCD1	P51178	Q14974	PLCD3	Q8N3E9	Q96JC9
PLCG1	P19174	P16234	PLCG2	P16885	O95997
PLCH1	Q4KWH8	P31946, P63104	PLCL2	Q9UPR0	P49914
PLD1	Q13393	P23528	PLD2	O14939	P23528
PLD3	Q8IV08	P19838	PLEC	Q15149	Q14145
PLEK	P08567	P51659	PLEKHA1	Q9HB21	Q9P0V3
PLEKHA2	Q9HB19	O14908	PLEKHA3	Q9HB20	O43148
PLEKHA5	Q9HAU0	Q9NWB1	PLEKHA7	Q6IQ23-2	Q15669
PLEKHB1	Q9UF11	P52306	PLEKHB2	Q96CS7	Q86T03
PLEKHF2	Q9H8W4	Q8IZD9	PLEKHG2	Q9H7P9	Q13526
PLEKHG3	A1L390	P31946	PLEKHG4	Q58EX7	Q6W0C5
PLEKHG7	Q6ZR37	P51149	PLEKHH3	Q7Z736	P31152
PLEKHM1	Q9Y4G2	P51149	PLEKHM3	Q6ZWE6	P63104
PLEKHO1	Q53GL0	Q9HCE7	PLIN2	Q99541	O76024
PLIN3	O60664	Q8TB40	PLK2	Q9NYY3	Q13177
PLK3	Q9H4B4	P14859	PLLP	Q9Y342	O43559
PLN	P26678	Q9H2K0	PLOD1	Q02809	Q9Y580
PLOD2	O00469	Q9Y230	PLP1	P60201-2	Q9H2K0

Input	UniProt Id	Interacts with	Input	UniProt Id	Interacts with
PLP2	Q04941	Q96QE2	PLRG1	O43660	P52272
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PLSCR1	O15162	Q9UBR2	PLSCR4	Q9NRQ2	P10745
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PON2	Q15165	Q12891	POP1	Q8WXC3	Q9ULZ3
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POT1	Q9NUX5	P50747	POU2AF1	Q16633	Q06124
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PPARD	Q03181-2	P40763	PPARG	P37231, P37231-2	Q9UBK2
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PRELP	P51888	P17707	PREX1	Q8TCU6	P08581
PREX2	Q70Z35	P15056	PRF1	P14222	P14222
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PRICKLE3	O43900	Q9UJK0	PRIM2	P49643	Q13257
PRIMPOL	Q96LW4	P25786	PRKAA1	Q13131	O75385
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PRKX	P51817	Q13451	PRLR	P16471-1, P16471-7	P01236
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PROS1	P07225	Q96CV9	PROSER2	Q86WR7	Q9Y316
PROSER3	Q2NL68	Q96ES7	PROX1	Q92786	P41235
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PRPF31	Q8WWY3	P05455	PRPF38A	Q8NAV1	Q96CW1
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PSMA6	P60900	Q15008, O43242, P28066, Q9Y244, P28065, Q9UNM6, P28070, Q99436, P62333, O14818, P20618, P25789, P25788, P49721, P49720, P25787, P25786	PSMA7	O14818	Q16665
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PTEN	P60484	P60484	PTGDR	Q13258	Q9NZ52
PTGDS	P41222	P00533	PTGER3	P43115	P27037
PTGES	O14684	O95070	PTGES2	Q9H7Z7	Q9UG63
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PVRL1	Q15223	O00560	PVRL2	Q92692-2	P07237
PVRL3	Q9NQS3-1, Q9NQS3	P15151, Q15223	PWP1	Q13610	P48431
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RAB11FIP5	Q9BXF6	P63104	RAB12	Q6IQ22	Q5S007
RAB13	P51153	P0DTC3	RAB14	P61106	P0DTC3
RAB15	P59190	P52306	RAB1A	P62820	Q92696
RAB1B	Q9H0U4	Q92696	RAB20	Q9NX57	P52306
RAB21	P35282	P17301	RAB24	Q969Q5	Q14353
RAB27A	P51159	Q92696	RAB27B	O00194	Q8IYJ3
RAB28	P51157	Q4G176	RAB29	O14966	Q92696
RAB2B	Q8WUD1-2	P06396	RAB30	Q15771	Q9HD47
RAB31	Q13636	Q92696, P53611	RAB33A	Q14088	Q92530
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RAB38	P57729	P14136	RAB39A	Q14964	Q08379
RAB39B	Q96DA2	P46059	RAB3A	P20336	Q9HD47
RAB3B	P20337	P46059	RAB3GAP1	Q15042	P15056
RAB3GAP2	Q9H2M9	P13569	RAB3IL1	Q8TBN0	Q96QF0
RAB3IP	Q96QF0	Q9NX63	RAB40A	Q8WXH6	Q15369
RAB40C	Q96S21	P53611	RAB4B	P61018	O75419
RAB5B	P61020	Q9Y324	RAB5C	P51148	P13569
RAB6A	P20340-2	P05067	RAB6B	Q9NRW1	Q92696, P53611
RAB6C	P20340-2	P05067	RAB7A	P51149	O75348, P21281, P61421
RAB7B	Q96AH8	Q92696, P53611	RAB8A	P61006	Q96QF0
RAB8B	Q92930	Q9UK39	RAB9B	Q9NP90	Q92696
RABAC1	Q9UI14	Q8TB40	RABEP1	Q15276	Q9NZ52
RABEPK	Q7Z6M1	Q15569	RABGAP1	Q9Y3P9	P40763
RABGAP1L	Q5R372-2	P06400	RABGGTA	Q92696	P53611
RABGGTB	P53611	O94955	RABIF	P47224	Q9NZU5
RABL2B	Q9UNT1-2	P06396	RABL3	Q5HYI8	Q9NRZ7
RABL6	Q3YEC7-3	P17252	RAC1	P63000	P52565
RAC2	P15153	P52306	RAC3	P60763	P52565
RAD18	Q9NS91	Q13416	RAD23A	P54725	Q01831
RAD51B	O15315	O43543, O75771, O43502	RAD51D	O75771	P54132
RAD52	P43351	Q06609	RAD54L	P46100	Q9UER7
RAD54L2	Q9Y4B4	P25788	RAD9A	Q99638	P40937
RAE1	P78406	P46937	RAG1	P15918	P52294
RAI1	Q7Z5J4	Q13526	RALA	P11233	P13569
RALB	P11234	P11182	RALGAPA1	Q6GYQ0	P63104
RALGDS	Q12967	P15056	RALGPS2	Q86X27	P31946, P63104
RALY	Q53GL6	Q969Y2	RALYL	Q86SE5	Q13895
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RANBP1	P43487	P18754	RANBP10	Q6VN20	P01275

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RANGAP1	P46060	P61769	RANGRF	Q9HD47-3	P06396
RAP1A	P62834	P52306	RAP1B	P61224	Q12967
RAP1GAP	P47736	P63104	RAP1GDS1	P52306	P17480
RAP2A	P10114	P53611	RAPGEF2	Q9Y4G8	P46937
RAPGEF3	O95398	Q13370	RAPGEF4	Q8WZA2	P05455
RAPGEF5	Q92565	Q12955	RAPGEF6	Q8TEU7	P31946
RAPH1	Q70E73	Q99962	RAPSN	Q13702-2	P50454
RARA	P10276	Q15648	RARB	P10826-2	Q9UBK2
RARG	P13631	Q08999	RARS	P54136	P13569
RASA1	P20936, P50904	Q13480	RASAL2	Q9UJF2	P48729
RASAL3	Q86YV0	Q9BUH8	RASD1	Q9Y272	O00308
RASD2	Q96D21	O75197	RASEF	Q8IZ41	P13569
RASGRP3	Q8IV61	Q8NBQ5	RASGRP4	Q8TDF6-2	Q12982
RASL10B	Q96S79	Q9H3L0	RASL12	Q9NYN1	Q14192
RASSF2	P50749	Q96CV9	RASSF3	Q86WH2	Q9UER7
RASSF4	Q9H2L5	Q13188, Q13043	RASSF5	Q8WWW0	Q8WY64
RASSF6	Q6ZTQ3	Q14160	RASSF7	Q02833	O43639
RASSF8	Q8NHQ8	Q13485	RAVER1	Q8IY67	O75791
RB1	P06400	O96017, O14757	RB1CC1	Q8TDY2	O75385
RBBP4	Q09028	P46531	RBBP7	Q16576	Q92769
RBBP8	Q99708	Q9UQ84	RBFA	Q8N0V3	Q96EB6
RBFOX1	Q9NWB1	Q92530	RBFOX2	O43251	Q9NWB1
RBFOX3	Q8BIF2	Q16665	RBL1	P28749	O43524
RBL2	Q08999	P11802	RBM10	P98175	Q92841
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RBM22	Q9NW64	O95926	RBM23	Q86U06	P14373
RBM26	Q5T8P6	Q9Y4Z2	RBM28	Q9NW13	P46087
RBM3	P98179	Q9Y6M1	RBM33	Q96EV2	O75152
RBM34	P42696	P40429	RBM39	Q14498-3	Q9UH92
RBM41	Q96IZ5	Q13077	RBM42	Q9BTD8	O00425
RBM47	A0AV96	P0DTC9	RBM4B	Q9BQ04	P07910
RBM5	P52756	P27797	RBM6	P78332	P09651
RBM7	Q9Y580	P31946	RBM8A	Q9Y5S9	Q92900
RBMS1	P29558	P84103	RBMS2	Q15434	O00425
RBMX	P38159	P09651	RBP1	P09455	Q9UBN6
RBP7	Q96R05	Q8WXH5	RBPJ	Q06330-6, Q06330	P46531
RBPMS	Q93062	Q53HV7	RBPMS2	Q6ZRY4	Q9BTD8
RBX1	P62877	P61081	RCAN3	Q9UKA8	P11802
RCBTB2	O95199	O60733	RCC1	P18754	O14980
RCE1	Q9Y256	Q96RK4	RCL1	Q9Y2P8	Q08379
RCN1	Q15293	Q9H4P4	RCN2	Q14257	Q13418
RCN3	Q96D15	Q9UMX1	RCOR1	Q9UKL0	Q92769
RCOR2	Q8IZ40	Q92769	RCOR3	Q9P2K3-2	Q9H668
RCSD1	Q6JBY9	P52907	RDH10	Q8IZV5	Q05329
RDH12	Q96NR8	P40855	RDH13	Q8NBN7	Q01523
RDX	P35241	P52565	REC8	O95072	Q9UNE7
RECK	Q6P9E2	Q96LC9	RECQL	P46063	P62195

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REEP1	Q9H902	P21291	REEP2	Q9BRK0	Q6NT55
REEP3	Q6NUK4	P31946, P63104	REEP6	Q96HR9-2, Q96HR9	O00560
REL	Q04864-2	P30622	RELB	Q01201	P49841
RELL1	Q8IUW5	P26572	RELL2	Q8NC24	P78382
RELN	Q60841	P98155	REPIN1	Q9BWE0	O15397
REPS2	Q8NFH8	Q96D71	RER1	O15258	A1L190
RERE	Q9P2R6	Q13547	REST	Q13127	Q9Y297
RET	P07949	P22455	REV1	Q9UBZ9	Q9UI95
REV3L	O60673	Q9UI95	REXO4	Q9GZR2	P40429
RFC1	P35251	P35250, P40938, P40937, P35249	RFC2	P35250	P40937, P35249
RFC3	P40938	P40937, Q9BVC3, P35249	RFC4	P35249	P35250, P0CG13, P40938, P40937, Q9BVC3, Q8WVB6
RFT1	Q9NWF4	Q16585	RFTN1	Q14699	P18827
RFTN2	Q52LD8	Q70CQ3	RFWD2	Q8NHY2	Q96RU7
RFX2	P48378	O75928-2	RFX3	P48380	Q12933
RFX7	Q2KHR2	P31946	RFXANK	O14593	P04049
RGAG1	Q8NET4	Q00613	RGL2	O15211	Q7Z3S9
RGP1	Q92546	O14543	RGPD8	O14715	P62195, Q13200
RGS2	P41220	P11142	RGS20	O76081-6, O76081	P48023
RGS3	P49796	P43155	RGS6	P49758	P08238
RGS8	P57771, P57771-2	Q08379	RHBDD1	Q8TEB9	Q96BA8
RHBDD2	Q6NTF9-3	P23434	RHBDD3	Q9Y3P4	Q6UWB1
RHBDF1	Q96CC6	Q16539	RHBDF2	Q80WQ6	P78536
RHEB	Q15382-PRO_0000082708	O43924	RHEBL1	Q8TAI7	P83876
RHNO1	Q9BSD3	Q9UBB9	RHOBTB1	O94844	P61081
RHOBTB2	Q9BYZ6	Q9Y4X5	RHOBTB3	O94955	P53611
RHOC	P08134	P52565	RHOD	O00212	Q9Y4D1
RHOG	P84095	P52565	RHOH	Q15669	Q92673
RHOJ	Q9H4E5	Q13177	RHOT1	Q8IXI2	Q9BXM7, O60260
RHOT2	Q8IXI1	P13569	RHOU	Q7L0Q8	Q14289
RHOV	Q96L33	O43639	RIC1	Q4ADV7	P20340
RIC3	Q7Z5B4-5	Q07817	RIC8A	Q9NPQ8-4	Q96QE2
RICTOR	Q6R327	P22891	RIF1	Q5T3J3	Q96A70
RILPL1	Q5EBL4-3	Q86T03	RIMS1	Q86UR5	P06241
RIMS3	Q9UJD0	P08670	RIMS4	Q9H426	Q96FA3
RIN1	Q13671	P00519	RIN3	Q8TB24	P46109
RING1	Q06587	P51668	RIOK1	Q9BRS2	P15880
RIOK2	Q9BVS4	Q13895	RIOK3	O14730	Q92851
RIPK1	Q13546	Q13546, Q9Y572	RIPK2	O43353	Q13114
RIPK3	Q9Y572	Q13546, Q9Y572	RIPK4	P57078	P98170
RIPPLY2	Q5TAB7	Q08117	RIPPLY3	P57055	Q13416
RIT2	Q5BJQ5, Q99578	Q01959	RITA1	Q96K30	Q9NPI6
RLF	Q13129	Q13185, P45973	RLTPR	Q6F5E8	P47756
RMDN1	Q96DB5	P05181	RMDN2	Q96LZ7	Q9Y4Z2
RMDN3	Q96TC7	Q9Y4Z2	RMI1	Q9H9A7	P27694, P54132
RMND1	Q9NWS8	Q8TDU9	RMND5A	Q9H871	P17844

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RMND5B	Q96G75	Q15008, O14818	RNASEH2A	O75792	Q99943
RNASEH2C	Q8TDP1	P11142, P04792	RNASEK	Q6P5S7	Q9H2C2
RNASEL	Q1RLL8	Q99081	RNASET2	O00584	Q14554
RND2	P52198	O43157	RND3	P61587	O43157
RNF10	Q8N5U6	P51668	RNF112	Q9ULX5	O14832
RNF113A	O15541	Q13123	RNF115	Q9Y4L5	P51668
RNF123	Q5XPI4	Q9BQ52	RNF125	Q96EQ8	O95786
RNF126	Q9BV68	P51668	RNF13	O43567	P24468
RNF135	Q8IUD6	Q13426	RNF138	Q8WVD3	P07550
RNF144B	Q7Z419	Q07812	RNF145	Q96MT1	P0DTC5
RNF149	Q8NC42	Q30201	RNF165	Q6ZSG1	P12755
RNF167	Q9H6Y7	Q9UNA3	RNF168	Q8IYW5	O76024
RNF169	Q8NCN4	Q08379	RNF181	Q9P0P0	P51668
RNF182	Q8N6D2	P51668	RNF185	Q96GF1	P51668
RNF187	Q5TA31	Q93034	RNF19A	Q9NV58	P00441
RNF19B	Q6ZMZ0	Q8TAF3	RNF20	Q5VTR2	O96017
RNF208	Q9H0X6	P49247	RNF213	Q63HN8	P18031
RNF216	Q9NWF9	Q96CV9	RNF217	Q8TC41	P62736
RNF220	Q9BWG1	Q96CV9	RNF25	Q96BH1	P51668
RNF26	Q9BY78	P62837	RNF34	Q969K3	Q14790
RNF38	Q9H0F5-2	O76024	RNF40	O75150	P06400
RNF41	Q9H4P4	P09681	RNF43	Q68DV7	P62837
RNF44	Q7L0R7	Q8TDS5	RNF5	Q99942	P51668
RNF7	Q9UBF6	Q8WXH5	RNF8	O76064	P62837
RNGTT	O60942	P52292	RNH1	P13489	Q12834
RNLS	Q5VYX0	P23634	RNMT	O43148	P06730
RNMTL1	Q9HC36	Q14162	RNPC3	Q96LT9	Q15696
RNPS1	Q15287	O43809, Q9P2I0, Q9C0J8, Q7Z6E9	ROBO1	Q9Y6N7	O94813
ROBO3	Q96HH0	Q9NRD5	ROCK1	Q13464	P61586
ROCK2	O75116	O15530	ROGDI	Q9GZN7	P21281
ROPN1L	Q96C74	P24588	ROR1	Q01973	P12931
ROR2	Q01974	P35548	RORA	P35398-4	Q9BZS1
RORB	Q92753-1	Q96GM5	RP9	Q8TA86	Q13895
RPA3	P35244	P23025	RPAIN	Q86UA6	O15315
RPAP1	Q9BWH6	Q9NUX5	RPAP2	Q8IXW5	Q15648
RPF1	P78424	O43592	RPGR	Q92834-2, Q92834	O43924
RPH3AL	Q9UNE2	P61289	RPL10	P27635	P12931
RPL10A	P62906	Q01831	RPL11	P62913	P40429
RPL12	P30050	P13569	RPL13	P26373	Q15361
RPL13A	P40429	P06748	RPL14	P50914	Q01831
RPL15	P61313	P06748	RPL17	P18621	Q12906
RPL18	Q07020	P40429	RPL18A	Q02543	Q15361
RPL19	P84098	P40429	RPL21	P46778	Q12906
RPL22	P35268	P06748	RPL23	P62829	Q99558
RPL23A	P62750	P06748	RPL24	P83731	Q99558
RPL26	P61254	P62249	RPL26L1	Q9UNX3	P42858
RPL27	P61353	Q99558	RPL27A	P46776	P63244
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RPL35A	P18077	P17252	RPL36	Q9Y3U8	Q8IY81, P62753
RPL36A	P83881	P63244	RPL36AL	Q969Q0	P03372
RPL37	P61927	P63244	RPL37A	P61513	P19338
RPL38	P63173	P63244	RPL4	P36578	P40429
RPL5	P46777	P40429	RPL6	Q02878	Q15843
RPL7	P18124	Q15843	RPL7A	P62424	P40429
RPL7L1	Q6DKI1	P40429	RPL8	P62917	Q15843
RPLP0	P05388	Q12906	RPLP1	P05386	O43809
RPLP2	P05387	P61981	RPN1	P04843	Q14165
RPN2	P04844	Q14165	RPP25	Q9BUL9	Q00403
RPP40	O75818	P04004	RPRD1B	Q9NQG5	Q9Y5B0
RPS10	P46783	P06748	RPS11	P62280	Q04637
RPS12	P25398	P09429	RPS13	P62277	Q99558
RPS14	P62263	Q15361	RPS15	P62841	P62937
RPS15A	P62244	Q15843	RPS16	P62249	Q8IZH2
RPS17	P08708	P15880, P62280	RPS18	P62269	Q99558
RPS19	P39019	P11309	RPS19BP1	Q86WX3	Q96EB6
RPS2	P15880	P05455	RPS20	P60866	Q15843
RPS21	P63220	P06748	RPS23	P62266	Q15843
RPS24	P62847	Q99558	RPS25	P62851	P62280
RPS26	P62854	P63244	RPS27	P42677	P61289
RPS27A	P62990	P51668	RPS27L	Q71UM5	P06748
RPS28	P62857	P06748	RPS29	P62273	P62280
RPS3	P23396	Q07157	RPS3A	P61247	P40429
RPS4X	P62701	Q00403	RPS5	P46782	P06748
RPS6	P62753	Q92900	RPS6KA3	P51812	Q12849
RPS6KA5	O75582	Q16539	RPS6KA6	Q9UK32	P27361
RPS6KB1	P23443-2	P08151	RPS6KB2	Q9UBS0	P67870
RPS6KL1	Q9Y6S9	Q96HR3	RPS7	P62081	Q15843
RPS8	P62241	Q14197	RPS9	P46781	P40763
RPSA	P08865	P05455	RPTOR	Q8N122	O75385
RPUSD2	Q8IZ73	Q9H2A2	RPUSD3	Q6P087	P84103
RPUSD4	Q96CM3	P01100	RQCD1	Q92600-3	P24522
RRAGA	Q7L523	Q9Y2Q5, Q0VGL1, Q6IAA8, O43504, Q8NBW4, Q9UHA4	RRAGB	Q5VZM2-2, Q5VZM2	Q9Y2Q5, Q0VGL1, Q6IAA8, O43504, Q8NBW4, Q9UHA4
RRAGC	Q9HB90	P17931	RRAS	P10301	Q9UBH0
RRAS2	P62070	Q92696, P53611	RRBP1	Q9P2E9	P13569
RREB1	Q92766	Q92769	RRM1	P23921	P31350
RRM2B	Q7LG56	O43929	RRN3	Q9NYV6	Q8NFM7
RRP1	P56182	Q13724	RRP12	Q5JTH9	P06748
RRP1B	Q14684	P40429	RRP36	Q12481	O14641
RRP7A	Q9Y3A4	Q9UBB9	RRP8	O43159	P19525
RRP9	O43818	Q9Y2X3	RRS1	Q15050	P40429
RSAD2	Q8WXG1	P55056, P02654	RSBN1L	Q6PCB5	P62249
RSF1	Q96T23	P09429	RSG1	Q9BU20	P61587
RSL1D1	O76021	P78563	RSL24D1	Q9UHA3	Q9UJW3
RSPO2	Q6UXX9-2	Q92504	RSPO4	Q2I0M5	O14964

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RTF1	Q92541	P49407	RTKN	Q9BST9, Q8C6B2	P61586
RTKN2	Q8IZC4	Q96B97	RTN1	Q16799-3	O00560
RTN3	O95197	P05919	RTN4	Q9NQC3	O43639
RTN4R	Q99M75	Q96FE5	RTN4RL1	Q86UN2	Q7Z3S9, P0DPK4
RTN4RL2	Q86UN3	P17931	RUFY1	Q96T51	Q96QF0
RUNDC3A	Q59EK9	Q96CW1	RUNDC3B	Q96NL0	P78368
RUNX1	Q01196	Q9H2X6	RUNX1T1	Q06455	P08047
RUNX2	Q13950	P17480	RUNX3	Q13761	Q09472
RUSC1	Q9BVN2	P42772	RUSC1-AS1	Q66K80	P99999
RUSC2	Q8N2Y8	P00533	RUVBL1	Q9Y265	O96019, O94805
RUVBL2	Q9Y230	O96019, O94805	RWDD2A	Q9UIY3	Q15052
RWDD2B	P57060	Q96HC4	RXRA	P19793	Q15648
RXRB	P28702	P37231	RXRG	P48443	P19793
RYBP	Q8N488	Q16143	RYK	P34925	Q68D85
RYR1	P11716	P62942	RYR2	Q92736	Q00987
S100A1	P23297	Q00994	S100A11	P31949	P04637
S100A12	P80511	P01375	S100A13	Q99584	P01375
S100A16	Q96FQ6	Q15915	S100A6	P06703	P35579
S100A8	P05109	P35228	S100A9	P06702	P35228
S100B	P04271	Q15109	S100PBP	Q96BU1	O43809
S100Z	Q8WYG8	Q00994	S1PR1	P21453	Q7Z2K8
S1PR3	Q99500	P21964	SACM1L	Q9NTJ5	Q9H244
SACS	Q9NZJ4	Q13304	SAE1	Q9UBE0	Q06710
SAFB	Q15424	P09651	SALL1	Q9NSC2	Q13547
SALL2	Q9Y467	P08138	SAMD1	Q6SPF0	Q13309
SAMD11	Q96NU1	Q96RU7	SAMD3	Q8N6K7-2	P04156
SAMD4A	Q9UPU9	P31946	SAMD4B	Q5PRF9	P62736
SAMHD1	Q9Y3Z3	P49840	SAMM50	Q9Y512	Q9BRQ6, Q9NX63, Q6UXV4
SAP18	O00422	Q07955	SAP30	O75446	Q00403
SAP30BP	Q9UHR5	Q9UBB9	SAPCD2	Q86UD0	O75496
SARAF	Q96BY9	Q96J02	SARM1	Q6SZW1	Q99836
SARNP	P82979	O76024	SARS	P49591	P38936
SARS2	Q9NP81	Q5T4S7	SART1	O43290	Q13573
SASH1	O94885	P31946, P63104	SASS6	Q6UVJ0	Q9HC77
SAT2	Q96F10	P21673	SATB1	Q01826	P63279
SAV1	Q9H4B6	P18509	SAXO1	Q8IYX7	Q9BUH8
SBF1	O95248	Q13614	SBF2	Q86WG5	Q13614
SCAF1	Q9H7N4	Q16629	SCAF11	Q99590	P61981
SCAF8	Q9UPN6	Q08379	SCAI	Q8N9R8	Q13526
SCAMP1	O15126	Q969Z0	SCAMP2	O15127	P13569
SCAMP3	O14828	P05919	SCAMP4	Q969E2	O76024
SCAMP5	Q8TAC9	P49638	SCAND1	P57086	Q6ZPD8
SCAP	Q9NP31	O14543	SCAPER	Q9BY12-3	P14136
SCARA3	Q6AZY7-2	P19404	SCARA5	Q6ZMJ2-2	Q96GM5
SCARB1	Q8WTV0	P11686	SCARF1	Q14162	Q96BA8
SCD	O00767	Q9H2K0	SCFD1	Q8WVM8	P16234
SCFD2	Q8WU76	P54920	SCG2	O00255-2, O00255	P35579

Input	UniProt Id	Interacts with	Input	UniProt Id	Interacts with
SCG3	Q8WXD2	O00560	SCGN	O76038	P15336
SCHIP1	P0DPB3-4	Q96GM5	SCIMP	Q6UWF3	P41240
SCIN	Q9Y6U3	P60709	SCLT1	Q96NL6	A1L190
SCMH1	Q96GD3	P35227	SCML1	Q9UN30-2	O00560
SCML2	Q9UQR0-1	P38936	SCN2B	O60939	O60313
SCN3B	Q9NY72	P21359	SCN4B	Q8IWT1	O95070
SCN5A	Q14524	Q99873	SCNM1	Q9BWG6	Q15669
SCNN1A	P37088	P46934	SCP2	P22307	Q03135
SCPEP1	Q9HB40	P27824	SCRIB	Q14160	P48065
SCRN1	Q12765	P10636-8	SCRN2	Q96FV2	P25963
SCUBE3	Q8IX30	P37173	SCYL1	Q96KG9-4	Q13526
SCYL2	Q6P3W7	O14640	SDC1	P18827	Q03468
SDC2	P34741	P21802	SDC3	A0A0S2Z4U3	P63252
SDC4	P31431	O15530	SDCCAG3	Q96C92-2	O00560
SDCCAG8	Q86SQ7	P61088	SDF2	Q99470	Q9UBS4
SDF4	Q9BRK5	P01106	SDHAF1	A6NFI7	Q9H1K1
SDHAF2	Q9NX18	P31040	SDHAF3	Q9NRP4	O14561
SDHAF4	Q5VUM1	Q9NRP4, P31040	SDHC	Q99643	P17252
SDK2	Q58EX2	P17931	SDPR	O95810	Q6NZI2
SDR16C5	Q8N3Y7	P49638	SEC11C	Q9BY50	P09601
SEC13	P55735	P48729	SEC14L5	O43304	P62333
SEC16A	O15027	Q9BY84	SEC22A	Q96IW7	P35232
SEC23A	Q15436	P50570	SEC23B	Q15437	P13569
SEC24B	O95487	Q15436	SEC24C	P53992	P13569
SEC24D	O94855, O94855-2	Q15436	SEC61A1	P61619	Q16288
SEC61B	P60468	P43307	SEC61G	P60059	P11686
SEL1L	Q9UBV2	Q16288	SELE	P16581	O76024
SELENBP1	Q13228	Q9UJ68	SELK	Q9Y6D0	Q9H2K0
SELL	P14151-2	Q9BRI3	SELM	Q8WWX9	P61604
SELO	Q9BVL4	Q9NRM6	SELP	P49908	P00533
SEMA3B	Q13214-2	Q9NRD5	SEMA3C	Q99985	P18827
SEMA3E	O15041, P70275	Q9Y4D7	SEMA4C	Q9C0C4	Q96PM5
SEMA4F	O95754	Q13485	SEMA4G	Q9NTN9-3	P99999
SENP1	Q9P0U3	P46060	SENP3	Q9H4L4	P06748
SENP5	Q96HI0	P06748	SENP7	Q9BQF6	Q13263
SEPHS1	P49903	Q99611	SEPP1	P49908	P00533
SEPSECS	Q9HD40	P04591	SERBP1	Q8NC51	P50454
SERF1B	O75920	A1L3X0	SERGEF	Q9UGK8	Q96LA8
SERINC1	Q9NRX5	Q06710	SERP1	Q9Y6X1	Q9H2K0
SERP2	Q8N6R1	Q9H2H9	SERPINA1	P01009	P08246
SERPINA5	P05154	P09958	SERPINB1	P30740	Q00403
SERPINE1	P05121	P04004	SERPINF1	P36955	Q9P2X3
SERPING1	E9PK97	Q9BRI3	SERPINH1	P50454	Q9NUD9
SERTAD2	Q14140	O60733	SERTM1	A2A2V5	Q9H2K0
SESN2	P58004	Q13501	SET	Q01105	P17480
SETD1A	O15047	P04637	SETD1B	Q9UPS6	Q9UER7
SETD2	Q9BYW2	P00740	SETD3	Q86TU7	P12004
SETD7	Q8WTS6	P26358	SETDB1	Q15047	Q6VMQ6
SETMAR	Q53H47	P11142	SETX	Q7Z333	P63279
SEZ6L2	Q6UXD5	P17931	SF1	Q13285	Q9NPJ4

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SF3A1	Q15459	P08047	SF3A2	Q15428	O43639
SF3A3	Q12874	P11142, P04792	SF3B1	O75533	O43809
SF3B2	Q13435	Q15696	SF3B3	Q15393	P53350
SF3B4	Q15427	O75177	SF3B5	Q9BWJ5	Q96FJ0
SF3B6	Q9Y3B4	O75533	SFMBT1	Q9UHH3	Q13547
SFPQ	P23246	Q13882	SFRP1	Q8N474	Q8N661
SFRP4	Q6FHJ7	Q96EB6	SFT2D1	Q8WV19	P26927
SFT2D2	O95562	Q99541	SFTPD	P35247, P35247-PRO_0000017465	P0DTC2
SFXN1	Q9H9B4	Q7KZN9	SFXN2	Q96NB2	Q7Z5P4
SFXN3	Q9BWM7	P46059	SFXN5	Q8TD22	Q92911
SGCB	Q16585	Q01453	SGCG	Q13326	O00501
SGCZ	Q08AT0	Q16585	SGK1	O00141	P49815
SGK2	Q9HBY8	Q13451	SGK223	Q86YV5	P46109
SGK3	Q96BR1	Q96J02	SGPP1	Q9BX95	Q8WTR4
SGSM1	Q2NKK1-4, Q2NKK1	P41208	SGSM3	Q96HU1	Q8NC56
SGTA	O43765	P01135	SH2B1	Q9NRF2	O60674
SH2B2	O14492-2	Q9UBH0	SH2D3C	Q8N5H7	P00533
SH2D4A	Q9H788	P02787	SH3BGR1	O75368	P00533
SH3BGR1	Q9H299	P00533	SH3BP1	Q9Y3L3	Q96B97
SH3BP4	Q9P0V3	O14908	SH3BP5	O60239	Q06187
SH3BP5L	Q7L8J4	P31946, P63104	SH3D19	Q5HYK7	P63104
SH3D21	A4FU49	P00533	SH3GL1	Q99961	O14672
SH3GL2	Q99962	Q99962	SH3GL3	Q99963	Q99962
SH3GLB1	Q9Y371	P01730	SH3GLB2	Q9NR46	Q9P0N9
SH3KBP1	Q96B97	P98082	SH3PXD2A	Q5TCZ1	P31946, P63104
SH3RF1	Q7Z6J0	P63000	SH3RF2	Q08AM8, Q8TEC5	Q8N9N5
SH3RF3	Q8TEJ3	Q13177	SH3YL1	Q96HL8	P14136
SHANK1	Q9Y566	P46937	SHANK2	Q9UPX8	P19174
SHANK3	Q9BYB0	P19174	SHARPIN	Q9H0F6	P08648
SHB	Q15464	P00519	SHC1	P29353	P35968
SHC3	Q92529	Q86U10	SHC4	Q6S5L8	P00533
SHCBP1	Q8NEM2	P05771	SHD	Q96IW2	O43639
SHH	Q62226, Q15465	Q4KMG0, Q96QV1	SHISA4	Q96DD7	O95406
SHISA5	Q8N114	Q9NZC7	SHISA6	Q6ZSJ9	P38936
SHKBP1	Q8TBC3	P61081	SHMT2	P34897	P13569
SHOC2	Q9UQ13	Q96RT1	SHOX2	O60902-3	Q8N2W9
SHPRH	Q149N8	Q09028	SHQ1	Q6PI26	Q9Y265
SHROOM2	Q13796	P61981	SHROOM3	Q8TF72	P31946, P63104
SIAH2	O43255	P35372	SIGLEC8	Q9NYZ4	P04155
SIGLEC9	Q9Y336	P11686	SIGMAR1	Q99720	P14416
SIK2	Q9H0K1	P61981	SIK3	Q9Y2K2	P31946, P63104
SIL1	Q9H173	P11021	SIPA1	Q96FS4	P38936
SIPA1L1	O43166	P31946, P63104	SIPA1L2	Q9P2F8	P31946, P63104
SIPA1L3	O60292	P31946, P63104	SIRPA	P78324, P97710	Q06124
SIRPB2	Q9P1W8	Q08722	SIRT1	Q96EB6	O15520
SIRT2	Q8IXJ6, Q8VDQ8, Q8IXJ6-2	Q9UM11, Q12834	SIRT3	Q9NTG7	O60313
SIRT4	Q9Y6E7	P10515	SIRT6	Q8N6T7	P01106
SIRT7	Q9NRC8	Q16656	SIVA1	O15304-2	P49639

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SIX1	Q15475	Q08117	SIX3	O95343	Q92570
SKAP1	Q86WV1	O15117	SKAP2	O75563	Q9Y2R2
SKI	P12755	P36941	SKIDA1	Q1XH10	Q15022
SKIL	P12757	P23025	SKIV2L2	P42285	P31946
SKP1	P63208-1	P53611	SKP2	Q13309	Q13415
SLA	Q13239-3	Q13882	SLA2	Q9H6Q3	P54646
SLAIN1	Q8ND83	P38646	SLAIN2	Q9P270	Q9Y6A5
SLAMF1	Q13291	Q06124	SLAMF6	Q96DU3	Q96DU3
SLAMF7	Q8BHK6	P06241	SLBP	Q14493	Q14493
SLC10A2	Q12908	Q01628	SLC10A6	Q3KNW5	Q86T03
SLC10A7	Q0GE19	P04114	SLC11A2	P49281-3	P00403
SLC12A2	P55011	P17931	SLC12A4	Q9UP95	P17931
SLC12A6	Q9UHW9	P17931	SLC12A7	Q9Y666	P43119
SLC12A8	A0AV02	P26367	SLC13A3	Q8WWT9	Q3SXY8
SLC13A4	Q9UKG4	P27352	SLC14A1	Q13336-2	Q5QGT7
SLC14A2	Q15849	P01375	SLC15A2	Q16348	Q92922
SLC15A3	Q05CH4	P14373	SLC15A4	Q8N697	Q9HAI6, Q13568
SLC16A2	P36021	P21926	SLC16A3	O15427	Q14160
SLC16A6	O15403	O76024	SLC16A7	O60669	P09601
SLC17A9	Q9BYT1	Q9NR28	SLC18A2	Q05940	Q969Z0
SLC19A1	P41440	Q03135	SLC19A2	O60779	Q92911
SLC19A3	Q9BZV2	O43561-2	SLC1A1	P43005	O60493
SLC1A2	P43004	P10636	SLC1A3	P43003	Q8WVV5
SLC1A5	Q15758	P0DTC5	SLC20A1	Q8WUM9	Q9UM47
SLC20A2	Q08357	Q8TDQ0	SLC22A15	Q8IZD6	Q86TM6
SLC22A16	Q86VW1	Q99808	SLC22A17	Q8WUG5	P07237
SLC22A23	A1A5C7-2	P27352	SLC22A5	O76082	Q9H015
SLC24A3	Q9HC58	Q86TM6	SLC25A1	P53007	P13569
SLC25A11	Q02978	P12830	SLC25A12	O75746	Q7Z434
SLC25A13	Q9UJS0	Q9Y5J6	SLC25A15	Q9Y619	Q15116
SLC25A16	P16260	Q15116	SLC25A21	Q9BQT8	P0DTC5
SLC25A22	Q9H936	P13569	SLC25A23	Q9BV35	Q8N695
SLC25A24	Q6NUK1	P13569	SLC25A29	Q8N8R3	P35613
SLC25A3	Q00325	Q7KZN9	SLC25A30	Q5SVS4	Q9UBS4
SLC25A4	P12235	P12236	SLC25A40	Q8TBP6	P02452
SLC25A41	Q8N5S1	P98082	SLC25A43	Q8WUT9	P10809
SLC25A46	Q96AG3	Q8IVP5	SLC25A48	Q6ZT89	Q13077
SLC25A5	P05141	P12236	SLC25A6	P12236	P0C0U1
SLC26A2	P50443	P17931	SLC26A4	O43511	Q13257
SLC26A6	Q9BXS9-3	P02787	SLC27A1	Q6PCB7	P40855
SLC27A4	Q6P1M0	Q16659	SLC27A6	Q9Y2P4	Q9Y2C2
SLC2A1	P11166	Q15904	SLC2A12	Q8TD20	Q7Z7B1
SLC2A13	Q96QE2	Q96CV9	SLC2A3	P11169	P17931
SLC2A4	P14672	Q9UER7	SLC2A5	P22732	Q3SXY8
SLC2A9	Q9NRM0-1	Q5J8M3	SLC30A1	Q9Y6M5	P43119
SLC30A4	O14863	P55010	SLC30A5	Q8TAD4	P21217
SLC30A6	Q6NXT4	Q8TAD4	SLC30A7	Q8NEW0	Q9Y2C2
SLC30A8	Q8IWU4	P63027	SLC30A9	Q6PML9	Q96BI1
SLC33A1	O00400	P13569	SLC35A1	P78382	Q9Y624
SLC35A2	P78381	P16885	SLC35A4	Q96G79	Q9UKG4

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SLC35B1	P78383	Q3SXY8	SLC35B2	Q8TB61	Q96BA8
SLC35B4	Q969S0	O43914	SLC35C1	Q96A29	P05090
SLC35E1	Q96K37	Q9Y366	SLC35F1	Q5T1Q4	P06576
SLC35F2	Q8IXU6	P06576	SLC35F5	Q8WV83	P61073
SLC35G1	Q2M3R5	Q15125	SLC36A1	Q7Z2H8	P05067
SLC37A3	Q8NCC5	Q9Y4D1	SLC38A1	Q9H2H9	Q9UKG4
SLC38A2	Q96QD8	P43657	SLC38A7	Q9NVC3	P62495
SLC38A9	Q8NBW4	Q9Y2Q5, Q0VGL1, Q6IAA8, O43504, Q9UHA4	SLC39A1	Q9NY26	Q96CV9
SLC39A11	Q8N1S5	Q9NZQ7	SLC39A13	Q96H72	O15173
SLC39A14	Q15043	P48960	SLC39A3	Q9BRY0	P32971
SLC39A4	Q6P5W5	Q09328	SLC39A6	Q13433	Q6ZMH5
SLC39A7	Q92504	Q9UBV7, O75063	SLC39A8	Q9C0K1	Q09328
SLC3A2	P08195	P21926	SLC40A1	Q9JHI9	O60674
SLC41A1	Q8IVJ1	P30518	SLC41A2	Q96JW4	P21217
SLC44A1	Q8WWI5	O14672	SLC44A2	Q8IWA5	P0DTC4
SLC44A3	Q8N4M1-3	Q8WTS1	SLC4A1	P02730	P18031
SLC4A1AP	Q9BWU0	Q13418	SLC4A3	P48751	Q12933
SLC4A7	Q9Y6M7	P17931	SLC50A1	Q9BRV3	O00560
SLC51B	Q86UW2	O43561-2	SLC52A3	Q9NQ40	P54252
SLC6A1	P30531	Q9H3Z4	SLC6A12	P48065	Q14160
SLC6A13	Q9NSD5-3	P04792	SLC6A15	Q9H2J7	Q9NRZ7
SLC6A20	Q9NP91	Q7Z3S9, P0DPK4	SLC6A8	P48029	P43119
SLC7A1	P30825	P27352	SLC7A11	Q9UPY5	P16070
SLC7A14	Q8TBB6	P15291	SLC7A2	P52569	P30825
SLC7A3	Q8WY07	Q9Y2B1	SLC7A6	Q92536	P04201
SLC7A8	Q9UHI5	O60906	SLC9A1	P19634	P43119
SLC9A3R1	O14745	P29762	SLC9A3R2	Q15599	O15439
SLC9B2	Q86UD5	O15173	SLCO1C1	Q9NYB5	Q14160
SLF1	Q9BQI6	P19838	SLF2	Q8IX21	P25786
SLFN12	Q8IYM2	P01019	SLIRP	Q9GZT3	P05919
SLIT1	O75093	P49639	SLIT2	O94813	O94813, Q9Y6N7
SLITRK4	Q8IW52	Q9BT09	SLITRK5	O94991	Q6PJG9
SLMAP	Q14BN4	Q96CV9	SLMO1	Q96N28	O43715
SLMO2	Q9Y3B1	O43715	SLPI	P03973	O43561-2
SLU7	O95391	P14136	SLX4	Q8IY92	P43246
SLX4IP	Q5VYV7	Q16539	SMAD2	Q15796	P98082
SMAD3	P84022	P98082	SMAD4	Q13485	Q6VMQ6
SMAD5	Q99717	Q9HAU4	SMAD6	O43541-2, O35182	O95630
SMAD7	O15105	Q9HAU4	SMAD9	O15198, O15198-2	Q13485
SMAP2	Q9H0E9	O96019	SMARCA1	P28370	Q09028
SMARCA2	P51531	Q8NFD5	SMARCA4	P51532	Q12778
SMARCB1	Q12824	Q92925, Q8NFD5, P51531, O96019, O75177, Q15532	SMARCC1	Q92922	Q92925, Q8NFD5, O96019, Q15532
SMARCC2	Q8TAQ2	Q92925, Q8NFD5, P51531, O96019	SMARCD1	Q96GM5	Q9H8M2
SMARCD3	Q6STE5-2, Q6STE5-1	P37231	SMARCE1	Q969G3	Q92925, Q8NFD5, P51531, O96019, O75177, Q15532
SMC1A	Q14683	Q7Z5K2	SMC2	O95347	P01106

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SMC3	Q9CW03, Q9UQE7	Q7Z5K2	SMC4	Q9NTJ3	P01106
SMC5	Q8IY18	Q9H668	SMCHD1	A6NHR9	Q96GD4
SMCO3	A2RU48	Q9H668	SMCO4	Q9NRQ5	Q96BA8
SMDT1	Q9H4I9	Q9BPX6, Q8NE86	SMG1	Q96Q15	Q92900
SMG5	Q9UPR3	Q92540	SMG6	Q86US8	P67870
SMG8	Q8ND04	Q9Y639	SMG9	Q9H0W8	Q9BRV3
SMIM3	Q9BZL3	Q16585	SMIM5	Q71RC9	Q08426
SMN2	Q16637	Q96FZ7	SMOC1	Q9H4F8	Q04721
SMPD3	Q9NY59	Q03135	SMPD4	Q9NXE4	Q9H1C4
SMTN	P53814-5	O76024	SMTNL2	Q2TAL5	P61024
SMUG1	Q53HV7-2	Q96CV9	SMURF1	Q9HCE7-2	O14757
SMURF2	Q9HAU4	Q9HAU4	SMYD2	Q9NRG4	P01160
SMYD3	Q9H7B4	Q15915	SNAI1	O95863	P49841
SNAI2	O43623	Q9NPB3	SNAP23	O00161	P13569
SNAP25	P60880-2	P63027	SNAP29	O95721	Q13190
SNAP47	Q5SQN1	P51809	SNAP91	O60641	P38398
SNAPC1	Q16533	Q8IYI6	SNAPC4	Q5SXM2	Q9UPP1
SNAPIN	O95295	P20700	SNCA	P37840	Q969S8
SNCAIP	Q9Y6H5, Q9Y6H5-2	P37840	SNCB	Q16143	O75925
SND1	Q7KZF4	P06748	SNF8	Q96H20	P55199
SNPH	O15079	P23025	SNRNP200	O75643	P01106
SNRNP25	Q9BV90	Q9Y2W6	SNRNP27	Q8WVK2	Q13177
SNRNP35	Q16560-2	P61024	SNRNP48	Q6IEG0	Q15696
SNRNP70	P08621	Q07955	SNRPA	P09012	Q15366
SNRPB	P14678-2	P52747	SNRPB2	P08579	Q13573
SNRPC	Q5TAL4	O75177	SNRPD1	P62316	P09651
SNRPD2	P62316	P09651	SNRPD3	P62318	Q13573
SNRPE	P62304	Q13573	SNRPF	P62306	Q13573
SNRPG	P62308	Q06547-3	SNRPN	P63162	O75563
SNTA1	Q13424	P60484	SNTB1	Q13884	P60484
SNTB2	Q13425	O14936	SNTG2	Q9NY99-2	Q8N3R9
SNU13	P55769	Q9Y2X3	SNW1	Q13573	Q86U42
SNX1	Q13596	Q8IVP5	SNX10	Q9Y5X0	P17252
SNX11	Q9Y5W9	P15514	SNX12	Q9UMY4	O60493
SNX13	Q9Y5W8	P61073	SNX14	Q9Y5W7	P61073
SNX16	P57768	O95619	SNX17	Q15036	Q13496
SNX18	Q96RF0	O14672	SNX2	O60749	Q5T8I9
SNX20	Q7Z614-3	P50454	SNX21	Q969T3	P17707
SNX24	Q9Y343	P48729	SNX27	Q96L92	P27037
SNX29	Q8TEQ0	Q13077	SNX3	O60493	Q8WVK2
SNX30	Q8WV41	Q05193	SNX31	Q8N9S9-2	P63000
SNX32	Q86XE0	Q06609	SNX33	Q8WV41	Q05193
SNX4	O95219	Q6ZPD8	SNX5	Q9Y5X3	Q9Y639
SNX7	Q9UNH6	O14964	SNX9	Q9Y5X1	Q07889
SOAT1	P35610	O75326	SOCS2	O14508	Q15369
SOCS3	O14543	Q15369, Q15370	SOCS4	Q8WXH5	Q8N3R9
SOCS5	O75159	Q93034, Q9UBF6	SOCS6	O14544	P46937
SOCS7	Q8WXH5	Q8N3R9	SOGA1	O94964	P31946, P63104
SORBS1	Q9BX66	Q13177	SORBS2	O94875	Q13177
SORBS3	O60504	Q13177	SORL1	Q92673	Q15669

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SORT1	Q99523	P05231	SOS1	Q62245	P62993-1
SOS2	Q07890	P19174	SOX10	P56693	O00712, Q12857
SOX12	O60248	O00712, Q12857	SOX13	Q9UN79	Q12933
SOX17	Q9H6I2	O00712, Q12857	SOX30	O94993	Q01543
SOX4	Q06945	P04637	SOX5	P35711	O00712, P08651, Q12857
SOX6	P35712	O00712, P08651, Q12857	SOX9	P48436	Q12857
SP1	P26951-1, P08047	P32519, P08700	SP100	P23497	Q9Y6K1
SP140	Q13342	Q13526	SP3	Q02447	P03372
SP4	Q02446	P14859	SP6	Q3SY56	P27797
SPACA6P	W5XKT8	Q8N9N5	SPAG8	Q99932-2	O43809
SPAG9	O60271	P19838	SPATA18	Q8TC71	O95995
SPATA2	Q9UM82	Q9UI12	SPATA22	Q8NHS9	Q00403
SPATA24	Q86W54-2	P06396	SPATA2L	Q8IUW3	O43639
SPATA33	Q96N06	P49753	SPATA5	Q8NB90	P25445
SPATA6	Q9NWH7-2	P55040	SPATC1L	Q9H0A9-2	Q9NPB3
SPATS2L	Q9NUQ6	Q9H9Z2	SPC25	Q9HBM1	P36941
SPCS2	Q15005	P0DTC3	SPCS3	P61009	P27824
SPECC1	Q5M775	P63104	SPEF1	Q9Y4P9	O96006
SPEG	Q15772	P31946	SPG11	Q96JI7	Q05193
SPG20	Q8N0X7	P19338	SPG21	Q9NZD8	P42772
SPG7	Q9UQ90	P61289	SPHK1	Q9NYA1-2	Q7Z4W1
SPHK2	Q9NRA0	Q96CV9	SPHKAP	Q2M3C7	Q9NYA1
SPI1	P17947	P78527	SPIB	Q01892	P42858
SPICE1	Q8N0Z3	O60341	SPIDR	Q14159	Q06609
SPIN2B	Q9BPZ2	Q01831	SPINT1	O43278-2	P02787
SPINT2	O43291	Q99640	SPIRE1	Q08AE8	P60709
SPIRE2	Q8WWL2	P31946	SPNS1	Q9H2V7	Q07817
SPNS3	Q6ZMD2-2	Q8WVV5	SPOPL	Q6IQ16	Q99836
SPP1	P10451	P16070	SPPL2A	Q8TCT8	O76024
SPPL2B	Q8TCT7	O75063	SPPL3	Q8TCT6	Q9H015
SPRED1	Q7Z699	P21359	SPRED2	Q7Z698	P14136
SPRY1	O43609	O43597	SPRY2	O43597	P22681
SPRY4	Q8WW59	P24534	SPRYD3	Q8NCJ5	P63208
SPRYD4	Q8WW59	P24534	SPSB1	Q96BD6	P08670
SPSB2	Q99619	Q96GM5	SPSB3	Q6PJ21	Q93034, Q9UBF6
SPSB4	Q96A44	P21926	SPTA1	P02549	Q9NQT4
SPTAN1	Q13813	P13569	SPTB	P11277	P42224
SPTBN1	Q01082	Q13813	SPTLC1	Q6NUL7	P60896
SPTY2D1	Q68D10	P78563	SQLE	Q14534	Q96BA8
SQSTM1	Q13501	P04629	SRA1	Q9HD15	Q92769
SRBD1	Q8N5C6	P49840	SRC	P12931	O15455
SRCIN1	Q9QWI6, Q9C0H9	P56945	SREBF1	P36956-PRO_0000314029	P20700, P02545-1
SREBF2	Q12772	P08047	SREK1	Q9JKL7	Q15424, Q92841
SRF	P11831	Q969V6	SRFBP1	Q8NEF9	Q13895
SRGAP1	Q7Z6B7	P31946, P63104	SRGAP2	O75044	Q9Y6N7
SRGAP2B	A0A286YFY3	Q8N3R9	SRGAP3	Q8IXS7	P04792
SRGN	P10124	P10144	SRI	P30626	P30626
SRM	P19623	P19623	SRP14	P37108	P67809
SRP68	Q9UHB9	P40429	SRP72	O76094	P48431

Input	UniProt Id	Interacts with	Input	UniProt Id	Interacts with
SRP9	P49458	O75533	SRPK1	Q96SB4, Q96SB4-2, Q96SB4-3	Q15424, P17844, P06748
SRPK2	P78362	P46087	SRPRB	Q9Y5M8	P09601
SRPX	P78539	Q9BRX2	SRR	Q9GZT4	Q9H4P4
SRRM4	A7MD48	Q15696	SRRT	Q9BXP5	P09651
SRSF1	Q07955	Q13573	SRSF11	Q05519	P61981
SRSF12	Q8WXF0	P61981	SRSF2	Q01130	Q9HD47
SRSF5	Q13243	Q13247, Q15696	SRSF6	Q13247	P05455
SRSF7	Q16629	O95639	SRSF8	Q9BRL6	P61981
SRSF9	Q13242	Q07955	SS18	Q15532	Q92925
SS18L1	O75177-5	Q06710	SS18L2	Q9UHA2	Q9H8M2
SSB	P05455	Q8WZA2	SSBP2	P81877	Q9UBH0
SSBP3	Q9BWW4	Q9UBH0	SSBP4	Q9BWG4	Q9UBH0
SSC5D	A1L4H1	Q13526	SSFA2	P28290	P22681
SSH1	Q8WYL5	P53667	SSH2	Q76I76	P31946, P63104
SSH3	Q8TE77	O43639	SSNA1	O43805	Q9P0N9
SSR2	P43308	Q86WV6	SSR3	Q9UNL2	P13569
SSR4	P51571	P13569	SSRP1	Q08945	Q9NTX7
SSSCA1	O60232	Q13459	SSTR3	P32745	P05067
SSU72	Q9NP77	O60216	SSUH2	Q9Y2M2-2	Q969R5
SSX2IP	Q8VC66	P12814	ST13	P50502	P02649
ST3GAL1	Q11201	Q9ULF5	ST3GAL2	Q16842	P11021
ST5	P78524	P63104	ST6GAL2	Q96JF0-2	Q9NPL8
ST6GALNAC1	Q9NSC7	P13569	ST8SIA1	Q92185	Q96CV9
ST8SIA3	O43173	Q9UBS3	ST8SIA4	Q92187	O75197
STAB1	Q9NY15	P02751	STAC	Q99469	Q08379, P62258
STAC2	Q6ZMT1	P54646	STAG1	Q8WVM7	Q7Z5K2
STAG2	Q8N3U4	Q7Z5K2	STAM	Q92783	O95630
STAM2	O75886	P02545, O76024	STAR	P49675	O60906
STARD10	Q9Y365	Q96LA8	STARD7	Q9NQZ5	P11245
STAT1	P42224	P40763	STAT2	P52630	O94782
STAT3	P40763	P35232	STAT4	Q14765	P42224
STAT5A	P42229	P10515	STAT5B	P51692	P00519
STAT6	P42226	Q9UHD2	STAU1	O95793	Q92900
STIL	Q15468	Q9HC77	STIM1	Q13586	P30101
STIM2	Q9P246	Q8NBK3	STIP1	P31948	P01106
STK11	Q15831	P26927	STK16	O75716	P49247
STK17B	O94768	P08173	STK19	B7ZLI8	Q969Z0
STK24	Q9Y6E0	P67775	STK25	O00506	Q3SXY8
STK26	Q9P289	P13569	STK3	Q13188	P18509
STK32C	Q86UX6	P37268	STK35	Q8TDR2	Q13451
STK38	Q15208	Q9Y5B0	STK39	Q9UEW8	P43119
STK4	Q13043	P18509	STK40	Q8N2I9	Q8NHY2
STKLD1	Q8NE28	P17980	STMN1	P16949	P46527
STMN2	Q93045	P20618	STMN3	Q9NZ72	Q96GM5
STMN4	Q9H169-2	Q9NR28	STOM	P27105	O43916
STOML2	Q9UJZ1	O76024	STON2	Q8WXE9	Q15843
STOX2	Q9P2F5	P61981	STPG2	Q8N412	P02787
STRA13	O14503	Q16613	STRADA	Q7RTN6-1, Q7RTN6	Q15831

Input	UniProt Id	Interacts with	Input	UniProt Id	Interacts with
STRADB	Q9C0K7	Q15831	STRAP	Q9Y3F4	P13569
STRBP	Q96SI9	P09651	STRIP1	Q5VSL9	P62195
STRN	O43815	P62195	STRN3	Q13033-2	P61981
STRN4	Q9NRL3	P05412	STS	P08842	Q8NBK3
STT3B	Q8TCJ2	Q14165	STUB1	Q9UNE7	P08138
STX11	O75558	P60880	STX12	Q86Y82	P60880
STX17	P56962	Q9BV40, O00161	STX18	Q9P2W9	Q13190, Q9NYM9
STX1A	Q16623	P60880	STX1B	P61266	O15554
STX2	P32856-2	Q13190	STX6	O43752	P51809
STX8	Q9UNK0	Q96PE7	STXBP1	P61764	P60880
STXBP3	O00186	Q12846	STXBP4	Q6ZWJ1	P53350
STXBP5	Q5T5C0	Q13277	STXBP6	Q8NFX7	Q13277
STYK1	Q6J9G0	P49841	SUB1	P53999	Q12906
SUCLG1	P53597	Q8IWL3	SUCLG2	Q96I99	Q8IWL3
SUCO	Q9UBS9	P07237	SUFU	Q9UMX1-2, Q9UMX1	P08151
SUGP1	Q8IWZ8	Q9NYY3	SUGP2	Q8IX01	P09651
SUGT1	Q9Y2Z0	Q8IW35	SULF1	Q8IWU6	P34741
SULT1B1	O43704	Q9NUX5	SULT4A1	Q9BR01	Q9NUX5
SUMF1	Q8NBK3	Q8NBK3, Q8NBJ7	SUMF2	Q8NBJ7	Q8NBK3
SUMO1	P63165	P19419	SUMO2	P61956	Q13569
SUMO3	P55854	P63279	SUN1	Q9D666, O94901	P02545
SUN2	Q9UH99	Q96PU5	SUOX	P51687	Q15669
SUPT16H	Q9Y5B9	Q9NTX7	SUPT3H	O75486	O75528, Q96ES7, Q92831
SUPT4H1	P63272	O00267	SUPT5H	O00267	P62937
SUPT6H	Q7KZ85	O00267	SUPT7L	O94864, O94864-2	Q96ES7
SURF2	Q15527	P05455	SURF4	O15260	P13569
SURF6	O75683	P78563	SUSD3	Q96L08	P27449
SUSD6	Q92537	Q96BN8	SUV39H1	O43463	P50454
SUV39H2	Q9H5I1	P45973	SUV420H2	Q86Y97	Q13185
SUZ12	Q15022	P01106	SV2A	Q02563	P0DPI0
SVIL	O46385	P35579	SWAP70	Q9UH65	Q8WZA2
SWSAP1	Q6NVH7	Q06609	SYDE1	Q6ZW31	P62258
SYDE2	Q5VT97	P31946, P63104	SYF2	O95926	Q13573
SYK	P43405	Q9ULZ3	SYMPK	Q92797	O95639
SYN1	P17600	P00533	SYN3	Q17R54	O75928-2
SYNC	Q9H7C4, Q9H7C4-2	P08670	SYNCRIP	O60506-4	Q00403
SYNDIG1	Q9H7V2	Q9H1C4	SYNE1	Q8NF91-1, Q8NF91	Q9UH99, O94901
SYNE2	Q8WXH0-1	P02545	SYNE3	Q6ZMZ3	Q9UH99
SYNGAP1	Q96PV0	Q6VMQ6	SYNGR1	O43759-2	P29274
SYNGR2	O43760	Q9Y6L6	SYNGR3	O43761	P49638
SYNJ1	Q62910, O43426	Q99962	SYNJ2	O15056	P31946
SYNPO	Q8N3V7	P04156	SYNPO2	Q9UMS6	Q13418
SYNPR	Q8TBG9	Q8IY26	SYP	P08247	Q96E11
SYPL1	Q16563	P63027	SYT1	P21579	P98155
SYT11	Q9BT88	Q9NQ11	SYT16	Q17RD7-3	P04792
SYT17	Q9BSW7	P02787	SYT2	Q8N9I0	P98155
SYT3	Q9BQG1	Q6XYQ8	SYT4	Q9H2B2	Q9UI09
SYT5	O00445	P21579	SYT6	Q5T7P8-2	Q12933

Input	UniProt Id	Interacts with	Input	UniProt Id	Interacts with
SYT7	O43581	Q8IUQ4	SYTL2	Q9HCH5-13	O95070
SYTL3	Q4VX76	P51159	SYTL4	Q96C24	P51159
SYTL5	Q8TDW5-2	O95070, O76024	SYVN1	Q86TM6	P04637
SZRD1	Q7Z422	Q92900	TAB1	Q15750	O43318
TAB2	Q9NYJ8	P52747	TACC1	O75410-7, O75410	Q9Y316
TACC2	B2RWP4	P22735	TACC3	Q9Y6A5	Q00994
TACO1	Q9BSH4	P21673	TACR1	P25103	P00395
TADA2B	Q86TJ2	Q96EB6	TADA3	O75528	Q8N2W9
TAF10	Q12962	P20226	TAF15	Q16514	P20226, Q15572
TAF1B	Q53T94	P27797	TAF1C	Q15572	P06748
TAF1D	Q9H5J8	P06746	TAF2	Q6P1X5	P35249
TAF3	Q5VWG9	P84243	TAF4	O00268	O43889
TAF5	Q15542	P20226	TAF5L	O75529	O43524
TAF6	P49848	P20226	TAF6L	Q9Y6J9	P11142
TAF7	Q15545	P20226	TAF8	Q7Z7C8-2	P54652
TAF9	Q16594	P62263	TAF9B	Q9HBM6	P35249
TAL1	P17542	Q01196	TALDO1	P37837	P42858
TAMM41	Q96BW9	P46059	TANC1	Q9C0D5	Q8NI35
TANC2	Q9HCD6	P31946	TANGO2	Q6ICL3	Q99962
TANK	Q92844	Q8NFX5	TAOK1	Q7L7X3	P61981
TAOK2	Q9UL54	P36941	TAOK3	Q9H2K8	Q12933
TAP1	Q03518	P30101	TAPBP	O15533	Q30201
TAPBPL	Q9BX59	P0DPK4	TARBP1	Q13395	O75496
TARBP2	Q15633	P03496	TARS	P26639	Q96CV9
TARS2	Q9BW92	Q9UBN6	TARSL2	A2RTX5	P18848
TAS2R5	Q9NYW4	Q9BRI3	TASP1	Q9H6P5	Q9H6P5
TATDN2	Q93075	Q03405	TAX1BP1	Q86VP1	Q99958
TBC1D1	Q86TI0	P31946, P63104	TBC1D10A	Q9BXI6	O43924
TBC1D13	Q9NVG8	O60664	TBC1D14	Q9P2M4	O75385
TBC1D15	Q8TC07	Q99941	TBC1D17	Q9HA65	Q8TC07
TBC1D20	Q96BZ9	P55056	TBC1D22A	Q8WUA7-2	P49821
TBC1D25	Q3MII6	P18848	TBC1D2B	Q9UPU7	Q9GZQ8
TBC1D30	Q9Y2I9-2	Q9UGI0	TBC1D4	O60343	P31946, P63104
TBC1D5	Q92609	Q9BZP6	TBC1D8	O95759	Q13077
TBC1D9B	Q66K14-2, A1L3A9	Q969R5	TBCB	Q99426	Q93009
TBCC	Q15814	P06396	TBK1	Q9UHD2	P59596
TBKBP1	A7MCY6	Q9UKV8	TBL1X	O60907-2	Q9Y618
TBL1XR1	Q9BZK7	P36404	TBL1Y	Q9BQ87	P25685
TBL2	Q9Y4P3	O15530	TBL3	Q12788	P06748
TBPL1	P62380	O76024	TBR1	Q16650	O00712, Q12857
TBRG1	Q3YBR2	Q8NB12	TBRG4	Q969Z0	Q9Y639
TBX15	Q96SF7	O94955	TBX18	O95935	Q7LC44
TBX19	O60806	P31273	TBX2	Q13207	Q6DKK2
TBX3	O15119	Q08117	TBXA2R	Q0VAB0	P29033
TC2N	Q8N9U0	Q96JC9	TCAF1	Q9Y4C2	P01275
TCAF2	A6NFK2	P80370	TCEA1	P23193	Q92917
TCEA2	Q15560	Q13158	TCEA3	O75764	Q8WVC0
TCEAL4	Q96EI5	O76024	TCEAL5	Q5H9L2	P43354
TCEAL8	Q8IYN2	P04792	TCEB1	Q15369	Q16665
TCEB2	Q15370	Q16665	TCEB3	Q14241	Q15742

Input	UniProt Id	Interacts with	Input	UniProt Id	Interacts with
TCF12	Q99081	P30626	TCF19	Q9Y242	Q15669
TCF25	Q9BQ70	P08311	TCF3	P15923	Q9Y4Z2
TCF4	Q9NQB0	Q13761	TCF7	P36402	P35222
TCF7L2	Q9NQB0	Q13761	TCHH	Q07283	P24385
TCIRG1	Q13488	Q15904	TCN2	P20062	Q9NPF0
TCOF1	Q13428	P32121, P49407	TCP1	P17987	Q13882
TCP11L1	Q9NUJ3	P50458	TCP11L2	Q8N4U5	P62736
TCTA	P57738	P78382	CTEX1D2	Q8WW35	P63172
TDG	Q13569	Q9GZX7	TDRD7	Q8NHU6	P42771
TEAD1	P28347	Q06609	TEAD2	Q15562-2, Q15562	P46937
TEAD3	Q99594	P46937	TEAD4	Q15561	P46937
TEC	Q92570	Q13526	TECPR1	Q7Z6L1	Q9H1Y0, Q9NT62, O94817
TECR	Q9NZ01	Q00059	TEF	Q10587	Q16236
TEK	Q02763	P29353	TELO2	Q9Y4R8	Q13315
TENM2	Q9NT68	P21926	TENM3	Q9P273	Q13217
TENM4	Q6N022	Q92608	TEP1	P60484	P60484
TERF1	P54274	O00267	TERF2	Q15554	Q13415
TERF2IP	Q9NYB0	P14550	TES	Q9UGI8	P13569
TESC	Q96BS2	Q9Y2W2	TESK1	Q15569	P31946
TESK2	Q96S53	P31946, P63104	TET1	Q8N7U7	O15294
TET2	Q6N021	P19544	TET3	Q8BG87, O43151	O15294
TEX15	Q9BXT5	P01008	TEX2	Q8IWB9	Q9NP91
TEX264	Q9Y6I9	Q13370	TEX29	Q8N6K0	Q02094
TEX9	Q8N6V9	O60341	TF	P02787	O00501
TFAP2A	P05549	Q09472	TFAP2C	Q92754	P63279
TFB1M	Q8WVM0	Q00059	TFDP1	Q14186	Q08999, P06400
TFDP2	Q14188	Q08999, P06400	TFE3	P19532	O75030
TFEB	P19484	P31946, P63104	TFIP11	Q9UBB9	Q6W0C5
TFPT	P0C1Z6	P55199	TFRC	P02786	Q9P0V3
TGDS	O95455	P42858	TGFA	P01135	P00533
TGFB1	P01137	O60551	TGFB1I1	O43294	Q14289
TGFB2	P61812	P05067	TGFB3	P10600	P37173
TGFBR1	P36897	P62942	TGFBR2	P37173	Q99741
TGFBR3	Q03167	P22455	TGIF1	Q15583	P53350
TGM2	P21980	P02751	TGS1	Q96RS0	Q9Y2X3
THADA	Q6YHU6	Q96CV9	THAP10	Q9P2Z0	P61289
THAP11	Q96EK4	Q13363	THAP3	Q8WTV1	P36941
THAP4	Q8WY91	P49638	THAP7	Q9BT49	P12830
THAP8	Q8NA92	Q9UBB9	THBD	P07204	Q9H2K0
THBS1	P07996- PRO_0000035842	P16671	THBS2	P35442	Q6Y288
THEM4	Q5T1C6	P11182	THEM6	Q8WUY1	Q8TC07
THEMIS	Q8N1K5-1	P43403	THEMIS2	Q5TEJ8	Q7Z3S9
THOC1	Q96FV9	Q07955	THOC2	Q8NI27	Q86V81, Q13838
THOC3	Q96J01	Q13838	THOC5	Q13769	P38919, Q86V81, Q13838
THOC6	Q86W42	Q13838	THOC7	Q6I9Y2	Q96GM5
THOP1	P52888	Q96CV9	THRA	P10827	Q15648
THRB	P10828	P60709	THRSP	Q92748	Q92748
THSD4	Q6ZMP0	P11142	THSD7A	Q9UPZ6	Q3SXY8

Input	UniProt Id	Interacts with	Input	UniProt Id	Interacts with
THSD7B	Q9C0I4	Q9H2K0	THUMPD2	Q9BTF0	O76024
THUMPD3	Q9BV44	Q9BPX6	THYN1	Q9P016	Q9Y4X5
TIA1	P31483	Q99497	TIAM1	Q13009	P18827
TICAM1	Q8IUC6	Q9UHD2	TIE1	P35590	P04629
TIGD1	Q96MW7	P42858	TIGD3	Q6B0B8	Q92466
TIGD5	Q53EQ6	Q6VN20	TIGIT	Q495A1	P15151, Q9NQS3
TIMM10	P62072	Q9Y5J6	TIMM13	Q9Y5L4	Q99558
TIMM17B	O60830	P06396	TIMM21	Q9BVV7	P04626
TIMM23	O14925	Q9BVV7	TIMM44	O43615	Q00403
TIMM50	Q3ZCQ8	Q99558	TIMM8A	O60220	Q99558
TIMM8B	Q9Y5J9	Q99558	TIMM9	Q9Y5J7	Q9Y5J6
TIMP1	P01033	P14780, P09238	TIMP3	P35625	P16035
TINAGL1	Q9GZM7	O75386	TINF2	Q9BSI4	P52565
TIPIN	Q9BVW5	Q9BRT9, Q14691, Q9BRX5	TIPRL	O75663	P67775
TJP1	Q07157	P59637	TJP2	Q9UDY2	Q07157
TKFC	Q3LXA3	O75695	TKT	Q16832	Q16288
TLCD1	Q96CP7	Q96BA8	TLDC1	Q6P9B6	O60551
TLE2	Q04725	P11142	TLE3	Q04726	Q08117
TLE4	Q04727	O95231	TLK1	Q9UKI8	P50458
TLK2	Q86UE8	Q92985	TLN1	Q9Y490, P26039	P05556
TLN2	Q9Y4G6	P00519	TLR1	Q15399	O60603
TLR10	Q9BXR5	O60603	TLR2	O60603	Q99836
TLR3	O15455	O15455	TLR4	O00206	Q99836
TLR5	O60602	O75911	TLR8	Q9NR97	Q9H1C4
TLX1	P31314	P14921	TLX2	O43763	O00712, Q12857
TLX3	O43711	Q8N0W3	TM2D1	Q9BX74	P05067
TM2D2	Q9BX73	P29033	TM2D3	Q9BRN9	P0DTC8
TM4SF18	Q96CE8	Q9NPF0	TM6SF1	Q9BZW5-2	P06396
TM9SF3	Q9HD45	Q9H244	TMA16	Q96EY4	Q96CW1
TMA7	Q9Y2S6	P10636-8	TMBIM1	Q969X1	Q8NCR0
TMBIM6	P55061	O00311	TMC6	Q7Z403	P08670
TMC7	Q7Z402	P50747	TMCC1	O94876	Q9BRI3
TMCC2	O75069	Q96T88	TMED1	Q13445	Q15363
TMED10	P49755	Q9Y3A6	TMED3	Q9Y3Q3	Q15363
TMED5	Q9Y3A6	Q15058	TMED9	Q9BVK6	Q9Y3A6
TMEFF1	Q8IYR6	Q13627	TMEFF2	Q9UIK5	Q9Y5Z9
TMEM100	Q9NV29	Q9Y624	TMEM101	Q96IK0	P63027
TMEM106A	Q96A25	Q16585	TMEM106B	Q9NUM4	Q6UX41
TMEM106C	Q9BVX2	P11686	TMEM108	Q6UXF1	Q9NYV6
TMEM109	Q9BVC6	Q9H2K0	TMEM11	P17152	Q16585
TMEM115	Q12893	Q86U10	TMEM120A	Q9BXJ8	P16410
TMEM120B	A0PK00	P61073	TMEM121	Q9BTD3	O43639
TMEM129	A0AVI4-2	P04792	TMEM130	Q8N3G9	P22732
TMEM131	Q92545	P0DTC3	TMEM132A	Q24JP5	Q01523
TMEM132D	Q14C87	P62136	TMEM133	A6NI28	P46108
TMEM139	Q8IV31	P48509	TMEM140	Q9NV12	O00501
TMEM143	Q96AN5	Q6ZPD8	TMEM144	Q7Z5S9	Q9NPL8
TMEM147	Q9BVK8	P25103	TMEM14A	Q9Y6G1	Q96BA8
TMEM14B	Q9NUH8	Q9Y584	TMEM14C	Q9P0S9	A1L3X0
TMEM150A	Q86TG1	P25490	TMEM154	Q6P9G4	Q15052

Input	UniProt Id	Interacts with	Input	UniProt Id	Interacts with
TMEM159	Q96B96	Q05329	TMEM160	Q9NX00	P30518
TMEM163	Q8TC26	Q9Y6M5, Q9BRI3	TMEM164	Q5U3C3	Q9H244
TMEM165	Q9HC07	P13569	TMEM168	Q9H0V1	P46109
TMEM169	Q96HH4	P08183	TMEM17	Q86X19	O95070
TMEM173	Q86WV6	Q96N66	TMEM176A	Q96HP8	Q9H2K0
TMEM176B	Q3YBM2	P17252	TMEM178A	Q8NBL3	Q12959
TMEM180	Q14CX5	P0DPK4	TMEM183A	Q8IXX5	O14879
TMEM184A	Q6ZMB5	Q4G176	TMEM184B	Q9Y519	O15357
TMEM185A	Q8NFB2	P01019	TMEM186	Q96B77	Q71RS6
TMEM187	Q14656	Q3SXY8	TMEM199	Q8N511	O15342, P27449
TMEM200A	Q86VY9	P16298	TMEM201	Q5SNT2-2	Q8N183
TMEM203	Q969S6	Q9NPL8	TMEM205	Q6UW68	P17152
TMEM206	Q9H813	Q92538	TMEM208	Q9BTX3	Q96BA8
TMEM219	Q86XT9	P17936	TMEM222	Q9H0R3	O00501
TMEM223	A0PJW6	Q6ZMH5	TMEM229B	Q8NBD8	Q3SXY8
TMEM231	Q9H6L2	Q6UX41	TMEM237	Q96Q45-2	P27352
TMEM243	Q9BU79	Q9H2K0	TMEM248	Q9NWD8	O95070
TMEM25	Q86YD3	P53611	TMEM254	Q8TBM7	Q9H2K0
TMEM255A	Q5JRV8	Q96J02	TMEM255B	Q8WV15	Q6ZPD8
TMEM258	P61165	P0DTC4	TMEM259	Q4ZIN3	P21453
TMEM30B	Q3MIR4	P21802	TMEM33	P57088	P13569
TMEM35	Q53FP2	O14653	TMEM38B	Q9NVV0	Q8WTS1
TMEM39B	Q9GZU3	O75326	TMEM42	Q69YG0	Q9H2K0
TMEM45A	Q9NWC5	P02652	TMEM47	Q9BQJ4	P42858
TMEM50A	O95807	Q8IWU4	TMEM50B	P56557	Q07817
TMEM51	Q9NWX97	P46934	TMEM54	Q969K7	Q3SXY8
TMEM55A	Q8N4L2	P00740	TMEM55B	Q86T03	P05919
TMEM57	Q8N5G2	Q96FE5	TMEM60	Q9H2L4	Q9UBD6
TMEM63A	O94886	P43119	TMEM63B	Q5T3F8	P43119
TMEM65	Q6PI78	Q3SXY8	TMEM67	Q5HYA8	P59665
TMEM68	Q96MH6	Q16581	TMEM74	Q96NL1	Q8NAT1
TMEM79	Q9BSE2	O00631	TMEM80	Q96HE8	O75063
TMEM86B	Q8N661	P17302	TMEM87A	Q8NBN3	Q9Y639
TMEM88	Q6PEY1	P17152	TMEM8A	Q9HCN3	P0DPK4
TMEM9	Q9P0T7	P51790	TMEM97	Q5BJF2	Q9UBD6
TMEM9B	Q9NQ34	P51790	TMF1	P82094	P53350
TMLHE	Q9NVH6	P10809	TMOD1	P28289	P06396
TMOD3	Q9NYL9	P60709	TMPO	P42167, P42166	P02545
TMPRSS2	O15393-2	P78382	TMPRSS3	Q9NRS4-3	Q9BXN2
TMSB10	P63313	Q9NUX5	TMSB4X	P20065	Q8WWQ8
TMTC1	Q8IUR5-4	P06396	TMTC3	Q6ZXV5	Q01523
TMTC4	Q5T4D3	Q9Y4Z2	TMUB1	Q9BVT8	O43353
TMUB2	Q71RG4	Q3SXY8	TMX1	Q9H3N1	Q9NPF0
TMX2	Q9Y320	P13500	TMX3	Q96JJ7-2	P27797
TNC	P24821	P0DTC8	TNFAIP1	Q13829	P12931
TNFAIP2	Q03169	O75604	TNFAIP3	P21580	Q14790
TNFAIP6	P98066	P10145	TNFAIP8	O95379	Q96JC9
TNFAIP8L1	Q8WVP5	Q6NZI2	TNFAIP8L3	Q5GJ75	O00560
TNFRSF10A	O00220	Q14790	TNFRSF10D	Q9UBN6	Q969Z0
TNFRSF11A	Q9Y6Q6-2, Q9Y6Q6	Q12933	TNFRSF11B	O00300	Q9Y6Q6

Input	UniProt Id	Interacts with	Input	UniProt Id	Interacts with
TNFRSF14	Q92956	Q7Z6A9	TNFRSF19	Q9NS68	P52565
TNFRSF1A	P19438	Q13546	TNFRSF1B	P20333	P11182
TNFRSF21	O75509	P08138	TNFRSF8	P28908	Q99741
TNFSF10	P50591	O15519-2, O15519-1	TNFSF12	O43508	P06493
TNFSF13	O75888-3	P05067	TNFSF13B	Q9Y275	O15397
TNFSF15	O95150	O43464	TNFSF4	P23510	P43489
TNFSF8	P32971	Q14118	TNIK	Q9UKE5	O43318
TNIP1	Q15025	Q00403	TNIP2	Q8NFZ5	P41279, P19838
TNK1	Q13470	P31946, P63104	TNKS	O95271	Q9NTX7
TNKS1BP1	Q9C0C2	O14965	TNKS2	Q9H2K2	Q9NTX7
TNNC1	P19429	O76024	TNNC2	P02585	Q9Y314
TNNI1	P19237	Q9NPB3	TNNI2	P48788	P62195
TNNI3	P19429	O76024	TNNT1	P13805	P62195
TNNT2	P45379-11	O60341	TNPO2	O14787	Q9Y639
TNRC18	O15417	Q9UER7	TNRC6A	Q8NDV7	P11940
TNRC6B	Q9UPQ9	P11940	TNRC6C	Q9HCJ0, Q9HCJ0-1	P11940
TNS1	Q9HBL0	P11532	TNS2	Q63HR2	Q9Y5Y2
TNS3	Q68CZ2	P12931	TNXB	P22105	P01137
TOB1	P50616	P11940	TOB2	Q14106	P11940
TOLLIP	Q9H0E2	Q9UM54	TOM1	O60784	Q9UM54
TOM1L1	O75674	P62993	TOM1L2	Q6ZVM7	Q9UM54
TOMM20	Q15388	P19367	TOMM22	Q9NS69	P0DOE7
TOMM40	O96008	P56705	TOMM7	Q75MR5	P50542
TOMM70A	O94826	P0DTD2	TONSL	Q96HA7	P49736
TOP1	P11387	P00519	TOP2B	Q02880	P46531
TOR1A	O14656-2	Q96FZ7	TOR1AIP1	Q5JTV8	P10599
TOR1B	O14657	P32971	TOR2A	Q5JU69	Q13363
TOX2	Q96NM4-3	O14561	TOX4	O94842	O95947
TP53BP1	Q12888	Q13315	TP53BP2	Q05BL1	P20700
TP53RK	Q96S44	Q9NS73, Q9BTK6	TP63	Q9H3D4	P46937
TPCN2	Q8NHX9	Q9UJK0	TPD52	P55327	Q86T03
TPD52L1	Q16890	P31946, P63104	TPD52L2	O43399	P13569
TPGS2	Q68CL5-3	P27797	TPI1	P60174	P13569
TPM2	P07951	P13569	TPM3	P06753	P62195
TPP1	Q96AP0	Q92900	TPP2	P29144	P06746
TPPP	O94811	P37840	TPRG1	Q6ZUI0	O00559
TPRKB	Q9Y3C4	P14373	TPRN	Q4KMQ1-2	O15247
TPSB2	P20231	Q9UBV2	TPST1	O60507	P61769
TPT1	P13693	P11940	TPX2	Q9ULW0	O75330
TRA2A	Q13595	O43592	TRABD2B	A6NFA1	P41221
TRAC	P01848	Q09328	TRAF1	Q13077	Q13546
TRAF2	Q12933	Q13233	TRAF3	Q60803	Q9Y2R2
TRAF3IP1	Q8TDR0	Q13114	TRAF3IP2	O43734	Q9Y4K3
TRAF3IP3	Q9Y228	P37173	TRAF4	Q9BUZ4	Q9P0N9
TRAF5	O00463	Q92956	TRAF7	Q6Q0C0	Q9NWB7
TRAFD1	O14545	P13569	TRAIP	Q9BWF2	O15160
TRAM1	Q9Y6Q9	Q09472	TRAM1L1	Q8N609	Q96BA8
TRAPPC1	Q9Y5R8	Q96QF0	TRAPPC10	P48553	Q96QF0
TRAPPC12	Q8WVT3	Q9P2M4	TRAPPC2	P0DI81	Q96QF0
TRAPPC2L	Q9UL33	Q96QF0	TRAPPC3	O43617	Q96QF0

Input	UniProt Id	Interacts with	Input	UniProt Id	Interacts with
TRAPPC4	Q9Y296	Q96QF0	TRAPPC5	Q8IUR0	Q9P2M4
TRAPPC6A	O75865	Q96QF0	TRAPPC6B	Q86SZ2	Q9P2M4
TRAPPC9	Q96Q05	Q96QF0	TRAT1	Q6PIZ9	P49748
TRDN	Q13061	A0PJW6	TREM2	Q9NZC2	O43914
TRERF1	Q96PN7	P46937	TRHDE	Q9UKU6	Q5SRI9
TRIB2	Q92519	Q8NHY2	TRIB3	Q96RU7	Q6W0C5
TRIL	Q7L0X0	O15455	TRIM11	Q96F44	Q07820
TRIM14	Q14142	P61024	TRIM21	P19474	P13569
TRIM22	Q8IYM9	Q9HC29	TRIM23	P36406	Q8NFF5
TRIM24	O15164	Q13263	TRIM25	Q14258	P03496
TRIM27	P14373	Q96E11	TRIM28	Q13263	Q13568
TRIM29	Q14134	O14641	TRIM3	O75382	P51668
TRIM32	Q13049	Q06124	TRIM33	Q9UPN9	P53611
TRIM35	Q9UPQ4-2	Q9NPB3	TRIM36	Q9NQ86	Q14451
TRIM38	O00635	Q13546	TRIM41	Q8WV44	P03466
TRIM44	Q96DX7	Q9H5J8	TRIM45	Q9H8W5	Q9BQP7
TRIM47	Q96LD4	Q9UPW6, Q01826	TRIM5	Q9C035	P13569
TRIM56	Q9BRZ2	P05388	TRIM65	Q6PJ69	P09917
TRIM69	Q86WT6-2	P06396	TRIM71	Q2Q1W2	A1L190
TRIM9	Q9C026	P50552	TRIO	O75962	Q9NV70
TRIOBP	Q9H2D6	P04004	TRIP10	Q15642-2	Q9Y4D1
TRIP11	Q15643	P31946	TRIP12	Q14669	Q8N726
TRIP13	Q15645	Q9UBX5	TRIP6	Q15654	Q9Y316
TRMT10C	Q7L0Y3	Q9BQ52	TRMT11	Q7Z4G4	P10809
TRMT112	Q9UI30	O43709	TRMT12	Q53H54	P11142
TRMT1L	Q7Z2T5	Q00403	TRMT2B	Q96GJ1	P07992
TRMT44	Q8IYL2	P00740	TRMT6	Q9UJA5	Q13451
TRMT61A	Q96FX7	Q14192	TRNT1	Q96Q11-3	O76024
TRO	Q12816	Q05655	TROVE2	P10155	P05455
TRPC1	P48995	Q03135	TRPM7	Q923J1	O00571
TRPS1	Q9UHF7	Q92769	TRPT1	Q86TN4-2	Q12933
TRPV4	Q9EPK8	Q96J02	TRPV5	Q9NQA5	P18846
TRRAP	Q9Y4A5	O96019	TRUB2	O95900	O00425
TSC2	P49815	Q9P0N9	TSC22D1	Q15714	Q15047
TSC22D4	Q9Y3Q8	Q8NG68	TSEN15	Q8WW01	Q9HD47
TSEN2	Q8NCE0	Q92989	TSEN54	Q7Z6J9	Q92989
TSFM	P43897	P46199	TSG101	Q99816	P21453
TSGA10	Q9BZW7	Q6P1J9	TSHR	P16473	Q9NPA3
TSHZ2	Q9NRE2	Q13077	TSHZ3	Q63HK5	Q12933
TSNARE1	A0AVG3	Q9NYP7	TSNAXIP1	Q2TAA8	O00311
TSPAN11	A1L157	P01189	TSPAN12	O95859	Q96AA3
TSPAN14	Q8NG11	P48509	TSPAN15	O95858	Q9H244
TSPAN17	Q96FV3	Q16585	TSPAN2	O60636	P17931
TSPAN3	O60637	Q99741	TSPAN33	Q86UF1	O14672
TSPAN5	P62079	P48509	TSPAN6	O43657	Q9NY25
TSPAN7	P41732	Q8TDQ0	TSPAN9	O75954	P21926
TSPO	P30536	P29762	TSPYL1	Q9H0U9	P61289
TSPYL2	Q9H2G4	Q9UGY1	TSPYL4	Q9UJ04	O00273
TSPYL5	Q86VY4	Q93009	TSR1	Q2NL82	O95453
TSR2	Q969E8	O75928-2	TSR3	Q9UJK0	Q9Y216

Input	UniProt Id	Interacts with	Input	UniProt Id	Interacts with
TSSC4	Q9Y5U2	Q13077	TSSK3	Q96PN8	Q08117
TSSK6	Q9BXA6	Q13451	TTBK2	Q6IQ55	Q9UPV0
TTC12	Q9H892-2	Q13153	TTC17	Q96AE7	Q8IV16
TTC19	Q6DKK2	Q16774	TTC21B	Q7Z4L5	P18206
TTC23	Q5W5X9-3	Q9H0D6	TTC27	Q6P3X3	Q5S007
TTC28	Q96AY4	O43639	TTC3	P53804	P31749
TTC30B	Q8N4P2	Q86T03	TTC33	Q6PID6	P20618
TTC5	Q8N0Z6	Q6NZI2	TTC8	A0A0C4DGX9, Q8TAM2-4, Q8TAM2	Q8IWX6, Q96RK4, Q8N3I7, Q3SYG4, Q8NFJ9, Q9BXC9
TTC9C	Q8N5M4	Q06187	TTF1	Q15361	P62263
TTF2	O00358	P19419	TLL12	Q14166	P68104
TLL7	I6L9D0	Q96CV9	TTN	Q8WZ42	Q96CV9
TTPA	P49638	Q6ZPD8	TTYH1	Q9H313	O15354
TTYH3	Q9C0H2	P21453	TUBA1A	Q71U36	Q9UBX3
TUBA1B	P68363	Q15398	TUBA1C	Q9BQE3	P08670
TUBA4A	P68366	P30622	TUBB	P07437	P36404
TUBB2A	Q13885	P36404	TUBB2B	Q9BVA1	P36404
TUBB4A	P04350	P22736	TUBB4B	P68371	Q9BTW9, P36404
TUBG1	P23258	Q96L34	TUBGCP2	Q9BSJ2	P08173
TUBGCP3	Q96CW5	P0DTC5	TUBGCP4	Q9UGJ1	Q9NTX7
TUFM	P49411, A0A384ME17	Q8IVP5	TUFT1	Q9NNX1	P14373
TULP3	O75386	P98160	TUSC1	Q2TAM9	O95749
TUSC2	O75896	P11142	TUSC3	Q13454	Q14165
TUSC5	A5PKU2	Q7Z2H8	TUT1	Q9H6E5	Q9P2I0, Q92797, Q10570
TWF2	Q6IBS0	P60709	TWIST1	Q15672	O60885
TWIST2	Q8WVJ9	Q99081	TXK	P42681	O14543
TXLNG	Q9NUQ3	Q9UNE7	TXN	P10599	P19883
TXN2	Q99757	P49247	TXNDC11	Q6PKC3	Q99633
TXNDC17	Q9BRA2	Q96T51	TXNDC9	O14530	O00628
TXNIP	Q9H3M7	P10599	TXNL4A	P83876	Q9BUQ8, O94906
TXNRD1	Q16881	O75695	TXNRD3	Q86VQ6	Q14192
TYMP	P19971	Q9BW85	TYRO3	Q06418	P62993
TYROBP	O43914	Q92538	TYRP1	P17643	O14908
TYW1	Q9NV66	Q9NRM6	TYW1B	Q6NUM6	P43115
TYW3	Q6IPR3	P01130	U2AF2	P26368	Q13573
U2SURP	O15042	Q16629	UACA	Q9BZF9	P04004
UBA1	P22314	Q9UNE7	UBA52	P62990	P51668
UBA6	A0AVT1	Q8WVK2	UBA7	P41226	P05161
UBAC1	Q9BSL1	Q09327	UBAC2	Q8NBM4	Q8TC07, Q9HA65
UBALD1	Q8TB05	O95822-2	UBAP2L	Q14157	P10636-8
UBASH3A	P57075-2	Q07889	UBB	P0CG47	O76075
UBC	P0CG48	Q9UDY8	UBE2A	P49459	Q8WZA2
UBE2D2	P62837	P22681	UBE2D3	P61077	P98170
UBE2D4	Q9Y2X8	Q9Y6K9	UBE2E1	P51965	Q96RU8
UBE2E2	Q96LR5	Q96EQ8	UBE2E3	Q969T4	Q9NS56
UBE2F	Q969M7	P26367	UBE2G1	P62253	P06307
UBE2G2	P60604	P0CG48	UBE2I	P63279	P19419
UBE2J1	Q9Y385	Q86TM6	UBE2J2	Q8N2K1	A1L3X0
UBE2K	P61086	Q15843	UBE2L3	P68036	Q9UNE7

Input	UniProt Id	Interacts with	Input	UniProt Id	Interacts with
UBE2L6	O14933	Q9NWF9, Q96EQ8	UBE2M	P61081	O94844
UBE2N	P61088	Q9UNE7	UBE2O	Q9C0C9	P04608
UBE2Q1	Q7Z7E8	Q9UNE7	UBE2Q2	Q8WVN8	P04792
UBE2S	Q16763	P13569	UBE2T	Q9NPD8	Q15078
UBE2V1	Q13404	P06396	UBE3A	Q05086	Q06413
UBE3C	Q15386	P60896	UBE3D	Q7Z6J8	P35813
UBE4A	Q14139	Q7Z434	UBIAD1	Q9Y5Z9	P04035
UBL4A	P11441	O15055	UBL5	Q9BZL1	Q9UL46
UBL7	Q96S82	P07992	UBN1	Q9NPG3, Q9NPG3-2	P54198, Q9Y294
UBN2	Q6ZU65	P54198, Q9Y294	UBOX5	O94941	P68036
UBQLN1	Q9UMX0	P23352	UBQLN2	Q9UHD9	P16035
UBQLN4	Q9NRR5	P10451	UBR1	Q8IWW7	P06850
UBR2	Q8IWW8	O14786	UBR3	Q6ZT12	Q96J84
UBR5	O95071	P49841	UBTD2	Q8WUN7	P07902
UBTF	P17480	P52306	UBXN1	Q04323-2	P04792
UBXN11	Q5T124	P06730	UBXN2A	P68543	P55072
UBXN2B	Q14CS0	Q13526	UBXN6	Q9BZV1, Q9BZV1-1, Q9BZV1-2	Q5VVQ6, P55072, Q96JH7
UCHL1	P09936	Q15843	UCHL5	Q9Y5K5	P60896
UCK1	Q9HA47-2	P06396	UCKL1	Q9NWZ5	Q9Y639
UCP2	P55851	P20930	UFD1L	Q92890	P55072
UFL1	O94874	P21462	UGCG	Q16739	P43119
UGGT1	Q9NYU2	P13569	UGGT2	Q9NYU1	P11021
UGT8	Q16880	P19526	UHMK1	Q8TAS1-2	Q9BU61
UHRF1BP1L	A0JNW5	P31946, P63104	UHRF2	Q96PU4	P09429
UIMC1	Q96RL1	P38398	ULBP1	Q9BZM6	Q6ZMH5
ULBP2	Q9BZM5	P32971	ULBP3	Q9BZM4	P01100
ULK1	O75385	O75385	ULK3	Q6PHR2	Q13451
UNC119	Q13432	Q13237	UNC13D	Q70J99	P51159
UNC45A	Q9H3U1	P48729	UNC93B1	Q8VCW4	Q96N66
UNG	P13051-2	P16671	UPF1	Q92900	Q14493
UPF2	Q9HAU5	Q92900	UPF3A	Q9H1J1	Q92900
UPF3B	Q9BZI7, Q9BZI7-2	Q92900	UPP2	O95045	Q8IUQ4
UQCC2	Q9BRT2	P62993	UQCR10	Q9UDW1	P31930, O14949
UQCRB	P14927	Q96T88	UQCRC1	P31930	P26641
UQCRC2	P22695	Q13573	UQCRFS1	P47985	Q9H1K1
UQCRH	P07919	P40855	UQCRQ	O14949	P31930, Q9UDW1, P14927, P22695
URB1	O60287	P06748	URI1	O94763	Q6P1J9
UROD	P06132	P00533	UROS	P10746	P48507
USB1	Q9BQ65	Q13573	USE1	Q9NZ43	O95249
USF1	P22415	Q13573	USF2	Q15853	Q15562
USH1G	Q495M9	Q9UER7	USO1	O60763	Q9Y6L6
USP1	O94782	Q8TAF3	USP11	P51784	Q86YC2
USP12	O75317	Q96B01	USP13	Q92995	O60216
USP14	P54578	P60896	USP15	Q9Y4E8	P60896
USP16	Q9Y5T5	O95714	USP18	Q9UMW8	P05161
USP19	O94966	P13569	USP2	O75604-3	Q00403
USP20	Q9Y2K6	Q99814	USP21	Q9UHP3	Q9UKG9
USP22	Q9UPT9-2, Q9UPT9	Q9NRN7	USP24	Q9UPU5	O75695

Input	UniProt Id	Interacts with	Input	UniProt Id	Interacts with
USP25	Q9UHP3	Q9UKG9	USP27X	A6NNY8	Q96B97
USP28	Q96RU2	Q12888	USP32	Q8NFA0-2	O43639
USP34	Q70CQ2	P48729	USP35	A2RRA6	O76024
USP36	Q9P275	P06748	USP37	Q86T82	Q9UM11
USP38	Q8NB14	P17252	USP39	Q53GS9	Q13523
USP4	Q13107	Q13546	USP44	Q9H0E7	P41208
USP46	P62068	Q96B01	USP47	Q96K76	P52198
USP5	P45974-2	Q9NWF9	USP53	Q70EK8	P46109
USP54	Q70EL1-9	O94955	USP6	EBI-11750223	Q9UBB9
USP8	P40818	Q16665, Q99814	USP9X	Q93008	P29353
UST	Q9Y2C2	Q8WVC6	UTP14A	Q9BVJ6	P06748
UTP14C	Q5TAP6	Q9NRX1	UTP20	O75691	Q9NZQ7
UTP23	Q9BRU9	Q96RU7	UTP3	Q9NQZ2	P02787
UTRN	P46939	Q14118	UTY	O14607	Q14686
UVRAG	Q9P2Y5	Q07812	UXS1	Q8NBZ7	P40855
UXT	Q9UBK9	Q7KZ85	VAC14	Q08AM6	Q9NUB1
VAMP1	P23763-3	A1L3X0	VAMP2	P63027	Q3SXY8
VAMP3	Q15836	P01730	VAMP4	O75379	Q13190
VAMP5	O95183	Q13190, Q9NYM9	VAMP7	P51809	P52594
VAMP8	Q9BV40	P54920	VANGL1	Q8TAA9	P18827
VANGL2	Q9ULK5	P11142	VARS	P26640	P26641, P68104
VARS2	P26640	P26641, P68104	VASH2	Q86V25	P61981
VASN	Q6EMK4	P78504	VASP	P50552	P22736
VAT1	P54219	Q7L8W6	VAT1L	Q9HCJ6	P42858
VAV1	P15498	P00519	VAV2	P52735-1, P52735	Q03135
VAX2	Q9UIW0	Q9Y3E7	VBP1	P61758	Q13164
VCAM1	P19320	P40429	VCAN	P13611	P14780
VCL	P18206	Q9Y490	VCP	P55072	P60880
VCPIP1	Q96JH7	Q9H063	VDAC1	P21796	P0C0U1
VDAC2	P45880	P04792	VDAC3	Q9Y277	P21796
VDR	P11473	Q15648	VEGFA	P15692-4, Q9MYV3-3, P15692	P35968, P17948
VEGFB	P49765	P15692	VEGFC	P97953-1	O60462
VENTX	O95231	P31273	VGf	O15240	P01160
VGLL3	A8MV65-2	Q92922	VGLL4	Q14135	P46937
VHL	P40337	Q6VMQ6	VILL	O15195-2	Q96CV9
VIM	P08670	Q12824	VIMP	Q9BQE4	P55072
VIPAS39	Q9H9C1	P12273	VIPR1	P32241	Q9BVG9
VIPR2	P41587	O60313	VIT	Q6UXI7	P11021
VKORC1	Q9BQB6	Q96BA8	VMP1	Q96GC9	Q07157
VNN2	O95498	Q8N8Q8	VPS11	Q9H270	P0DTC3
VPS13B	Q7Z7G8	P49674, P48730	VPS13D	Q5THJ4	P31946, P63104
VPS16	Q9H269	P0DTC3	VPS18	Q9P253	P0DTC3
VPS25	Q9BRG1	Q96FZ7	VPS26B	Q4G0F5	O60493
VPS28	Q548N1	Q99081	VPS29	Q9UBQ0	O60493
VPS33A	Q96AX1	Q14790	VPS35	Q96QK1	O60493
VPS36	Q86VN1	P0CG47	VPS37A	Q8NEZ2	P06396
VPS37D	Q86XT2	Q99816	VPS39	Q96JC1	P51149
VPS41	P49754	P0DTC3	VPS4A	Q9UN37	P13569
VPS50	Q96JG6	Q96EV8	VPS51	Q9UID3	Q9Y639

Input	UniProt Id	Interacts with	Input	UniProt Id	Interacts with
VPS54	Q9P1Q0-4	P06396	VPS72	Q15906	O96019
VPS8	Q8N3P4	P21462	VPS9D1	Q9Y2B5	Q13200
VRK1	Q99986	P48431	VRK2	Q86Y07-2, Q86Y07-1	Q13469
VSIG4	Q9Y279	Q9P0V3	VSIG8	P0DPA2	Q92911
VSNL1	P62760	P04155	VSTM4	Q8IW00	Q16623
VTI1A	Q96AJ9	P54920	VTI1B	Q9UEU0	Q8TC07
VWA2	Q5GFL6	Q12933	VWA5B2	Q8N398	P20336
VWA8	A3KMH1	Q12824	VWC2L	B2RUY7	Q92570
VWF	P04275	Q76LX8	WAC	Q9BTA9	Q13077
WAPAL	Q7Z5K2	Q7Z5K2	WARS	P23381	P13569
WAS	P42768	Q05209	WASF1	Q92558	P00519, Q9UQB8
WASF2	Q9Y6W5	P00519, Q9UQB8	WASH1	A8K0Z3	Q8IYI6, Q9NV70
WASL	O00401	Q13882	WBP1	P97764, Q96G27	P46937
WBP11	Q9Y2W2	Q92841	WBP1L	Q9NX94	P06396
WBP2	Q969T9	P46937	WBP2NL	Q6ICG8	Q9BYE7
WBSCR27	Q8N6F8	Q9NQ94	WDFY1	Q8IWB7	Q8WY64
WDFY2	Q96P53	O95070	WDFY3	Q8IZQ1	P40763
WDFY4	Q6ZS81-2	Q96CV9	WDR1	O75083	P60709
WDR12	Q9GZL7	P06748	WDR13	Q9H1Z4	P21709
WDR18	Q9BV38	Q5SY16	WDR20	Q8TBZ3	Q8TAF3
WDR24	Q96S15	P55735	WDR27	A2RRH5	P08243
WDR3	Q9UNX4	P06748	WDR33	Q9C0J8	O95639
WDR34	Q96EX3	P84243	WDR35	Q9P2L0	O75386
WDR36	Q8NI36	P03372	WDR37	Q9Y2I8	Q6VY07
WDR4	P57081	P39748	WDR43	Q15061	Q13901, Q9H8H0
WDR44	Q5JSH3	P62993	WDR45B	Q5MNZ6	P51813
WDR46	O15213	P06748	WDR5	P61964	O00255
WDR54	Q9H977	Q9Y316	WDR55	Q9H6Y2	P06748
WDR59	Q6PJI9	Q9HD47	WDR6	Q9NNW5	P03496
WDR60	Q8WVS4	Q16629	WDR61	Q9GZS3	Q16143
WDR7	Q9Y4E6	P21281	WDR70	Q9NW82	Q06330
WDR73	Q6P4I2	P43353	WDR74	Q6RFH5	Q96PM5
WDR77	Q9BQA1	P03418	WDR82	Q6UXN9	Q9P0U4
WDR83	Q9BRX9	O75695	WDR88	Q6ZMY6-2	P01100
WDR91	A4D1P6	P42858	WDTC1	Q8N5D0	Q13371
WFDC2	Q14508	Q3SXY8	WFS1	O76024	Q9BRX2
WHSC1	O96028	P17096	WIBG	Q9BRP8	Q9NUX5
WIPF1	O43516-4, O43516	O43639	WIPI2	Q9Y4P8-4	Q96E11
WISP1	O95388-2	P06396	WISP2	O76076	P49639
WIZ	O95785	P08311	WLS	Q5T9L3	P41221
WNK1	Q9H4A3	Q15149	WNK4	Q80UE6	O00141
WNT10B	O00744	O75197	WNT2B	Q93097	Q70UQ0
WNT3	P56703	P56704	WNT4	P56705	Q5T9L3
WNT5A	P41221	Q5T9L3	WNT7A	O00755	Q5T9L3
WNT7B	P56706	P56705	WRAP53	Q9BUR4	Q14814
WRAP73	Q9P2S5	Q9Y266	WRB	O00258	P02652
WRN	Q14191	P39748	WRNIP1	Q96S55	O95786
WSB2	Q9NYS7	Q07817	WWC1	Q8IX03	Q05513
WWC2	Q6AWC2	P31946, P63104	WVOX	Q9NZC7-5	P04792

Input	UniProt Id	Interacts with	Input	UniProt Id	Interacts with
WWP1	Q9H0M0	P84022	WWTR1	Q9EPK5	Q07157
XAB2	Q9HCS7	P23025	XAF1	Q6GPH4	P40937
XBP1	P17861	Q16665	XIAP	P98170	P51668
XIRP1	Q702N8	P50552	XKR8	Q9H6D3	P43119
XPC	Q01831	P41208	XPNPEP1	Q9NQW7-3	Q15669
XPNPEP3	Q9NQH7	P08670	XPO1	O14980	P04618, P18754
XPO4	Q9C0E2	Q9NZQ7	XPO5	Q9HAV4	P01106
XPO7	Q9UIA9	Q9NZQ7	XPOT	O43592	P13569
XPR1	Q9UBH6	P21802	XRCC1	P18887	Q9NTX7
XRCC3	O43542	Q86YC2	XRCC4	Q13426	P19838
XRCC5	P13010	Q14191	XRCC6	P12956	Q01831
XRCC6BP1	Q9Y6H3	O14867	XRN1	Q8IZH2	P09651
XYLT2	Q9H1B5	Q9Y2C2	YAE1D1	Q9NRH1	P08670
YAF2	Q8IY57-5	Q6NZI2	YAP1	P46937	Q8N3R9
YARS	P54577	O75925	YBEY	P58557	Q9BQ52
YBX1	P67809	Q99697	YBX2	Q9Y2T7	Q14197
YBX3	P16989	P59595	YES1	P07947	P46109
YIF1A	O95070	Q9NQ11	YIF1B	Q5BJH7	O75888
YIPF1	Q9Y548	O00501	YIPF2	Q9BWQ6	P62736
YIPF3	Q9GZM5	Q9H244	YIPF4	Q9BSR8	O95070
YKT6	O15498	Q92696, P53611	YOD1	Q5VVQ6	O14818
YPEL2	Q96QA6	Q9NRD5	YRDC	Q86U90	Q5VWX1
YTHDC2	Q9H6S0	P40429	YTHDF1	Q9BYJ9	P08311
YTHDF2	Q9Y5A9	Q99558	YTHDF3	Q7Z739	P13569
YWHAB	P31946	P21359	YWHAE	P62258	Q86UK7
YWHAG	P61981	Q9BRX2	YWHAH	Q04917	P21359
YWHAQ	P27348	P40818	YWHAZ	P63104	P61587
ZADH2	Q8N4Q0	P11182	ZAP70	P43403	Q9Y2R2
ZBTB1	Q9Y2K1	P11309	ZBTB10	Q96DT7	O15397
ZBTB11	O95625	P06748	ZBTB12	Q9Y330	O43639
ZBTB16	Q05516	P05231	ZBTB17	Q13105	P04198
ZBTB21	Q9ULJ3	P06748	ZBTB22	O15209	Q9Y316
ZBTB24	O43167	P62263, P62241	ZBTB25	P24278	Q96GX9
ZBTB33	Q86T24	P57764	ZBTB34	Q8NCN2	Q9Y2W6
ZBTB38	Q8NAP3	Q96EB6	ZBTB39	O15060	Q13200
ZBTB4	Q9P1Z0	Q08379	ZBTB40	Q9NUA8	Q99435
ZBTB44	Q8NCP5	P60900	ZBTB45	Q96K62	O95429
ZBTB46	Q86UZ6	Q9UF56	ZBTB47	Q9UFB7	Q9UGI0
ZBTB48	P10074	Q96CV9	ZBTB5	O15062	O43602
ZBTB7A	O95365	Q13547	ZBTB7B	O15156, O15156-2	O43639
ZBTB9	Q96C00	Q04721	ZC2HC1A	Q96GY0	O43639
ZC3H12A	Q5D1E8	Q14694	ZC3H14	Q6PJT7	Q16629
ZC3H18	Q86VM9	P0DTC4	ZC3H7B	Q9UGR2	Q04637
ZC3H8	Q8N5P1	Q9H9Z2	ZC3HAV1	Q8K3Y6	Q92841
ZC3HAV1L	Q96H79	P53611	ZC3HC1	Q86WB0	P53350
ZC4H2	Q9NQZ6	P55040	ZCCHC11	Q5TAX3	Q9H9Z2
ZCCHC12	Q6PEW1	Q13158	ZCCHC17	Q9NP64	P06396
ZCCHC3	Q9NUD5	P63244	ZCCHC6	Q5VYS8-5	Q96CV9
ZCCHC7	Q8N3Z6	P63279	ZCCHC8	Q6NZY4	P38398
ZCRB1	Q8TBF4	P04792	ZDBF2	Q9HCK1	P48431

Input	UniProt Id	Interacts with	Input	UniProt Id	Interacts with
ZDHHC16	Q969W1	O95070	ZDHHC17	Q8IUH5	Q8IVB4
ZDHHC18	Q9NUE0	Q01113	ZDHHC20	Q5W0Z9-4	O76024
ZDHHC21	Q8IVQ6	Q16581	ZDHHC22	Q8N966	Q96BA8
ZDHHC24	Q6UX98	Q15125	ZDHHC5	Q9C0B5	Q8TDQ0
ZDHHC9	Q9Y397	P24394	ZEB1	P37275	Q92769
ZEB2	O60315	Q92769	ZER1	Q7Z7L7	Q6IN84
ZFAND2A	Q8N6M9	Q15008	ZFAND2B	Q8WV99	Q9Y5K5
ZFAND5	O76080	P0CG48	ZFAND6	Q6FIF0	Q12933
ZFAT	Q9P243	Q9UER7	ZFC3H1	O60293	Q14493
ZFHX2	Q9C0A1	P10636	ZFHX3	Q15911-2	P14373
ZFHX4	Q86UP3	P48431	ZFP1	Q6P2D0	O15397
ZFP2	Q6ZN57	Q7Z5W3	ZFP36	P26651	Q9NPI6
ZFP36L1	Q07352	Q16539	ZFP36L2	P47974	P31946, P63104
ZFP62	Q8NB50	O43159	ZFP90	Q8TF47	Q9Y2W6
ZFPL1	O95159	Q96BA8	ZFPM1	Q8IX07	P49841
ZFPM2	Q8WW38	P14136	ZFY	P08048	Q969R5
ZFYVE1	Q9HBF4	Q8NC69	ZFYVE16	Q7Z3T8	P62136
ZFYVE19	Q96K21	P04792	ZFYVE26	Q68DK2-5	Q13077
ZFYVE27	Q5T4F4	O95070	ZFYVE9	O95405	P08100
ZGPAT	Q8N5A5-2	Q96CV9	ZGRF1	Q86YA3	Q06609
ZHX1-C8orf76	Q96EF9	P12830	ZIC1	Q15915	P30740
ZIC2	Q62520	P10071	ZKSCAN1	P17029	Q13164
ZKSCAN4	Q969J2	Q9UBH0	ZKSCAN5	Q9Y2L8	P14136
ZMAT2	Q96NC0	Q96EB6	ZMAT3	Q9HA38	Q9UKV8
ZMAT4	Q9H898-2	Q15633	ZMIZ2	Q8NF64-3, Q8NF64-2, Q8NF64	Q13526
ZMPSTE24	O75844	P02545-1	ZMYM2	Q9UBW7	Q16822
ZMYM3	Q14202	Q92769	ZMYM4	Q5VZL5-4	Q96CV9
ZMYM5	Q9UJ78-2	Q13158	ZMYND11	Q15326	P36941
ZMYND19	Q96E35	Q6UN15	ZMYND8	Q9ULU4	Q92769
ZNF100	Q8IYN0	Q9Y4Z2	ZNF106	Q9H2Y7	Q96FJ0
ZNF114	Q8NC26	Q9UMX1	ZNF124	Q15973	Q13077
ZNF134	P52741	P55081	ZNF138	P52744	Q96EB6
ZNF14	P17017	Q13263	ZNF148	Q9UQR1	Q8IUQ4
ZNF169	Q14929	Q96RR1	ZNF175	Q9Y473	P13196
ZNF195	O14628	Q13263	ZNF197	O14709	P05067
ZNF200	P98182	Q14192	ZNF202	O95125	Q9H2K0
ZNF205	O95201	Q9NRD5	ZNF211	Q13398	Q13263
ZNF212	Q9UDV6	O95995	ZNF213	O14771	Q9UMX1
ZNF217	O75362	Q92769	ZNF219	Q9P2Y4	Q04721
ZNF22	P17026	P61247	ZNF224	Q9NZL3	P98164
ZNF225	P18146	Q8N726	ZNF232	Q9UNY5	P12931
ZNF235	Q14590	Q13263	ZNF239	Q16600	O76024
ZNF248	Q8NDW4	Q13263	ZNF250	P15622-3	O75496
ZNF263	O14978	Q96EB6	ZNF264	O43296	O15397
ZNF266	Q14584	Q9HD47	ZNF267	Q14586	Q13263
ZNF268	Q14587	O15431	ZNF275	Q9NSD4	Q9H9Z2
ZNF277	G5E9M4, Q8WWA6	P63211	ZNF280C	Q8ND82	Q9H2G4
ZNF280D	Q6N043, Q6N043-2	Q13185, P45973	ZNF281	Q9Y2X9	P40763
ZNF287	Q9HBT7	Q13263	ZNF292	O60281	P83916

Input	UniProt Id	Interacts with	Input	UniProt Id	Interacts with
ZNF296	Q8WUU4	Q09028	ZNF3	P17036	Q08379
ZNF302	Q9NR11-2	Q96CV9	ZNF316	A6NFI3	Q04721
ZNF319	Q9P2F9	Q9BUH8	ZNF32	P17041	Q7Z3S9
ZNF320	A2RRD8	Q13263	ZNF324	O75467	Q04721
ZNF324B	Q6AW86	Q9NS56	ZNF329	Q86UD4	Q96CV9
ZNF331	Q9NQX6	Q9UBX5	ZNF333	Q96JL9-2	P06396
ZNF34	Q8IZ26	Q13263	ZNF345	Q14585	P02545
ZNF35	P13682	Q99836	ZNF358	Q9NW07	Q04721
ZNF366	Q8N895	Q13363	ZNF382	Q96SR6	O15397
ZNF384	EBI-11743072	P14373	ZNF395	Q9H8N7	P31946, P63104
ZNF407	Q9C0G0	Q04721	ZNF408	Q9H9D4	Q99611
ZNF410	Q86VK4-3	Q9Y314	ZNF414	Q96IQ9	Q9NYA4
ZNF416	Q9BWM5	Q13263	ZNF420	Q8TAQ5	Q8N726
ZNF428	Q96B54	P09429	ZNF430	Q9H8G1	Q9Y4Z2
ZNF431	Q8TF32	P17643	ZNF436	Q9C0F3	P06396
ZNF438	Q7Z4V0	P52747	ZNF44	M0R160	O14964
ZNF444	Q8N0Y2-2	Q96CV9	ZNF445	P59923	P46087
ZNF446	Q9NWS9	Q9BTV6	ZNF451	Q9Y4E5	P63279
ZNF460	Q14592	O15397	ZNF462	Q96JM2	Q8N465
ZNF467	Q7Z7K2	Q969R5	ZNF471	Q9BX82	Q8N9N5
ZNF473	Q8WTR7	P05067	ZNF48	Q96MX3	Q13895
ZNF483	Q8TF39	P04004	ZNF484	Q5JVG2	Q13263
ZNF490	Q9ULM2	Q13077	ZNF496	Q96IT1	Q9Y4Z2
ZNF507	Q8TCN5	O94901	ZNF512	Q96ME7	P61247
ZNF512B	Q96KM6	P52747	ZNF513	Q8N8E2	Q9Y6K1
ZNF514	Q96K75	Q13263	ZNF516	Q92618	Q92769
ZNF517	Q6ZMY9	Q9H4P4	ZNF518A	Q6AHZ1	Q13185, P45973
ZNF521	Q96K83	Q13547	ZNF526	Q8TF50	Q04721
ZNF536	O15090	P48431	ZNF541	Q9H0D2	Q96IF1
ZNF544	Q6NX49	O15228	ZNF552	Q9H707	Q13263
ZNF554	Q86TJ5	O15160	ZNF558	Q96NG5	P57764
ZNF561	Q8N587	Q13263	ZNF563	Q8TA94	Q92673
ZNF566	Q969W8	Q13263	ZNF574	Q6ZN55-2	Q12933
ZNF575	Q86XF7	Q9UGI0	ZNF576	Q9H609	Q7Z6E9
ZNF577	Q9BSK1	O95373	ZNF579	Q8NAF0	P04608
ZNF580	Q9UK33	P14373	ZNF581	Q9P0T4	Q92922
ZNF584	Q8IVC4	P05067	ZNF587	Q96SQ5	P19883
ZNF593	O00488	Q9Y4Z2	ZNF596	Q8TC21	P60900
ZNF598	Q86UK7	P63244	ZNF599	Q96NL3	Q9UGI0
ZNF606	Q8WXB4	P05067	ZNF608	Q9ULD9	Q13547
ZNF611	Q8N823	Q13263	ZNF619	Q8N2I2	P05067
ZNF620	Q6ZNG0	O75093	ZNF622	Q969S3	Q6NZI2
ZNF627	Q7L945	Q9UBX5	ZNF628	Q5EBL2	Q8IYI6
ZNF629	Q9UEG4	P40429	ZNF638	Q14966	Q9Y4Z2
ZNF639	Q9UID6	Q99633	ZNF641	Q96N77-2	P04792
ZNF644	Q9H582	P48431	ZNF646	O15015	A8MZ59
ZNF648	Q5T619	Q9BUH8	ZNF668	Q96K58-2, Q96K58	Q9UGY1, Q13895
ZNF669	Q96BR6	Q9UBX5	ZNF671	ZNF671	Q8IUQ4
ZNF672	Q499Z4	Q04721	ZNF684	Q5T5D7	Q13263
ZNF687	Q8N1G0	Q13547	ZNF688	P0C7X2	O43639

Input	UniProt Id	Interacts with	Input	UniProt Id	Interacts with
ZNF689	Q96CS4	Q13263	ZNF691	Q5VV52	Q9NRD5
ZNF692	Q9BU19	Q92993	ZNF696	Q9H7X3	Q9UBX5
ZNF697	Q5TEC3	O76024	ZNF703	Q9H7S9	Q8N695
ZNF704	Q6ZNC4	Q96ST3, Q9H160	ZNF707	Q96C28	Q16623
ZNF71	Q9NQZ8	Q8IY81, Q13895	ZNF711	Q9Y462	Q9UPP1
ZNF738	Q8NE65	Q13263	ZNF74	Q16587	Q8N2S1
ZNF746	Q6NUN9	O60260	ZNF763	Q0D2J5	Q9UMX1
ZNF764	Q96H86	Q8N2S1	ZNF768	Q9H5H4	Q04721
ZNF771	Q7L3S4	P40429	ZNF775	Q96BV0	Q9H2G4
ZNF777	Q9ULD5	P04004	ZNF785	A8K8V0	Q9NPB3
ZNF786	Q8N393	Q9NS56	ZNF800	Q2TB10	P06748
ZNF808	Q8N4W9	Q13263	ZNF830	Q96NB3	Q13573
ZNF835	Q9Y2P0	Q9BUH8	ZNF837	Q96EG3	Q9BUH8
ZNF843	Q8N446	P49639	ZNF865	P0CJ78	P36941
ZNFX1	Q9P2E3	Q9Y3F4	ZNHIT1	O43257	Q14451
ZNHIT2	Q9UHR6	Q9Y265	ZNHIT3	Q15649	Q9Y265
ZNHIT6	Q9NWK9	P00740	ZNRD1	Q9P1U0	P06396
ZNRF1	Q8ND25	P51668	ZNRF2	Q8NHG8	P61088
ZPR1	O75312	P68104	ZRANB2	O95218	P32121, P49407
ZRANB3	Q5FWF4	P12004	ZSCAN1	Q8NBB4-2	Q00403
ZSCAN12	O43309	Q9BYJ1	ZSCAN16	Q9H4T2	O60437
ZSCAN18	Q8TBC5	Q9UMX1	ZSCAN21	Q9Y5A6	Q6NZI2
ZSCAN22	P10073	Q00403	ZSCAN31	Q96LW9	Q9NYA4
ZSCAN32	Q9NX65	P11142	ZSCAN5A	Q9BUG6	O15397
ZSCAN9	O15535	P14136	ZSWIM3	Q96MP5	Q12933
ZSWIM7	Q19AV6	P14927	ZSWIM8	A7E2V4	P49674, P48730
ZWINT	O95229	P36941	ZXDA	P98168	P33076
ZYX	Q15942	P02751	ZZEF1	O43149	Q8NBL1
ZZZ3	Q8IYH5	Q12933			

Input	ChEBI Id	Interacts with	Input	ChEBI Id	Interacts with
ACBD6	Q9BR61	15530	ACPP	A0ABF7PH85	29105
ANGPTL4	Q9BY76	28815	APP	P05067-PRO_0000000092	28815
ARHGEF7	Q14155	15996	BOLA3	Q53S33	33737
CDC42	P60953	15996	CFH	P08603	18132
CHP2	O43745	29108	CLEC14A	Q86T13	28815
CRY1	P97784	29105	DNAJC27	Q8CFP6	15996
DNM1	Q05193	P50570, 15996	FGF1	P05230	29036
GCC2	Q8IWJ2	15996	GLRX	P35754	16856
GOLPH3	Q9H4A6	17526	HADH	P40939	28494
HADHA	P40939	28494	IDH1	O75874	16810
IL11	P20809	29108	LRP1	Q07954	29108
LRRK2	Q5S007	17552	MACROD2	A1Z1Q3	16960
MMADHC	Q9H3L0, Q99LS1	16856	MUT	P22033	18408
NMT2	O60551	15530	NPC1	O15118	16113
NPC2	P61916	16113	PAK1	Q16512, Q13153	15996, P05067
PITPNC1	Q9UKF7-1	17526	PLEKHM1	Q9Y4G2	15996
RAB1A	P62820	15996	RAB38	P57729	15996
RAB5B	P61020	15996	RAB8A	P61006	17552
RAC1	P63000	P53365, 15996, P05067	RAN	P62826	17552
RCC1	P18754	17552	RHOG	P84095	16152

Input	ChEBI Id	Interacts with	Input	ChEBI Id	Interacts with
RILPL2	Q969X0	15996	S100A12	P80511	29108
S100A13	P97352	29036	SNCA	P37840	17552
SP1	Q9UKF7-1	17526	SRC	P12931	15530
SYT1	P21707	29036	THBS1	P07996	28815
TLL1	O43897	29108			

7. Identifiers not found

These 4378 identifiers were not found neither mapped to any entity in Reactome.

A1BG-AS1	A2M-AS1	AACSP1	AADACL2-AS1	AASDH	AB015752.3	ABCA17P	ABHD12B
ABHD13	ABHD17AP6	ABHD2	ABHD8	ABI3BP	ABRA	ABRACL	AC000068.10
AC000068.5	AC000068.9	AC002066.1	AC002117.1	AC002127.4	AC002310.13	AC002429.5	AC002451.3
AC002454.1	AC002467.7	AC002539.1	AC003006.7	AC003075.4	AC003088.1	AC003090.1	AC003101.1
AC003991.3	AC004053.1	AC004076.5	AC004076.7	AC004381.7	AC004471.9	AC004538.3	AC004540.4
AC004862.6	AC004893.11	AC004901.1	AC004980.7	AC005062.2	AC005076.5	AC005154.6	AC005237.4
AC005307.3	AC005355.1	AC005537.2	AC005592.2	AC005606.14	AC005609.16	AC005609.18	AC005618.6
AC005682.5	AC005682.6	AC005780.1	AC005943.2	AC006026.13	AC006369.2	AC006538.4	AC006547.14
AC006548.28	AC006946.15	AC006946.16	AC006946.17	AC006994.2	AC007038.7	AC007040.8	AC007091.1
AC007128.1	AC007193.6	AC007204.2	AC007246.3	AC007255.8	AC007285.6	AC007292.3	AC007292.6
AC007318.5	AC007319.1	AC007383.3	AC007405.6	AC007563.5	AC007566.10	AC007682.1	AC007743.1
AC007750.5	AC007796.1	AC007879.2	AC007879.4	AC007879.5	AC007966.1	AC008074.3	AC008753.6
AC009014.3	AC009120.4	AC009120.6	AC009229.5	AC009229.6	AC009237.8	AC009299.3	AC009495.4
AC009784.3	AC009950.2	AC009961.3	AC010084.1	AC010127.3	AC010148.1	AC010226.4	AC010729.1
AC010729.2	AC010761.6	AC010883.5	AC010907.5	AC010967.2	AC010980.2	AC011239.1	AC011288.2
AC011530.4	AC011899.9	AC012074.2	AC012123.1	AC012146.7	AC012354.6	AC012358.8	AC012363.4
AC012442.5	AC012501.2	AC012507.3	AC013463.2	AC016735.2	AC016747.3	AC016757.3	AC016768.1
AC016831.7	AC016907.3	AC016910.1	AC016995.3	AC018359.1	AC018462.2	AC018647.3	AC018755.16
AC018816.3	AC018890.6	AC019117.1	AC019186.1	AC022182.1	AC024084.1	AC025171.1	AC025627.7
AC026150.8	AC026188.1	AC027612.4	AC027612.6	AC058791.1	AC061992.2	AC064875.2	AC067956.1
AC067959.1	AC068286.1	AC069213.1	AC069368.3	AC069513.4	AC072062.1	AC073283.4	AC073869.1
AC074117.10	AC074289.1	AC074363.1	AC074391.1	AC079154.1	AC079305.10	AC079305.11	AC079305.8
AC079610.1	AC079630.4	AC079807.4	AC083843.1	AC083864.4	AC084219.4	AC084809.2	AC087294.2
AC091133.1	AC091729.9	AC091814.2	AC091878.1	AC092155.4	AC092171.2	AC092198.1	AC092535.3
AC092614.2	AC092675.3	AC092684.1	AC092835.2	AC093323.3	AC093388.3	AC093609.1	AC093627.9
AC093642.5	AC093673.5	AC093838.4	AC096559.1	AC097468.7	AC097662.2	AC097721.2	AC097724.3
AC098617.1	AC098973.2	AC099850.1	AC104655.3	AC104820.2	AC105053.3	AC105402.4	AC105760.2
AC105760.3	AC106869.2	AC107218.3	AC108142.1	AC108448.2	AC109826.1	AC113189.5	AC114271.2
AC114498.1	AC114765.1	AC116609.3	AC125232.1	AC129778.2	AC131056.3	AC133680.1	AC136289.1
AC137723.5	AC138969.4	AC140542.2	AC141928.1	AC144449.1	AC144450.1	AC144652.1	AC144831.1
AC144835.1	AC241585.2	AC245100.1	AC253572.1	ACAP2	ACAP3	ACRC	ACSL6_2
ACTG1P17	ACVR2B-AS1	ACYP1	ACYP2	ADAM29	ADAMTS7P1	ADAMTS7P4	ADAMTS9-AS1
ADAMTS9-AS2	ADAMTSL4-AS1	ADAP2	ADARB2	ADCK1	ADD3-AS1	ADGRA1-AS1	ADGRB1
ADGRB2	ADGRD1	ADGRE4P	ADGRG2	ADGRG7	ADGRL1	ADGRL4	ADIRF-AS1
ADORA2A-AS1	ADPRH	AE000661.37	AEBP1	AF001548.5	AF003625.3	AF038458.3	AF064858.8
AF064860.5	AF127936.3	AF127936.5	AF127936.9	AF131216.1	AF131217.1	AF146191.4	AF165138.7
AF196970.3	AFAP1	AFAP1L1	AFAP1L2	AFF2	AFTPH	AGAP1	AGAP11
AGAP2-AS1	AHSA2	AIF1L	AIFM3	AJAP1	AKIRIN1	AKNA	AL035610.2
AL132709.8	AL163953.3	AL365181.2	AL450992.2	AL591893.1	ALG1L	ALKBH6	ALKBH7
ALLC	ALMS1P	ALOX12-AS1	ALOX12P2	ALPK3	AMER2	AMZ1	AMZ2
ANGEL2	ANGPTL1	ANGPTL5	ANKAR	ANKDD1A	ANKDD1B	ANKFN1	ANKMY1
ANKRD12	ANKRD13B	ANKRD18A	ANKRD18CP	ANKRD19P	ANKRD20A1	ANKRD20A11P	ANKRD20A4
ANKRD20A5P	ANKRD20A7P	ANKRD20A9P	ANKRD22	ANKRD26P3	ANKRD30B	ANKRD30BL	ANKRD31
ANKRD33B	ANKRD34A	ANKRD34C	ANKRD35	ANKRD36BP2	ANKRD36C	ANKRD37	ANKRD39

ANKRD65	ANKUB1	ANOS2P	ANTXRLP1	ANXA11	ANXA7	AOAH	AOC4P
AP000221.1	AP000235.2	AP000251.3	AP000320.7	AP000350.5	AP000473.5	AP000473.8	AP000476.1
AP000479.1	AP000662.4	AP000688.14	AP000704.5	AP000892.6	AP001062.7	AP001172.2	AP001258.4
AP001347.6	AP001350.4	AP001412.1	AP001505.10	AP001604.3	AP001626.2	AP002884.2	AP003068.12
AP005530.1	AP006621.5	AP006621.8	AP3B2	AP3S2	APCDD1L-AS1	APOBEC3D	APOL4
APOLD1	APOPT1	APTR	AQP4-AS1	AQP7P1	AQP7P2	ARHGAP19-SLIT1	ARHGEF26-AS1
ARHGEF7-AS2	ARIH2OS	ARL17B	ARL5A	ARMC2	ARMC4P1	ARMCX2	ARMCX4
ARMCX6	ARPP21	ARRDC1-AS1	ARRDC3-AS1	ARRDC5	ASAH2B	ASB16-AS1	ASH1L-AS1
ASIP	ASPHD1	ASTN2	ATAD2B	ATE1-AS1	ATF7IP2	ATHL1	ATOH7
ATOH8	ATP1A1-AS1	ATP2A1-AS1	ATP6APIL	ATP6V0E2-AS1	ATXN7L3B	AVIL	AVL9
AXDND1	AZIN1-AS1	B4GALNT3	B4GALNT4	BAALC	BACE1-AS	BAIAP2-AS1	BAIAP3
BARHL2	BBOX1-AS1	BCAS1	BCDIN3D-AS1	BCRP2	BCRP3	BCYRN1	BDNF-AS
BEND6	BEX4	BHLHE40-AS1	BICC1	BLOC1S5-TXNDC5	BMP3	BMP6	BMP8A
BMP8B	BMS1P1	BMS1P14	BMS1P20	BNC2	BOLA3-AS1	BRSK1	BSDC1
BTBD10	BTBD16	BTBD17	BTBD19	BTBD7	BTBD8	BUD13P1	BVES
BVES-AS1	BX322557.10	BZRAP1-AS1	BZW1	C10orf10	C10orf105	C10orf11	C10orf111
C10orf128	C10orf131	C10orf32	C10orf53	C10orf76	C10orf90	C11orf21	C11orf58
C11orf63	C11orf70	C11orf71_1	C11orf80	C11orf95	C11orf98	C12orf42	C12orf43
C12orf54	C12orf56	C12orf60	C12orf79_1	C14orf132	C14orf28	C14orf79	C15orf40
C15orf52	C15orf59-AS1	C15orf61	C15orf65	C16orf13	C16orf52	C16orf72	C16orf89
C16orf91	C16orf93	C16orf95	C16orf96	C17orf100_1	C17orf102	C17orf51	C17orf53
C17orf58	C17orf75	C18orf21	C18orf32	C18orf42	C19orf12	C19orf24	C19orf33
C19orf48	C19orf73	C19orf81	C1QTNF3-AMACR	C1QTNF4	C1QTNF7	C1RL	C1RL-AS1
C1orf101	C1orf115	C1orf123	C1orf132	C1orf145	C1orf159	C1orf162	C1orf174
C1orf194	C1orf21	C1orf233	C1orf43	C20orf166-AS1	C20orf194	C20orf197	C20orf203
C20orf96	C21orf62-AS1	C21orf91	C22orf23	C22orf34	C22orf42	C22orf46	C2CD2
C2CD4B	C2CD4C	C2orf27A	C2orf48	C2orf73	C2orf74	C2orf76	C2orf80
C2orf81	C2orf88	C3orf20	C3orf35	C3orf58	C3orf80	C4orf19	C4orf36
C4orf48	C5orf17	C5orf38	C5orf42	C5orf49	C5orf56	C5orf63	C5orf66
C5orf67	C6orf1	C6orf136	C6orf223	C6orf3	C6orf48	C6orf52	C7orf26
C7orf61	C7orf73	C8orf34	C8orf34-AS1	C8orf37-AS1	C8orf4	C8orf44	C8orf44-SGK3
C8orf58	C8orf82	C8orf88	C9orf116	C9orf129	C9orf153	C9orf163	C9orf171
C9orf3	C9orf69	C9orf91	C9orf92	CA5BP1	CAAP1	CACHD1	CACNA2D4
CADPS2	CALB2	CALHM2	CAMK2N1	CAMK2N2	CAMTA1	CAPN10-AS1	CAPS2
CARD16	CARD6	CARM1P1	CASC1	CASC15	CASC18	CASC2	CASC21
CASC4	CASC8	CASC9	CASD1	CASZ1	CBLN1	CBLN2	CBR3-AS1
CCBE1	CCDC110	CCDC122	CCDC126	CCDC144B	CCDC144CP	CCDC144NL	CCDC144NL-AS1
CCDC149	CCDC15	CCDC157	CCDC159	CCDC170	CCDC171	CCDC173	CCDC177
CCDC18	CCDC181	CCDC3	CCDC30	CCDC39	CCDC61	CCDC64	CCDC65
CCDC66	CCDC69	CCDC73	CCDC74A	CCDC78	CCDC7_1	CCDC7_2	CCDC80
CCDC81	CCDC91	CCER2	CCL8	CCNT2-AS1	CCPG1	CCSAP	CCSER1
CCT5P2	CCT6P3	CD163L1	CD27-AS1	CD276	CD81-AS1	CD99P1	CDADC1
CDC123	CDH20	CDH22	CDH26	CDHR1	CDKN2B-AS1	CDNF	CDR1-AS
CDV3	CECR6	CECR7	CELF2	CELF3	CELF4	CELSR1	CELSR3
CENPBD1	CENPBD1P1	CENPVP1	CEP120	CEP128	CEP83-AS1	CERS3-AS1	CFAP43
CFAP44	CFAP54	CFAP69	CFAP70	CFAP97	CFLAR-AS1	CGREF1	CH17-125A10.2

CH17-140K24.8	CH17-189H20.1	CH17-340M24.3	CH17-437K3.1	CH17-472G23.1	CH17-472G23.2	CH507-210P18.4	CH507-24F1.1
CH507-39O4.2	CH507-513H4.1	CH507-513H4.5	CH507-528H12.1	CH507-9B2.3	CHD2	CHI3L2	CHIC1
CHKB-AS1	CHL1-AS2	CHMP1B2P	CHURC1	CHURC1-FNTB	CIDCEP	CKLF-CMTM1	CKMT2-AS1
CLCN3P1	CLDN10-AS1	CLEC2L	CLEC9A	CLECL1	CLIC6	CLMN	CLN8
CLRN1-AS1_1	CLSTN2	CLYBL	CMAHP	CNBD1	CNTLN	CNTN4-AS2	CNTNAP3B
CNTNAP3P1	CNTNAP3P2	COLCA1	COLCA2	CORO2B	CORO7	COX10-AS1	CPAMD8
CPB2-AS1	CPEB1-AS1	CPEB2	CPEB2-AS1	CPEB3	CPED1	CPQ	CPSF1P1
CPT1C	CPXM1	CPXM2	CRAT37	CREG2	CRIM1	CRISPLD1	CRNDE
CROCCP2_2	CROCCP3	CRTAC1	CRYBB2P1	CRYBG3	CRYGN	CRYM-AS1	CRYZ
CSDC2	CSMD2	CSMD3	CSNK2B_1	CSPG4P10	CSPG4P11	CSPG4P12	CSPG4P13
CTA-204B4.2	CTA-212A2.1	CTA-292E10.6	CTA-29F11.1	CTA-384D8.36	CTA-38K21.2	CTA-445C9.14	CTA-796E4.4
CTB-107G13.1	CTB-111F10.1	CTB-113D17.1	CTB-113I20.2	CTB-129O4.1	CTB-12O2.1	CTB-131B5.5	CTB-131K11.1
CTB-152G17.6	CTB-161M19.4	CTB-171A8.1	CTB-179K24.3	CTB-1I21.1	CTB-25B13.5	CTB-31O20.2	CTB-35F21.1
CTB-55O6.12	CTB-55O6.8	CTB-58E17.1	CTB-58E17.5	CTBP1-AS2	CTC-228N24.2	CTC-231O11.1	CTC-232P5.1
CTC-235G5.3	CTC-236F12.4	CTC-241N9.1	CTC-246B18.8	CTC-254B4.1	CTC-255N20.1	CTC-273B12.10	CTC-277H1.6
CTC-281F24.1	CTC-281F24.3	CTC-301O7.4	CTC-321K16.1	CTC-329D1.2	CTC-338M12.4	CTC-338M12.9	CTC-340A15.2
CTC-340D7.1	CTC-347C20.1	CTC-347C20.2	CTC-348L5.1	CTC-351M12.1	CTC-378H22.2	CTC-428H11.2	CTC-436P18.1
CTC-436P18.3	CTC-444N24.11	CTC-444N24.6	CTC-448D22.1	CTC-458G6.2	CTC-467M3.1	CTC-471J1.11	CTC-487M23.8
CTC-490E21.12	CTC-498M16.4	CTC-512J12.6	CTC-523E23.4	CTC-525D6.1	CTC-529P8.1	CTC-535M15.2	CTC-543D15.8
CTC-550B14.7	CTC-559E9.6	CTC-575N7.1	CTC-756D1.2	CTC-786C10.1	CTC-806A22.1	CTD-2001C12.1	CTD-2005H7.1
CTD-2005H7.2	CTD-2007H13.3	CTD-2008L17.1	CTD-2015G9.2	CTD-2017F17.2	CTD-2020K17.3	CTD-2021H9.3	CTD-2021J15.1
CTD-2023M8.1	CTD-2023N9.1	CTD-2023N9.2	CTD-2024P10.1	CTD-2024P10.2	CTD-2026D20.2	CTD-2026D20.3	CTD-2026G6.3
CTD-2034I4.1	CTD-2047H16.4	CTD-2049O4.1	CTD-2054N24.2	CTD-2058B24.2	CTD-2066L21.3	CTD-2089N3.1	CTD-2089N3.2
CTD-2091N23.1	CTD-2104P17.1	CTD-2126E3.4	CTD-2135J3.3	CTD-2140G10.2	CTD-2143L24.1	CTD-2147F2.1	CTD-2147F2.2
CTD-2184D3.5	CTD-2186M15.3	CTD-2201G3.1	CTD-2201I18.1	CTD-2203K17.1	CTD-2206N4.4	CTD-2215E18.1	CTD-2223O18.1
CTD-2228K2.7	CTD-2231E14.2	CTD-2231H16.1	CTD-2235C13.3	CTD-2240J17.1	CTD-2245F17.3	CTD-2256P15.2	CTD-2267D19.3
CTD-2269F5.1	CTD-2270L9.4	CTD-2270P14.5	CTD-2281E23.2	CTD-2287O16.5	CTD-2288O8.1	CTD-2292P10.4	CTD-2303H24.2
CTD-2307P3.1	CTD-2308L22.1	CTD-2314G24.2	CTD-2319I12.4	CTD-2320G14.2	CTD-2336H13.2	CTD-2337A12.1	CTD-2341M24.1
CTD-2349B8.1	CTD-2349P21.11	CTD-2353F22.1	CTD-2368P22.1	CTD-2369P2.10	CTD-2371O3.3	CTD-2378E12.1	CTD-2384B11.2
CTD-2507G9.1	CTD-2516F10.2	CTD-2523D13.2	CTD-2535L24.2	CTD-2538C1.2	CTD-2542C24.14	CTD-2542L18.1	CTD-2544M6.1
CTD-2547L16.1	CTD-2553L13.10	CTD-2554C21.3	CTD-2555K7.2	CTD-2555O16.2	CTD-2561J22.2	CTD-2561J22.3	CTD-2561J22.5
CTD-2562J15.6	CTD-2562J17.2	CTD-2562J17.4	CTD-2562J17.6	CTD-2572N17.1	CTD-2583A14.10	CTD-2583A14.11	CTD-2583A14.9
CTD-2587H24.14	CTD-2587M2.1	CTD-2619J13.17	CTD-2619J13.19	CTD-2619J13.3	CTD-2619J13.9	CTD-2620I22.1	CTD-2649C14.3
CTD-2651B20.3	CTD-2651B20.4	CTD-2666L21.1	CTD-3001H11.2	CTD-3006G17.2	CTD-3014M21.1	CTD-3037G24.3	CTD-3064C13.1

CTD-3065J16.6	CTD-3065J16.9	CTD-3088G3.8	CTD-3092A11.1	CTD-3105H18.16	CTD-3105H18.18	CTD-3113P16.11	CTD-3131K8.2
CTD-3162L10.1	CTD-3179P9.1	CTD-3184A7.4	CTD-3193K9.3	CTD-3193O13.8	CTD-3220F14.1	CTD-3222D19.12	CTD-3222D19.2
CTD-3252C9.4	CTDSPL	CTXN1	CUX2	CXorf23	CXorf36	CXorf40B	CYP17A1-AS1
CYP1B1-AS1	CYP20A1	CYP21A1P	CYP2T1P	CYP4A22-AS1	CYP4F24P	CYP4F29P	CYP4F35P
CYP4X1	CYP4Z1	CYP51A1-AS1	CYTL1	CYYR1	CYYR1-AS1	D4S234E	DAAM2
DACH2	DACT2	DACT3	DACT3-AS1	DANCR	DANT2	DBIL5P	DCDC1
DCDC2C	DCHS1	DDT	DDX26B	DDX43	DDX60	DDX60L	DENND6A-AS1
DEPDC7	DGCR5	DHRS12	DHRS4L2	DICER1-AS1	DIO3OS	DIP2C	DIRC2
DIRC3	DISC1FP1	DKFZP434I0714	DLEC1	DLEU2	DLEU7	DLEU7-AS1	DLGAP1-AS1
DLGAP1-AS2	DLGAP4-AS1	DLK2	DMTF1	DNAAF3	DNAH1	DNAH11	DNAH12
DNAH14	DNAH5	DNAH6	DNAH7	DNAI1	DNAJC21	DNAJC22	DNAJC25
DNAJC27-AS1	DNAJC3-AS1	DNAJC9	DNASE1	DNASE1L3	DNHD1	DNM1P46	DNM1P47
DNM3OS	DOC2B	DOPEY1	DPH6-AS1	DPP10	DPP6	DPT	DPY19L1P1
DPY19L1P2	DPY19L2	DPY19L2P2	DPY19L2P3	DPY19L2P4	DPY19L4	DPYD-AS1	DRAIC
DRAM2	DRGX	DTD1	DTHD1	DTX2P1	DTX2P1-UPK3BP1-PMS2P11	DUS1L	DUSP18
DUSP28	DUXAP8_1	DUXAP9	Dec-01	ECHDC3	EEF1A1P11	EEF1A1P13	EFCAB1
EFCAB13	EFCAB8	EFCC1	EFHB	EFR3A	EFR3B	EFTUD1	EGFEM1P
EGFLAM	EID2	EID2B	EIF1B-AS1	EIF3FP3	EIF3J-AS1	EIF5AL1	ELAC1
ELF4	ELFN1	ELK4	ELMOD1	ELOF1	ELOVL2-AS1	EMC3-AS1	EMCN
EML5	EML6	EMP2	EMX1	EMX2OS	EN2	ENDOU	ENDOV
ENHO	ENOSF1	ENOX2	ENPP5	ENPP7P4	ENTHD1	ENTPD1-AS1	ENTPD3-AS1
EPB41L4A	EPB41L4A-AS1	EPB41L4A-AS2	EPB41L4B	EPG5	EPHA1-AS1	EPHX4	EQTN
ERCC6L2	ERICH3	ERMARD	ERMN	ERV3-1	ETAA1	ETV2	EWSAT1
EXD3	EXO5	EXPH5	F11-AS1	FAM101A	FAM102B	FAM104B	FAM109B
FAM120AOS	FAM127B	FAM131A	FAM131C	FAM133B	FAM13A-AS1	FAM149A	FAM151B
FAM153A	FAM153B	FAM153C	FAM155A	FAM155B	FAM157C	FAM159A	FAM160A1
FAM160A2	FAM160B2	FAM162B	FAM173B	FAM174B	FAM178B	FAM179A	FAM179B
FAM180A	FAM181B	FAM182B	FAM184B	FAM185BP	FAM193A	FAM196B	FAM198A
FAM198B	FAM19A2	FAM19A3	FAM19A5	FAM200A	FAM200B	FAM212B-AS1	FAM216A
FAM220A	FAM224B	FAM227A	FAM227B	FAM228B	FAM229A	FAM229B	FAM27E3
FAM43B	FAM46C	FAM47E-STBD1_1	FAM47E-STBD1_2	FAM53A	FAM57A	FAM57B	FAM58A
FAM63B	FAM65A	FAM66A	FAM66B	FAM66C	FAM66D	FAM69C	FAM71E1
FAM71F2	FAM76A	FAM84A	FAM85B	FAM89A	FAR2P1	FAR2P3	FAT2
FAT3	FAT4	FBLL1	FBLN7	FBXL19-AS1	FBXO39	FBXO8	FCF1P2
FCGR1C	FCGR2C	FER1L5	FER1L6	FER1L6-AS1	FER1L6-AS2	FGD5-AS1	FGF10-AS1
FGF12-AS3	FGGY	FHAD1	FIBIN	FIGNL2	FILIP1L	FIRRE	FJX1
FKBP11	FLG-AS1	FLJ12825	FLJ20021	FLJ27354	FLJ33360	FLJ37453	FLJ46284
FLVCR1-AS1	FLVCR2	FMO4	FNBP1P1	FNDC1	FNDC5	FOCAD	FOXDI
FOXDI2-AS1	FOXDI3-AS1	FOXI3	FOXJ3	FOXP4-AS1	FP236383.5	FP236383.7	FP325317.1
FPGT-TNNI3K	FREM1	FREM2	FRG1HP	FRG1JP	FRMD4B	FRMD6-AS2	FRMD7
FRMD8	FRMPD2	FRZB	FSD1L	FSIP1	FSIP2-AS1	FTCDNL1	FTX
FZDI10-AS1	GAGE1	GAL3ST3	GAL3ST4	GANC	GAPDHP21	GAPT	GARNL3
GAS5	GAS6-AS1	GATA2-AS1	GATS	GBAP1	GBGT1	GBP1P1	GCC2-AS1
GDAP1L1	GDF10	GDF11	GDF7	GDPD2	GEMIN8P4	GFOD2	GGNBP2
GGT8P	GGTA1P	GHRLOS_1	GIMAP4	GIMAP6	GIMAP7	GIMAP8	GIN1

GIPC3	GJC3	GK5	GLDN	GLIDR	GLIS1	GLMP	GLOD4
GLRA4	GLT1D1	GLTPD2	GMDS-AS1	GNAS-AS1	GNG12-AS1	GOLGA2P10	GOLGA2P7
GOLGA6D	GOLGA8H	GOLGA8M	GOLGA8N	GOLGA8S	GPALPP1	GPATCH3	GPATCH4
GPC5-AS1	GPR135	GPR137	GPR137B	GPR139	GPR146	GPR149	GPR153
GPR157	GPR158	GPR171	GPR173	GPR180	GPR19	GPR34	GPR75-ASB3
GPR82	GPR85	GPRIN3	GPX1P1	GREB1L	GRINA	GRM5-AS1	GRM7-AS1
GRTP1	GRWD1	GS1-114I9.3	GS1-124K5.11	GS1-124K5.12	GS1-124K5.2	GS1-124K5.3	GS1-124K5.4
GS1-166A23.2	GS1-259H13.11	GS1-358P8.4	GS1-57L11.1	GS1-72M22.1	GSAP	GSDMB	GSDMC
GSG1L	GTDC1	GTF2H2B	GTF2IP1	GTF2IP14	GTF2IP4	GTF2IP7	GTF2IRD2
GTPBP6	GTSE1-AS1	GULOP	GULP1	GUSBP2	GUSBP4	GUSBP9	GVINP1
GXYLT1P4	GZMA	GZMK	H1FX-AS1	H3F3AP6	H6PD	HAGHL	HAGLR
HAMP	HAND2-AS1	HARBI1	HAS2-AS1	HCFC1R1	HCG11	HCG18	HCG22
HCG23	HCP5	HDGFRP3	HECA	HEIH	HELLPAR	HEPACAM	HERC2P10
HERC2P3	HERC2P4	HERC2P9	HEXDC	HFM1	HHIP-AS1	HHIPL1	HIATL2
HIF1A-AS2	HILS1	HLA-DRB6	HLA-DRB9	HLA-F-AS1	HLA-K	HLA-L	HMGB1P10
HMGB1P5	HMGNI	HMGNI2P46	HMGNI3	HMGNI4	HMP19	HMX1	HNRNP1P13
HNRNPKP3	HNRNPU-AS1	HORMAD1	HOTAIRM1	HOXA-AS2	HOXB-AS1	HOXC10	HOXD-AS2
HPN-AS1	HPS3	HPS5	HPS6	HRK	HSD17B7P2	HSDL2	HSPD1P8
HSPE1-MOB4	HTATSF1P2	HYDIN2	ICOSLG_1	IDNK	IDS_1	IER5L	IFI27L1
IFNG-AS1	IFRD1	IGDCC3	IGFBP7-AS1	IGIP	IGLON5	IGSF10	IGSF11
IGSF21	IGSF5	IKZF2	IL12A-AS1	IL17B	IL17D	ILDR2	ILF3-AS1
IMMP1L	IMMP2L	IMPDH1P10	INAFM1	INAFM2	INMT-FAM188B	INSM2	INTS4P1
INTS6-AS1	INVS	IPO5P1	IPO9-AS1	IQCH	IQCH-AS1	IQCJ-SCHIP1	IQCK
IRAK1BP1	IRX4	IRX5	ISOC2	ITGA9-AS1	ITGB2-AS1	ITIH5	JAZF1-AS1
JHDM1D-AS1	JMJD8	JPX	JTB	KANSL1L	KANTR	KAZALD1	KB-1125A3.11
KB-1208A12.3	KB-1458E12.1	KB-1507C5.4	KB-1517D11.4	KB-1836B5.1	KB-68A7.1	KBTBD11	KBTBD12
KBTBD3	KC6	KCNJ2-AS1	KCNMB2-AS1	KCNQ1OT1	KCTD2	KCTD8	KIAA0513
KIAA0895	KIAA0895L	KIAA0922	KIAA1024L	KIAA1107	KIAA1109	KIAA1210	KIAA1407
KIAA1462	KIAA1614	KIAA1755	KIAA2018	KIAA2026	KIF9-AS1	KIRREL3-AS1	KLF3-AS1
KLHDC1	KLHDC8B	KLHL1	KLHL29	KLHL4	KLRAP1	KLRC4-KLRK1	KMT2E-AS1
KNCN	KNDC1	KRCC1	KRT222_1	KRT222_2	KRT8P12	KRTAP5-AS1	KRTCAP2
KTN1-AS1	LA16c-306E5.2	LA16c-306E5.3	LA16c-312E8.2	LA16c-380H5.5	LA16c-444G7.1	LA16c-OS12.2	LAMP5
LAMTOR5-AS1	LBH	LBX1-AS1	LCNL1	LCTL	LDAH	LDLRAD3	LEF1-AS1
LEKR1	LEMD1-AS1	LETM2	LGALS12	LGALS8-AS1	LHFP	LHFPL3	LINC00032
LINC00092	LINC00094	LINC00106	LINC00159	LINC00174	LINC00205	LINC00222	LINC00242
LINC00271	LINC00276	LINC00278	LINC00282	LINC00299	LINC00304	LINC00324	LINC00327
LINC00339	LINC00342	LINC00346	LINC00377	LINC00403	LINC00407	LINC00461	LINC00462
LINC00463	LINC00466	LINC00472	LINC00475	LINC00476	LINC00486	LINC00487	LINC00504
LINC00506	LINC00511	LINC00534	LINC00535	LINC00536	LINC00540	LINC00571	LINC00574
LINC00578	LINC00596	LINC00598	LINC00599	LINC00607	LINC00609	LINC00616	LINC00623
LINC00630	LINC00632	LINC00634	LINC00639	LINC00641	LINC00643	LINC00645	LINC00648
LINC00649	LINC00652	LINC00654	LINC00662	LINC00665	LINC00667	LINC00672	LINC00674
LINC00675	LINC00680	LINC00693	LINC00702	LINC00836	LINC00839	LINC00840	LINC00843
LINC00847	LINC00854	LINC00856	LINC00862	LINC00863	LINC00865	LINC00871	LINC00880
LINC00882	LINC00883	LINC00886	LINC00888	LINC00891	LINC00893	LINC00894	LINC00899
LINC00907	LINC00908	LINC00909	LINC00910	LINC00922	LINC00923	LINC00924	LINC00936
LINC00937	LINC00950	LINC00957	LINC00959	LINC00964	LINC00968	LINC00969	LINC00970
LINC00973	LINC00976	LINC00977	LINC00982	LINC00987	LINC00989	LINC00996	LINC00997

LINC01002	LINC01003	LINC01004	LINC01006	LINC01012	LINC01014	LINC01018	LINC01021
LINC01031	LINC01036	LINC01037	LINC01057	LINC01060	LINC01073	LINC01088	LINC01090
LINC01091	LINC01094	LINC01098	LINC01105	LINC01115	LINC01116	LINC01117	LINC01119
LINC01123	LINC01125	LINC01126	LINC01128	LINC01133	LINC01135	LINC01137	LINC01138
LINC01141	LINC01146	LINC01182	LINC01184	LINC01193	LINC01197	LINC01201	LINC01212
LINC01215	LINC01224	LINC01225	LINC01229	LINC01237	LINC01239	LINC01250	LINC01252
LINC01266	LINC01268	LINC01278	LINC01285	LINC01290	LINC01296	LINC01301	LINC01310
LINC01311	LINC01314	LINC01315	LINC01340	LINC01344	LINC01346	LINC01351	LINC01355
LINC01358	LINC01359	LINC01362	LINC01372	LINC01374	LINC01376	LINC01399	LINC01410
LINC01411	LINC01422_1	LINC01422_2	LINC01426	LINC01435	LINC01446	LINC01456	LINC01471
LINC01473	LINC01481_1	LINC01482	LINC01483	LINC01497	LINC01500	LINC01504	LINC01505
LINC01508	LINC01515	LINC01521	LINC01524	LINC01535	LINC01547	LINC01550	LINC01560
LINC01563	LINC01578	LINC01580	LINC01592	LINC01600	LINC01604	LINC01605	LINC01606
LINC01608	LINC01611	LINCR-0001	LINGO3	LINGO4	LIPE-AS1	LL09NC01-139C3.1	LL0XNC01-116E7.2
LL0XNC01-237H1.2	LL0XNC01-240C2.1	LL22NC03-80A10.6	LL22NC03-86G7.1	LLNLF-173C4.2	LLNLF-187D8.1	LLNLR-268E12.1	LLfos-48D6.2
LMBR1	LMBR1L	LMCD1-AS1	LMLN	LMNTD1	LMO7-AS1	LMO7DN	LMOD3
LOH12CR1	LOH12CR2	LRGUK	LRP3	LRP5L	LRRC10B	LRRC16B	LRRC17
LRRC2	LRRC24	LRRC26	LRRC28	LRRC29	LRRC3	LRRC34	LRRC37A11P
LRRC37A2	LRRC37A3	LRRC37A4P	LRRC37A6P	LRRC37B	LRRC38	LRRC3B	LRRC4
LRRC42	LRRC48	LRRC53	LRRC55	LRRC63	LRRC69	LRRC75A	LRRC75A-AS1
LRRC75B	LRRC9	LRRC1	LRRIQ1	LRRN3	LRRN4CL	LSG1	LURAP1L-AS1
LUZP2	LY6G5C	LY9	LYPD6	LYPLAL1	LYPLAL1-AS1	LYSMD2	LYSMD3
LYSMD4	MACC1	MAFG-AS1	MAFIP	MAFTRR	MAGI2-AS3	MALAT1_1	MALRD1
MAMDC2	MAMDC2-AS1	MAN1B1-AS1	MANEAL	MAP10	MAP3K10	MAP3K13	MAP6
MAP6D1	MAP9	MAPKAPK5-AS1	MARVELD1	MARVELD3	MAST4	MATR3_1	MBL1P
MBLAC1	MBNL2	MCF2L-AS1	MCM3AP-AS1	MCMDC2	MCTP1	MCTP2	MCTS1
MDFIC	MDM1	MED14OS	MEF2BNB-MEF2B	MEF2C-AS1	MEG3	MEG8	MEG9
MEGF10	MEGF9	METAP1D	METTL11B	METTL21EP	METTL24	METTL25	METTL2A
METTL2B	METTL4	METTL5	METTL7B	METTL8	MEX3A	MEX3D	MFAP3L
MFSD1	MFSD12	MFSD7	MGAM2	MGC16275	MGC27382	MGEA5	MGP
MIA	MIA-RAB4B	MIAT	MIATNB	MIF-AS1	MIPEPP3	MIR1-1HG	MIR100HG
MIR124-2HG	MIR137HG	MIR143HG	MIR155HG	MIR17HG	MIR181A1HG	MIR181A2HG	MIR2052HG
MIR210HG	MIR222HG	MIR22HG	MIR31HG	MIR325HG	MIR381HG	MIR4435-2HG	MIR4458HG
MIR4500HG	MIR4697HG	MIR646HG	MIR7-3HG	MIR762HG	MIR99AHG	MIRLET7BHG	MIS18A-AS1
MKLN1-AS	MLLT4-AS1	MMEL1	MMP23B	MMP24-AS1	MMP25-AS1	MNX1	MOAP1
MOB3A	MORN2	MPEG1	MPV17L2	MREG	MRGPRE	MRGPRF	MRGPRX1
MRGPRX3	MRO	MROH7-TTC4	MRPS31P4	MRT04	MS4A4E	MS4A6A	MSANTD1
MSANTD2	MSANTD3-TMEFF1	MSC	MSC-AS1	MTATP6P1	MTFR1L	MTHFSD	MTMR11
MTMR9LP	MTND2P28	MTND4P12	MTRNR2L1	MTRNR2L12	MTRNR2L8	MTX3	MUM1L1
MXRA5	MXRA7	MYCBP2-AS1	MYCBPAP	MYLK-AS1	MYO16	MYO16-AS1	MYO1A
MYO1F	MYOF	MYRFL	MZF1	Mar-01	Mar-02	Mar-03	Mar-04
Mar-07	Mar-08	Mar-09	Metazoa_SRP_137	Metazoa_SRP_138	N4BP2L1	NAA25	NAALADL1
NAALADL2	NACC2	NAGPA	NAIF1	NAIP	NALCN-AS1	NANOS1	NAP1L6
NAPSB	NAT16	NAT6	NAT9	NAV3	NBEAP1	NBEAP3	NBPF10
NBPF11	NBPF12	NBPF14	NBPF19	NBPF26	NBPF3	NCAM2	NCBP2-AS2

NCMAP	NDFIP2	NDNF	NDUFA6-AS1	NDUFB2-AS1	NDUFV2-AS1	NDUFV2P1	NEAT1
NEK10	NEK5	NELL1	NEMP2	NEUROD2	NEUROG2	NFIA-AS2	NFKBIL1
NFKBIZ	NIM1K	NIT1	NKAIN1	NKAIN2	NKAIN3	NKAIN4	NKAPP1
NKPD1	NKX6-2	NMD3	NME1-NME2	NME6	NNAT	NOP14-AS1	NOVA1
NOVA2	NPHP3-ACAD11	NPHP3-AS1	NPIPA1	NPIPB4	NPIPB5	NPM1P50	NPR3
NPTX2	NPTXR	NR2F1-AS1	NR2F2-AS1	NRAP	NRAV	NREP	NRIR
NRK	NSFP1	NSUN5	NSUN5P1	NSUN5P2	NSUN7	NT5DC1	NT5DC2
NT5DC3	NUP210L	NUS1P1	NUS1P2	NUTM2A	NUTM2A-AS1	NUTM2B-AS1	NUTM2D
NWD2	OCEL1	OCIAD2	ODF3B	OGFR	OIP5-AS1	OLFML1	OLFML3
ONECUT2	OPA3	OR2A1-AS1	OR7E14P	OR7E47P_1	OSCP1	OSER1-AS1	OSGIN2
OSMR-AS1	OTUD6B	OTUD6B-AS1	OVCH1-AS1	P4HTM	PABPC1P4	PACRG	PACRG-AS1
PAMR1	PANO1	PAPD4	PAPL	PAPLN	PAPPA-AS1	PAPPA-AS2	PAQR4
PAQR9	PARGP1	PARP15	PARP4P2	PARBPB	PART1	PATL2	PAXBP1-AS1
PAXIP1-AS1	PAXIP1-AS2	PCAT1	PCAT19	PCAT29	PCAT6	PCAT7	PCBP1-AS1
PCDH1	PCDH11X	PCDH11Y	PCDH12	PCDH17	PCDH8	PCDH9	PCDHA10
PCDHA12	PCDHA6	PCDHB10	PCDHB12	PCDHB15	PCDHB2	PCDHGA7	PCDHGB4
PCDHGB6	PCDHGC4	PCED1B-AS1	PCIF1	PCNX	PCNXL3	PCOLCE-AS1	PCP4
PDCD6IPP2	PDDC1	PDLIM2	PDLIM3	PDXDC2P_1	PDXP	PEPD	PEX5L
PGM5	PGM5P2	PGM5P4	PHACTR1	PHEX	PHLDA2	PHLDA3	PHTF2
PHYHIPL	PI15	PI4K2B_1	PI4K2B_2	PI4KAP2	PIEZO2	PIGBOS1	PIK3CD-AS2
PIK3IP1-AS1	PIK3R3_1	PIK3R3_2	PIM2	PIM3	PINK1-AS	PIP5KL1	PIRT
PITHD1	PITPNA-AS1	PKD1L1	PKD1L2	PKD1L3	PKD1P5	PKD1P6_2	PKHD1
PKHD1L1	PKI55	PKIA-AS1	PKMP3	PKN2-AS1	PLA2G12AP1	PLAC9	PLAG1
PLBD1-AS1	PLBD2	PLCL1	PLCXD1	PLCXD2	PLEK2	PLEKHA8P1	PLEKHD1
PLEKHF1	PLEKHH2	PLEKHJ1	PLEKHM1P	PLEKHM2	PLGLA	PLGRKT	PLS3-AS1
PLXDC1	PMFBP1	PMS2CL	PNCK	PNISR	PNMA3	PNMA6A	PNMAL1
PODN	POLR2J2_1	POM121B	POM121L9P	POMZP3	POPDC3	POSTN	POT1-AS1
PP14571	PP7080	PPAPDC3	PPDPF	PPIAP22	PPIAP29	PPIEL	PPM1M
PPP1R14B	PPP1R14C	PPP1R17	PPP1R26-AS1	PPP1R35	PPP1R36	PPP1R3G	PPP3CB-AS1
PPP4R1L	PPP4R4	PPP5D1	PPT2-EGFL8	PQLC2L	PQLC3	PRB2	PRDM12
PRELID2	PRH1	PRICKLE2	PRIMA1	PRKAG2-AS1	PRKXP1	PRKY	PROB1
PROX1-AS1_1	PRR12	PRR34-AS1	PRR36	PRRG3	PRRT3-AS1	PRRT4	PRRX1
PRRX2	PRSS12	PRSS27	PRSS35	PRSS42	PRTG	PSD2	PSMD5-AS1
PSMD6-AS2	PSORS1C1	PSPC1P1	PSPHP1	PTCHD1-AS	PTCHD2	PTCHD4	PTER
PTGES3L	PTGES3L-AARSD1	PTGES3P2	PTMAP5	PTMS	PTPN20CP	PTPRVP	PUS10
PVRL3-AS1	PVT1	PWRN1	PWRN4	PXDC1	PXDNL	PXK	PYHIN1
PYROXD1	PYROXD2	PYY2	QSER1	R3HCC1	R3HCC1L	R3HDM4	RAB30-AS1
RAB6C-AS1	RABL2A	RAD51-AS1	RAD51L3-RFFL	RAD54B	RAMP2-AS1	RANBP17	RAP2C-AS1
RAPGEFL1	RARA-AS1	RARRES1	RASGEF1B	RASGEF1C	RASIP1	RASSF8-AS1	RASSF9
RAVER2	RBM12B	RBM14-RBM4	RBM18	RBM20	RBM43	RBMS3	RBMS3-AS3
RBPM5-AS1	RC3H1	RCAN2	RCBTB1	RCN1P2	REM2	RERG	RERGL
REXO1	RFPL1S	RFPL2	RFX3-AS1	RFX8	RFXAP	RGAG4	RGS5_1
RGS7BP	RHBDL3	RIC8B	RIIAD1	RIMBP2	RIMS2	RMDN2-AS1	RMST
RN7SK	RN7SKP203	RN7SKP255	RN7SKP71	RN7SL1	RN7SL181P	RN7SL471P	RN7SL4P
RNA5SP508	RNASE4	RNASEH1-AS1	RNASEH2B	RNASEH2B-AS1	RNF144A-AS1	RNF157	RNF175
RNF180	RNF212	RNF216P1	RNF217-AS1	RNF219-AS1	RNFT2	RNPEP	RNU4-2
RNU5B-1	ROBO4	ROM1	ROPN1B	RORA-AS1	RORB-AS1	RP1-102E24.8	RP1-111C20.4
RP1-122P22.2	RP1-138B7.6	RP1-149A16.12	RP1-151F17.1	RP1-151F17.2	RP1-152L7.5	RP1-153P14.8	RP1-156L9.1

RP1-167A14.2	RP1-167F1.2	RP1-168L15.5	RP1-179N16.6	RP1-184J9.2	RP1-193H18.3	RP1-198K11.5	RP1-209A6.1
RP1-223E5.4	RP1-228H13.5	RP1-239B22.5	RP1-241P17.4	RP1-251M9.2	RP1-261D10.2	RP1-266L20.4	RP1-267D11.6
RP1-269M15.3	RP1-270F5.3	RP1-280I10.1	RP1-292L20.3	RP1-301L19.1	RP1-34H18.1	RP1-39G22.7	RP1-40E16.12
RP1-40E16.9	RP1-41C23.1	RP1-43E13.2	RP1-66C13.4	RP1-69D17.4	RP1-78O14.1	RP1-80N2.3	RP1-8B1.4
RP1-90G24.11	RP1-90G24.6	RP11-1000B6.3	RP11-1000B6.8	RP11-1007O24.3	RP11-1016B18.1	RP11-1018N14.5	RP11-101E3.5
RP11-1020A11.2	RP11-1023L17.1	RP11-1024P17.1	RP11-1029J19.5	RP11-102K13.5	RP11-102N12.3	RP11-103J8.1	RP11-104O19.4
RP11-1055B8.3	RP11-1055B8.9	RP11-105N13.4	RP11-106M3.2	RP11-1072A3.3	RP11-1079K10.3	RP11-1080G15.1	RP11-1084E5.1
RP11-108L7.15	RP11-1094H24.4	RP11-1094M14.11	RP11-1096G20.5	RP11-109N23.4	RP11-109N23.6	RP11-10K16.1	RP11-10K17.6
RP11-100I7.3	RP11-1100L3.7	RP11-1101K5.1	RP11-1102P16.1	RP11-110G21.1	RP11-111A21.1	RP11-111F5.4	RP11-111F5.5
RP11-111M22.2	RP11-111M22.3	RP11-1129I3.1	RP11-112J3.16	RP11-1134I14.8	RP11-113C12.4	RP11-1148L6.8	RP11-114F10.2
RP11-114G11.5	RP11-114G22.1	RP11-114H21.2	RP11-114H23.1	RP11-114J13.1	RP11-114N19.3	RP11-115C21.2	RP11-115J23.1
RP11-115N4.1	RP11-1166P10.6	RP11-116N8.1	RP11-116O18.3	RP11-118B22.4	RP11-119J2.2	RP11-1198D22.1	RP11-11H9.2
RP11-11N5.1	RP11-11N7.5	RP11-120D5.1	RP11-120J1.1	RP11-120K24.3	RP11-120M18.2	RP11-1212A22.1	RP11-121G22.3
RP11-121M22.1	RP11-122C21.1	RP11-122K13.7	RP11-122L9.1	RP11-123O10.4	RP11-123O22.1	RP11-1246C19.1	RP11-124B13.1
RP11-124O11.1	RP11-125O18.1	RP11-1260E13.4	RP11-1263C18.1	RP11-126K1.6	RP11-126O1.4	RP11-1277A3.1	RP11-1277A3.2
RP11-127B20.3	RP11-127I20.7	RP11-1299A16.3	RP11-12A20.7	RP11-12G12.7	RP11-12J10.3	RP11-1319K7.1	RP11-131L23.1
RP11-132E11.2	RP11-133F8.2	RP11-134L10.1	RP11-134O21.1	RP11-136C24.3	RP11-136H19.1	RP11-136K7.2	RP11-138A9.1
RP11-138E2.1	RP11-138I1.4	RP11-138I17.1	RP11-138J23.1	RP11-138M12.1	RP11-139I7.1	RP11-139H14.5	RP11-13A1.1
RP11-13A1.3	RP11-13K12.1	RP11-13N12.1	RP11-140A10.3	RP11-141M1.3	RP11-142A5.1	RP11-143A22.1	RP11-145A3.2
RP11-146D12.2	RP11-146I2.1	RP11-146N18.1	RP11-147I3.1	RP11-147L13.11	RP11-147L13.13	RP11-147L13.15	RP11-148I19.1
RP11-148L24.1	RP11-150C16.1	RP11-151A10.3	RP11-152K4.2	RP11-152N13.5	RP11-153F1.1	RP11-153M7.3	RP11-154D6.1
RP11-154H12.3	RP11-154H23.4	RP11-156E6.1	RP11-156K13.1	RP11-156L14.1	RP11-158M2.3	RP11-158M2.4	RP11-158M2.5
RP11-159D12.5	RP11-159D12.6	RP11-159D12.8	RP11-159F24.3	RP11-159G9.5	RP11-159H10.3	RP11-15A1.2	RP11-15B17.1
RP11-15B24.5	RP11-15E1.5	RP11-15E18.1	RP11-15H20.6	RP11-15H20.7	RP11-15K19.2	RP11-15K3.1	RP11-15N24.4
RP11-160C18.2	RP11-160O5.1	RP11-161M6.2	RP11-162D16.2	RP11-162P23.2	RP11-164J13.1	RP11-164N3.2	RP11-165E7.1
RP11-166B2.8	RP11-166D19.1	RP11-166O4.6	RP11-167B3.3	RP11-167H9.4	RP11-168O16.1	RP11-169E6.1	RP11-16B13.1
RP11-16D22.2	RP11-16P20.3	RP11-16P20.4	RP11-16P6.1	RP11-172E9.2	RP11-173D3.1	RP11-173M1.4	RP11-173M1.5
RP11-174G6.1	RP11-175K6.1	RP11-175O19.4	RP11-175P13.3	RP11-176H8.1	RP11-176N18.2	RP11-177G23.1	RP11-177H2.1
RP11-179A16.1	RP11-179B2.2	RP11-17A1.3	RP11-180I4.2	RP11-180I4.4	RP11-180P8.1	RP11-181G12.2	RP11-182J1.12
RP11-182J1.3	RP11-182L21.6	RP11-182N22.9	RP11-184E9.1	RP11-185E8.2	RP11-186B7.7	RP11-187E13.1	RP11-187E13.2
RP11-187O7.3	RP11-190P13.2	RP11-191G24.1	RP11-191N8.2	RP11-192P3.4	RP11-196G18.23	RP11-19O2.2	RP11-1D12.2
RP11-1M18.1	RP11-200A1.1	RP11-201K10.3	RP11-202G18.1	RP11-206L10.8	RP11-206L10.9	RP11-209K10.2	RP11-20B7.1

RP11-20D14.6	RP11-20E24.1	RP11-20I20.2	RP11-20I23.13	RP11-20I23.8	RP11-210M15.2	RP11-212D19.4	RP11-212P7.2
RP11-213G2.3	RP11-214N9.1	RP11-215A19.2	RP11-215H22.1	RP11-216B9.6	RP11-216L13.17	RP11-216L13.19	RP11-218M22.1
RP11-219A15.1	RP11-219G17.4	RP11-21A7A.2	RP11-21B21.4	RP11-21J7.1	RP11-21L19.1	RP11-21L23.2	RP11-223I10.1
RP11-225H22.7	RP11-227G15.8	RP11-228B15.4	RP11-229P13.27	RP11-22N19.2	RP11-22P4.1	RP11-230B22.1	RP11-231C18.3
RP11-234B24.6	RP11-235E17.6	RP11-236F9.2	RP11-236L14.2	RP11-239H6.2	RP11-23D24.2	RP11-23J9.4	RP11-23P13.6
RP11-243A14.1	RP11-244H3.1	RP11-244O19.1	RP11-245D16.4	RP11-245P10.4	RP11-246A10.1	RP11-247C2.2	RP11-248J23.6
RP11-248J23.7	RP11-24I21.1	RP11-250B2.3	RP11-252A24.2	RP11-253E3.3	RP11-253I19.3	RP11-254F19.4	RP11-255M2.3
RP11-256I9.3	RP11-259K15.2	RP11-259O18.5	RP11-259O2.1	RP11-25I15.3	RP11-25K19.1	RP11-25K21.6	RP11-25L3.3
RP11-25O3.1	RP11-260M2.1	RP11-260O18.1	RP11-261N11.8	RP11-262A12.1	RP11-262A16.1	RP11-262H14.4	RP11-262H14.5
RP11-263K19.4	RP11-263K19.6	RP11-264B14.1	RP11-266A24.1	RP11-266L9.5	RP11-266O8.1	RP11-267C16.1	RP11-267M23.4
RP11-268J15.5	RP11-269C23.5	RP11-26J3.1	RP11-271K11.5	RP11-272L13.3	RP11-273B20.1	RP11-273G15.2	RP11-274B21.1
RP11-274B21.10	RP11-274B21.13	RP11-274B21.14	RP11-274H2.3	RP11-275H4.1	RP11-275I4.1	RP11-276H19.1	RP11-276M12.1
RP11-277P12.20	RP11-278H7.1	RP11-280K24.4	RP11-281A20.2	RP11-281O15.4	RP11-281P23.2	RP11-282I11.1	RP11-283G6.4
RP11-284M14.1	RP11-285F16.1	RP11-285F7.2	RP11-285G1.9	RP11-286B14.1	RP11-286E11.1	RP11-286N22.8	RP11-287D1.4
RP11-290D2.6	RP11-290F24.3	RP11-290F5.1	RP11-290H9.2	RP11-290O12.2	RP11-292E2.1	RP11-293M10.6	RP11-295K2.3
RP11-295P9.3	RP11-296O14.2	RP11-296O14.3	RP11-297D21.4	RP11-298D21.3	RP11-298I3.1	RP11-298I3.4	RP11-298J20.4
RP11-299G20.2	RP11-299J3.8	RP11-29H23.5	RP11-2B6.3	RP11-2E11.9	RP11-2G1.1	RP11-302B13.1	RP11-302M6.4
RP11-304F15.3	RP11-304F15.7	RP11-305B6.3	RP11-305F18.1	RP11-305L7.3	RP11-305L7.6	RP11-305P22.9	RP11-307B6.3
RP11-307C12.11	RP11-307P5.1	RP11-308D16.2	RP11-308N19.1	RP11-309L24.4	RP11-30J20.1	RP11-30K9.6	RP11-310E22.5
RP11-310N16.1	RP11-312B8.1	RP11-313J2.1	RP11-314B1.2	RP11-314C9.1	RP11-314N13.10	RP11-315O6.1	RP11-316N24.2
RP11-317J19.1	RP11-317N12.1	RP11-318A15.2	RP11-319G6.1	RP11-320H14.1	RP11-321A17.6	RP11-321N4.5	RP11-323I15.5
RP11-324C10.1	RP11-324I22.4	RP11-325F22.2	RP11-326A19.4	RP11-326E22.1	RP11-326I19.2	RP11-326K13.4	RP11-327P2.5
RP11-32B5.1	RP11-32B5.2	RP11-32B5.7	RP11-332H14.2	RP11-332H18.5	RP11-332J15.1	RP11-338N10.1	RP11-338N10.2
RP11-33B1.1	RP11-33N16.3	RP11-340I6.8	RP11-340L3.1	RP11-341D18.4	RP11-341G23.4	RP11-343H5.4	RP11-344B5.2
RP11-345F18.1	RP11-345J4.5	RP11-345P4.9	RP11-346C4.3	RP11-347C12.1	RP11-347E10.1	RP11-348F1.2	RP11-348F1.3
RP11-348J24.2	RP11-348P10.2	RP11-34P13.15	RP11-34P13.16	RP11-350N15.6	RP11-351A11.1	RP11-351D16.3	RP11-351J23.1
RP11-351M16.3	RP11-351M8.2	RP11-352D13.6	RP11-352M15.1	RP11-352M15.2	RP11-353N4.5	RP11-355B11.2	RP11-355F16.1
RP11-356I2.4	RP11-356J5.12	RP11-356K23.1	RP11-357D18.1	RP11-357K6.1	RP11-359D14.3	RP11-359P5.1	RP11-35G9.5
RP11-35N6.1	RP11-360L9.4	RP11-360P21.2	RP11-362F19.1	RP11-362K2.2	RP11-363E7.4	RP11-363J17.1	RP11-363J20.2
RP11-364L4.3	RP11-366L20.2	RP11-368L12.1	RP11-369K17.1	RP11-370I10.6	RP11-371E8.4	RP11-371F15.3	RP11-373N22.3

RP11-374M1.4	RP11-375N15.2	RP11-377D9.3	RP11-378A13.1	RP11-379B18.5	RP11-379B18.6	RP11-379B8.1	RP11-379H18.1
RP11-379L18.1	RP11-37B2.1	RP11-37N22.1	RP11-380B4.3	RP11-380N8.7	RP11-380P13.1	RP11-381K20.2	RP11-382A20.5
RP11-384F7.2	RP11-384G23.1	RP11-384K6.2	RP11-384K6.6	RP11-384P7.7	RP11-387H17.4	RP11-389C8.2	RP11-389O22.1
RP11-38H17.1	RP11-38L15.3	RP11-390E23.6	RP11-390F4.6	RP11-392A14.8	RP11-392O1.4	RP11-392P7.6	RP11-394B2.1
RP11-394O2.3	RP11-394O4.5	RP11-395B7.4	RP11-395I14.2	RP11-395I6.3	RP11-396B14.2	RP11-396O20.1	RP11-397O8.7
RP11-398C13.6	RP11-398E10.1	RP11-398G24.2	RP11-398K22.12	RP11-399E6.1	RP11-399J13.2	RP11-399K21.11	RP11-399O19.9
RP11-400F19.6	RP11-400K9.4	RP11-400L8.2	RP11-400N13.1	RP11-401P9.4	RP11-403A3.3	RP11-403B2.7	RP11-403I13.5
RP11-403I13.8	RP11-403P17.5	RP11-404I7.2	RP11-404P21.8	RP11-405A12.2	RP11-406A9.2	RP11-407B7.1	RP11-407P15.2
RP11-408P14.1	RP11-410D17.2	RP11-410E4.1	RP11-410L14.2	RP11-411B6.6	RP11-415A20.1	RP11-416N2.4	RP11-417L19.4
RP11-418I22.2	RP11-418J17.1	RP11-418J17.3	RP11-419I17.1	RP11-420A23.1	RP11-420L9.5	RP11-420N3.3	RP11-421L21.3
RP11-422P24.10	RP11-422P24.11	RP11-422P24.12	RP11-426C22.5	RP11-427J23.1	RP11-427M20.1	RP11-428O18.6	RP11-429J17.7
RP11-429O1.1	RP11-42I10.1	RP11-430B1.2	RP11-430C1.1	RP11-430C7.4	RP11-430H10.4	RP11-432J22.2	RP11-434C1.4
RP11-434D11.4	RP11-434D9.1	RP11-436D23.1	RP11-436K8.1	RP11-438J1.1	RP11-440L14.4	RP11-442N1.1	RP11-443P15.2
RP11-444A22.1	RP11-444D3.1	RP11-445F12.1	RP11-445F6.2	RP11-445N18.5	RP11-446H18.5	RP11-446J8.1	RP11-447D11.3
RP11-449J1.1	RP11-449P15.2	RP11-44F14.6	RP11-452D12.1	RP11-452K12.4	RP11-454P21.1	RP11-455F5.3	RP11-455G16.1
RP11-455O6.8	RP11-455P21.3	RP11-457M11.5	RP11-458D21.1	RP11-458F8.1	RP11-458F8.4	RP11-458J1.1	RP11-45A16.4
RP11-460B17.3	RP11-460N11.2	RP11-461A8.4	RP11-463J10.3	RP11-465B22.3	RP11-465L10.10	RP11-466C23.5	RP11-466L17.1
RP11-467H10.2	RP11-467J12.4	RP11-467L24.1	RP11-46C24.7	RP11-46J23.1	RP11-470C13.4	RP11-472N13.3	RP11-473M20.9
RP11-474C8.7	RP11-475E11.9	RP11-475O6.1	RP11-478C19.2	RP11-479G22.8	RP11-479O9.4	RP11-47P18.1	RP11-480C22.1
RP11-480D4.1	RP11-480I12.5	RP11-481J2.4	RP11-482M8.3	RP11-483P21.2	RP11-485F13.1	RP11-486A14.2	RP11-486G15.2
RP11-486O13.4	RP11-488L18.4	RP11-489O18.1	RP11-48B3.5	RP11-490G2.2	RP11-490M8.1	RP11-491F9.1	RP11-491F9.5
RP11-496B10.3	RP11-496D24.2	RP11-496H1.2	RP11-496I9.1	RP11-496N12.9	RP11-497E19.1	RP11-497H16.5	RP11-498P14.5
RP11-499F3.2	RP11-499P20.2	RP11-49K24.4	RP11-4G2.1	RP11-4L24.4	RP11-504A18.1	RP11-504P24.3	RP11-505E24.2
RP11-505K9.1	RP11-505P4.7	RP11-506H21.5	RP11-507B12.1	RP11-507B12.2	RP11-507K2.2	RP11-507K2.3	RP11-508N22.12
RP11-508N22.9	RP11-50D16.4	RP11-511B23.2	RP11-513H8.1	RP11-514O12.4	RP11-517B11.7	RP11-517C16.2	RP11-517I3.2
RP11-519G16.2	RP11-51F16.1	RP11-51J9.5	RP11-521B24.3	RP11-521O16.1	RP11-521O16.2	RP11-522B15.3	RP11-522I20.3
RP11-524D16__A.3	RP11-524N5.1	RP11-525K10.3	RP11-526D8.11	RP11-526I2.5	RP11-527F13.1	RP11-527H14.4	RP11-527H14.6
RP11-529E10.6	RP11-529H2.2	RP11-530C5.1	RP11-531A24.3	RP11-531H8.1	RP11-532F6.4	RP11-532N4.2	RP11-535M15.1
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RP11-541P9.3	RP11-543D5.1	RP11-545G3.1	RP11-545I5.3	RP11-545J16.1	RP11-546B8.6	RP11-546K22.1	RP11-549B18.1

RP11-54A4.2	RP11-54A9.1	RP11-54O7.16	RP11-54O7.3	RP11-550I24.2	RP11-551L14.4	RP11-552D4.1	RP11-553L6.5
RP11-554A11.7	RP11-554A11.9	RP11-554D14.2	RP11-554D15.1	RP11-554D15.3	RP11-554F20.1	RP11-556E13.1	RP11-556K13.1
RP11-557H15.4	RP11-561C5.4	RP11-563J2.2	RP11-563J2.3	RP11-563K23.1	RP11-563N12.2	RP11-563P16.1	RP11-566E18.1
RP11-566H8.3	RP11-566K19.6	RP11-567J20.1	RP11-567M16.1	RP11-568K15.1	RP11-569G13.3	RP11-572M11.4	RP11-573G6.4
RP11-573J24.1	RP11-574K11.32	RP11-575F12.3	RP11-576D8.4	RP11-576N17.5	RP11-57A19.2	RP11-57H12.3	RP11-57H14.3
RP11-57H14.4	RP11-580I1.2	RP11-580I16.2	RP11-582E3.4	RP11-582J16.4	RP11-583F2.1	RP11-584P21.2	RP11-586K2.1
RP11-588K22.2	RP11-589F5.4	RP11-58A18.1	RP11-58C22.1	RP11-594C13.1	RP11-596D21.1	RP11-597D13.7	RP11-597D13.9
RP11-599J14.2	RP11-5C23.1	RP11-5P18.10	RP11-5P4.1	RP11-600F24.7	RP11-605B16.1	RP11-608O21.1	RP11-609N14.4
RP11-60A8.1	RP11-612B6.2	RP11-613M10.6	RP11-615I2.2	RP11-616M22.12	RP11-617F23.1	RP11-618P17.4	RP11-61A14.4
RP11-61I13.3	RP11-61J19.5	RP11-61K12.2	RP11-61K9.3	RP11-61L23.2	RP11-620J15.3	RP11-622C24.2	RP11-624C23.1
RP11-624L4.1	RP11-624M8.1	RP11-627G23.1	RP11-62H20.1	RP11-630C16.2	RP11-632K20.2	RP11-632K20.7	RP11-634B7.4
RP11-634H22.1	RP11-637O19.3	RP11-638L3.1	RP11-63A11.1	RP11-63E9.1	RP11-644C3.1	RP11-644F5.11	RP11-648K4.2
RP11-649A16.1	RP11-649A18.12	RP11-649G15.2	RP11-64J4.2	RP11-650L12.2	RP11-651L5.3	RP11-653J6.1	RP11-654A16.3
RP11-656D10.3	RP11-656G20.1	RP11-657O9.1	RP11-658F2.8	RP11-65B7.2	RP11-65D24.2	RP11-65F13.3	RP11-65J21.3
RP11-65M17.3	RP11-660L16.2	RP11-661A12.5	RP11-662G23.1	RP11-664H17.1	RP11-665G4.1	RP11-669M16.1	RP11-66B24.4
RP11-66B24.5	RP11-66B24.8	RP11-66N11.8	RP11-66N24.3	RP11-671P2.1	RP11-672F9.1	RP11-672L10.3	RP11-672L10.6
RP11-673E1.1	RP11-673E1.4	RP11-678G14.2	RP11-679C8.2	RP11-680G24.4	RP11-680G24.6	RP11-686D22.5	RP11-686D22.7
RP11-686D22.8	RP11-686O6.2	RP11-689K5.3	RP11-68I18.10	RP11-690J15.1	RP11-692D12.1	RP11-693N9.2	RP11-696N14.1
RP11-699L21.1	RP11-69E11.4	RP11-69H14.6	RP11-69I8.3	RP11-6C14.1	RP11-6I2.3	RP11-6N13.1	RP11-6N17.4
RP11-6O2.3	RP11-705C15.2	RP11-705C15.3	RP11-706F1.2	RP11-706O15.1	RP11-706O15.3	RP11-706O15.5	RP11-706O15.7
RP11-707A18.1	RP11-707M3.3	RP11-707P17.1	RP11-70P17.1	RP11-710C12.1	RP11-711K1.8	RP11-712B9.2	RP11-713P17.5
RP11-714G18.1	RP11-71E19.2	RP11-71L14.4	RP11-71N10.1	RP11-722G7.1	RP11-723O4.2	RP11-723O4.9	RP11-726G1.1
RP11-727A23.10	RP11-727A23.5	RP11-728K20.1	RP11-729M20.1	RP11-72M17.1	RP11-733O18.1	RP11-734I18.1	RP11-734K2.4
RP11-735G4.1	RP11-736K20.5	RP11-736K20.6	RP11-739G5.1	RP11-739L10.1	RP11-73O6.3	RP11-742B18.1	RP11-745L13.2
RP11-74E22.8	RP11-74J13.8	RP11-74M11.2	RP11-750B16.1	RP11-750H9.5	RP11-756G20.1	RP11-756H20.1	RP11-757O6.1
RP11-757O6.6	RP11-758H9.2	RP11-759A24.3	RP11-760H22.2	RP11-761N21.1	RP11-763B22.4	RP11-766N7.3	RP11-767C1.2
RP11-768F21.1	RP11-76E17.3	RP11-76G10.1	RP11-770E5.1	RP11-770J1.3	RP11-774D14.1	RP11-776H12.1	RP11-779P15.2
RP11-77H9.2	RP11-77K12.9	RP11-781A6.1	RP11-783K16.13	RP11-783K16.5	RP11-788M5.4	RP11-78F17.1	RP11-792A8.4
RP11-796E2.4	RP11-797M17.1	RP11-798K3.2	RP11-798K3.4	RP11-798M19.3	RP11-799D4.4	RP11-79P5.9	RP11-7F18.2
RP11-800A3.4	RP11-802E16.3	RP11-803B1.8	RP11-804N13.1	RP11-807H22.7	RP11-80I15.4	RP11-813I20.2	RP11-815M8.1

RP11-817O13.8	RP11-817O13.9	RP11-818F20.5	RP11-819C21.1	RP11-81H14.2	RP11-81K2.1	RP11-830F9.6	RP11-831H9.11
RP11-834C11.4	RP11-83A24.2	RP11-83N9.6	RP11-844P9.2	RP11-848G14.5	RP11-848P1.9	RP11-849F2.5	RP11-849H4.2
RP11-850A17.1	RP11-855O10.2	RP11-861A13.3	RP11-861E21.1	RP11-861E21.2	RP11-862G15.1	RP11-863K10.4	RP11-863P13.4
RP11-867G2.8	RP11-867G23.3	RP11-867G23.8	RP11-872D17.8	RP11-872J21.3	RP11-873E20.1	RP11-875O11.1	RP11-876N24.2
RP11-876N24.3	RP11-876N24.4	RP11-879F14.1	RP11-87C12.5	RP11-87G24.3	RP11-881M11.4	RP11-884K10.7	RP11-885B4.1
RP11-886D15.1	RP11-88E10.4	RP11-893F2.14	RP11-894J14.5	RP11-89K10.1	RP11-89K11.1	RP11-89K21.1	RP11-89N17.1
RP11-900F13.3	RP11-902B17.1	RP11-909N17.2	RP11-90C4.1	RP11-91G21.1	RP11-91H12.4	RP11-91J19.2	RP11-91P24.7
RP11-923I11.8	RP11-927P21.6	RP11-932O9.7	RP11-933H2.4	RP11-93H24.3	RP11-93I21.3	RP11-945C19.4	RP11-95D17.1
RP11-95H3.1	RP11-95O2.1	RP11-963H4.3	RP11-968O1.5	RP11-96C21.2	RP11-96C23.14	RP11-96H19.1	RP11-977G19.5
RP11-97O12.7	RP11-981G7.1	RP11-981P6.1	RP11-983G14.1	RP11-983P16.4	RP11-98D18.17	RP11-98D18.3	RP11-98D18.9
RP11-98J23.2	RP11-996F15.2	RP11-999E24.3	RP11-99E15.2	RP11-99J16__A.2	RP11-99L13.1	RP11-9G1.3	RP13-1032I1.10
RP13-1032I1.11	RP13-1032I1.7	RP13-1039J1.4	RP13-104F24.2	RP13-131K19.6	RP13-188A5.1	RP13-20L14.6	RP13-349O20.2
RP13-395E19.3	RP13-492C18.2	RP13-512J5.1	RP13-516M14.1	RP13-516M14.2	RP13-52K8.1	RP13-578N3.3	RP13-726E6.1
RP13-726E6.2	RP13-735L24.1	RP13-766D20.4	RP13-895J2.7	RP13-941N14.1	RP13-977J11.2	RP3-331H24.5	RP3-337H4.9
RP3-355L5.5	RP3-380B8.4	RP3-395M20.8	RP3-399L15.3	RP3-402G11.25	RP3-402G11.26	RP3-405J10.5	RP3-414A15.10
RP3-414A15.12	RP3-414A15.2	RP3-422G23.4	RP3-428L16.1	RP3-436N22.3	RP3-449H6.1	RP3-460G2.2	RP3-461P17.10
RP3-467N11.1	RP3-475N16.1	RP3-476K8.3	RP3-508I15.14	RP3-510D11.1_1	RP3-510D11.2	RP3-510D11.4	RP3-510L9.1
RP3-522P13.3	RP4-535B20.1	RP4-535B20.4	RP4-545C24.1	RP4-549L20.3	RP4-555D20.2	RP4-555D20.4	RP4-564M11.2
RP4-565E6.1	RP4-570O12.3	RP4-605O3.4	RP4-608O15.3	RP4-612B15.3	RP4-613B23.1	RP4-614O4.12	RP4-635E18.7
RP4-639F20.1	RP4-646N3.1	RP4-647C14.2	RP4-651E10.4	RP4-655J12.4	RP4-663N10.1	RP4-669P10.20	RP4-676L2.1
RP4-683M8.2	RP4-714D9.5	RP4-717I23.3	RP4-739H11.3	RP4-756G23.5	RP4-756H11.5	RP4-758J18.10	RP4-758J18.13
RP4-758J18.2	RP4-765H13.1	RP4-769N13.6	RP4-782L23.1	RP4-798A10.7	RP4-798P15.3	RP4-816N1.6	RP5-1011O1.2
RP5-1021I20.1	RP5-1021I20.4	RP5-1031D4.2	RP5-1039K5.12	RP5-1039K5.19	RP5-1044H5.1	RP5-1061H20.4	RP5-1070A16.1
RP5-1074L1.1	RP5-1086K13.1	RP5-1092A3.4	RP5-1112D6.8	RP5-1119A7.17	RP5-1126H10.2	RP5-1136G13.2	RP5-1142A6.2
RP5-1142A6.8	RP5-1186N24.3	RP5-827C21.2	RP5-827C21.4	RP5-832C2.5	RP5-857K21.4	RP5-857K21.6	RP5-858B6.3
RP5-858L17.1	RP5-864K19.4	RP5-877J2.1	RP5-884C9.2	RP5-903G2.2	RP5-908M14.9	RP5-921G16.1	RP5-943J3.2
RP5-944M2.3	RP5-965F6.2	RP5-965G21.3	RP5-966M1.6	RP5-968J1.1	RP5-972B16.2	RP5-991G20.1	RP6-159A1.4
RP6-65G23.3	RP6-65G23.5	RP9P	RPARP-AS1	RPL23AP49	RPL23AP53	RPL23AP7	RPL23AP82
RPL34-AS1	RPL41_2	RPP14_1	RPS10-NUDT3	RPS11P6	RPS2P32	RPS2P5	RPS6KC1
RPSAP52	RPSAP9	RPUSD1	RRN3P1	RRN3P2	RRNAD1	RRS1-AS1	RSAD1
RSPH1	RSPH4A	RTBDN	RTN2	RTP4	RTTN	RUFY3	RUNDC1
RUNDC3A-AS1	SAC3D1	SAMD12	SAMD12-AS1	SAMD13	SAMD14	SAMD15	SAMD5
SAMD9	SAMD9L	SAMSN1	SAP30L-AS1	SASH3	SATB2-AS1	SAYSD1	SBDSP1
SBF2-AS1	SBK1	SBNO2	SCAMP1-AS1	SCAND2P	SCART1	SCG5	SCGB1B2P
SCGB2B2	SCML4	SCRGI	SCRT1	SCRT2	SDCBP2-AS1	SDHAP1	SDR39U1

SDR42E1	SEC14L2	SEC14L6	SEC24B-AS1	SECISBP2L	SECTM1	SEMA3D	SEMA3G
SEMA6A-AS1	SEMA6B	SEMA6C	SENP6	SEPT7P3	SEPT7P9	SEPW1	SERF2
SERHL2	SERPINA3_1	SERPINB9P1	SERPINI1	SETBP1	SETD4	SEZ6	SEZ6L
SFMBT2	SFRP5	SFSWAP	SH2D5	SH3BGR	SH3TC1	SH3TC2	SHE
SHF	SHISA2	SHISA3	SHISA7	SHISA9	SHROOM4	SIAE	SIDT1
SIX3-AS1	SIX4	SIX5	SKINTL	SLAMF8	SLC10A4	SLC14A2-AS1	SLC16A1-AS1
SLC16A12	SLC16A4	SLC16A5	SLC17A4	SLC22A10	SLC22A14	SLC22A20	SLC25A25
SLC25A25-AS1	SLC25A39	SLC25A45	SLC25A5-AS1	SLC25A53	SLC2A1-AS1	SLC35E2B	SLC35F3
SLC35F4	SLC35G5	SLC38A11	SLC38A6	SLC45A1	SLC46A3	SLC6A16	SLC9A7P1
SLCO5A1	SLFN11	SLFN12L	SLFN13	SLFN5	SLFNL1-AS1	SMC5-AS1	SMCR2
SMG1P1	SMG1P2	SMG1P3	SMG1P5	SMG1P6	SMIM10	SMIM10L1	SMIM10L2A
SMIM10L2B	SMIM15	SMIM18	SMIM19	SMIM7	SMIM8	SMKR1	SMOC2
SMPDL3A	SMPX	SMYD5	SNCG	SNED1	SNHG1	SNHG10	SNHG11
SNHG14	SNHG15	SNHG17	SNHG18	SNHG19	SNHG20	SNHG22	SNHG23
SNHG24	SNHG4	SNHG5	SNHG6	SNHG7	SNHG8	SNHG9	SNN
SNORD13	SNRK	SNX18P13	SNX19	SNX25	SNX29P2_1	SOGA3_1	SOHLH2
SORCS1	SOSTDC1	SOWAHA	SOX11	SOX2-OT	SOX8	SOX9-AS1	SP110_1
SP140L	SP2-AS1	SPA17	SPAG1	SPAG16	SPAG17	SPAG7	SPANXA2-OT1
SPATA1	SPATA13_1	SPATA6L	SPATA7	SPATA9	SPEF2	SPG20-AS1	SPIN3
SPOCK2	SRGAP2-AS1_1	SRGAP2C	SRGAP2D	SRP54-AS1	SRPX2	SRRD	SRRM3
ST18	ST20	ST20-AS1	ST20-MTHFS	ST3GAL6-AS1	ST7-AS1	ST7L	ST8SIA6-AS1
STARD4-AS1	STARD9	STC1	STEAP1B	STEAP2	STEAP2-AS1	STK17A	STK31
STK32A	STK32B	STK33	STK38L	STOML1	STON1-GTF2A1L	STX16-NPEPL1	STX18-AS1
STXBP5L	STYX	SUCLG2-AS1	SUGCT	SUSD1	SUSD2	SUSD4	SUZ12P1
SVEP1	SVIL-AS1	SVILP1	SVOP	SWT1	SYCE1L	SYCP2L	SYNDIG1L
SYNM	SYPL2	SYT13	SYT14	Sep-01	Sep-02	Sep-03	Sep-05
Sep-06	Sep-08	Sep-09	Sep-10	TAF1A-AS1	TAGLN	TAGLN3	TANGO6
TAPT1	TAPT1-AS1	TARDBP_1	TARID	TATDN1	TBC1D12	TBC1D19	TBC1D23
TBC1D31	TBC1D32	TBCEL	TBCK	TCAF1P1	TCAIM	TCEA1P1	TCEAL2
TCEAL3	TCEAL6	TCEAL7	TCERG1L	TCTE1	TDRD10	TDRD3	TDRD5
TDRG1	TDRP	TEKT2	TEKT4P2	TESPA1	TEX14	TEX38	TEX41
TFCP2L1	TFP1	TGIF2-C20orf24	THA1P	THAP7-AS1	THEM7P	THSD4-AS1	THUMPD3-AS1
TIGD2	TIMM10B_2	TIMP4	TINAG	TINCR	TIPARP	TJAP1	TLCD2
TLE6	TLX1NB	TM4SF1	TM7SF3	TMC3	TMC5	TMC8	TMCC3
TMCO1	TMCO4	TMED7-TICAM2	TMEM116	TMEM117	TMEM119	TMEM123	TMEM127
TMEM132B	TMEM132C	TMEM134	TMEM135	TMEM136	TMEM141	TMEM145	TMEM147-AS1
TMEM150B	TMEM150C	TMEM151A	TMEM151B	TMEM155	TMEM156	TMEM158	TMEM161B
TMEM161B-AS1	TMEM167A	TMEM175	TMEM179	TMEM191A	TMEM196	TMEM198	TMEM200B
TMEM200C	TMEM204	TMEM209	TMEM215	TMEM220	TMEM220-AS1	TMEM232	TMEM233
TMEM236	TMEM238	TMEM240	TMEM241	TMEM245	TMEM249	TMEM253	TMEM254-AS1
TMEM256	TMEM257	TMEM26	TMEM260	TMEM263	TMEM37	TMEM39A	TMEM41B
TMEM44	TMEM51-AS1	TMEM52	TMEM53	TMEM56-RWDD3	TMEM59L	TMEM62	TMEM64
TMEM69	TMEM70	TMEM71	TMEM72	TMEM72-AS1	TMEM74B	TMEM81	TMEM8B
TMEM9B-AS1	TMLHE-AS1	TMPRSS4-AS1	TMPRSS5	TMSB15A	TMSB15B	TMSB4Y	TMTC2

TNRC6C-AS1	TOLLIP-AS1	TONSL-AS1	TOPORS-AS1	TOX	TP53I11	TP53INP2	TP73-AS1_1
TPBGL	TPRG1L	TPT1-AS1	TPTE2P5	TPTEP1	TRAF3IP2-AS1	TRAM2	TRAM2-AS1
TRANK1	TRAPPC13P1	TRAPPC2B	TRAPPC3L	TRBC2	TRGC2	TRHDE-AS1	TRIM52
TRIM52-AS1	TRIM66	TRIM67	TRIQK	TRNP1	TRPV1_1	TRPV1_2	TSC22D2
TSHZ1	TSKU	TSNAX-DISC1	TSPAN13	TSPAN32	TSPAN8	TSPY26P	TSSC2
TSSK1A	TSTD3	TTBK1	TTC14	TTC22	TTC28-AS1	TTC38	TTC39A
TTC39B	TTC39C	TTC3P1	TTC6	TTC7A	TTC7B	TTC9B	TTN-AS1
TTPAL	TTTY10	TTTY14	TTTY15	TUBA3FP	TUBB3_1	TUBB3_2	TUBBP1
TUBBP5	TUBD1	TUBE1	TUG1	TUNAR	TVP23A	TVP23C-CDRT4	TWSG1
TXLNGY	TXNDC12	TXNDC16	U47924.31	U52111.14	U91319.1	UBA6-AS1	UBALD2
UBAP1L	UBBP4	UBE2F-SCLY	UBE2Q2P1	UBE2Q2P2	UBE2Q2P6_1	UBE2QL1	UBFD1
UBL3	UBR5-AS1	UBR7	UBXN10-AS1	UEVLD	UFC1	UFL1-AS1	ULK4
UMAD1	UNC13C	UNC5B-AS1	UNCX	UQCRFS1P1	URGCP	USP2-AS1	USP3-AS1
USP40	USP46-AS1	USP50	USP9Y	VLDLR-AS1	VPS13A	VPS13C	VSIG10L
VSTM2A	VSTM2B	VSTM2L	VSTM5	VTCN1	VWA5B1	VWA7	VWDE
VWFP1	WAC-AS1	WASH6P	WASH7P_1	WDPCP	WDR11-AS1	WDR17	WDR47
WDR49	WDR63	WDR72	WDR76	WDR78	WDR83OS	WDR86	WDR89
WDSUB1	WFDC1	WFDC3	WFIKKN1	WFIKKN2	WHAMMP2	WHAMMP3	WI2-1896O14.1
WSCD1	WSCD2	WWC2-AS2	WWC3	XACT	XG	XIST	XKR4
XKR6	XKR7	XKR9	XKRX	XRRA1	XXbac-BPG154L12.5	XXbac-BPG181B23.7	XXbac-BPG181M17.5
XXbac-BPG294E21.9	XXbac-BPG299F13.17	XXbac-BPG55C20.7	XXbac-BPGBPG55C20.2	YDJC	YIPF7	YJEFN3	YPEL1
YPEL3	YTHDF3-AS1	YWHAEP7	ZBBX	ZBED3-AS1	ZBED4	ZBED9	ZBTB20
ZBTB37	ZBTB41	ZBTB7C	ZC3H12B	ZC3H12C	ZC3H12D	ZCCHC18	ZCCHC2
ZCCHC24	ZCCHC4	ZCWPW1	ZCWPW2	ZDHHC11B	ZDHHC12	ZDHHC14	ZDHHC23
ZDHHC4	ZDHHC8P1	ZEB1-AS1	ZFAND1	ZFAND3	ZFAND4	ZFAS1	ZFHX4-AS1
ZFP92	ZFPM2-AS1	ZFR2	ZFX	ZFYVE28	ZGLP1	ZHX2	ZIC4
ZIM2-AS1	ZKSCAN2	ZMAT1	ZMIZ1-AS1	ZMYM1	ZMYND15	ZNF107	ZNF117
ZNF132	ZNF146	ZNF174	ZNF185	ZNF204P	ZNF213-AS1	ZNF236	ZNF251
ZNF252P	ZNF283	ZNF286A_1	ZNF300P1	ZNF337-AS1	ZNF350-AS1	ZNF365	ZNF37BP
ZNF385B	ZNF385D	ZNF436-AS1	ZNF469	ZNF487	ZNF491	ZNF503-AS1	ZNF503-AS2
ZNF525	ZNF527	ZNF528-AS1	ZNF532	ZNF542P	ZNF559-ZNF177	ZNF561-AS1	ZNF623
ZNF630	ZNF652	ZNF653	ZNF66	ZNF667-AS1	ZNF670_1	ZNF674	ZNF674-AS1
ZNF709_2	ZNF728	ZNF747_1	ZNF780B	ZNF784	ZNF787	ZNF788	ZNF789
ZNF790-AS1	ZNF793-AS1	ZNF804A	ZNF812	ZNF814	ZNF821	ZNF83	ZNF831
ZNF876P	ZNF90	ZNF91	ZNF98	ZNRD1-AS1	ZNRF2P2	ZRANB2-AS2	ZSCAN2
ZSCAN20	ZSCAN29	ZSWIM1	ZSWIM4	ZSWIM6	bP-21201H5.1	bP-21264C1.1	bP-21264C1.2
bP-2189O9.2	gene						